PART 8

Answers and explanations

All 800 answers with explanations and source references, plus a complete answer-key grid. Every question number links back to the question.

IN THIS PART

Answers — Block A — History, risk assessment and medical considerations (Q1–28) 829
Answers — Block B — Examination, occlusion and space (Q29–48) 833
Answers — Block C — Investigations, imaging, guides and the diagnostic wax-up (Q49–70) 836
Answers — Block D — Surgical anatomy (Q71–86) 840
Answers — Block E — Osseointegration, bone biology and healing (Q87–106) 842
Answers — Block F — Materials, implant design and surfaces (Q107–132) 845
Answers — Block G — Treatment planning, SAC, options, consent and referral (Q133–160) 849
Answers — Block H — Prescribing, local anaesthesia, drug interactions and sedation (Q161–180) 853
Answers — Block I — Surgical placement, flaps, timing, loading, suturing and asepsis (Q181–218) 856
Answers — Block J — Minor hard and soft tissue augmentation (Q219–246) 862
Answers — Block K — Restoring implants, occlusion and the laboratory (Q247–280) 866
Answers — Block L — Complications, peri-implant disease, maintenance and governance (Q281–300) 871
Answers — Block M — History, risk assessment and medical considerations, extended (Q301–340) 874
Answers — Block N — Examination, occlusion and space, extended (Q341–375) 879
Answers — Block O — Investigations, imaging, guides and the wax-up, extended (Q376–410) 884
Answers — Block P — Surgical anatomy, extended (Q411–440) 889
Answers — Block Q — Osseointegration, bone biology and healing, extended (Q441–475) 894
Answers — Block R — Materials, implant design and surfaces, extended (Q476–515) 899
Answers — Block S — Treatment planning, SAC, consent and referral, extended (Q516–560) 904
Answers — Block T — Prescribing, local anaesthesia, drug interactions and sedation, extended (Q561–590) 911
Answers — Block U — Surgical placement, extended (Q591–650) 916
Answers — Block V — Minor hard and soft tissue augmentation, extended (Q651–700) 924
Answers — Block W — Restoring implants, occlusion and the laboratory, extended (Q701–760) 932
Answers — Block X — Complications, peri-implant disease, maintenance and governance, extended (Q761–800) 940
Answer key grid 946
Disclaimer 947

PART 8 — ANSWERS AND EXPLANATIONS

Each answer gives the correct option, a short explanation of the reasoning, and a source reference so you can read further. **Read the explanation for every question in a block, not only the ones you answered incorrectly** — the reasoning is often more examinable than the fact. Each question number is a link back to the question itself.

A complete answer-key grid for rapid marking follows at the end of this Part.

Answers — Block A — History, risk assessment and medical considerations (Q1–28)

1 A Residual pockets of 6 mm that still bleed are active sites, and they matter twice over: they are a reservoir of periodontal pathogens that will colonise a new implant surface, and they mark a patient whose disease is not fully resolved, which is the group in whom peri-implantitis rates are consistently highest. Treated periodontitis with a stable, well-maintained dentition carries a much smaller excess risk than treated periodontitis with residual disease. Option D is the attractive error: 40 per cent bone loss describes the *severity of past destruction*, which classifies the disease but is already accounted for and, being stable, does not predict future peri-implant breakdown. The plaque and bleeding scores at 18 and 22 per cent are acceptable rather than alarming, and grade B is average progression. Treat the residual sites to resolution before placing. (*2017 World Workshop; Lindhe; ITI Treatment Guide vol 13*)

2 B SDCEP separates a low-risk group — patients taking an oral bisphosphonate for osteoporosis without other complicating factors — from a higher-risk group, and systemic corticosteroid therapy taken alongside the antiresorptive is one of the named features that moves a patient across. Prednisolone 10 mg daily for two years impairs wound healing, suppresses the inflammatory and remodelling response, and compounds the antiresorptive effect. Option A is the one most candidates choose, and it is the right criterion applied at the wrong number: duration counts, but the threshold quoted is more than five years, and she is at three. C is reassuring rather than alarming, since uneventful healing after previous surgery is a favourable sign. The weekly regimen and her age do not alter the stratification. (*SDCEP MRONJ*)

3 A Monthly intravenous zoledronic acid for malignancy carries an MRONJ risk one to two orders of magnitude above that of oral bisphosphonates for osteoporosis, and elective implant placement — surgery that is not necessary and that creates a transmucosal wound in loaded bone — is not justified in that group. The answer is to say no, explain why, and provide a well-designed removable prosthesis with careful attention to denture trauma. Option B is the trap set by the competence rule: referral is the right answer when a case is *complex*, but this case is not complex, it is contraindicated, and passing it on does not make it safer. C and E reduce surgical magnitude but not the underlying risk, and D repeats the drug-holiday fallacy for a drug with prolonged skeletal retention. (*SDCEP MRONJ; AAOMS position paper*)

4 D Two thresholds are being tested and candidates usually remember only one. The value must be below 4.0 — his is 3.8, which just qualifies — but the reading must also be *current*: SDCEP accepts a sample taken up to 72 hours beforehand only in patients with a stable INR, and 24 hours otherwise. Four days is outside even the generous window, and at 3.8 he sits close enough to the ceiling that a small drift matters. Option A is the attractive answer because it applies the number everyone knows while ignoring the date on the result. B and C describe interruption or reversal that guidance explicitly advises against for dental surgery, and E exposes him to unnecessary risk; bleeding here is managed with local haemostatic measures. (*SDCEP anticoagulants and antiplatelets*)

5 B Implant placement with flap elevation is a higher bleeding-risk procedure in the SDCEP classification, and for a patient on a twice-daily direct oral anticoagulant the advice is to miss the morning dose, treat early in the day, and resume with the evening dose — a single missed dose, not a suspension of therapy. Option A is the answer candidates give because they have correctly learned that anticoagulants are not interrupted for dental treatment, and for lower bleeding-risk procedures it would be right; the distinction it misses is the procedure category. C is an over-long interruption with thromboembolic risk in a patient with recent venous thromboembolism, D applies a warfarin test to a drug that does not usefully alter the INR, and E is hospital-level bridging with no place here. (*SDCEP anticoagulants and antiplatelets*)

6 A Clopidogrel is a thienopyridine whose active metabolite binds the platelet P2Y12 ADP receptor covalently and irreversibly. The inhibited platelet never recovers function; haemostasis returns only as the marrow replaces the circulating pool, over roughly seven to ten days. A 24-hour omission therefore removes one dose while leaving essentially the whole platelet population inhibited — all risk of stent thrombosis, no bleeding benefit. Option C is the most attractive distractor because its *conclusion* is correct — recovery does depend on new platelets — but the mechanism belongs to aspirin, which acetylates

cyclo-oxygenase-1; candidates who have merged the two antiplatelet stories choose it. B and D confuse plasma pharmacokinetics, which are short, with the duration of effect, and E describes warfarin. (*SDCEP anticoagulants and antiplatelets; BNF*)

7 A The diagnostic threshold for diabetes is an HbA1c of 48 mmol/mol, with 42 to 47 mmol/mol defining the at-risk or pre-diabetic band. At 50 mmol/mol he is above the diagnostic line and, on this history, undiagnosed; the finding belongs to his general practitioner for confirmation and management, and that referral is part of your duty of care rather than an aside to implant planning. Option C is the near miss that catches most candidates, because 50 mmol/mol *feels* like a borderline number and pre-diabetes is the label they expect; it is two units above the cut-off, not below it. B is simply wrong on the threshold, D sets a target below the pre-diabetic band, and E defers a diagnosis. (*NICE NG28; Lindhe*)

8 A Smoking roughly doubles implant failure overall, but the effect is markedly amplified where the outcome depends on periosteal and mucosal microcirculation: nicotine-induced vasoconstriction, carbon monoxide-related hypoxia and impaired fibroblast and neutrophil function fall hardest on the augmented site, and wound dehiscence with membrane exposure is the commonest way a regenerative case is lost in a smoker. That is the risk this particular plan runs, and it is the reason to consider staging or to reconsider grafting altogether. Option B is true as far as it goes and is what candidates recall, but it is the smaller of the two effects and it is not what governs this case. There is no evidence that halving consumption abolishes the excess risk, and the effect is not site-restricted. (*Lindhe; Misch; ITI Treatment Guide vol 7*)

9 D A dose of 60 Gy delivered to the site is well above the level at which the risk of osteoradionecrosis becomes material, and the radiographic change described is consistent with radiation-induced alteration of the bone. Osseointegration in irradiated bone is possible but failure rates are higher, the consequence of a failure is far worse than usual, and the decision requires the oncology and restorative team who know the field, the dose distribution and the surveillance plan — this is outside the scope of a general dentist with a special interest. Option A is the seductive answer, because the Marx protocol is widely taught; hyperbaric oxygen is not established as effective prophylaxis for implants in irradiated bone and the evidence remains insufficient. C overstates the position, and B and E ignore the dose. (*Misch; ITI Treatment Guide vol 2*)

10 C Work it through: 7 mg/kg in a 55 kg patient allows 385 mg of articaine. A 4 per cent solution contains 40 mg/ml, so a 2.2 ml cartridge delivers 88 mg. Dividing 385 by 88 gives 4.4, and since cartridges are used whole the safe answer is four, which supplies 352 mg. Option D is the commonest error and it is instructive: six cartridges is 528 mg, comfortably over the limit, and it is chosen by candidates who half-remember that articaine is a concentrated solution but never do the division — the same candidates who would give six cartridges to a small adult in a long appointment. Small, light and elderly patients are exactly the group in whom the maximum is reached sooner than expected, and the calculation should be done before the first injection. (*BNF; SDCEP*)

11 C Propranolol is a non-selective beta blocker, so it occupies the beta-2 receptors that normally mediate adrenaline's vasodilatation in skeletal muscle as well as the beta-1 receptors of the heart. Exogenous adrenaline then acts almost purely on alpha-1 receptors: systemic vascular resistance rises, blood pressure climbs, and the baroreceptor reflex slows a heart that cannot mount a compensatory tachycardia because beta-1 is blocked. The combination of hypertension with bradycardia is the signature. Option E is the distractor that catches candidates, because anxiety genuinely does release catecholamines and the picture would look similar in origin — but an endogenous surge in an unblocked patient produces tachycardia, not a pulse of 52. The practical lesson is to limit the adrenaline dose and aspirate carefully in patients on non-selective beta blockers. (*BNF; SDCEP*)

12 D An implant behaves as an ankylosed tooth: the alveolus continues to develop around it and the restoration is progressively left in infraocclusion, a change that can appear years later and is difficult to correct. Cessation of growth must therefore be demonstrated rather than assumed, and the two practical methods are radiographic — superimposition of cephalograms separated by at least six to twelve months showing no vertical change — and anthropometric, a plateau in standing height. Option E fails on item C alone, and that item is the one candidates accept without argument: chronological age is a poor proxy, since growth continues into the late teens in most girls and well into the twenties in some men. Delay, with a well-made provisional, costs far less than infraocclusion. (*Misch; ITI Treatment Guide vol 1*)

13 E Each of the other four names a real consideration, which is what makes this discriminating, but none of them is the *reason*. The governing principle is that elective care is deferred in pregnancy because the benefit can be obtained just as well afterwards, so no exposure of any kind — radiographic, pharmacological or physiological — needs to be justified at all. Option A is the answer most candidates give and it is the weakest: the fetal dose from a small-field dental CBCT with the beam directed away from the pelvis is extremely small, and if a radiograph were genuinely necessary it would be taken. Supine hypotension

is managed by positioning, amide anaesthetics are used routinely, and paracetamol is available. She is comfortable in a denture; there is nothing to gain and something to explain. (*SDCEP; Misch*)

14 C The pattern — preoccupation with a defect others cannot see, meticulous self-measurement, serial consultation and repeated complaint — is characteristic of body dysmorphic disorder, and the essential point is that surgery does not treat it: satisfaction is not achieved, the complaint transfers to the new clinician, and the patient is harmed. Declining kindly, explaining that the tissues are healthy, and offering a route to psychological assessment is the answer that protects him. Option E is the sophisticated wrong answer and deserves comment, because it looks like defensive good practice; a disclaimer does not create valid consent, cannot bind a dissatisfied patient, and treats a medico-legal symptom rather than the clinical problem. A and B are irreversible interventions for a non-existent defect, and D delegates the same error. (*GDC Standards; SAC Assessment Tool*)

15 B *Montgomery* replaced the professional standard with a patient-centred one: the clinician must disclose material risks, material meaning those to which a reasonable person in the patient's position would attach significance, together with any risk this particular patient would obviously regard as significant, and the reasonable alternatives — including doing nothing. Her occupation is the reason the singer is in the stem: altered lower lip sensation or a change in speech would matter to her more than to most, and that specific knowledge places a duty on you. Option A is the most attractive distractor because it was the law before *Montgomery* and is still how many clinicians think; the *Bolam* professional-standard test no longer governs consent. C sets an arbitrary numerical filter, D reverses the duty, and E is not a consent framework. (*Montgomery; GDC Standards*)

16 D Capacity is presumed, and it is decision-specific and time-specific; a diagnosis of dementia does not remove it. The correct first step is the functional assessment — can she understand the information, retain it long enough to use it, weigh it, and communicate her decision? Retention need only last long enough to make the choice, so forgetting later in the appointment does not by itself demonstrate incapacity. Option B is the trap, because a best-interests decision is exactly what you may end up making — but only *after* capacity has been shown to be lacking, and taking that route first bypasses her autonomy. A is legally wrong in England and Wales, where no adult can consent for another without a registered authority; C misunderstands who assesses capacity; E defers care that may be entirely within her competence to choose. (*Mental Capacity Act; GDC Standards*)

17 C Both surviving statements rest on the same pharmacology. Serotonin has a role in bone metabolism, and retrospective cohorts have reported a modest increase in implant failure among SSRI users; the association is worth knowing and worth mentioning, but it is observational and confounded by depression itself, by smoking and by oral hygiene. Platelets do not synthesise serotonin — they take it up — so an SSRI depletes the platelet granule store and can produce a mild prolongation of bleeding, occasionally relevant when other haemostatic drugs are also in play. Statement (ii) is the one that makes E and D wrong: a psychotropic maintaining stable mental health is not stopped for an elective dental procedure, and suggesting it exceeds your remit. Proceed, and record the discussion. (*BNF; Lindhe*)

18 B Two separate issues arise and only one of them is bleeding. Elective surgery is conventionally postponed in the months following a myocardial infarction and coronary stenting, when the risk of a further cardiac event is highest, and the timing should be agreed with the cardiologist rather than assumed. Option C is the strong distractor and the one that reveals half-knowledge: its premise is correct, since dual antiplatelet therapy must not be interrupted and bleeding after implant placement is controlled locally, but it answers the haemostatic question while ignoring the cardiac one. A carries a real risk of stent thrombosis, a catastrophic event, and is not a decision for a dentist to take. D invents an indication — stents are not covered with antibiotics — and E defers longer than the cardiac risk requires. (*SDCEP anticoagulants and antiplatelets; NICE CG64*)

19 A NICE removed routine antibiotic prophylaxis for dental procedures, and a prosthetic heart valve does not reinstate it; what remains is that patients at increased risk — prosthetic valves, previous endocarditis, certain congenital lesions — form a special-consideration group whose circumstances and wishes are discussed with them and, where appropriate, with their cardiologist. Item B is what makes D wrong and it is the attractive one, because a mouthrinse feels like a harmless compromise for a nervous clinician; NICE specifically advises that chlorhexidine mouthwash should not be offered as prophylaxis. Item C is the regimen every candidate memorised before the guidance changed and is no longer recommended, so E fails on two counts. Her warfarin, separately, is managed as for any higher bleeding-risk procedure. (*NICE CG64; SDCEP*)

20 C Routine steroid supplementation for dental treatment under local anaesthesia is no longer advised; the physiological stress of a forty-minute implant placement with effective anaesthesia is modest, and adrenal crisis in this setting is vanishingly rare. What matters is that he takes his normal dose, that

anaesthesia and post-operative analgesia are good, and that he is monitored. Option A is the answer taught for years and still widely given, which is why it is the distractor to beat; doubling the dose exposes him to the harms of extra glucocorticoid for no demonstrated benefit. B belongs to major surgery under general anaesthesia. D would risk precipitating exactly the insufficiency being feared. Note separately that his long-term steroid impairs healing and would raise MRONJ risk were he also on an antiresorptive. (*SDCEP; BNF*)

21 A Randomised evidence summarised in systematic review supports a single dose of amoxicillin, commonly 2 g, given shortly before placement: it produces a small but statistically significant reduction in early failures, with a large number needed to treat, and it is the only antibiotic intervention around straightforward implant placement with a defensible evidence base. Option B is the practice most candidates follow and therefore the distractor most select: continuing antibiotics after surgery has not been shown to add benefit, and it contributes to resistance, to gastrointestinal harm and to occasional anaphylaxis for nothing. C confuses an outcome that antibiotics do not influence, D reverses the population in which the trials were done, and E overstates a stewardship argument that is otherwise sound. Prescribe once, before, or not at all. (*Esposito et al., Cochrane review; SDCEP Drug Prescribing*)

22 B Methotrexate is eliminated largely unchanged by the kidney, through glomerular filtration and active tubular secretion. Non-steroidal anti-inflammatory drugs reduce renal blood flow through prostaglandin inhibition and compete for tubular transport, so plasma methotrexate rises; the consequences — myelosuppression, mucositis, hepatic and renal toxicity — are serious and disproportionate to the analgesia gained. Paracetamol, with codeine if needed, is the safe choice. Option A is the most attractive alternative because it sounds like sophisticated implant physiology and there is a real, if unsettled, literature on NSAIDs and bone healing; but that concern is speculative at analgesic doses and is not why this particular patient must avoid ibuprofen. D is true of NSAIDs generally and is a lesser hazard, and C and E describe interactions that do not occur. (*BNF; SDCEP Drug Prescribing*)

23 D Parafunction is a mechanical risk factor: it threatens screws, ceramic and components rather than osseointegration, and the loads generated during sleep bruxism are not controlled by anything you do to the occlusal surface, because the patient is not guiding his mandible in a functional pattern at the time. A hard full-coverage splint is the intervention that removes the implant restoration from the parafunctional load, and it must be made and reviewed as part of the plan rather than offered after the first fracture. Option A is the good wrong answer — narrowing the occlusal table genuinely reduces the lever arm of off-axis contacts and is worth doing — but it modifies a load it cannot abolish. C swaps one brittle interface for another, E enlarges the vertical cantilever, and B increases the number of loaded contacts. (*Misch; ITI Treatment Guide vol 12*)

24 E All three hold, and together they are the whole of what osteoporosis means for implant planning. Densitometry at the hip and spine is a poor predictor of the trabecular density a drill will meet in a posterior mandible; the site is assessed by the CBCT appearance and, above all, by the tactile feedback and insertion torque at surgery. Osseointegration in osteoporotic patients is reported at survival rates approaching those of healthy controls, so the diagnosis modifies technique — under-preparation, slower drilling, an extended healing period — rather than prohibiting treatment. And the clinically dangerous variable is not the disease but any antiresorptive prescribed for it, which she has declined. Candidates most often reject (i), because a bone density scan intuitively feels as though it should describe all bone. (*Misch; SDCEP MRONJ*)

25 C A general statement of fitness is not a clinical answer, and it does not transfer responsibility: you remain the person deciding whether to operate. Effective liaison asks named questions — his current eGFR and whether it alters drug dosing, whether he is on an antiresorptive or immunosuppressant, his bleeding tendency and any transplant plans — because a specific question is what generates a usable reply. Option A is the answer that feels safe and is the commonest real-world error; the word "clearance" is doing work it cannot do, and a court would ask what you had asked. B is a useful adjunct but patients report their own medicine imperfectly, D abandons a patient who may well be treatable in practice, and E arranges investigations you did not identify a need for. (*GDC Standards; NICE CG64*)

26 E All three are free of porcine and bovine material. Autogenous bone from the adjacent ramus or the osteotomy itself is the obvious answer; beta-tricalcium phosphate is wholly synthetic; and allograft is human in origin, which is precisely why option D is the trap — candidates exclude it on the assumption that "graft" means animal, when the objection raised is to porcine and bovine sources, not to donated human tissue. She may of course have her own view on allograft, and that is a matter for the consent conversation rather than for you to decide. Note that the barrier matters equally: most resorbable collagen membranes are porcine or bovine, so a synthetic dense PTFE membrane would be needed. Origin must be disclosed for every material used. (*GDC Standards; ITI Treatment Guide vol 7*)

27 B Withdrawal from supportive care is the strongest behavioural predictor of peri-implant disease: cohorts followed for five years show substantially higher incidence of peri-implantitis in patients not enrolled in regular maintenance, and his plaque score of 62 per cent is the visible consequence of three years without professional reinforcement. Option D is the distractor with real substance, and it should be argued rather than dismissed — residual excess cement is a well-documented cause of peri-implant inflammation, and it may yet be found here — but cement retention is a site-specific technical hazard, not the factor that best explains simultaneous, symmetrical breakdown at both implants in a patient who has stopped cleaning and stopped attending. Five cigarettes daily is a genuine but lesser contributor. (*2017 World Workshop; Lindhe; ITI Treatment Guide vol 13*)

28 E Chronological age is not a contraindication; outcomes in older patients are good and the mandibular two-implant overdenture is the option most likely to transform his daily life, which is the substance of the McGill consensus. What must be assessed instead is manual capability and its likely trajectory, since a tremor that already prevents denture control will make interdental and abutment cleaning harder, and Parkinson's disease progresses. Design should follow from that assessment: two separate attachments are cleaned by wiping around each abutment, whereas a bar creates a covered space beneath it that demands dexterity he does not have. Candidates most often reject (iii), assuming a bar is always the more retentive and therefore the better choice; retention is not the constraint here, maintenance is. (*McGill consensus; ITI Treatment Guide vol 4*)

Answers — Block B — Examination, occlusion and space (Q29–48)

29 B The controlling measurement is at the crest, not at the contact points, and the requirement is at least 1.5 mm of bone between implant and adjacent root so that the crestal remodelling around the implant does not consume the interproximal bone supporting the neighbouring papilla. A 3.3 mm implant in 6.5 mm leaves 1.6 mm on each side and satisfies it. Option A is the answer most candidates give, because 4.1 mm is the default premolar diameter and 6.5 mm sounds generous; it leaves only 1.2 mm each side and is the arithmetic error that produces interproximal bone loss and a black triangle. C creates orthodontic burden for a space that is already adequate, D discards a straightforward implant site, and E worsens the same problem as A. (*Tarnow; Misch; ITI Treatment Guide vol 1*)

30 D Two mechanisms are at work. Adjacent implants that sit closer than about 3 mm at the crest share a single zone of remodelling, and the lateral bone loss around each summates, dropping the crest that must support the papilla; Tarnow's data show mean crestal loss roughly doubling below that separation. Avoiding the inter-implant papilla altogether by cantilevering a pontic from a single implant is better still, because the soft tissue then rests on a crest supported by a natural tooth on one side. Item C is what makes E wrong and it is the intuitively appealing one: deeper placement does not build papilla, it simply moves the remodelling apically and increases crestal bone loss, deepening the sulcus rather than filling the embrasure. (*Tarnow; ITI Treatment Guide vol 1*)

31 A Tarnow's measurements between natural teeth are the reference: at 5 mm from contact point to bone crest the papilla is present in essentially every case, at 6 mm in about 56 per cent, and at 7 mm in only about 27 per cent. At 7 mm the odds are firmly against complete fill, and the honest conversation happens before treatment, not at the fit appointment. The remedy is prosthetic — lengthening and broadening the contact area, moving it apically, and accepting a slightly longer clinical crown. Option D is the most attractive distractor because implant depth is genuinely under your control and feels like the lever to pull; papilla height, however, is dictated by the bone level on the adjacent teeth, which no amount of implant repositioning changes. (*Tarnow; ITI Treatment Guide vol 1*)

32 C Vertical restorative space is measured from the mucosa to the occlusal plane and consumed from the bottom up: attachment, denture base, and the teeth themselves. A bar assembly with its housing needs roughly 12 to 15 mm before an acceptable base thickness remains, whereas low-profile stud attachments will work in the region of 8 to 10 mm. At 9 mm a bar leaves an unacceptably thin acrylic base over the housing, and the denture fractures across the midline within a year or two. Option A is the design many candidates default to, on the reasonable belief that splinting distributes load better; here the space measurement excludes it. D is not supported on two implants in this position, and E gives poor retention and wears. (*McGill consensus; Misch; ITI Treatment Guide vol 4*)

33 E The freeway space is the difference between rest and occluded dimensions: 68 minus 62 gives 6 mm, well above the usual 2 to 4 mm and consistent with a dentition that has worn and over-closed. That surplus is the reservoir from which restorative space can be borrowed, and reorganising the occlusion upwards by two or three millimetres will gain height at every site at once. The change must nonetheless be tested — a removable appliance or provisional restorations worn for several weeks — because the freeway space is a physiological range, not a licence. Option D is the distractor with surgical appeal and it is exactly

backwards: removing crestal bone gains space by shortening the ridge you are about to put an implant into. A worsens the over-closure. (*Misch; prosthodontic texts*)

34 A The radiographic detail is the whole question. Over-eruption here is dento-alveolar rather than simple tooth movement: the bone and gingival margin have migrated with the tooth, so grinding the occlusal surface can only remove the crown height, leaving an unaltered gingival level, a short clinical crown and, at 3 mm, a real risk of pulp exposure. Intrusion moves the entire dento-alveolar complex back and corrects the gingival level as well as the occlusal one. Option B is the answer chosen most often because it is quick and free — a millimetre of enameloplasty is reasonable adjunctively, but three millimetres of it is not. C sacrifices a vital unrestored tooth, D restores in inadequate space, and E removes a sound tooth. (*Misch; ITI Treatment Guide vol 6*)

35 E All three belong to implant-protective occlusion and each has a mechanical rationale. Light centric contact, verified so that a thin foil is held only in firm closure, allows the natural teeth to intrude under load and share it, so the implant is not the first and hardest contact in every cycle. Excursive contact on a posterior implant crown applies precisely the off-axis force the joint tolerates least, and canine guidance is retained instead. Narrowing the occlusal table brings contacts closer to the implant axis and shortens the lever arm of any that stray from it. Candidates most often reject (i), thinking it produces over-eruption of the opposing tooth; the difference intended is a few tens of micrometres, which is a load-sharing measure, not an infraocclusion. (*Misch; ITI Treatment Guide vol 12*)

36 B A natural tooth is suspended in a periodontal ligament and depresses vertically by something of the order of 25 to 100 micrometres under functional load; an osseointegrated implant, having no ligament, moves only about 3 to 5 micrometres, and that movement is deformation of bone rather than displacement in a socket. Contacts that are equal at rest therefore become unequal the moment force is applied, with the implant taking a disproportionate share, which is why the crown is left fractionally clear in light closure and comes into contact only when the neighbouring teeth have been depressed. Option A is the seductive half-truth: proprioception is reduced, and the raised tactile threshold matters, but the reason for this specific adjustment is the difference in axial mobility, not in sensation. (*Misch; Lindhe*)

37 C The question a fixed prosthesis must answer in an edentulous maxilla is whether the lip can be supported without a flange, and the only reliable way to find out is to take the flange away and let the patient see and feel the result. A duplicate denture trimmed to the level a fixed prosthesis would occupy tests the appearance, the lip profile and the speech in one appointment, before anything irreversible has happened. Option B is the answer that sounds most thorough and is genuinely part of planning, but a CBCT tells you where the bone is, not whether the resulting prosthesis will look acceptable; ridge resorption in this pattern often means implants can be placed and the aesthetic result still be unsatisfactory. Loss of lip support is a leading cause of dissatisfaction after conversion to fixed work. (*ITI Treatment Guide vol 4; Misch*)

38 D Papilla height at an implant site is governed by the interproximal bone on the neighbouring teeth, and that bone is the one thing the implant cannot restore: 4 mm of loss on the mesial of the UL2 guarantees an open embrasure on that side whatever is done surgically or prosthetically, and it is the finding that should shape both the consent discussion and the crown design. Option B is the strong distractor and is a genuine high-risk feature in every aesthetic risk assessment — thin tissue shows the abutment, recedes more readily and heals less forgivingly — but it modifies the quality of a result that adjacent bone loss has already limited. The high smile line raises the stakes without itself creating the defect, rectangular crowns are the favourable form, and three weeks is a normal early placement interval. (*ITI Treatment Guide vol 1; Tarnow*)

39 A Patient report is unreliable, since sleep bruxism is unwitnessed and daytime clenching is habitual and unnoticed, so the diagnosis rests on signs: hypertrophic masseters that firm visibly on clenching, a linea alba, tongue scalloping, and wear facets that oppose each other in excursion and are not explained by diet. Recession is the item that makes D wrong, and it is the one candidates accept: recession is multifactorial — thin phenotype, brushing technique, tooth position and previous periodontal treatment all produce it — and it has no established relationship with parafunction. Cervical lesions carry the same problem in stronger form, since the abfraction hypothesis that links them to occlusal loading remains contested and erosion explains most of them. Confirming parafunction changes the design; assuming it from soft-tissue signs does not. (*Misch; ITI Treatment Guide vol 12*)

40 E Resorption of the posterior mandible proceeds downwards and lingually, and the honest response is to put the implant in the bone that exists and design the occlusion around it — a crossbite relationship, with the upper palatal cusp occluding into the central fossa of the implant crown, is functional, cleansable and stable, and it keeps the load along the implant axis. Option B is the most attractive alternative because it preserves the appearance of a normal relationship; a buccal cantilever converts every occlusal contact

into an off-axis bending moment on the abutment screw, which is exactly the loading pattern implant components tolerate least. A courts a buccal perforation and off-axis loading together, C proposes a block graft that this case does not need, and D mutilates a sound tooth. (*Misch; ITI Treatment Guide vol 12*)

41 D A baseline does two jobs that nothing else can. It establishes whether the alteration is new at all — a proportion of patients have pre-existing deficits they have never noticed — and, if it is new, it gives the reference against which recovery is measured over the following weeks, which is what determines whether you observe, remove the implant early, or refer. Mapping with light touch, sharp discrimination and a marked diagram takes two minutes. Option A is the answer most candidates choose, and it is a genuine and necessary protection, but it is protection of a different kind: it shows the risk was disclosed, and says nothing about whether this deficit was caused by the surgery. B and C are useful supporting evidence of technique, and E records the soft tissue only. (*GDC Standards; FGDP guidance*)

42 C A BPE of 4 means probing depths of 6 mm or more, which is a diagnosis of periodontitis until proved otherwise, and it obliges detailed charting of the affected sextants, plaque and bleeding scores, radiographs where indicated, non-surgical therapy and re-evaluation. Placing an implant into an untreated dentition puts a susceptible surface into a mouth full of pathogens and a patient whose behaviour has not yet changed; the periodontal response also tells you, better than anything else, whether he will maintain an implant. Option B is the near miss and the sequencing error: radiographs are part of the assessment, but they are prescribed on the basis of the charting rather than before it, and they will not alter the immediate need for therapy. The holiday is not a clinical consideration. (*BSP implementation of the 2017 classification; Lindhe*)

43 B Access is a hard constraint and it is measured, not estimated: 32 mm of opening cannot accommodate an assembly needing 38 mm, and no change of technique creates the six missing millimetres. The LL6, needing 27 mm, is straightforward, and a shortened arch to the first molar is a well-established functional endpoint that most patients tolerate without complaint. Option A is the trap, and it inverts the mechanics: a surgical guide adds the height of the sleeve and the guided drill to the assembly, so it makes the access problem worse rather than better. C saves no vertical space at all, since the constraint is the handpiece rather than the flap. D is the commonest wrong instinct — general anaesthesia relaxes muscle but does not change a fibrotic opening — and E treats a stable old injury. (*SAC Assessment Tool; Misch*)

44 A The concavity means the long axis of the available bone is not the long axis of the ridge crest, and the drill must be angled buccally, sometimes markedly, with the depth limited so that it stops above the level at which the fossa begins — a shorter implant in the correct axis is far safer than a longer one in the wrong one. Item B is what makes D wrong and it is the item that reads as harmless diligence: a panoramic radiograph is a two-dimensional projection and cannot show buccolingual morphology at all, so it adds nothing to a CBCT that has already shown the undercut. Item C is worse than useless here, since raising a flap and palpating the lingual plate is one of the few direct safeguards available. Perforation risks the sublingual space and a bleed that threatens the airway. (*Lindhe; ITI Treatment Guide vol 11*)

45 B An implant will not integrate against the soft tissue contents of the nasopalatine canal, so the site must be found labial to it; with 3 mm of bone available that is feasible, but the resulting position leaves a thin labial plate and simultaneous contour augmentation is prudent, both for stability of the buccal profile and because the plate is only 1 mm thick to begin with. Option A is the sophisticated distractor and deserves respect: enucleation of the canal with simultaneous grafting is a described technique, but it carries a risk of altered palatal sensation, demands a delayed placement in most hands, and belongs to a specialist rather than to a general dentist with an interest. C places the implant into the canal by another route, D discards a treatable site, and E is mechanically implausible. (*ITI Treatment Guide vol 10; Misch*)

46 D The conventional guideline limits the distal cantilever to about one and a half times the anteroposterior spread in favourable circumstances, so 10 mm of spread supports roughly 15 mm of cantilever — and the reason is geometric rather than arbitrary: the spread determines the length of the lever arm resisting the cantilever, so a wide anteroposterior distribution buys cantilever length while implants placed in a straight line buy almost none. Option E is the attractive error, chosen by candidates who remember that a cantilever is permissible without remembering what governs its length; 20 mm doubles the spread and loads the terminal implants heavily. Note also that the guideline assumes favourable conditions, and an opposing complete denture is favourable, whereas an opposing fixed prosthesis or parafunction would argue for less. (*Misch*)

47 E A facebow and a semi-adjustable articulator earn their cost when the occlusion is being *reorganised* — when the vertical dimension is changed, the intercuspal position is being moved to a retruded contact position, or the anterior guidance is being redesigned — because in those cases the arc of closure and the condylar guidance determine where the new contacts will fall, and a hinge-articulated cast will predict them wrongly. Option C is the plausible alternative and is where many candidates draw the line, assuming

that multiple units justify full mounting; two premolar crowns conformed to an existing stable occlusion can be made accurately to the patient's own guidance. A, B and D are all conformative single units. Mount when you are changing the occlusion, not merely when you are restoring within it. (*Misch; prosthodontic texts*)

48 C The association between a narrow band of keratinised mucosa and higher plaque scores, more bleeding and greater discomfort on brushing is consistently reported, and in this patient the mechanism is visible: it hurts, so he does not clean it, so it bleeds. That is the argument for augmenting the band with a free gingival or connective tissue graft where hygiene is being compromised. Statement (ii) is what makes E and D wrong, and it overstates the evidence in the direction candidates expect: keratinised width has not been shown to determine implant survival, many implants function for years without it, and grafting is indicated by symptoms and hygiene rather than by a measurement alone. Treat the inflammation, then decide whether the graft is needed. (*2017 World Workshop; Lindhe*)

Answers — Block C — Investigations, imaging, guides and the diagnostic wax-up (Q49–70)

49 B The regulations separate the duties deliberately. The referrer supplies sufficient medical data to allow the exposure to be justified but does not justify it; the practitioner, entitled under the employer's procedures and adequately trained, decides that this exposure carries a net benefit for this individual; the operator carries out the practical aspects; and the employer provides the framework of procedures and referral criteria within which all three work. Your colleague reviewed the referral, decided the scan was warranted and selected the field of view — that is justification, and she was also the operator, since one person commonly holds more than one role. Option A is the answer most candidates give, because it feels as though whoever wants the scan must be the one who justifies it; supplying the indication is a referrer's duty, and a distinct one. The radiologist's report is a clinical evaluation, a further duty again. (*IR(ME)R 2017*)

50 C Voxel size sets how finely the volume is sampled. A smaller voxel resolves smaller detail, but fewer photons contribute to each one, so noise rises and the exposure must be increased to hold image quality — which is exactly why the 0.125 mm protocol runs longer. That is the cost. The benefit has to be judged against the diagnostic task, and the task here is to locate the corticated roof of the mandibular canal, a high-contrast structure a few millimetres across that is displayed perfectly adequately at 0.2 or 0.3 mm. Buying resolution that cannot change management is a failure of optimisation, which IR(ME)R requires as separately as it requires justification. Option A is the physics reversed and is the commonest error: for a fixed dose the finer voxel is the noisier image, not the quieter one. (*IR(ME)R 2017; SEDENTEXCT guidelines*)

51 D Cone beam grey values are not Hounsfield units. Multidetector CT numbers are calibrated against water on a fan-beam geometry with effective scatter rejection; a cone beam image is reconstructed from a divergent beam with a large scattering volume, and the resulting value for the same bone differs between machines, between fields of view, and between the centre and periphery of the same volume. A figure of 320 therefore carries no transferable meaning. What can be read is the pattern — cortical thickness and the coarseness and spacing of the trabeculae — and that is then confirmed by tactile feedback and insertion torque at surgery. Option B is the attractive distractor because it is a true statement about attenuation measurement, and it sounds like a sophisticated objection; it is simply not the reason the number is unusable. (*SEDENTEXCT guidelines; Misch*)

52 A This is beam hardening. The X-ray beam is polychromatic; as it passes through a dense, high atomic number restoration the softer photons are absorbed preferentially, the mean energy of the emerging beam rises, and the reconstruction algorithm — which assumes a constant attenuation coefficient — undervalues the material along that path. Between two dense objects the effect is at its greatest, producing the classic dark band, with bright streaks at the edges. Option B is the strong distractor and deserves credit, because scatter genuinely does degrade cone beam images and contributes to streaking; but the specific sign described, a band between two dense objects, is the signature of beam hardening. Practically, the crestal bone level at that site cannot be read reliably, and a metal artefact reduction algorithm, a change of patient position, or an intraoral radiograph is needed. (*SEDENTEXCT guidelines*)

53 E Doubled contours with generalised unsharpness are motion artefact, and motion corrupts the projection data themselves: no reconstruction parameter recovers the information, so option B fails at the level of the physics. The examination has not answered the question that justified it, and the correct response is to address the cause — head and chin support, a shorter scan protocol, a seated position with the back supported, an explanation rehearsed with the patient — and to justify the repeat as a fresh exposure, recording why it was needed. Option C is the thoughtful wrong answer, and it attracts candidates who correctly value dose reduction: retreating to a panoramic abandons the cross-sectional information that

prompted the referral, and an arbitrarily enlarged margin substitutes guesswork for measurement. Measuring from a double outline, as in A, invites a systematic error of a millimetre or more. (*IR(ME)R 2017; SEDENTEXCT guidelines*)

54 B The description is that of a pseudocyst: a dome-shaped accumulation of fluid beneath the lining, arising from the floor, homogeneous, without a corticated wall of its own and without bony expansion. It is a common incidental finding, it is usually asymptomatic, it fluctuates over time, and it does not by itself contraindicate sinus floor elevation, though its presence should be recorded, discussed and taken into account by whoever performs the surgery. Option A is the distractor most often chosen, because any soft-tissue opacity in an antrum reads as "sinusitis" at first glance; chronic rhinosinusitis produces circumferential mucosal thickening, symptoms and often an obstructed ostium, none of which are present here. A mucocele expands and remodels bone, odontogenic sinusitis is anchored to a diseased tooth or an oro-antral communication, and an antrolith is a discrete calcification. (*ITI Treatment Guide vol 5; SEDENTEXCT guidelines*)

55 D The site, the shape and, above all, the lingual concavity with an intact buccal cortex are diagnostic of a Stafne bone cavity — a developmental depression of the lingual surface, usually containing salivary or adipose tissue, that requires no treatment and does not affect an implant plan at the LL6. The item worth taking from this question is not the diagnosis but the duty that produced it: the whole irradiated volume must be examined and a report recorded, not merely the region of interest, and incidental findings are the price and the point of that obligation. Option C is the seductive answer because it looks cautious; a repeat examination is a further exposure that must be justified by a question it can answer, and there is no question here. Biopsy and urgent referral, A and B, would subject an asymptomatic man to investigation of a normal variant. (*SEDENTEXCT guidelines; IR(ME)R 2017*)

56 D Justification asks a single question: will this examination produce information that changes what is done, which cannot be obtained at a lower dose? At the LR6 the answer is plainly yes, because the margin above the canal decides the implant length and whether the case is done at all. At the UR2 it is also yes, because ridge width is invisible on any two-dimensional projection and a knife-edged ridge decides between placement, simultaneous augmentation and a staged approach. The UL5 has adequate height, adequate width on clinical examination and no pathology, and imaging it in three dimensions adds dose without adding a decision — which is precisely what "CBCT should not be used routinely" means. Option E is the answer chosen by candidates who treat cone beam imaging as a standard of care rather than a justified exposure, and it fails on the third site alone. (*SEDENTEXCT guidelines; FGDP/RCR Selection Criteria*)

57 B Work through it. The ball bearing gives the true vertical magnification at that point: 6.3 divided by 5 is 1.26. The apparent 13 mm therefore represents 13 divided by 1.26, or about 10.3 mm of real bone. Subtracting the 2 mm safety margin leaves 8.3 mm, so an 8 mm implant is the longest that fits, and a 10 mm implant would sit within the margin. Option D is the commonest error and the instructive one: 11 mm is 13 minus 2, the calculation of a candidate who applies the safety margin but forgets that a panoramic image is magnified by roughly a fifth to a quarter. Note that vertical magnification is reasonably constant on a panoramic while horizontal magnification is not, and that this radiograph still says nothing at all about ridge width. (*Misch; FGDP/RCR Selection Criteria*)

58 C Bone level around an implant is measured from the platform to the first bone-to-implant contact, and both landmarks are only reproducible when the central ray is perpendicular to the long axis of the implant. The visible test of that geometry is the threads: when the beam is orthogonal they are resolved crisply and symmetrically on both proximal surfaces, and when it is not they blur and overlap, shifting the apparent first bone contact by a millimetre or more between visits. Option A is the good wrong answer and it is worth arguing, because a paralleling holder is how one usually achieves an orthogonal beam and is undoubtedly good practice; but an implant placed at an angle to the adjacent teeth can be imaged badly with a holder and well without one. The holder is the means; thread resolution is the criterion. (*Lindhe; 2017 World Workshop*)

59 A The one-year film has a specific purpose that no later film can serve: crestal remodelling occurs mainly in the first year, so the level recorded then, rather than at fitting, is the reference against which subsequent loss is judged and against which the case definitions of peri-implant disease are applied. Item B is the reason candidates most often add, and it is why D is the popular wrong answer — excess cement is a genuine and well-documented cause of peri-implant inflammation, and radiographs do sometimes reveal it, but this crown is screw-retained and there is no cement to find. Item C fails on principle: radiography demonstrates the absence of a peri-implant radiolucency, not the presence of integration, which is a clinical judgement resting on the absence of mobility and of symptoms. (*2017 World Workshop; FGDP/RCR Selection Criteria*)

60 E Mapping gives a set of point measurements at whatever heights the operator chooses to pierce, and the shape of the ridge between and apical to those points is inferred rather than seen. That matters because the features that cause perforations — a buccal concavity beneath a broad crest, a lingual undercut, a fenestration over a prominent root — lie precisely in the territory the caliper does not reach. Option A is the attractive distractor because it describes the error mapping was designed to eliminate: piercing to bone under anaesthesia measures bone to bone, so the mucosal thickness is subtracted rather than added, and a candidate who chooses A has confused mapping with palpation. Density is not obtainable from cone beam grey values either, so C is no discriminator, and a stent relates mapping to the tooth position perfectly well. (*Misch*)

61 B The figures quoted for static guided surgery are means: entry deviations of roughly a millimetre, apical deviations of the order of a millimetre and a half, angular deviations of a few degrees. The distribution matters more than the mean, because the maximum values in the same series are considerably larger, and a single case at the tail of that distribution consumes a 2 mm margin entirely. That is the reason the margin is retained under guidance rather than abandoned because of it, and the reason a guide is not a substitute for knowing where the canal is. Option D is the answer the patient has already half-formed and the one candidates choose to reassure her; it quotes an accuracy no systematic review supports and would justify reducing the very margin that protects her. A and C each isolate one component of a cumulative error chain. (*Tahmaseb et al. systematic review; ITI Treatment Guide vol 11*)

62 E Guided surgery is a chain of transformations, and every link adds error. The registration of the surface scan to the volume is the first, and it is degraded exactly where restorations scatter and harden the beam, which is why fiducial markers or a scan appliance are used in restored mouths. Manufacture and seating of the guide are the second: a guide that rocks on compressible mucosa, or that is not fully seated because of a soft-tissue undercut, displaces every subsequent step. The sleeve-to-drill interface is the third, since a drill that cannot move within its sleeve cannot be inserted at all, and the clearance that permits insertion also permits tipping. Candidates who reject item (iii) do so because software registration feels like arithmetic rather than measurement; it is measurement, and artefact corrupts it. (*Tahmaseb et al. systematic review; ITI Treatment Guide vol 11*)

63 A Two geometrical facts govern angular error. First, the drill can tip inside its sleeve through an angle set by the ratio of the clearance to the length over which it is constrained, so a longer sleeve holds it closer to the planned axis. Second, whatever angulation error escapes the sleeve is amplified over the unguided span between the sleeve and the bone, so a short offset limits the lateral displacement at the entry point and, in turn, at the apex. The two act together, and A optimises both. Option D is the mirror image and is chosen by candidates who reason about mouth opening alone. The trade-off is real, however: sleeve length plus offset plus drill length must fit the interarch space, which is why access is checked at the planning stage and not at surgery. (*Tahmaseb et al. systematic review; ITI Treatment Guide vol 11*)

64 C Dense cortical bone, a long guided path and a sleeve that shields the cutting flutes from external irrigation combine to produce the one complication that is silent at the time and irreversible afterwards. Bone necrosis follows heating to about 47 °C for one minute, and thermal injury of that kind presents weeks later as a mobile implant surrounded by sequestrated bone. Copious pre-cooled irrigation delivered into the sleeve, intermittent drilling with frequent withdrawal to clear debris, sharp single-use drills and full use of the intermediate diameters are the countermeasures. Option B is the hazard candidates name first, and rightly so in general, because a flapless approach conceals a perforation — but the stem has excluded it, describing a broad ridge with no concavity, and noticing that datum is the discrimination being tested. (*Eriksson & Albrektsson; ITI Treatment Guide vol 11*)

65 A A buccal plate of 1 mm over the planned implant surface is the datum that decides this question. Fenestration and dehiscence are common at such sites, and without a flap the defect is neither seen nor grafted; the first sign may be a recession or a fistula a year later, by which time simultaneous augmentation is no longer possible. That is the price of the flapless approach, and it is why it suits broad ridges with generous plates. Item B is the plausible addition that makes D attractive: primary stability is a function of bone quality, osteotomy preparation and implant design, and direct vision contributes nothing to it. Item C fails because a guide supported on two sound adjacent teeth is rigidly seated without pins; fixation is a requirement of mucosa-supported guides in edentulous arches. (*ITI Treatment Guide vol 11*)

66 E This is the wax-up doing the work it exists to do: converting a vague statement that the ridge is deficient into a defined prosthetic problem with only two honest solutions, both of which must be discussed before any surgery. Either the missing vertical tissue is rebuilt — the least predictable form of augmentation, and a decision that takes this case beyond a general dentist with an interest — or it is replaced with pink ceramic or acrylic, which is a legitimate plan the patient must consent to knowingly. Option B is the comfortable answer and the one that produces the unhappy result: with 2 mm of gingiva

showing at rest of the smile, a crown 3 mm longer than its neighbour will be seen every time she speaks. A connective tissue graft addresses thickness and contour, not 3 mm of lost vertical height. (*ITI Treatment Guide vol 1; Misch*)

67 B Six millimetres cannot accommodate an abutment, a matrix housing and a thickness of acrylic that will survive function, whatever attachment is chosen — and a bar needs more room than a stud, not less, which disposes of D. The space is created where it can be created, by reducing the crest at the time of placement, and the amount is calculated from the mounted set-up beforehand and transferred with a reduction guide, not judged by eye at surgery. Option E is the choice that reflects what actually happens in practice and is the most attractive distractor for that reason: a base thinned over the housings fractures, usually within a year and usually in the midline. Option C creates space by a route the patient's muscles and freeway space will not tolerate; the vertical dimension is determined by her physiology, not by the prosthesis. (*Misch; ITI Treatment Guide vol 4*)

68 C The wax-up is a prosthetic instrument, and its two great contributions are dimensional. It measures the space that an overerupted opponent has left, which decides whether enameloplasty, an onlay, orthodontic intrusion or a change of plan is required before anything is placed. And it shows the relationship between the planned tooth and the ridge beneath it, which is what distinguishes a straightforward case from one needing augmentation or an angulated, cement-retained solution. Item (iii) is the one that must be rejected, and E is chosen by candidates who have come to regard the wax-up as a general planning tool: bone density cannot be inferred from a cast, is not reliably obtained from cone beam grey values either, and is finally judged by the resistance felt during preparation and by the insertion torque. (*Misch; ITI Treatment Guide vol 2*)

69 D A bifid canal carries neurovascular tissue in both branches, and the branch that matters surgically is whichever lies nearer the crest — here the accessory canal, 3 mm above the main one, which moves the effective ceiling of the osteotomy 3 mm coronally and may shorten the implant or change the plan altogether. Reading the volume slice by slice, rather than accepting the panoramic reconstruction, is what reveals this. Option A is a true statement and the reason bifid canals are best known: they are a recognised cause of incomplete inferior alveolar anaesthesia, and a supplementary infiltration may well be needed. It answers a different question. B ignores the anatomy it has just described, C substitutes a lateral shift for a vertical constraint, and E orders a further exposure to confirm something already visible. (*SEDENTEXCT guidelines; Misch*)

70 A Consent is purpose-specific. Agreement that photographs may be taken and kept as part of the clinical record is not agreement that they may be shown to an audience, and the patient must be asked again, told who will see the images and in what setting, and told that she may refuse or later withdraw without any effect on her care. That she cannot be contacted is an obstacle to obtaining consent, not a reason to dispense with it. Option D is the answer most candidates choose, on the reasonable-sounding view that dentists are bound by confidentiality anyway and that a study evening is a professional setting; professional audiences are still audiences, and the guidance draws no such exemption. Item C is wrong in the mouth above all others: an intraoral view with distinctive restorations, alongside a case history, remains identifiable. (*GDC Standards; UK GDPR*)

Answers — Block D — Surgical anatomy (Q71–86)

71 D An anterior loop exists where the inferior alveolar nerve runs forward beyond the level of the foramen and then turns posteriorly and superiorly to exit, and the only way to see that turn is to follow the canal slice by slice on cross-sections and identify where its most anterior point lies. Option A is the answer the image invites and the one most candidates take: a panoramic reconstruction superimposes the mandibular incisive canal, which continues forward from the same trunk in most patients, on to the loop, and consistently over-reads the anterior extent. That over-reading is not conservative in the way it seems, because it also breeds disbelief in the measurement. Adding a fixed 2 mm, as in B, assumes what should be measured, and a two-dimensional image, as in C, cannot show a structure that turns in three planes. (*Greenstein & Tarnow; ITI Treatment Guide vol 11*)

72 C The margin is measured from the nerve, not from the foramen: 3 mm of loop plus 2 mm of clearance gives 5 mm anterior to the radiographic foramen as the most posterior acceptable position on that side. Option A is the commonest error, and it is the right number applied to the wrong landmark by a candidate who has learned the 2 mm rule without asking what it is measured from. Option D is the figure often quoted as a blanket precaution when no cross-sectional imaging is available, and it is defensible in that situation; here the loop has been measured, and measuring is what makes 5 mm safe and 8 mm unnecessarily restrictive in a mandible where interforaminal space is precious. Note that the incisive canal continues forward, so "anterior to the loop" does not mean "free of neurovascular structures". (*Greenstein & Tarnow; ITI Treatment Guide vol 11*)

73 B The nerve turns back on itself: having run forward within bone beyond the level of the foramen, it emerges directed posteriorly and superiorly before fanning into its labial, mental and gingival branches. Two consequences follow. The anterior loop is a direct corollary of that geometry, which is why bone anterior to the foramen is not automatically safe. And surgically, a relieving incision placed anterior to the foramen with the flap raised from anterior to posterior, and a retractor held against bone rather than pulled coronally, keeps the instrument behind the emerging trunk. Option A is chosen by most candidates, because the nerve supplies the lip and chin, which lie anteriorly, so an anterior course feels intuitive; the branches travel forward, but the trunk leaves the foramen going the other way. (*Greenstein & Tarnow*)

74 E The osteotomy has almost certainly entered the mandibular incisive canal, the intraosseous continuation of the inferior alveolar nerve that runs forward from the mental foramen supplying the incisors, canine and labial gingiva. It is a small, entirely intrabony bundle: injury produces bleeding into the osteotomy that stops with tamponade by the implant, and most patients notice nothing, some report a patch of altered sensation over the labial gingiva, and a minority lose the vitality of an anterior tooth — which is why C is a real but less likely outcome rather than an absurd one. Option A is the answer chosen by candidates who see "mandible", "nerve" and "bleeding" together; lip and chin sensation is carried by the mental nerve, which leaves the bone posterior to this injury and is untouched. Option B belongs to lingual plate perforation. (*Greenstein & Tarnow; ITI Treatment Guide vol 11*)

75 E All three compound the same anatomical trap, and a candidate who stops at two has understood the concavity but not the case. The crest of a posterior mandible overlying a submandibular fossa is typically inclined lingually, so a drill following its apparent long axis is aimed straight into the undercut — the reason cross-sections must be read before the handpiece is picked up. Working without a flap removes the two safeguards that remain, direct vision of the lingual contour and a finger placed against the plate. And length is the multiplier: the concavity deepens apically, so every additional millimetre of osteotomy travels further into it, which is why a shorter implant placed more buccally is the usual solution here. Option D is the popular answer, chosen by candidates who treat length as a purely biomechanical decision. (*Lindhe; ITI Treatment Guide vol 11*)

76 A Perforation of the lingual plate into the sublingual space produces a haematoma that expands within a closed compartment bounded by mylohyoid, elevating the tongue and obstructing the airway, and the crucial feature of the natural history is that it progresses over minutes to hours — a patient who is comfortable and saturating well now may be in extremis in the car park. Pressure is correct and should be applied at once, but it is the accompaniment to summoning help, not an alternative, and definitive care may require imaging, ligation and a secured airway. Option B is the trap and the one that has caused real harm: observing in a dental surgery places the patient in the one location without the means to rescue him. Seating an implant does not compress a soft-tissue space, and E treats a life-threatening event as an oozing socket. (*Lindhe; ITI Treatment Guide vol 11*)

77 C The mylohyoid nerve leaves the inferior alveolar trunk above the lingula, before the parent nerve enters the mandibular foramen, so a block deposited at the foramen never reaches it. It runs in the mylohyoid groove on the medial surface of the ramus, supplies mylohyoid and the anterior belly of digastric, and in a proportion of patients sends fine sensory branches through small lingual foramina to the molar

region — the classic cause of persistent sensitivity in a mandible whose lip is undeniably numb. Lingual infiltration or an intraosseous supplement solves it. Option A is the reflex answer and is excluded by the stem: a block that has anaesthetised the lip to profound numbness has reached the nerve. A bifid canal, D, is a genuine cause of failed anaesthesia but usually leaves the lip incompletely numb as well. (*Misch*)

78 B In the retromolar and third molar region the lingual nerve lies directly against the lingual plate in a large proportion of people and at or above the level of the alveolar crest in roughly one in ten, with nothing but periosteum between it and a blade. A vertical incision there transects it, and lingual nerve injury is among the most disabling complications in dentistry. The safe technique is a crestal incision, blunt subperiosteal elevation with the instrument kept hard on bone and directed inferiorly, and a broad retractor holding the flap; releasing incisions for advancement are made buccally, by scoring the periosteum. Option D is the attractive distractor because it contains a truth — the mylohyoid attachment does limit how far a lingual flap will advance — but it is an argument about effectiveness, not about the injury that must be avoided. (*Pogrel et al.; ITI Treatment Guide vol 7*)

79 A The canal contains the nasopalatine nerve and vessels within loose fibrous connective tissue lined by epithelium, and an implant surface presented to that tissue does not osseointegrate; the result is a partially integrated implant that is painful, mobile or slowly failing, and the canal enlarges as the anterior maxilla resorbs, so this is a problem of the atrophic ridge in particular. The remedies are to move the implant labially or apically within available bone, to accept a different angulation, or — a specialist decision — to enucleate the canal contents and graft it before staged placement. Option B is the distractor worth arguing: nasopalatine nerve injury does occur, but altered palatal sensation after it is usually transient and well tolerated because of overlap with the greater palatine nerve, so it is a consequence rather than the consequence. (*Misch; ITI Treatment Guide vol 1*)

80 B Resorption of the labial plate brings the crest closer to a canal that itself widens with age and edentulism, which is why the structure that was irrelevant when the incisor was present becomes the limiting factor when it is replaced. Its arterial content is the nasopalatine artery, the terminal branch of the sphenopalatine artery from the maxillary artery, which anastomoses on the palate with the greater palatine artery — anatomy that explains both the bleeding encountered when the canal is instrumented and the survival of a palatal flap raised across it. Statement (iii) is the one that fails, and D is the tempting choice for candidates who recall that the canal is well seen on an upper standard occlusal or periapical: it is well seen, but its buccopalatal diameter and its relationship to the planned implant axis are three-dimensional questions that only cross-sections answer. (*Misch; SEDENTEXCT guidelines*)

81 D The greater palatine artery and nerve run forward from the foramen in the groove between the alveolar process and the horizontal palate, and the deeper the vault the further that groove lies from the gingival margin. A shallow palate brings the bundle close, and the safe donor zone — commonly described as extending some 5 to 7 mm apical to the gingival margin, between the distal of the canine and the mesial of the first molar — shrinks accordingly. Judging that distance from the individual palate, not from a textbook figure, is the point. Option A is the strong distractor because mucosal thickness is genuinely critical and 3.5 mm is a favourable measurement; it governs how thick a graft can be raised and whether a partial-thickness flap can be dissected, which is a different dimension from the one the question asks about. (*Lindhe; ITI Treatment Guide vol 7*)

82 A The posterior superior alveolar artery, a branch of the maxillary artery, gives an intraosseous branch that runs forward within the lateral antral wall and anastomoses with a branch of the infraorbital artery; it is seen on a substantial proportion of cone beam volumes, typically some 15 to 20 mm above the crest, and a channel of 2 mm carries a vessel large enough to bleed briskly and obscure the field. Its identification changes the design of the window, whose superior border is kept below the vessel or the vessel deliberately managed — one of several reasons this operation is planned by whoever performs it. Option D is the distractor that catches candidates who know the infraorbital canal is nearby; it lies in the roof and anterior wall of the antrum, not in its lateral wall at the level of a window. (*ITI Treatment Guide vol 5*)

83 E The maxillary ostium opens high on the medial wall into the middle meatus, so the antrum drains upwards, against gravity, and depends entirely on the mucociliary escalator sweeping secretions towards it. A few millimetres of post-operative oedema at that narrow opening obstructs the escalator, secretions stagnate, and bacterial sinusitis follows — the reason a pre-operative volume is examined for ostial patency and why decongestion and early ENT involvement matter after grafting. Option A is a true statement and a good distractor: the antral floor does lie below the nasal floor in the dentate adult, which is why root apices project into it, but it describes the anatomy of the floor rather than of drainage. Option B reverses the direction of ciliary beat, which is towards the ostium, and C names the wrong meatus. (*ITI Treatment Guide vol 5*)

84 A Septa most commonly rise from the inferior wall and cross the antrum buccopalatally, dividing the floor into compartments; that orientation is what makes them dangerous during elevation, because the membrane must be lifted over a bony crest running across the direction of dissection, and it is also why they are missed on projection imaging that superimposes them. Statement (ii) fails for the same reason as most panoramic claims: a reconstruction from a volume carries the superimposition problems of the original technique, and membrane thickness is read on cross-sections. Statement (iii) is the one candidates accept most readily, and C or D follow: the Schneiderian membrane is a thin, ciliated, pseudostratified columnar respiratory epithelium continuous with that of the nose, which is exactly why its swelling obstructs drainage and why it is repaired rather than simply patched. (*ITI Treatment Guide vol 5*)

85 C The posterior maxilla is the classic type 4 site: a thin, sometimes barely present cortex over a loose lattice of widely spaced trabeculae, and it accounts for the lowest insertion torques and the highest early failure rates of any region. The response is technique rather than abandonment — under-preparation, omitting the final drill, osteotome condensation, avoiding immediate loading, allowing a longer healing period and accepting that a torque of 12 Ncm forbids anything but a submerged or unloaded protocol. Option A is the well-informed wrong answer: pneumatisation is real, progresses after extraction and is the reason posterior maxillary sites so often need grafting, but it explains a shortage of bone height, and this site has 11 mm. Bundle bone, B, is a property of the socket wall and disappears within months of extraction. (*Lekholm & Zarb; Misch*)

86 D Two anatomical facts make this a specialist case, and both are visible on the film. Eight millimetres of total height means the osteotomy occupies almost the whole cross-section of the bone, and a severely atrophic mandible can fracture during preparation or weeks afterwards under function — a complication requiring open reduction in an elderly patient. And foramina at the crest place the emerging nerve where the incision, the flap and the drill all pass, with no bone above them to shelter it. Item C is what makes E wrong and it is the one candidates add without thinking: chronological age is not a contraindication, older patients do well, and the two-implant overdenture is the very treatment most likely to transform her daily life. She should be referred for it, not refused it. (*McGill consensus; Misch; SAC Assessment Tool*)

Answers — Block E — Osseointegration, bone biology and healing (Q87–106)

87 A At 0.7 mm the buccal wall of an upper central incisor is composed very largely of bundle bone — the tooth-dependent bone into which Sharpey's fibres insert and which is supplied through the periodontal ligament. Extraction removes the fibres and their vascular supply together, so that tissue is resorbed as part of normal modelling, and Araújo and Lindhe showed that placing an implant into the socket does not arrest the process. A connective tissue graft and a grafted gap may camouflage or reduce the change, but neither prevents it, and she must be consented on that basis. Option B is the attractive error because it is a perfectly true statement about immediate placement, and candidates recognise it as such; it simply says nothing about buccal contour. Option E reverses the phenotype relationship — thin walls lose proportionately more. (*Araújo & Lindhe; ITI Treatment Guide vol 3*)

88 B Woven bone is the tissue of emergency repair: collagen is deposited quickly and in disorganised bundles, mineral content is lower, and the result is mechanically weak but arrives fast enough to bridge the healing chambers between implant and osteotomy wall within days. Statement (ii) follows from it — secondary stability rises as that scaffold mineralises and is progressively replaced by orientated lamellar bone, a process that continues for months after the implant is loaded. Statement (iii) is what makes C and E wrong, and it is the item candidates accept because they know lamellar bone is the definitive tissue: lamellar bone is never the first bone formed at any healing site, it can only follow woven bone, and at one week the interface contains fibrin, granulation tissue and the earliest woven trabeculae. (*Lindhe; Davies*)

89 C Contact osteogenesis, in Davies's terminology, means bone formed *de novo* on the implant itself: osteogenic cells migrate along the retracting fibrin scaffold onto the surface, differentiate there, and lay down a cement line matrix directly against the titanium, so that new bone is seen on the implant with no old bone behind it. That is precisely what a moderately rough surface buys — it retains the fibrin during clot retraction so the cells reach the surface at all. Option A is the distractor that catches the well-read candidate, because it describes real and healthy healing; but bone advancing from the osteotomy wall towards the implant is *distance* osteogenesis, the pattern typical of machined surfaces. Option B describes the precondition rather than the event, and D is normal removal of surgically damaged bone. (*Davies; Lindhe*)

90 D Sixty newton centimetres in dense bone compresses the crestal cortex beyond what it tolerates; the compressed and microdamaged bone is devitalised, must be removed by osteoclasts before it can be replaced, and it is that resorption which the falling ISQ measures. A high starting value therefore has the furthest to fall, and the classic stability dip at roughly two to four weeks is deepest in exactly this situation.

Option B is the strong distractor because it is true and relevant — woven bone filling the healing chambers is why the LR4 rose — but it explains the ascending curve, not the descending one, and the question asks about the difference. Transducer orientation alters absolute values a little, not a 14-point fall, and neither implant is mobile. (*Raghavendra et al.; ITI Treatment Guide vol 2; Misch*)

91 E What destroys osseointegration during early healing is relative movement between implant and bone. Above roughly 50 to 150 micrometres of micromotion the fibrin scaffold is disrupted and repair proceeds to fibrous tissue rather than bone; below it, healing is undisturbed. Her torque and ISQ are sufficient to restore but not to load, and with a 4 mm overbite the provisional would otherwise be struck in protrusion as well as in centric, so removing every contact is the intervention that matters. Option C is the answer that tempts the experienced operator, and splinting is not absurd — but composite splints debond unpredictably, create a plaque trap in the aesthetic zone, and do nothing about the excursive contact that would still load the tooth-implant unit. A soft diet helps but cannot be relied upon. (*Szmukler-Moncler et al.; Brunski; ITI Treatment Guide vol 2*)

92 A Drill sharpness is the dominant determinant of the heat generated during osteotomy preparation: a worn drill burnishes rather than cuts, work is converted into heat instead of chips, and no volume of irrigation compensates. Bone is killed at about 47 °C sustained for one minute — the threshold established by Eriksson and Albrektsson and the reason for graded, sharp, irrigated drilling — and a necrotic cylinder of bone around the whole implant produces exactly this picture of mobility with a circumferential radiolucency. Option D is the distractor with real substance, because 55 Ncm in D1 bone genuinely risks compression necrosis; but pressure necrosis is a crestal phenomenon that costs marginal bone, not an implant. External irrigation is adequate with sharp drills, and 1200 rpm is within accepted limits. (*Eriksson & Albrektsson; Misch*)

93 C The difference is one of starting position, not of healing rate. A thin cortex with sparse trabeculae offers far less mineralised tissue in contact with the implant at the moment of placement, so a greater proportion of the eventual stability has to be generated by new bone formation in the healing chambers — and that takes time, which is why conventional protocols allowed three to six months in the posterior maxilla and six to eight weeks in the dense mandible. Option A is the most attractive wrong answer because it sounds like sound physiology and produces the right conclusion for the wrong reason: trabecular bone is in fact the *more* vascular and the faster remodelling of the two, which is why grafts consolidate faster in it. B is a true statement about cortical bone that argues the opposite case. (*Misch; ITI Treatment Guide vol 2*)

94 D Twelve newton centimetres and an ISQ of 52 are both well below the values usually required before an implant is restored early or immediately — around 35 Ncm and an ISQ in the region of 70 — so this implant depends almost entirely on secondary stability, and the answer is to protect it while that develops. Submerged healing removes transmucosal loading and the risk of the healing abutment being knocked, and D4 bone justifies extending healing towards six months. Option B is the answer most candidates give, because six weeks feels like a compromise between his wishes and caution; it loads an implant during the very period when primary stability is falling and secondary stability has not yet compensated. C discards an implant that is stable and well placed, and E adds surgery to solve a healing problem. (*Misch; ITI Treatment Guide vol 2*)

95 B The peri-implant supracrestal attachment — the old "biologic width" — occupies about 3 to 4 mm in total, roughly 2 mm of barrier epithelium over 1 to 1.5 mm of connective tissue, and it will establish itself at the expense of bone if it is not given room. An implant seated 3 mm subcrestally, then repeatedly disconnected so that the attachment re-forms more apically each time, produces exactly this: symmetrical loss to the first thread, ending when the dimension is satisfied, in tissues that are otherwise healthy. Option A is the distractor to argue with rather than dismiss, since retained cement is a genuine and common cause of peri-implant bone loss — but it produces an inflamed site that bleeds and often suppurates, and it rarely destroys bone with this symmetry. C and D would not localise in this pattern. (*Berglundh & Lindhe; 2017 World Workshop*)

96 A Only the first is true, and it is the one that matters: with no cementum there is nothing for fibres to insert into, so the collagen of the peri-implant connective tissue runs circumferentially and parallel to the surface, forming a cuff rather than an attachment, and the seal is correspondingly easier for an inflammatory lesion to bypass. Item B is the one that makes D attractive, and it is false — the barrier epithelium attaches to titanium by hemidesmosomes and a basal lamina, much as junctional epithelium attaches to enamel. Item C is false in the opposite direction: without a periodontal ligament plexus the peri-implant connective tissue is *less* vascular, more collagenous and poorer in fibroblasts, which is precisely why its defence is weaker. (*Berglundh & Lindhe; Lindhe*)

97 C At six days the wound is early in the proliferative phase: granulation tissue is forming, epithelium is migrating, and the tensile strength of the flap is still negligible, which is why the correct response to a

small, clean dehiscence is meticulous local hygiene and supervision while it epithelialises by secondary intention. Most of the graft is retained if the site is kept clean, even though an exposed native collagen membrane will degrade early. Options A and E are the distractors that appeal to the instinct to close the wound, and E in particular looks like sound surgery — but sutures tear through tissue that has no strength, a second flap sacrifices vestibular depth and blood supply, and both convert a manageable exposure into a larger wound. Antibiotics treat an infection that is not present. (*Wang & Boyapati PASS; ITI Treatment Guide vol 7*)

98 **E** At 64 mmol/mol — roughly 8 per cent in the old units, and above the figure of about 58 mmol/mol usually accepted for elective implant surgery — chronic hyperglycaemia drives non-enzymatic glycation of proteins. The resulting advanced glycation end-products cross-link collagen so that it resists normal turnover, bind receptors on osteoblasts and macrophages to shift the balance towards resorption and inflammation, and reduce the quality of the matrix the healing chamber depends on; microvascular change compounds it. Option C is the most attractive alternative because impaired neutrophil function in diabetes is real and well taught, and it explains his susceptibility to infection and to periodontal breakdown — but it is a mechanism of infection rather than of impaired bone formation, and the question asks about osseointegration. (*Lindhe; NICE NG28; Misch*)

99 **B** Nitrogen-containing bisphosphonates act on the osteoclast, inhibiting farnesyl pyrophosphate synthase and reducing resorption; osteoblastic bone formation is not directly blocked, which is why implants placed in these patients generally integrate perfectly well and why integration is a poor test of safety. What is lost is the capacity to remodel — to remove microdamage, to turn over bone around a chronically infected or traumatised site, and to repair. Add infection and the jaws, with their high turnover and thin mucosal cover, are where that deficit shows. Option A is the commonest misconception and the reason candidates expect early failure rather than late necrosis; it inverts the drug's mechanism. C is untrue, and D and E have no basis. (*SDCEP MRONJ; AAOMS position paper*)

100 **D** Even faultless preparation kills a thin rim of bone at the cut surface, and this devitalised layer, recognised histologically by its empty lacunae, is resorbed over the following weeks by osteoclasts recruited to the site. That resorption is one of the two structural reasons — the other being remodelling of any compressed bone — why primary mechanical stability falls to its nadir at around two to four weeks before secondary stability overtakes it. Option A is the trap, because dead bone reads as overheating to most candidates; but thermal necrosis at 47 °C and above produces a far thicker zone and, in a specimen prepared with sharp drills at low speed under irrigation, is not the explanation. Option C misapplies Davies's terminology to a resorptive, not a formative, event. (*Davies; Roberts; ITI Treatment Guide vol 2*)

101 **B** Mobility is the one unequivocal sign of failed osseointegration, and rotation of the implant when the cover screw is engaged means the interface is fibrous, not bony. The continuous peri-implant radiolucency corroborates it; unlike the localised crestal changes of remodelling or peri-implantitis, a uniform line around the whole fixture is the radiographic signature of fibrous encapsulation. A mobile implant does not recover, so it is removed, the site allowed to heal, and replacement considered later. Option A is the thoughtful distractor and would be right in a different case, since projection geometry genuinely creates spurious radiolucent lines — but no radiographic technique alters a clinical finding of mobility. D risks placing a wider implant into a compromised bed, and E would guarantee failure. (*Albrektsson & Zarb; Lindhe*)

102 **C** Statements (i) and (iii) are the mechanical basis of most prosthetic design advice. Bone is a strain-adapted tissue: within a physiological range it remodels and maintains itself, below it — disuse — it is resorbed, and above it microdamage accumulates faster than remodelling can repair. Finite element and histological work consistently locates the highest strain at the crest, because that is where the stiff implant is most rigidly held and where a lateral force acts through the longest lever. Statement (ii) is what makes D and E wrong, and it is the item most candidates accept: the 2017 World Workshop concluded that there is no evidence that occlusal overload alone, without plaque-induced inflammation, causes peri-implantitis. Overload breaks screws, ceramic and components. (*Frost; 2017 World Workshop; Misch*)

103 **E** This is creeping substitution. A cancellous or particulate graft has an open architecture that capillaries and osteogenic cells enter within days, so vital bone appears throughout it early. A cortical block is dense; revascularisation must follow osteoclastic cutting cones that resorb their way through it before osteoblasts can lay down new bone behind them, and the process takes months, during which the graft is weaker than at the time of fixation. Option A is the attractive answer because osteoinduction is a term candidates associate with autogenous bone; but fresh autogenous bone is osteogenic, osteoinductive and osteoconductive whatever its form, and particle size does not change that. Note also that block grafting sits beyond the scope of most general dentists with an interest. (*Lindhe; ITI Treatment Guide vol 7*)

104 **B** The criteria proposed by Albrektsson and colleagues allow up to 1.5 mm of marginal bone loss during the first year of loading and less than 0.2 mm annually thereafter, alongside absence of mobility, absence of a peri-implant radiolucency, and absence of pain, infection or neuropathy. This implant meets every one of them: 1.3 mm is inside the first-year allowance and 0.1 mm is inside the annual figure. Option C is the answer that catches candidates who remember the distinction between survival and success but not the numbers attached to it, and who read any bone loss as a departure from success. There is no bleeding, so D is excluded, and peri-implantitis requires both inflammation and progressive loss beyond initial remodelling. (*Albrektsson & Zarb; 2017 World Workshop*)

105 **A** Glucocorticoids suppress the proliferative phase of healing: fibroblast replication and collagen synthesis fall, angiogenesis is reduced, and the inflammatory signalling that recruits cells to the wound is blunted. The practical consequences here are a flap that gains tensile strength slowly and a raised risk of dehiscence over the membrane, which is why meticulous tension-free closure matters more in this patient than in most. Item B is what makes D attractive, and it inverts the pharmacology: corticosteroids increase RANKL expression and reduce osteoprotegerin, so resorption rises and bone is lost — the mechanism of steroid-induced osteoporosis — rather than being suppressed as with a bisphosphonate. Item C is unfounded. Steroid therapy is separately relevant as a factor that raises MRONJ risk when an antiresorptive is also taken. (*BNF; SDCEP MRONJ; Lindhe*)

106 **E** All three are established, and holding them together is what separates a considered plan from a slogan. Marginal defects of about 2 mm or less between implant and socket wall predictably resolve with bone during healing, provided the implant is stable and the site is not infected. Grafting the gap, typically with a slowly substituted xenograft, nevertheless reduces the horizontal contour change measured at the ridge, which is why it is recommended in the aesthetic zone despite the first point. And the buccal wall, being largely bundle bone, undergoes modelling regardless — grafting the gap modifies the outcome without abolishing it. Candidates most often reject A, because they have been taught to graft every gap and infer that unfilled gaps do not heal. (*Botticelli et al.; Araújo & Lindhe; ITI Treatment Guide vol 3*)

Answers — Block F — Materials, implant design and surfaces (Q107–132)

107 **A** Narrow-diameter implants fail mechanically rather than biologically: the section modulus falls steeply as diameter is reduced, and fracture of the implant or of its screw is the recognised risk at 3.3 mm and below. The titanium–zirconium alloy, containing roughly 13 to 17 per cent zirconium, was developed for precisely this problem, offering appreciably greater tensile and fatigue strength than grade 4 commercially pure titanium while retaining its osseointegration behaviour. Option B is the thoughtful wrong answer, because modulus mismatch is a genuine theme in implant biomechanics and a bone-like modulus would be desirable; the alloy's modulus is essentially that of titanium, and no metallic implant approaches bone. Option D borrows a property of zirconia — the alloy is a metal, not a ceramic, and ageing does not apply to it. (*ITI Treatment Guide vol 2*)

108 **B** The four grades of commercially pure titanium are all more than 99 per cent titanium and are distinguished only by their permitted interstitial and impurity content — principally oxygen, with iron, carbon, nitrogen and hydrogen. Those small atoms sit within the lattice, impede dislocation movement, and progressively raise tensile and yield strength while reducing ductility, so grade 4 is the strongest and least ductile of the four and is the grade generally chosen for implant bodies. Option A is the distractor that most candidates take, and it confuses grade with alloy: aluminium and vanadium define Ti-6Al-4V, which is grade 5 and an alloy, not commercially pure titanium — which is exactly the distinction the colleague's question exposes. The oxide layer described in D is a few nanometres thick whatever the grade. (*Misch*)

109 **C** The screw is the weakest and most highly stressed part of the assembly, and its job is to be stretched: tightening elongates it elastically and stores a tensile clamping force, the preload, that holds the joint closed against functional separation. Preload can safely be taken to a large fraction of the yield strength, so a material with a higher yield strength permits a higher preload and a joint less likely to open — Ti-6Al-4V has roughly double the yield strength of commercially pure titanium. Option D is the sophisticated error: elongation is indeed the mechanism, so a lower modulus sounds helpful, but the alloy's modulus is close to that of the pure metal and it is strength, not compliance, that has been bought. Option E is a category mistake, since zirconia is not a conductor. (*Misch*)

110 **D** Stainless steel is harder than titanium, so a steel instrument does not polish the surface — it ploughs it, leaving scratches that are more plaque-retentive than the original topography and smearing steel onto the titanium, which alters the surface chemistry against which cells and biofilm interact. That is the reason for titanium, carbon-fibre or PEEK instruments and glycine air polishing. Option A is the attractive answer and is half right: the passive layer is indeed disrupted, but titanium dioxide re-forms spontaneously within milliseconds wherever oxygen is present, so the damage is topographic and chemical contamination rather

than permanent loss of passivation. Option C invents a coating this implant does not have, and there is no galvanic couple of clinical significance from momentary contact. (*ITI Treatment Guide vol 13*)

111 E All three are documented, and the honest position is that particles arise from ordinary clinical events rather than from misadventure alone. Insertion through cortical bone shears particles from the threads, which is why they are found in the tissues of implants that have never been instrumented; metal instruments and rotary implantoplasty generate them in far greater numbers; and fretting at the microgap under cyclical load releases them from the joint, particularly where the connection permits micromovement. Candidates most often exclude A and choose D, because insertion is the one source that involves no error and therefore feels as though it ought not to count. Whether these particles cause peri-implantitis or merely accompany it is unresolved, and the sensible clinical response is to avoid the avoidable sources. (*ITI Treatment Guide vol 13; Misch*)

112 C In the classification of Albrektsson and Wennerberg, surfaces are smooth below 0.5 micrometres, minimally rough from 0.5 to 1.0, moderately rough from 1.0 to 2.0, and rough above 2.0. Bone-to-integration data plateau in the moderately rough band: increasing Sa beyond about 2 micrometres yields no reliable further gain, while the additional topography retains biofilm and is markedly harder to decontaminate once threads become exposed. In a patient who has already lost an implant to peri-implantitis, that trade-off decides the question. Option A is the seductive half-truth, because roughness genuinely does increase surface area and early bone contact; it simply ignores the second half of the trade, which is the half that matters here. (*Albrektsson & Wennerberg; ITI Treatment Guide vol 13*)

113 A The modification is chemical, not topographic. Preparing and storing the surface so that it remains hydroxylated and free of atmospheric hydrocarbon contamination gives it a very high surface energy: blood wets it immediately, plasma proteins adsorb and spread, the fibrin clot adheres firmly during retraction, and osteogenic cells reach the surface earlier, which is the basis for shortening the stability dip and permitting earlier loading. Item B is the one that makes D attractive and it is false — the roughness is deliberately identical to that of the untreated surface, which is what allowed the effect of chemistry to be isolated. Item C is a marketing claim rather than a finding; the surface does not prevent infection and it does not remove the need for primary stability. (*ITI Treatment Guide vol 2*)

114 D Plasma-sprayed hydroxyapatite is an additive coating bonded to the titanium substrate, and its weakness is that interface. Coatings of variable crystallinity dissolve at unpredictable rates and can separate from the implant in sheets; once a coated surface is exposed to the peri-implant environment, loss of the coating leaves a rough, contaminated surface that is exceptionally difficult to decontaminate, and the pattern of rapid progressive loss seen in these implants follows. This is why subtractive treatments displaced additive coatings for routine use. Option B is the plausible near-miss: these surfaces are indeed rough, but the figure is wrong by an order of magnitude — plasma-sprayed surfaces sit in the low single figures of micrometres. Hydroxyapatite is radiopaque, not radiolucent, so A is doubly wrong. (*Misch; ITI Treatment Guide vol 13*)

115 B Where a horizontal defect leaves the implant surface permanently exposed to the mouth, the durable solution is to change the surface itself: implantoplasty removes the threads and reduces the roughness towards that of a machined surface, and the resective approach with surface modification has the better medium-term data for suprabony defects of this shape. Option A is the strong distractor and is not wrong as maintenance — glycine air polishing is an effective and safe method of decontamination, and it should be used — but it removes biofilm without altering the topography that keeps re-accumulating it. Chemical agents and local or systemic antimicrobials are adjuncts to debridement whose benefit is short-lived. Note the trade-off: implantoplasty releases titanium particles and weakens a narrow implant. (*ITI Treatment Guide vol 13; 2017 World Workshop*)

116 E Yttria-stabilised tetragonal zirconia is metastable: energy delivered at the surface — by a crack tip, or by a bur — converts tetragonal grains to the monoclinic form with a volume increase of some four per cent. That is transformation toughening, and it is why the immediate effect of grinding is a compressive surface layer that may even raise strength. The reason to polish anyway is that grinding also introduces deep flaws, leaves grains already transformed and therefore unable to toughen a future crack, and produces a surface that abrades opposing enamel. Option A is the attractive error and would be correct for a layered, glazed restoration; a monolithic zirconia crown has no feldspathic veneer to expose. Options C and D describe processes that do not occur. (*Misch*)

117 A This is low-temperature degradation, or ageing. In a warm wet environment the metastable tetragonal grains at the surface transform spontaneously to monoclinic over years; the associated expansion produces microcracking, roughening and grain pull-out, and the transformed surface layer advances inwards, so the toughening mechanism that protects the ceramic is progressively spent before it is needed. The result is delayed fracture under loads the abutment carried without difficulty when new. Option B

is the distractor with genuine mechanical logic, and screw fatigue does occur — but the stem states the screw is intact, and a fatigued screw loosens or fractures rather than splitting the ceramic. Ceramics do not creep at oral temperatures, and the stabiliser is not hydrolysed. (*Misch*)

118 C Zirconia is very much harder than titanium, so a ceramic abutment engaging a titanium implant wears the softer component: retrieval and laboratory studies show deformation of the internal connection, wear debris and an increasing microgap under cyclical loading, and the effect is greatest in narrow-platform implants where the walls are thinnest — which is exactly this case. A titanium base restores a titanium-to-titanium joint while keeping the aesthetic advantage of ceramic above it. Option E is the distractor that appeals because it sounds like practical prosthodontics, and it is simply untrue; zirconia abutments can be contoured subgingivally. Zirconia is radiopaque, so B is wrong, its flexural strength exceeds that of lithium disilicate, and it bonds predictably after appropriate surface conditioning. (*Misch; ITI Treatment Guide vol 12*)

119 D Both surviving items are real constraints of a one-piece design and both bear on this case. With the abutment integral to the implant, the emergence is determined entirely by the surgical axis, so a 20-degree discrepancy cannot be corrected prosthetically and would have to be accepted or compensated by the crown at the expense of contour. And because the implant is transmucosal from the day of surgery it cannot be submerged, so it must be protected from load while integration proceeds — a real difficulty in the aesthetic zone. Item C is what makes E wrong and it is the assumption candidates carry into the question: zirconia osseointegrates in experimental studies at levels comparable with titanium, and the legitimate reservation is the shorter clinical evidence base, not a failure to integrate. (*ITI Treatment Guide vol 12*)

120 B Torque is a means, not an end. Turning the screw stretches it within its elastic range, and the tension stored in that stretched shank — the preload — is what holds abutment and implant together and resists the separating forces of function. A screw joint fails when applied load exceeds preload, which is why correct torque matters and why the screw, not the abutment, is the sacrificial component. Option C is the near-miss that catches careful candidates, and it is nearly true: most of the applied torque, commonly quoted as around 90 per cent, is consumed overcoming friction, which is why contamination and reused screws matter so much. But if it were consumed *entirely* there would be no preload at all, and the joint could not exist. (*Misch; ITI Treatment Guide vol 12*)

121 A Two correctable faults are being repeated. A screw that has been tightened and loosened several times has deformed threads and altered friction, so the same applied torque no longer generates the same preload; a new screw restores it. And a screw tightened once loses a few per cent of its preload within minutes as the mating asperities flatten — embedment relaxation — so it must be re-torqued after a short interval, which has not been done here. Option D is the distractor with clinical instinct behind it, and occlusal reduction is the right instinct in the wrong case: the stem specifically excludes excursive contact and cantilever. Thread-locking adhesive and cementing the joint both make retrieval difficult without addressing preload. (*Misch; ITI Treatment Guide vol 12*)

122 B A stiff implant rigidly held in a much more compliant material does not distribute load evenly along its length: strain concentrates where the two first meet, which is the crestal bone, and this is one of the standard explanations for the crestal location of load-related bone change. That mismatch, together with the absence of the ligament's damping and proprioception, is why compliant abutment materials and ligament-like interfaces have been investigated for decades without a clinically successful result. Statement (iii) is the item that makes C and E wrong, and it is chosen by candidates who assume a ceramic must be more forgiving than a metal: zirconia's elastic modulus is about 210 GPa, roughly twice that of titanium and further still from bone. (*Misch*)

123 E In an external hexagon the abutment sits on a flat platform and the screw is effectively the only element resisting the separating and bending forces of function; the joint's stability depends almost entirely on preload being maintained. A conical connection wedges the abutment into the implant, and the friction fit over a long tapered wall carries much of the lateral load itself, so the screw is loaded less, preload is preserved and loosening is less frequent — with the incidental benefits of a smaller microgap and less bacterial leakage. Option D is the strong distractor because conical connections are very often platform switched as well, and candidates conflate the two; platform switching is a separate design feature affecting crestal bone, not screw mechanics. (*Misch; ITI Treatment Guide vol 12*)

124 A At any implant–abutment junction there is a microgap harbouring bacteria and a small chronic inflammatory infiltrate, and crestal bone recedes to a distance from it. Platform switching moves that junction horizontally towards the axis of the implant, so the infiltrate sits over the platform rather than at its outer edge and the bone at the periphery is spared — the effect measured in systematic reviews as roughly 0.3 to 0.5 mm less crestal bone loss. Option B is the distractor worth arguing with rather

than dismissing: a narrower abutment does alter stress distribution at the neck, and this biomechanical explanation is genuinely proposed in the literature, but it is a secondary contributor and does not account for the effect being confined to the crest. (*Lazzara & Porter; ITI Treatment Guide vol 12*)

125 **C** Occlusal load reaches an implant largely as a bending moment resisted by its cross-section, and the relevant geometry rises with a high power of the radius, so a millimetre of diameter buys far more mechanical resistance than a millimetre of length. Length principally adds surface area and primary stability; in a molar site with dense cortical bone and a canal to respect, the wider, shorter implant is the mechanically sounder choice and preserves a safer margin. Option E is the attractive overstatement of a true idea — length contributes proportionately less than diameter, but it certainly contributes, and a candidate who believes length is irrelevant will place short implants in soft bone. Option D reverses the evidence, since crown-to-implant ratio correlates poorly with bone loss. (*Misch; ITI Treatment Guide vol 3*)

126 **D** Bone is strongest in compression, considerably weaker in tension and weakest in shear, so a thread geometry that presents a face perpendicular to the implant axis transmits axial load to the surrounding bone predominantly as compression, which is the argument for the square or power thread. A V-thread, with its inclined faces, resolves the same load into a mixture of compressive and shear components, and a buttress thread is designed the other way round, to resist tensile or pull-out forces. Option C is the distractor for the candidate who has learned that there are three thread forms but not which does what, and it describes the buttress thread. The threefold surface area figure in B is invented, and square threads generally require more insertion torque, not less. (*Misch*)

127 **B** The property required here is volume stability, not osteogenic potency: a scaffold that resists resorption for long enough to hold the contour while bone forms around and between its particles. Deproteinised bovine bone mineral is substituted very slowly and particles are demonstrable in biopsies years later, which is a criticism of it in other contexts and precisely its virtue in this one. Option A is the answer most candidates give because autogenous bone is the material they have learned to call the gold standard — and it is, for regenerating bone in a defect — but it is also the fastest to resorb, so it is the worst choice when maintaining a dimension is the object. Beta-tricalcium phosphate and calcium sulphate resorb faster still. (*ITI Treatment Guide vol 7; Araújo & Lindhe*)

128 **E** Around a quarter of the volume as new vital bone is the expected histological finding in a xenograft-augmented sinus at this interval; published series report vital bone fractions typically in the region of 20 to 30 per cent, with residual particles making up much of the remainder. The features that indicate healthy incorporation are qualitative rather than numerical — new bone in direct contact with the particles, osteoid with active osteoblasts, and no inflammatory infiltrate — and all are present. Option C is the trap for the candidate who assumes that graft must convert to bone in proportion, and who reads 25 per cent as a failing mark; waiting a further six months would not materially change the figure, because the particles are not designed to disappear. (*ITI Treatment Guide vol 5*)

129 **A** Native collagen membranes are degraded enzymatically by macrophage and neutrophil collagenase, typically over about four to eight weeks, which is adequate for a small contained defect but at the short end of what a larger regeneration needs; cross-linking prolongs the barrier at the cost of poorer tissue integration and more exposures. Item B is the one that makes D tempting, and it is false: collagen membranes are limp and collapse onto the graft, so space must be maintained by the graft particles themselves, by tenting screws or by a titanium-reinforced barrier — the commonest reason a regeneration under-performs. Item C describes a non-resorbable membrane; a resorbable one is not retrieved, which is the point of using it. (*ITI Treatment Guide vol 7; Wang & Boyapati PASS*)

130 **E** High-density PTFE has a pore size below about 0.3 micrometres, well under the dimension of oral bacteria, so the membrane cannot be colonised through its thickness. That single property changes the surgical problem: the membrane may be left deliberately exposed and removed at three to four weeks without loss of the graft, which is why it is chosen when closure is doubtful — quite unlike expanded PTFE, whose larger pores become contaminated once exposed. Option A is the attractive answer because cross-linking is a genuine strategy for prolonging a collagen barrier and candidates associate it with resistance to breakdown; but a cross-linked collagen membrane is still resorbable, still degrades once exposed, and carries a higher rate of exposure complications. (*ITI Treatment Guide vol 7*)

131 **C** Only the third addresses the quantity of cement that reaches the tissues. Seating the crown first on an abutment replica or a copy abutment expels most of the excess outside the mouth, so that far less is expressed subgingivally when the crown is finally placed, and it is the single most effective technique described for a deeply placed margin. Item A is the one that makes D attractive, because a low film thickness is a genuine requirement of a luting agent and is what allows complete seating; it says nothing about how much excess escapes into a 2 mm sulcus. Item B aids detection after the event rather than prevention.

Where the margin is this deep, a screw-retained design remains the safer answer. (*ITI Treatment Guide vol 13; Misch*)

132 D Radiopacity varies enormously between luting agents. Zinc phosphate and several resin-modified glass ionomer cements are clearly visible, whereas a number of resin cements approximate the radiopacity of dentine or less and cannot be identified against bone and abutment, particularly in a thin buccal or lingual layer superimposed on metal. A negative radiograph therefore does not exclude retained cement, and the history here — inflammation beginning within months of cementation at a margin 2 mm subgingivally — is stronger evidence than the film. Option A is the conclusion most candidates draw and it is the one that leads to a peri-implantitis being treated with debridement while the cause is left in the sulcus. Cone beam imaging is degraded by scatter from the implant. (*ITI Treatment Guide vol 13*)

Answers — Block G — Treatment planning, SAC, options, consent and referral (Q133–160)

133 D The classification is applied separately to the two aspects and the two need not agree. Surgically, immediate placement in the aesthetic zone is classified as complex: the margin for error in three-dimensional position is small, the facial wall at 1.1 mm is close to the threshold below which it cannot be relied upon, and a thin phenotype punishes any deviation — this is not work for a general dentist with a special interest. Restoratively, a single unit in the aesthetic zone with a high lip line is advanced, since the emergence profile and shade must be developed through provisionalisation but the occlusal scheme and vertical dimension are unchanged. Option B is the answer most candidates give, having correctly learned that the aesthetic zone is never straightforward while forgetting that immediacy raises the surgical grade again. (*SAC Assessment Tool; ITI Treatment Guide vol 3*)

134 D The SAC framework separates site-specific factors from general modifiers, and general modifiers apply to every case in that patient irrespective of how favourable the anatomy is. Smoking is one of the named general modifiers, and the periodontal status of the remaining dentition is another; residual bleeding 5 mm pockets mark disease that is not fully resolved and a patient whose peri-implantitis risk is raised, so both items count. Option E is the choice to beat, and it fails on item C alone: 8 mm of crestal width is generous, not limiting, since a regular-diameter implant of about 4 mm needs roughly 6 mm of ridge to leave a millimetre of bone on each side. Twelve millimetres above the canal is likewise ample. The anatomy here is straightforward; the patient is not. (*SAC Assessment Tool; 2017 World Workshop*)

135 A Do the arithmetic before anything else. At least 1.5 mm of bone must separate an implant from each adjacent root if the interproximal crest, and with it the papilla, is to be maintained, so even the narrowest conventional implant of about 3.3 mm requires something over 6 mm of space. At 5.5 mm at the crest she does not have it, and no augmentation creates mesiodistal room; only orthodontics would. Option C is the attractive error because it is the right answer to the opposite question — virgin adjacent teeth are the classic argument *for* an implant and against a conventional bridge — but a cantilever resin-bonded design removes almost no tooth tissue, so the objection largely disappears. The ridge width can be augmented and the lip line raises risk without prohibiting treatment. (*Misch; ITI Treatment Guide vol 1*)

136 B The shortened dental arch concept holds that around twenty teeth in occluding pairs as far back as the premolars meet the functional and aesthetic demands of most patients, and that further replacement should be justified by a stated benefit rather than assumed. His own history is the strongest evidence available: eight years without drifting, over-eruption or symptoms is a demonstration that the arch is stable in him. Item (iii) is what makes C and E wrong, and it is the item candidates accept because it sounds mechanically obvious; the epidemiological evidence does not support a predictable link between molar loss and temporomandibular disorder. No treatment is a legitimate option here, and it must be presented with its consequences rather than as a failure to treat. (*Käyser shortened dental arch; GDC Standards; Montgomery*)

137 C With two-thirds of the buccal plate already destroyed there is no bony envelope to contain a graft around an immediately placed implant, and the site carries active infection; placing now converts a manageable staged case into an aesthetic failure that is very difficult to retrieve. Grafting the socket at extraction and returning to a healed, contained ridge at about four months allows the implant to be positioned to the planned crown rather than to the bone that happens to remain. Option D is the strong distractor: early placement at six to eight weeks is exactly right when the facial wall is thin *but intact*, and the soft tissue gain in that interval is real — the stem removes the wall that protocol depends on. E allows the ridge to collapse before you intervene. (*ITI Treatment Guide vol 3; Chen & Buser*)

138 B Materiality under *Montgomery* has two limbs: what a reasonable person in the patient's position would attach significance to, and what this clinician should know that this patient would. Her embouchure depends on lower lip sensation, so a complication that most patients would find unpleasant would end her career; the 3.5 mm clearance is workable but the canal is near enough that the risk is genuine

rather than theoretical. Option A is the attractive answer because it is far the most frequent of the five, and candidates equate materiality with frequency; a loose screw is detected at review and retightened in ten minutes. That she has not asked about complications does not reduce the duty — the disclosure is owed whether or not it is requested. (*Montgomery; GDC Standards*)

139 E She is awake, competent and able to make a decision now, so the additional procedure can be consented to contemporaneously and recorded — including the animal origin of both the particulate and the membrane, which some patients refuse and which must be disclosed for every material used. Option D is the sophisticated wrong answer and worth arguing: harvesting locally does side-step the materials discussion, and autogenous bone is the ideal graft, but it still performs an augmentation she has not agreed to, and the volume obtainable from a single osteotomy will not reliably cover a 4 mm fenestration. A is treatment without consent. C discards an otherwise well-placed implant for a problem solved by a two-minute conversation. (*GDC Standards; Montgomery*)

140 C A best-interests decision must consider whether the purpose can be achieved in a way that is less restrictive of his rights and freedom, and an eleven-year-old denture on a resorbed ridge has never been remade. A new denture may solve the problem outright; if it does not, the case for implants is then made on evidence rather than assumption, having also weighed his capacity to clean around abutments and the lifelong maintenance a care home must support. Option A is the trap for candidates who know the Mental Capacity Act only in outline: a registered health and welfare attorney can indeed consent, but the attorney is bound by the same best-interests framework and cannot authorise treatment that fails it. B is unnecessary where an attorney exists, and D abandons him. (*Mental Capacity Act; GDC Standards*)

141 D The hard part of this case is not the graft but the plan: two aesthetic sites in a high lip line, a previously orthodontically managed space of uncertain dimension, and an outcome that depends on where the crowns must sit before anyone decides where bone is needed. Restorative-led assessment sets that target and then commissions the surgery to meet it. Option A is the most attractive because augmentation is the obvious technical requirement and referring only the surgery keeps the restorative work — and the fee — with you; but it reverses prosthetically driven planning, and implants placed to the available bone in the anterior maxilla produce restorations that cannot be made to look right. B addresses a later stage, C proposes a technique before a diagnosis, and E does not resolve the complexity. (*GDC Standards; SAC Assessment Tool*)

142 A Every parameter given is favourable, and the examinable skill is recognising that as readily as recognising complexity. Eleven millimetres of height accommodates a standard implant with the customary 2 mm of clearance above the canal, 7.5 mm of width takes a regular platform, a treated diabetic at 48 mmol/mol is controlled, and mild anxiety in a woman who has already tolerated surgery is managed with time and explanation. Option D is the seductive error and the one that reveals a misread threshold: 42 mmol/mol is the *lower* limit of the pre-diabetic band, a value no treated diabetic reaches, and the figure usually quoted as adequate control for elective surgery is nearer 58. Referring a straightforward case is not a safe default; it is a failure to practise. (*NICE NG28; SAC Assessment Tool; GDC Standards*)

143 E Registrants must give patients a clear written estimate of costs before treatment starts and an updated one whenever the plan or the price changes, so that the patient can decide whether to proceed, defer or decline — a decision she cannot make properly once the graft is done. Option C is the answer that describes what most practices actually do, and it is much better than silence, which is exactly why it is the distractor to beat; a note that you explained the fee is evidence of your recollection, whereas a revised estimate she has seen and agreed to is evidence of hers. B confuses generosity with obligation and would be unsustainable, A invoices for work she never agreed to, and D asks her to endorse a document that no longer describes the treatment. (*GDC Standards*)

144 B The two-implant mandibular overdenture is prosthetically successful and mechanically busy: the nylon or plastic matrices that grip the studs wear, lose retention and are replaced at intervals of months to a couple of years, and that is the commonest maintenance event by a wide margin. Saying so before treatment converts a future irritation into an expected, budgeted appointment. Option C is the good wrong answer and deserves mention in the same conversation — the acrylic is thinned where the housings sit and midline fracture of a lower overdenture is genuinely common — but it is less frequent than matrix change. Survival of interforaminal implants exceeds 95 per cent at ten years, so A is unlikely, and D and E are uncommon in a compliant, well-maintained patient. (*McGill consensus; ITI Treatment Guide vol 4*)

145 D An implant restoration is a maintained prosthesis, not a repair, and the two true items are the ones patients are least often told. Supportive care is lifelong, chargeable and the single best predictor of a healthy peri-implant outcome; technical events — screw loosening, ceramic chipping, decementation, attachment wear — affect a substantial minority over ten years, so the components, if not the implant, may well be worked on. Option E fails on item C alone, and that item is the one candidates wave through

because "ten-year survival" is the figure they have memorised. Ten-year survival of around 95 per cent means the great majority of implants are still in service, not that they are exchanged on a schedule; nothing about osseointegration expires. (*Pjetursson et al. systematic reviews; ITI Treatment Guide vol 12; Montgomery*)

146 A The whole design question turns on one fact not yet established: whether his lip can be supported by teeth alone. An 8 mm discrepancy between crest and incisal position means a fixed prosthesis must carry a large amount of prosthetic material in exactly the region a flange currently occupies, and only a trial set-up made without a flange shows him, and you, what that looks like. It costs little, is reversible, and converts an argument into a demonstration. Option B is very likely to be the eventual answer and is the distractor with real merit — an overdenture with a flange is the classic solution to this discrepancy — but choosing the design before testing the requirement is planning backwards. D is an exposure without a question to answer. (*ITI Treatment Guide vol 4; IR(ME)R*)

147 C A facial wall of 0.7 mm sits below the 1 mm generally required before immediate placement is contemplated; a wall that thin is substantially bundle bone, loses its periodontal supply with the tooth and will resorb whatever is put into the socket. Early placement at four to eight weeks allows the soft tissue to thicken and the socket to remain a contained defect for simultaneous contour augmentation, which is the compromise the protocol exists to provide. Option A is the tempting choice because the stem supplies the two arguments usually advanced for it — 6 mm of apical bone gives primary stability, and she does not want a denture — but neither addresses the wall. Her thick phenotype and medium lip line reduce the consequences of error; they do not change the biology. (*ITI Treatment Guide vol 3; Araújo & Lindhe*)

148 E Subcrestal caries is the finding that makes the tooth unrestorable rather than merely compromised: there is no ferrule, and creating one means either crown lengthening, which on a molar with an existing grade II furcation will expose it further and worsen what is already the hardest area to clean, or orthodontic extrusion, which shortens an already reduced root. Option B is the answer many candidates choose, and it is a genuine predictor of tooth loss, but a grade II furcation in a compliant patient with an otherwise stable mouth is maintainable for years and is not by itself a reason to extract. Bone loss of 45 per cent describes past destruction, an old root filling with no periapical change is not a problem, and a defective crown margin is remade. (*Lindhe; Misch*)

149 B An osseointegrated implant has no periodontal ligament and moves only a few micrometres under load, almost all of it elastic deformation of bone, whereas a tooth is suspended in a ligament that permits displacement of the order of 25 to 100 micrometres. Connect the two rigidly and the tooth depresses while the implant does not, so the implant becomes a fulcrum carrying a disproportionate share of the load; the recognised consequences are cement failure at the tooth retainer, intrusion of the natural abutment, and screw loosening or fracture at the implant. Option E is the distractor with genuine substance, because retrievability is a real objection to any cemented implant prosthesis and is worth raising with him — but it is a property of the retention method, not of the connection. (*Misch; ITI Treatment Guide vol 12*)

150 D He is entitled to hear about the treatment he has asked for and to be assessed for it by someone who provides it, and he is equally entitled to hear about the option he has not asked for, since a two-implant overdenture is transformative, far cheaper and may be all he wants once the difference is explained. Item C is what makes E wrong, and it is the one that sounds responsible: mentorship is a legitimate route to new competence, but it is not a substitute for competence at the time of treatment, and an immediately loaded full-arch reconstruction — where a single implant failure in the first weeks can destabilise the entire prosthesis — is not the case on which to start. Complexity is recognised before it is encountered. (*GDC Standards; SAC Assessment Tool*)

151 A The McGill and York consensus statements are unusually direct: the evidence no longer supports a conventional complete lower denture as the first-choice treatment for the edentulous mandible, and two implants retaining an overdenture should be considered the minimum standard offered. Item (ii) is the one candidates accept, because more implants intuitively means more satisfaction; comparative studies show satisfaction gains that plateau, with no consistent advantage of four over two in the mandible, which is why option C is the commonest wrong answer. Item (iii) fails because the maxilla behaves differently — poorer bone quality, greater leverage and the question of palatal coverage mean more implants are usually needed, and no equivalent two-implant standard exists. (*McGill consensus; ITI Treatment Guide vol 4*)

152 C Seven millimetres of height accommodates a 6 mm implant, and the accumulated trial evidence shows short implants achieving survival comparable to longer implants placed in augmented sinuses over the medium term, with fewer biological complications, less morbidity, less cost and less chair time. The higher crown-to-implant ratio that results has not translated into greater marginal bone loss in these studies. Option B is the strong distractor and is entirely defensible practice — a transcrestal elevation with 7 mm of residual bone, an intact membrane and no septum is within competence and is done every day —

but it introduces perforation risk and a graft in exchange for length that the evidence does not show she needs. A is disproportionate, and D and E do not meet the request. (*ITI Treatment Guide vol 5; Esposito et al., Cochrane review*)

153 E Papilla fill between a tooth and an implant is determined by the attachment on the adjacent tooth, and the governing measurement is the vertical distance from the contact point to the interproximal crest: at 5 mm or less the papilla is present almost always, and beyond that the proportion falls steeply, so at 7 mm a black triangle should be predicted and discussed rather than discovered. No soft tissue procedure reliably rebuilds interproximal height. Option C is the distractor with real merit and is worth arguing: triangular crowns place the contact point more incisally and widen the embrasure, genuinely increasing risk — but the effect is expressed through the same distance, which the stem has already measured. The lip line decides whether it will be seen and her expectations whether it will be tolerated. (*Tarnow; ITI Treatment Guide vol 1*)

154 B Between two adjacent implants there is no periodontal attachment on either side to support tissue, and the crest between them remodels both vertically and horizontally, so the achievable papilla height is only around 3 to 4 mm and an open embrasure in a high lip line is the predictable result. A pontic borrows support from the papilla of the neighbouring implant on one side and the natural tooth on the other. Option A is the intuitive choice — replace each tooth with an implant — and is precisely the design that produces the complaint. D reverses the biomechanics, cantilevering the larger central incisor from an implant in the narrower lateral site. C shares load but does nothing for the interimplant crest, and E modifies bone loss at one implant, not the distance between two. (*Tarnow; ITI Treatment Guide vol 1*)

155 C Biology sets this timetable, not the diary: the peri-implant soft tissue matures for months after provisionalisation, and a definitive crown made at eight to ten weeks in the aesthetic zone will be remade when the margin migrates. The examinable behaviour is to name the constraint before treatment starts and then to make a provisional good enough to be photographed, which is entirely achievable. Option A is the answer given by clinicians who want to please and is where aesthetic complaints begin; the patient hears a promise, not an estimate. B is a sensible clinical sequence that answers a question he did not ask and leaves the wedding unaddressed. D and E defer or divert care he can reasonably have. (*ITI Treatment Guide vol 3; GDC Standards*)

156 A Consent is a process, and what is examined afterwards is the evidence that a conversation took place with this patient about her options, the material risks and the likely costs, recorded when it happened. Option B is the answer most candidates give and it is the weakest protection on the list: a signature proves that a form was signed, not that anything was understood, and a catch-all phrase such as "with grafting if required" is exactly the generic wording that *Montgomery* renders inadequate, since it discloses nothing specific about her case. A leaflet shows what the practice publishes rather than what she was told, a recollection obtained two years later carries little weight, and payment is not agreement to a discussion that never occurred. (*GDC Standards; Montgomery*)

157 B The tipped second molar is the problem from which the others follow. Its crown has rotated into the space, narrowing it occlusally to 6 mm while the roots have moved distally and left 8 mm at the crest, so an implant could be placed but the crown made on it would be too narrow, poorly contoured and a plaque trap, with a compromised contact against a tilted tooth. Uprighting the LR7 orthodontically restores the space, improves the mesial bone contour and often addresses the over-eruption at the same time. Option E is the number most candidates seize on, and it is a measurement of the tipping rather than an independent finding — it describes the consequence while B identifies the cause and, with it, the treatment. (*Misch; ITI Treatment Guide vol 1*)

158 A Only the first statement holds, and it is the point of the tool: a posterior single-tooth case may be straightforward surgically and straightforward restoratively, while an anterior immediate case may be complex surgically and advanced restoratively, and the two grades drive different decisions about who does what. Item (ii) is what makes C the popular wrong answer, because the classification is often taught as a list of procedures; in fact general modifiers — medical status, smoking, growth, aesthetic expectations, compliance and the periodontal condition of the remaining dentition — attach to the patient and can raise the grade of an anatomically simple site. Item (iii) fails because the classification is provisional: new findings, at re-evaluation or at surgery, may raise it, and the plan must change with it. (*SAC Assessment Tool*)

159 A Every cone beam volume must be clinically evaluated in its entirety and the evaluation recorded, whatever the reason it was taken; a finding cannot be ignored because it lies outside the region of interest. Option D is the trap, and it fails on item B: a smooth, dome-shaped antral opacity arising from the floor in an asymptomatic patient has the classic appearance of a mucous retention pseudocyst, an incidental finding of no consequence to a mandibular implant, and an urgent cancer referral would alarm the patient

and waste a pathway. Documented observation, with an ear, nose and throat opinion only if it enlarges, becomes symptomatic or the sinus is to be operated upon, is proportionate. Item C fails on justification: a further exposure must answer a question, and this one has none. (*IR(ME)R; ITI Treatment Guide vol 5*)

160 C Plaque at 55 per cent, smoking and an explicit refusal of supportive care are the three strongest behavioural predictors of peri-implant disease, and the honest position is that the case is not yet plannable — not that it is impossible. Treat the gingivitis, demonstrate a sustained change in his scores, cost the maintenance into the treatment plan so it is not experienced as an extra, and revisit the decision with evidence rather than promises. Option A is the answer that feels defensively sound and is the one to reject: a record that a patient declined an obligation does not make the treatment appropriate, and it would not answer the complaint that follows a peri-implantitis at year four. D forecloses too early and E passes the same problem sideways. (*2017 World Workshop; GDC Standards; ITI Treatment Guide vol 13*)

Answers — Block H — Prescribing, local anaesthesia, drug interactions and sedation (Q161–180)

161 A A 1:80,000 solution is 1 g in 80,000 ml, which is 12.5 micrograms per millilitre, so a 2.2 ml cartridge contains 27.5 micrograms of adrenaline. One cartridge sits comfortably inside the 40 microgram ceiling; two deliver 55 micrograms and exceed it. Option B is the commonest error and is instructive, because "two cartridges for a cardiac patient" is a genuine rule of thumb — it belongs to 1:100,000 solutions, which contain 10 micrograms per millilitre and therefore 22 micrograms per cartridge, so two give 44 micrograms and are themselves marginal. The lidocaine is nowhere near its own limit at 44 mg per cartridge; it is the vasoconstrictor that constrains this appointment, and the practical answer is to use the lower concentration and plan the injections. (*BNF; SDCEP*)

162 D Systemic toxicity is a function of peak plasma concentration, not of the amount deposited in the tissue. Lidocaine is itself a vasodilator, so a plain solution is carried away quickly and produces a high, early peak; alpha-1 mediated vasoconstriction from adrenaline holds the drug at the injection site, which lowers that peak, prolongs the block and improves the surgical field. Option C is the attractive distractor because tissue pH genuinely matters in local anaesthesia — acidity in an infected site increases the ionised fraction and reduces the unionised base available to cross the nerve membrane, and adrenaline-containing cartridges are indeed more acidic for stability. That affects onset and efficacy, not the systemic maximum. B and E describe hepatic handling that adrenaline does not alter. (*BNF; SDCEP*)

163 A Articaine is a hybrid molecule: an amide link, but also a methyl ester group on its thiophene ring which is rapidly hydrolysed by plasma cholinesterase to inactive articainic acid. The consequence is a short plasma half-life of roughly twenty to thirty minutes and much less dependence on hepatic clearance than lidocaine, which is metabolised almost entirely by hepatic cytochrome P450 and whose half-life is measured in hours. Option D is the tempting choice because it pairs the true statement with a plausible elaboration; B is wrong, since a proportion of articaine is still handled hepatically and it is not excreted unchanged renally. C is wrong in the other direction: prilocaine and benzocaine are the classic causes of methaemoglobinaemia, and articaine at high dose has itself occasionally been implicated. The 7 mg/kg maximum still applies. (*BNF; SDCEP*)

164 B Prilocaine is metabolised to o-toluidine, which oxidises the ferrous iron of haemoglobin to the ferric form; methaemoglobin cannot carry oxygen and shifts the dissociation curve leftwards, and the topical benzocaine adds to the load. The signature is cyanosis with a saturation that plateaus around 85 per cent, is unresponsive to oxygen because the pigment rather than the ventilation is at fault, and dark blood that does not redden in air. Management is to stop the drug, give oxygen, arrange urgent transfer and, on specialist advice, intravenous methylene blue. Option A is the strong distractor: agitation is a real early feature of systemic toxicity and seven cartridges is a large dose, but toxicity does not produce oxygen-refractory desaturation in a patient who is ventilating normally, and the ninety-minute delay fits a metabolite, not the parent drug. (*BNF; SDCEP*)

165 C Circumoral tingling, a metallic taste, tinnitus and light-headedness are the classic prodrome of rising plasma concentrations. The excitatory phase comes first because inhibitory cortical pathways are blocked before excitatory ones, and progression is to muscle twitching, then seizure, then central and cardiovascular depression; the response is to stop injecting, give oxygen, monitor and be ready to manage a convulsion. Option E is the good wrong answer, because an intravascular adrenaline injection produces palpitations, tremor and a sense of dread within seconds and would be high on anyone's list — but it drives the pulse up, and hers is 88 and regular with a normal blood pressure. A vasovagal reaction gives pallor, sweating and bradycardia; hyperventilation gives peripheral rather than circumoral tingling with carpopedal spasm. (*BNF; SDCEP*)

166 E Felypressin is a vasopressin analogue and retains weak oxytocic activity, so it is the vasoconstrictor conventionally avoided in pregnancy; lidocaine with adrenaline is the standard choice and is used routinely at every stage of pregnancy. Option A is the answer most candidates give and its premise sounds physiological — high intravenous doses of adrenaline do reduce uterine blood flow — but the microgram quantities in one or two dental cartridges, correctly aspirated, are trivial beside the endogenous catecholamine released by an inadequately anaesthetised patient in pain. C sacrifices duration and haemostasis and raises the peak plasma concentration for a given dose. Treat her, treat her comfortably, and keep the appointment short with her positioned to avoid supine hypotension. (*BNF; SDCEP*)

167 B Gastrointestinal upset without any mucocutaneous, respiratory or cardiovascular feature is a predictable adverse effect of the drug, not an immune-mediated reaction; a genuine type I reaction produces urticaria, angioedema, wheeze or collapse, usually within an hour. The great majority of "penicillin allergy" labels do not survive scrutiny, and the label causes measurable harm by forcing broader-spectrum, less effective and more toxic alternatives. Correct the record and explain why. Option C is the attractive answer because it is what most practices do and it feels like the cautious course; substituting a macrolide on an unverified thirty-year-old label perpetuates the error and brings its own hazards of interaction and Clostridioides difficile. E answers a question that no longer arises, and D is disproportionate for a history that resolves itself. (*BNF; SDCEP Drug Prescribing*)

168 C Clarithromycin inhibits both CYP3A4 and P-glycoprotein, and this patient takes two drugs that depend on them. Simvastatin is metabolised by CYP3A4 and the combination is contraindicated because plasma statin concentrations rise steeply, with a real risk of myopathy and rhabdomyolysis. Apixaban is a substrate of both CYP3A4 and P-glycoprotein; inhibiting both raises its concentration and, with a half-life of about twelve hours and no routine monitoring test, the excess anticoagulation is neither measured nor easily reversed. Option E is the distractor worth arguing, and for a different reason: cefalexin has no relevant interaction, so it is not the answer to the question asked, but a documented anaphylaxis to amoxicillin makes any beta-lactam an unwise choice. Metronidazole or clindamycin fits his spectrum and his drug list. (*BNF; SDCEP Drug Prescribing*)

169 D Warfarin is a racemic mixture and the S-enantiomer is several times more potent than the R form; it is cleared by CYP2C9, which metronidazole inhibits. The active enantiomer therefore accumulates, and the INR climbs over a few days — exactly the interval described. Option B is the distractor that catches most candidates, and it is half-right in principle: broad-spectrum antibiotics can reduce vitamin K production by colonic flora, the mechanism is widely taught, and metronidazole's anaerobic spectrum makes it sound especially plausible here. It is a minor contributor and does not account for a rise of this magnitude. C is wrong because warfarin is cleared hepatically, not renally, and A describes an interaction of little clinical consequence. Choose a non-interacting antibiotic or arrange monitoring. (*BNF; SDCEP anticoagulants and antiplatelets*)

170 D Patient (i) is the classic triple combination: an angiotensin-converting enzyme inhibitor, a diuretic and a non-steroidal anti-inflammatory drug together remove the prostaglandin-dependent afferent arteriolar dilatation that a failing kidney relies on, and acute kidney injury on an already reduced glomerular filtration rate is a well-documented consequence. Patient (ii) has aspirin-exacerbated respiratory disease, in which cross-reactivity between non-steroidal drugs is the rule rather than the exception, so ibuprofen may reproduce the bronchospasm. Option E is the trap, and it fails on item (iii): non-steroidal drugs do raise blood pressure modestly and blunt some antihypertensives, but that is a caution requiring advice and a short course, not a contraindication in a man whose pressure is controlled on a calcium channel blocker alone. Paracetamol, with or without codeine, covers those excluded. (*BNF; SDCEP Drug Prescribing*)

171 B A non-steroidal drug combined with paracetamol is more effective for acute post-surgical dental pain than either component alone, and taking it regularly for the first 48 hours, with the first dose while the anaesthetic is still working, controls pain far better than waiting for it to return. Options A and E are attractive because adding an opioid feels like the correct next rung of an analgesic ladder, and that reasoning is where over-prescribing starts: codeine is a prodrug requiring CYP2D6 activation to morphine, a substantial minority of patients convert it poorly, its number needed to treat as a single agent in acute pain is far worse than ibuprofen's, and it adds nausea, constipation and drowsiness to a patient who has just had surgery. C abandons the regular dosing that makes the combination work. (*SDCEP Drug Prescribing; BNF*)

172 E Glucocorticoids increase hepatic gluconeogenesis and induce peripheral insulin resistance, and a single perioperative dose reliably raises capillary glucose for 24 to 48 hours. In a diabetic patient that is the consideration that governs the decision, but it governs it by prompting a conversation rather than a prohibition: at 52 mmol/mol she is well controlled, so warn her, advise her to monitor, and reconsider entirely if her control were poor. Option A is the distractor most candidates take, because adrenal sup-

pression is the standard anxiety about corticosteroids and it is a real phenomenon — it requires sustained administration, not one pre-operative tablet. B overstates a theoretical effect, C describes an interaction that does not occur, and D invents an indication for antibiotics. (*BNF; SDCEP*)

173 C Patient assessment, not operator enthusiasm, decides the environment in which sedation is provided. Morbid obesity with treated obstructive sleep apnoea and an exercise tolerance of one flight of stairs places him at least in ASA class III, with an airway that is likely to obstruct as soon as he is sedated, a reduced tolerance of any apnoea and a rescue that would be difficult in a dental chair; that combination belongs where anaesthetic support is immediately available. Option B is the thoughtful wrong answer and deserves credit, since inhalation sedation has a far wider margin of safety and does not depress the airway — it may well form part of the eventual plan — but it is frequently insufficient for this degree of anxiety through lengthy surgery, and it does not answer the assessment question the stem is asking. (*IACSD standards; SDCEP*)

174 A Flumazenil has a shorter elimination half-life than midazolam — of the order of an hour against roughly one and a half to two and a half hours for midazolam, and longer still in an elderly patient in whom clearance falls and sensitivity rises. The antagonist can therefore be cleared while active agonist remains, and re-sedation is the recognised hazard; she must be observed for a prolonged period, discharged only when fully recovered, and sent home with a responsible escort and clear written instructions. Option C is the distractor with a grain of truth, since flumazenil can precipitate seizures — but that is described in patients on long-term benzodiazepines or with tricyclic antidepressant overdose, not after a single sedation dose. D overstates a requirement that prolonged recovery in the practice usually meets. (*IACSD standards; BNF*)

175 B Benzodiazepines and opioids depress ventilation by different mechanisms, and their effects on respiratory drive and airway tone are synergistic rather than merely additive; the practical consequences are smaller increments, longer pauses between them, a lower end-point and continuous pulse oximetry with real attention to respiratory rate. Option A is the attractive answer because tolerance to opioids is genuine and sedationists do meet patients who need more midazolam than expected — but tolerance to morphine does not confer tolerance to a benzodiazepine, and the dose-response that matters here is respiratory, not hypnotic. C is simply wrong and dangerous to believe: flumazenil antagonises benzodiazepines at the GABA-A receptor and has no effect on opioids, for which naloxone is the antagonist. (*IACSD standards; BNF*)

176 E All three hold, and each for a different reason, which is what makes the combination defensible rather than obvious. Nitrous oxide is far more soluble in blood than nitrogen, so it diffuses into any closed gas-filled space faster than nitrogen leaves it; pressure rises in the middle ear, and after recent middle ear surgery that is a real hazard. Nitrous oxide irreversibly oxidises the cobalt atom of vitamin B12, inactivating methionine synthase — of no consequence in a replete patient, but capable of precipitating neurological harm where B12 is already deficient. And the gas is delivered through a nasal hood, so a blocked nose simply defeats the technique and drives mouth breathing. Option D is the choice most candidates make, dismissing item C as a nuisance rather than a genuine contraindication to this method. (*IACSD standards; BNF*)

177 D Prosthetic joints are not an indication for antibiotic prophylaxis before dental procedures in current United Kingdom guidance: the evidence does not support a causal link between dental treatment and late prosthetic joint infection, and the harms of prescribing — anaphylaxis, Clostridioides difficile and resistance — are real. Explain that to him, and record it. Option A is the distractor to beat, because deference to a consultant's letter feels both courteous and safe; a generic instruction written at discharge is not a clinical decision about today's procedure, and the prescribing responsibility remains yours. Note the separate point that a single 2 g dose of amoxicillin before implant placement has its own modest evidence base for reducing early failure — but that is a decision about the implant, not about his hip, and the two must not be confused. (*SDCEP; NICE CG64; Esposito et al., Cochrane review*)

178 C Chlorhexidine is an established cause of immediate, IgE-mediated hypersensitivity, and the regulator has warned about it precisely because it is so widely and invisibly used — mouthwashes, skin preparations, lubricants, impregnated dressings and catheters. Sensitisation requires prior exposure, so a history of uneventful use is not reassurance; it is the expected background to a first reaction. Option B is the distractor with surface plausibility, and it fails because mucosal and cutaneous exposure sensitises perfectly well. Management is the same as for any anaphylaxis: adrenaline 500 micrograms intramuscularly without delay, oxygen, lay flat with legs raised, call for help — which is why D is dangerous, since antihistamines and corticosteroids are adjuncts that must never postpone adrenaline. (*MHRA; BNF*)

179 E Cocaine blocks the reuptake of noradrenaline at the synapse and sensitises the myocardium to catecholamines; an adrenaline-containing solution given within about 24 hours of use risks severe hy-

pertension, coronary vasospasm and arrhythmia, and the accepted advice is to defer elective treatment. Option A is the sophisticated wrong answer, and it deserves argument: omitting the vasoconstrictor does remove the pharmacological interaction, and it is what many clinicians instinctively reach for — but it leaves an intoxicated, hypertensive and tachycardic man undergoing elective surgery in a sympathetically driven state, with poorer haemostasis and, importantly, without the capacity to give reliable consent. B is not a workaround, since felypressin is itself a vasoconstrictor with coronary effects. Postpone, explain why without judgement, and see him sober. (*BNF; SDCEP; GDC Standards*)

180 **A** Antibiotics are an adjunct to the removal of the cause, not a substitute for it, and the indications are spreading infection, systemic involvement or a host who cannot contain the infection. Situation (i) has all the features that matter — diffuse spread beyond the alveolus, trismus and fever — and needs an antibiotic alongside urgent attention to the source. Option C is the popular wrong answer and fails on item (ii): a localised collection that has been drained has had its treatment, and adding an antibiotic to effective drainage in a well patient confers no benefit while contributing to resistance and adverse effects. Item (iii) describes expected post-operative discomfort, which is an analgesia and reassurance problem; prescribing for it is one of the commonest unnecessary courses in dentistry. (*SDCEP Drug Prescribing; BNF*)

Answers — Block I — Surgical placement, flaps, timing, loading, suturing and asepsis (Q181–218)

181 **C** Crestal remodelling around a two-piece implant is not only vertical: it extends laterally, roughly 1.3 to 1.4 mm from the implant surface, so when two platforms sit closer than about 3 mm the two zones of resorption overlap and the bone between them is lost rather than merely reduced — this is the observation behind the 3 mm interimplant rule, and the reason the interimplant papilla averages only about 3.4 mm in height against 4.5 mm beside a tooth. Option A is the trap and it is a correct rule applied to the wrong interface: papilla height beside a *natural tooth* does follow that tooth's attachment level, which is why the 4 mm figure in the stem is reassuring for the outer papillae, but between two implants there is no periodontal attachment to support and the interimplant crest alone determines the outcome. (*Tarnow; ITI Treatment Guide vol 10*)

182 **A** The 3 mm figure exists to buy running room: the distance from the platform to the mucosal margin is the space in which the restoration widens from a 4 mm platform to the 7 to 8 mm cervical dimension of a central incisor. Compress that transition into 1 mm and the emergence becomes a shoulder rather than a curve, the mucosa is unsupported, and the abutment–crown junction sits close enough to the margin to shine through a thin phenotype. Option B is the answer for the opposite error and is the one candidates reach for because "too deep" is the mistake they have been warned about; a 1 mm sulcus is if anything easier to clean of cement than a normal one. The orofacial position quoted is correct, the bone is intact, and 40 Ncm is unremarkable. (*Buser et al.; ITI Treatment Guide vol 10*)

183 **B** A buccal wall thinner than about 1 mm is largely non-viable after elevation and drilling: it has little cancellous component and depends on a periosteal supply that surgery interrupts, so it resorbs, and in a thin phenotype with a high smile line the resulting mid-facial recession is visible. The working target is at least 2 mm of buccal bone at the shoulder, which here means simultaneous contour augmentation rather than moving the implant palatally out of its prosthetic position. Option A is the attractive error and it rewards doing the arithmetic badly: 7 mm minus 4.1 mm leaves 2.9 mm of bone in total, so the implant does fit — the problem is that the 2.9 mm is unevenly distributed, not that it is absent. C is false, since the position given is prosthetically correct. (*ITI Treatment Guide vol 10; Lindhe*)

184 **D** The papilla beside a single implant is supported by the attachment apparatus of the *adjacent tooth*, so the measurement that matters is the distance from the contact point to that tooth's interproximal bone crest. Tarnow's data give complete fill in almost every case at 5 mm, in about 56 per cent at 6 mm and in only about 27 per cent at 7 mm, and the same relationship has been shown to hold at implant–tooth sites. Here the UL3 side is at 5 mm and should fill; the UL1 side is at 7 mm and probably will not. Option B is simply the two sides transposed and catches candidates who read the numbers without checking which neighbour is mesial to a UL2. C misstates the rule as though the implant crest governed, which is true only between two implants. (*Tarnow; Choquet et al.; ITI Treatment Guide vol 10*)

185 **E** All three hold, and together they are the whole argument for conservative flap design. A mucoperiosteal flap is an axial-pattern flap fed from its base: the supraperiosteal vessels run apico-coronally, so the base must never be narrower than the free margin and reliefs must diverge apically. Reliefs are placed at line angles over intact bone, away from the membrane, and extended beyond the mucogingival junction so that a periosteal release can be made. Option D is the attractive answer because candidates accept A and C readily and doubt B, reliefs being so routine — but a vertical incision does sever the lateral collateral

supply, which is precisely why a single relief is preferred to two where access allows, and why a flap with two reliefs and a narrow base necroses at its margin. (*Lindhe; ITI Treatment Guide vol 7*)

186 **A** Raising a full-thickness flap over a thin buccal plate strips the periosteum from bone that has no cancellous reserve; transient surface resorption follows, and where the plate is already thin or dehisced the mucosal margin recedes with it. The lesson is that sulcular incisions around aesthetically critical, thin-phenotype neighbours are not free: reliefs are better placed one tooth away and the flap kept off those roots, or a split-thickness approach used over them. Option B is the thoughtful wrong answer — tension genuinely causes marginal ischaemia and is the commonest reason a grafted site dehisces — but tension produces early necrosis and opening of the incision line within days, not a stable 1.5 mm recession on an untouched tooth at six weeks. E is excluded by the stem. (*Lindhe; ITI Treatment Guide vol 7*)

187 **C** The periosteum is the inelastic layer; the mucosa above it stretches freely. Release is therefore achieved by scoring the periosteum alone, horizontally, at the base of the flap and just apical to the mucogingival junction, and then advancing the flap by blunt dissection within the loose submucosa — a technique that gains substantial advancement without perforating the flap. Option A is the error that causes the harm the stem is set up for: a blade taken full thickness at the base of a mandibular premolar flap perforates the flap, bleeds, and threatens the mental neurovascular bundle, which here lies only 9 mm below the crest. D reverses the layers, B places the release over the graft where it achieves nothing, and E describes an unnecessary muscular detachment. (*Lindhe; ITI Treatment Guide vol 7*)

188 **D** The two real costs of a flapless technique are informational and technical: you cannot see the bone, so a buccal fenestration or a knife-edge crest goes undetected, and having not raised a flap you are not in a position to graft what you would have found. Both are reasons the approach demands generous bone width, an ample band of keratinised mucosa and accurate three-dimensional planning before the first drill. Option E fails on item C alone, and that item is the one candidates accept because it sounds like a cost — in fact preservation of the periosteal supply to the crest is one of the *advantages* of not elevating a flap, and is the mechanism behind the reduced early crestal bone loss reported for flapless placement. (*Misch; ITI Treatment Guide vol 11*)

189 **B** A submerged concavity is exactly the anatomy that a flapless technique cannot manage: the crest feels and measures reassuringly wide, but the osteotomy passes into a narrowing corridor and may fenestrate the buccal plate below the level at which any of it can be seen or palpated. A guide constrains the axis but does not tell you whether the wall has been perforated, and if it has you can neither inspect it nor graft it. Option A is the distractor for candidates applying a memorised width rule: 7 mm accommodates a 4.1 mm implant with roughly 1.4 mm each side, thin but workable, and it is the contour below the crest rather than the width at it that decides this case. The 6 mm of keratinised mucosa and 13 mm of height both favour the approach. (*Misch; ITI Treatment Guide vol 11*)

190 **E** Two independent classifications are being tested and candidates routinely blur them. Placement is timed from *extraction*: type 1 is immediate, type 2 is early placement with soft tissue healing at four to eight weeks, type 3 is early placement with partial bone healing at twelve to sixteen weeks, and type 4 is late placement in a healed ridge. Six weeks with mucosal closure but no appreciable bone fill is type 2. Loading is timed from *placement*: immediate is within one week, early is from one week to two months, conventional is beyond two months. Twelve weeks is therefore conventional loading. Option B is the commonest error, choosing the correct placement type and then calling any interval short of three months "early". (*Hämmerle, Chen & Wilson consensus; Gallucci et al.; ITI Treatment Guide vol 3*)

191 **C** Immediate *restoration* and immediate *loading* both describe a prosthesis connected within one week of placement; the discriminator between them is occlusal contact, not appearance, not the patient's use of the tooth and not whether the restoration is provisional or definitive. A crown deliberately taken out of contact in intercuspation and in excursion is an immediate restoration; the same crown in occlusion is immediate loading. Placement on the day of extraction is type 1 throughout. Option E is the seductive answer because the crown is plainly doing work — it supports the mucosa, it is chewed around and the patient is wearing it — but "function" in this vocabulary means occlusal loading, and a restoration out of contact does not qualify. (*Gallucci et al.; ITI Treatment Guide vol 2*)

192 **D** Seven months into a healed ridge is late placement, type 4; the type 3 window closes at about sixteen weeks and requires a socket still showing partial bone healing, which the stem excludes. Six weeks after placement falls inside the one-week-to-two-month early loading window, so this is type 4 placement with early loading — a combination that is well documented in a healed mandibular premolar site with 40 Ncm of primary stability. Option B is the answer most candidates give, and the reasoning behind it is worth naming: they see a seven-month wait before surgery and assume the case must therefore be "conventional". The two clocks are independent. A long delay before placement says nothing about when the implant may be loaded afterwards. (*Gallucci et al.; ITI Treatment Guide vol 2*)

193 E Every published protocol for immediate loading excludes uncontrolled parafunction, and for a mechanically coherent reason: the criterion that matters is not the peak force but the *number and duration* of loading cycles applied to an interface that has no bone-to-implant contact yet, and sleep bruxism delivers exactly that, at night, unguided, and beyond the patient's control. Torque of 45 Ncm and an ISQ of 76 are comfortably above the usual thresholds of about 35 Ncm and 70, so primary stability is not the objection. Option B is the strong distractor because splinting is a genuine criterion in multi-unit work; single unsplinted immediate loading is nonetheless documented and acceptable where stability is high. A 10 mm crown height and a maxillary site are modest additional risks, not disqualifying ones. (*Gallucci et al.; ITI Treatment Guide vol 2; Misch*)

194 A Insertion torque measures one thing: the peak rotational friction between implant and osteotomy wall as the implant is seated. That is a serviceable surrogate for mechanical primary stability, and 45 Ncm clears the 35 Ncm figure customarily required before an immediate restoration is contemplated. It is not a measurement of bone, because the same torque can be produced by an aggressive thread form, a tapered body or a deliberately undersized preparation in mediocre bone — which is why item B fails and why option D, the answer most candidates choose, is wrong. Item C fails for a different reason: torque describes the mechanical situation at time zero and has no consistent relationship with the biological outcome, and very high torque may in fact provoke crestal compression necrosis. (*Misch; ITI Treatment Guide vol 2*)

195 B Resonance frequency analysis measures the stiffness of the implant–bone–transducer unit, and stiffness is not the same in every plane: buccolingual readings are characteristically lower than mesiodistal ones, particularly where the crestal bone is thin, so the direction of measurement must be standardised if serial values are to be compared. A rise from 55 to 68 across eight weeks is the expected signature of secondary stability, mechanical interlock being progressively replaced by bone contact; the trend is far more informative than any single value. Statement (iii) is what makes D and E wrong, and it is the one candidates accept because 55 sits in the "low" band below 60: low stability rules out immediate and early loading, but conventional loading after undisturbed healing remains entirely appropriate. (*Osstell; ITI Treatment Guide vol 2*)

196 C The two tests interrogate different quantities. Torque is the friction generated as threads cut and compress the osteotomy wall, and undersizing a preparation in soft bone raises that friction substantially — which is the whole point of doing it. Resonance frequency analysis measures how stiffly the implant is held by the surrounding bone as a structure, and a thin cortex over sparse trabeculae is not stiff however tightly the implant was screwed in. High torque with low ISQ is therefore a coherent and common finding in the posterior maxilla, not an artefact. Option A is simply the relationship inverted and is chosen by candidates who remember that undersizing matters without remembering which index it moves. D and E describe measurement errors that the stem gives no reason to invoke. (*Misch; Osstell*)

197 D This is the physiological stability dip, and knowing that it exists is what stops a clinician panicking or, worse, intervening. Primary stability is purely mechanical, generated by bone that the implant has compressed and cut; that bone is largely devitalised and is resorbed by osteoclasts over the first weeks, so mechanical stability falls. Secondary stability, from woven bone forming in the healing chambers and later remodelling to lamellar bone, has not yet replaced it. The nadir sits at roughly two to four weeks, which is precisely why early loading protocols specify either within the first week or after six to eight weeks. Option B is the plausible alternative and is genuinely happening, but remodelling to the first thread is a fraction of a millimetre at this stage and would not move the ISQ by nine points. (*Raghavendra, Wood & Taylor; ITI Treatment Guide vol 2*)

198 A The healing interface tolerates a limited amount of relative movement; beyond a threshold in the region of 100 micrometres — figures between about 50 and 150 are quoted — the differentiating mesenchymal cells form fibrous tissue rather than bone, and the implant heals as a fibrous-encapsulated body. A continuous peri-implant radiolucency with a mobile restoration is the radiographic and clinical expression of that. The number to notice in the stem is 25 Ncm: primary stability below the customary 35 Ncm threshold was never adequate to hold an occlusally loaded provisional. Option B is the reflex answer of candidates who associate implant failure with over-compression, and it is a real entity — but it follows torque of 50 Ncm and above, produces crestal saucerisation, and is excluded twice over here. (*Brunski; Misch; ITI Treatment Guide vol 2*)

199 B Eriksson and Albrektsson's work established that bone exposed to 47 degrees Celsius for one minute undergoes irreversible injury, with resorption and fat cell infiltration replacing the expected bone formation; that pairing of temperature and time is the figure implant surgery is designed around, and it is why irrigation must be continuous, drills sharp, and deep osteotomies in cortical bone made in short passes rather than one long one. Option C is the distractor with real provenance and it is the reason this question discriminates: 56 degrees is the temperature at which alkaline phosphatase is denatured and cell

death is immediate, a number from the same body of work — but it is the *higher* threshold, and a protocol built around it would permit sustained heating that is already destructive. A, D and E are not established thresholds. (*Eriksson & Albrektsson; Misch*)

200 A Only the irrigation helps. Its job is not merely to cool but to carry away the bone chips that otherwise pack the flutes and convert a cutting instrument into a rubbing one, which is why the flow must reach the cutting tip and why deep osteotomies need the drill withdrawn intermittently to let coolant down. Item B is the one that makes option D attractive: jumping straight to final diameter feels efficient and shortens the procedure, but it asks a single drill to remove the whole volume of bone, greatly increasing the work done and the heat produced — graded sequences exist for this reason. Item C is wrong in the same direction; higher rotational speed increases frictional heating for a given feed, and osteotomy drilling is conventionally done in the hundreds rather than thousands of revolutions per minute. (*Eriksson & Albrektsson; Misch*)

201 C In type 4 bone the objective is to leave as much bone as possible for the implant to compress and engage. Under-preparation does exactly that: the final drill is omitted or a narrower one substituted, the countersink is not used so the crestal cortex is preserved intact, tapping is avoided, and the implant is allowed to condense the trabeculae as it advances. Option D is the good wrong answer and deserves argument, because a wider implant does engage more crestal cortex and is a legitimate adjunct — but shortening the implant discards the one thing this site has, 11 mm of usable height, and length contributes more than diameter to stability in soft bone. A increases heat without helping, B removes the very interference that generates stability, and E abandons a treatable site. (*Misch; ITI Treatment Guide vol 2*)

202 D Torque above roughly 50 Ncm in dense bone stops being reassuring and starts being harmful: the crestal cortex, which has the poorest vascularity and the least capacity to accommodate strain, is compressed beyond its tolerance, the osteocytes there die, and the bone is resorbed — producing early marginal bone loss around an implant that felt magnificently stable on the day. The correct manoeuvre is to back the implant out, relieve the coronal part of the osteotomy with a countersink or crestal drill so that the platform can seat without crushing the cortex, and reseat. Option A is the belief this question exists to correct, and it is widely held because torque and stability are conflated with prognosis; beyond a point more torque buys damage, not integration. (*Misch; ITI Treatment Guide vol 2*)

203 A A spinning implant has no primary stability, and without it there is nothing to prevent the micromotion that produces fibrous repair — this is not a situation to be managed by waiting. The first-line rescue is to increase the interference between implant and bone, and the stem supplies everything needed to do it: a 9 mm ridge will accept a wider implant, and a wider body of the same length engages fresh, uncut bone throughout the osteotomy. Option B is the honest fallback and is correct if no wider implant achieves stability, which is why it attracts candidates who have learned the failure pathway but not the sequence; it commits the patient to a second surgery and four months that may be unnecessary. C would place the apex within 0.5 mm of the canal, and D and E do not create stability. (*Misch; ITI Treatment Guide vol 2*)

204 B The working minimum between an implant surface and an adjacent root is 1.5 mm, which allows for the lateral component of crestal remodelling and for the periodontal ligament's own blood supply; 0.9 mm is below that, and widening the osteotomy on the present axis will only reduce it further. The moment to correct an angulation error is now, when only a 2 mm pilot channel exists: a side-cutting instrument such as a Lindemann bur, braced against the far wall or a guide sleeve, will translate the osteotomy mesially without the drill simply falling back into the existing track. Option A is the reflex of candidates who confuse the vertical and horizontal problems — shortening the implant does nothing to a convergence that begins at the crest. (*Misch; ITI Treatment Guide vol 11*)

205 E Perforation of the lingual plate in the anterior mandible opens the sublingual space, where the sublingual and submental arteries anastomose; the resulting haematoma expands into loose tissue with no fascial restraint, elevates the tongue and can obstruct the airway within an hour. An altered voice is an ominous sign. This is an emergency in which the only useful actions are local pressure, keeping the patient sitting up, oxygen, and rapid transfer to a unit that can secure an airway — not a wound to be managed in a dental surgery. Option A is the plausible wrong answer, and it has some intuitive force since the implant would compress the osteotomy; it does nothing for bleeding that is already outside the bone, and it wastes the minutes that matter. (*Lindhe; Misch; ITI Treatment Guide vol 11*)

206 C Twelve and a half millimetres of available height less a 2 mm safety margin leaves 10.5 mm, so a 10 mm implant is the longest that fits the plan and an 11.5 mm implant leaves only 1 mm. The margin is not arbitrary padding: it absorbs the measurement error inherent in identifying the canal roof on a cross-sectional image, the discrepancy between the drill stop and the actual apical extent of the cutting flutes, and the fact that drills tend to advance slightly beyond their marked depth in soft cancellous bone.

Option D is chosen by candidates who deduct nothing, or who assume the safety margin is somehow built into the implant length; 1 mm of bone above an inferior alveolar nerve is not a margin, and permanent paraesthesia is one of the few implant complications that cannot be repaired. (*Misch; ITI Treatment Guide vol 11*)

207 D Two properties decide this. The first is capillarity: a braided multifilament wicks saliva and bacteria along its interstices into the wound, and a suture lying over a membrane is a direct conduit to the graft, so a monofilament is required — polytetrafluoroethylene is soft, inert, non-capillary and holds a knot well. The second is retention of strength: a closure under residual tension must be held for ten to fourteen days. Option B is the answer most candidates give, and it is a reasonable general-purpose material, but polyglactin 910 is braided and its coating does not abolish capillarity. A is the worst choice for the same reason in greater degree, and the plain and chromic gut in C and E lose tensile strength unpredictably and provoke a marked inflammatory response. (*Lindhe; ITI Treatment Guide vol 7*)

208 A Tension is the problem, and no suturing strategy overcomes a flap that is too short: blanched margins mean the sutures are pulling the flap against elastic recoil, capillary perfusion pressure is exceeded, the margin becomes ischaemic and it separates. The remedy is to lengthen the flap by releasing the periosteum, then to close in two layers — horizontal mattress sutures placed 4 to 5 mm from the incision to evert the margins and carry the residual tension away from them, with interrupted sutures between to adapt the edges precisely. Option D is the instinct this question is designed to expose, since a gaping wound looks like an under-tied one; tighter knots accelerate exactly the ischaemia that caused the dehiscence. B and C strangulate more tissue, and E prolongs a failed closure. (*Wang & Boyapati PASS; ITI Treatment Guide vol 7*)

209 B Epithelial continuity across an approximated incision is achieved within a few days, but it confers almost no mechanical strength; strength comes from fibroblast proliferation and collagen deposition through the proliferative phase, and a wound at one week holds only a small fraction of its eventual tensile strength. Where a flap has been advanced over a graft there is residual elastic recoil pulling the margins apart, so the sutures are still the structure holding the closure together at the point where a simple, tension-free wound could manage without them. Option A is a true statement and is precisely why sutures are removed *early* elsewhere — it is the argument for the opposite practice, which makes it the most attractive distractor. C and D misdescribe how membranes and grafts behave, and E overstates the ischaemia. (*Lindhe; ITI Treatment Guide vol 7*)

210 C A type N steriliser relies on steam displacing air downwards through an open chamber; it cannot evacuate air from a narrow lumen, so steam never reaches the internal surface and the item is not sterilised however long the cycle runs. HTM 01-05 therefore requires a vacuum (type B) steriliser for hollow, lumened, wrapped and cannulated instruments, with cycle records retained. Option E is the distractor with commercial appeal and it is worth dismantling: buying single-use drills removes one lumened item, but the handpiece and the hollow ratchet shaft remain and still require type B processing, so the problem is not solved. A misunderstands the physics, B substitutes disinfection for sterilisation at a surgical site, and D omits the sterilisation stage altogether. (*HTM 01-05*)

211 D An air turbine vents its exhaust at the head. Directed into an open bony wound, that air is forced along tissue planes and produces surgical emphysema, which at best causes alarming swelling and crepitus and at worst carries oral flora into the deep spaces of the neck, with mediastinitis described. It is the objection that concerns catastrophic harm rather than convenience, and it is why electric surgical motors that vent away from the field are used. Option B is the strong distractor because it is entirely true — a turbine runs at some 300,000 revolutions per minute and cannot be brought down to the few hundred used for osteotomy — but that is a limitation of control which better equipment addresses, whereas the exhaust problem is intrinsic. A is also true and also secondary. (*Misch; HTM 01-05*)

212 A Only the first argument survives. A healing abutment crosses the mucosa immediately over a non-resorbable membrane, and the peri-abutment sulcus provides bacteria with a direct path to a device that has no vascular supply and cannot resist infection — membrane contamination is the commonest way a regenerative case is lost, so burying the implant is justified here. Item B is what makes option D attractive, and it is a plausible-sounding claim that is not true: submergence does not create or preserve keratinised tissue, and the second-stage incision is the opportunity to reposition it, most usefully with an apically positioned flap. Item C fails because randomised comparisons of one-stage and two-stage protocols in native bone show equivalent survival; submergence is chosen for local reasons, not for a survival advantage. (*ITI Treatment Guide vol 7; Lindhe*)

213 A A mucosa-borne saddle transmits every masticatory load directly through the flap onto a graft and membrane that need absolute mechanical quiet; the result is ischaemia at the incision line, dehiscence and exposure. The instruction that works is to keep the denture out entirely for the first day or two, then

to grind the fitting surface generously clear of the site — not to a light relief but to an unmistakable gap — and to line the remaining contact with a soft material, reviewing it as swelling subsides. Option C is the reasoned wrong answer, and the principle behind it is sound, since tooth support is preferable to mucosal support; but adding clasps to an existing acrylic denture does not convert it into a tooth-borne prosthesis, and the saddle still rests on the flap. (*Misch; ITI Treatment Guide vol 7*)

214 **B** A horizontal gap of 2 mm or more at the shoulder is grafted, and the material of choice is a low-substitution xenograft precisely because it resorbs slowly: its function is not to become bone but to occupy volume and resist the buccal contour collapse that follows loss of the bundle bone. At 2.4 mm this site qualifies. Option E is the error that does the most harm and it feels like good engineering — a wider implant would obliterate the gap, but only by bringing the implant surface against a 0.8 mm plate that will then resorb, converting a manageable defect into a dehiscence with visible recession. A is true of osseointegration and false of aesthetics: small gaps do fill with bone, but the buccal contour is lost meanwhile. C resorbs too quickly to maintain volume. (*ITI Treatment Guide vol 10; Araújo & Lindhe*)

215 **C** A drill entering an extraction socket will follow the path of least resistance and skid buccally down the sloping socket wall, and the two things that most reliably ruin an immediate anterior implant are a buccally positioned shoulder and contact between the implant and the thin buccal plate. Starting palatally, and taking primary stability from the 3 to 5 mm of native bone apical and palatal to the socket, controls both. Option D is the sophisticated distractor and it is not wrong as a statement — palatal positioning does move the screw channel towards the cingulum and make screw retention feasible — but that is a welcome consequence, not the reason for the manoeuvre. B inverts the anatomy: the incisive canal lies palatal, and is itself a hazard of over-palatal placement. (*ITI Treatment Guide vol 10; Buser et al.*)

216 **E** Immediate placement is decided on aesthetic risk rather than on surgical feasibility, and the two modifiers that most reliably predict mid-facial recession are a thin mucosal phenotype and a high smile line — the first because there is no tissue thickness to mask the underlying change, the second because the change is displayed. Together they push this case towards early placement at four to eight weeks with the soft tissue matured. Option B is the strong distractor: 1.1 mm is thin, and thin plates do resorb, but a thin *intact* plate is the normal finding in the anterior maxilla and is managed by palatal positioning and gap grafting rather than abandoned. A is the classic trap — a chronic, non-suppurative periapical lesion is not a contraindication provided the socket is thoroughly debrided; acute suppuration is. (*ITI Treatment Guide vol 10; Chen & Buser*)

217 **E** All three are established, and together they replace the marketing account of immediate placement with the biological one. The buccal wall of an anterior socket is composed substantially of bundle bone, a tooth-dependent tissue supplied through the periodontal ligament; it loses its blood supply when the tooth is removed and it resorbs, and Araújo and Lindhe showed that an implant in the socket does not arrest that modelling. Grafting the gap with a slowly resorbing material limits the contour collapse and is worth doing, but the ridge still changes. And wall thickness matters: sockets with more than about 1 mm of buccal bone, which have a cancellous component and a vascular supply independent of the ligament, lose far less than the thin-walled majority. Candidates most often reject (i). (*Araújo & Lindhe; Chen & Buser; ITI Treatment Guide vol 10*)

218 **D** Traceability applies to implantable medical devices, and it applies to all of them: the implant itself, and equally the xenograft and the membrane, each of which is a regulated device with a batch or lot number that must be recoverable if a product is recalled or if a transmissible-agent question arises about material of animal origin. Recording only the implant is the commonest omission. Option E fails on item C, and that item is the one candidates include because it is genuinely important — insertion torque should certainly be in the notes, since it justifies the loading protocol chosen and is invaluable if the implant later fails — but it is a clinical finding, not a device identifier, and no traceability requirement rests on it. (*GDC Standards; FGDP*)

Answers — Block J — Minor hard and soft tissue augmentation (Q219–246)

219 C Autogenous bone is the only graft that is osteogenic, because it alone transfers surviving osteoblasts and osteoprogenitor cells into the defect; it is also osteoinductive and osteoconductive, which is why it remains the reference material. Chips harvested from the osteotomy or with a bone scraper cost nothing and add no second surgical site, and mixing them with a slowly resorbing xenograft combines cellularity with volume stability. Option A is the distractor that catches most candidates, because osteoconduction is a real and important property of the autogenous chips — but it is shared by every particle in the mixture, including the bovine mineral, so it cannot justify adding one component in particular. B describes the xenograft, D describes demineralised allograft, and E is the reverse of the truth: the deproteinised particles are the harder of the two. (*Lindhe; ITI Treatment Guide vol 7*)

220 E Contour augmentation on the labial surface is not a contained defect and the regenerate remodels towards the underlying bone; what preserves the profile in the long term is the persistence of particles that are barely substituted, which is exactly why a deproteinised bovine mineral is the conventional choice for this indication. Option C is the attractive answer because a resorption rate matched to bone formation is the textbook ideal for a graft, and it is the property a fast-resorbing beta-tricalcium phosphate is marketed on; but a material that disappears takes the contour with it in a site with no bony walls to define the volume. A is a genuine biological virtue that is inversely related to volume stability, B belongs to demineralised allograft, and D is a minor handling variable. (*ITI Treatment Guide vol 7; Lindhe*)

221 A Osteoinduction means the recruitment of host mesenchymal cells and their differentiation into bone-forming cells, mediated principally by bone morphogenetic proteins held in the organic matrix. Demineralising an allograft removes the mineral phase and exposes those proteins, so demineralised freeze-dried bone allograft carries some osteoinductive potential, although it varies considerably between donors and processing batches. Option D is the trap and it is chosen often: deproteinised bovine bone mineral has, by definition, had its organic phase removed by heat or chemical treatment, so no protein remains to induce anything — it is purely osteoconductive, and its virtue is volume stability rather than biological signalling. Beta-tricalcium phosphate is a synthetic ceramic and is likewise osteoconductive only. Only autogenous bone is osteogenic. (*Lindhe; ITI Treatment Guide vol 7*)

222 B Removing the mineral phase is what exposes the matrix proteins, and it is also what makes the particles soft. A demineralised allograft therefore trades space maintenance for osteoinductive potential: under the pressure of an overlying flap, and with a collagen membrane that has no stiffness of its own, the graft compresses and the augmented width is lost. Where the volume must be held, a mineralised allograft or a low-substitution xenograft is the more sensible choice, alone or in a composite. Option D is the good wrong answer, because particle size genuinely affects handling, porosity and revascularisation and is a real variable — but it is a second-order effect beside the loss of the mineral skeleton itself. A and E do not occur with processed allograft, and C misstates the time course. (*Lindhe; ITI Treatment Guide vol 7*)

223 D This is the first letter of PASS. Primary wound closure matters because a flap held together under tension is a flap whose marginal circulation is compromised: the edges blanch, the wound line breaks down in the first two weeks, and the membrane and graft are exposed to the oral environment. Releasing the periosteum, which is the inelastic layer, allows the mucosa to advance and the margins to lie passively together. Option A is the attractive alternative because tension really does press the flap onto the graft, and candidates reason from the mechanics they can see; but compression of a particulate graft is not what loses these cases, whereas dehiscence over the barrier is the commonest complication of guided bone regeneration and the one that most reduces the bone gained. (*Wang & Boyapati PASS; ITI Treatment Guide vol 7*)

224 A Stability is the second S of PASS, and its rationale is cellular. The differentiation of mesenchymal cells within a healing wound is strain-dependent: an immobile clot and graft permit angiogenesis and osteoblastic differentiation, whereas repeated micromovement disrupts the fragile new vessels and drives the same cells down a fibroblastic route, so the defect fills with fibrous tissue rather than bone. This is the identical principle to the micromotion threshold in osseointegration. Option C is the strongest distractor, because fixation does help hold the barrier out over the defect and space maintenance is a genuine PASS element — but space can be lost without any movement at all, and it is motion specifically that changes what the cells become. E inverts the requirement: vascular invasion is wanted, not prevented. (*Wang & Boyapati PASS; Lindhe*)

225 B Collagen membranes behave differently from non-resorbable barriers when they become exposed: the exposed portion degrades quickly through enzymatic breakdown, the site epithelialises over, and in the absence of infection most such wounds heal by secondary intention with some reduction in bone gain but without loss of the graft. Topical chlorhexidine, no brushing at the site and frequent review

is the whole of the treatment. Option A is the answer that would be right for an exposed expanded PTFE membrane, where the porous structure is colonised by bacteria and early removal is mandatory, and candidates who have learned that rule apply it here. Re-suturing an ischaemic margin at day twelve will not hold, D adds trauma, and E discards a graft that will probably still take. (*ITI Treatment Guide vol 7; Lindhe*)

226 **C** Dense PTFE has a pore size well below one micrometre, far smaller than a bacterium, so unlike the expanded form it can be left open to the mouth without being colonised through its thickness. That is what permits the open-barrier socket technique: the membrane is tucked under the flap margins, left exposed, and lifted off at about three to four weeks with forceps and no flap, because tissue does not grow into it. Option D is the sophisticated wrong answer — cell occlusion is exactly what a barrier does, and it is the reason the technique works at all, but a native collagen membrane is equally cell-occlusive, so this property does not explain choosing PTFE over it. A is false, since PTFE does not resorb at all, and B is the opposite of the truth. (*ITI Treatment Guide vol 7*)

227 **B** Exposure is the commonest complication of guided bone regeneration and it has a consistent effect: sites that dehisce regenerate measurably less bone than sites that heal closed, even when the outcome is otherwise uneventful and the implant integrates normally. That is the honest thing to record and to tell the patient, and it is a reason to plan tension-free closure rather than to rely on rescue. Option D is the attractive choice, because prescribing an antibiotic feels like doing something useful for an exposed graft; there is no evidence that systemic antibiotics preserve graft volume once a barrier is exposed, and the prescription treats the operator's anxiety rather than the wound. Item A is the rule for an exposed non-resorbable membrane, not a resorbable one. (*ITI Treatment Guide vol 7; Lindhe*)

228 **A** Do the arithmetic before reaching for a graft: 8 mm of height above the canal, less the 2 mm safety margin that is conventionally kept above it, leaves 6 mm of usable bone, and 6 mm implants of a modern moderately rough surface, splinted and restored with a controlled crown height, show survival comparable with longer implants placed in vertically augmented bone — with far fewer complications and no second surgical site. Option B is the answer that rebuilds the anatomy and is the one many candidates choose, because the 4 mm vertical discrepancy invites correction; but vertical augmentation carries the highest complication rate of any regenerative procedure and is not required to restore two teeth here. C refers a case that does not need referring, and D ignores the safety margin. (*ITI Treatment Guide vol 7; Misch*)

229 **E** All three are established, and together they define what the procedure is honestly for. Grafting a socket roughly halves the horizontal dimensional change but does not abolish it, because the bundle bone of the socket wall loses its periodontal ligament blood supply whatever is placed inside. It is a preservation technique and not a regenerative one, so a socket whose buccal plate has already gone needs guided bone regeneration with a barrier, not a graft dropped into an open defect. And the histology is unflattering: grafted sockets contain residual particles and a smaller proportion of vital bone at four to six months than sockets left to heal, which is an argument for allowing more healing time rather than less. Candidates most often reject statement (iii). (*Araújo & Lindhe; Lindhe; ITI Treatment Guide vol 7*)

230 **C** The systematic reviews of unassisted socket healing are consistent: at six months the horizontal reduction of the ridge is around 3 to 4 mm and the vertical reduction around 1 mm, so the loss is predominantly in width and predominantly from the buccal aspect, where the plate is thin and largely bundle bone. That asymmetry is the whole reason implants drift palatally in unplanned cases. Option E is the distractor to beat, and it is half right: the figure of roughly 50 per cent of ridge width lost in the first year is correct and widely quoted, but the timing is inverted — about two-thirds of the change happens within the first three months, and the curve is flattening by six. B reverses the two dimensions and D grossly understates both. (*Tan et al. systematic review; Schropp; Araújo & Lindhe*)

231 **D** Containment is the variable that governs regeneration. A defect enclosed by bony walls stabilises the clot, holds the graft where it was placed, and supplies osteoprogenitor cells and blood vessels from several directions, so it needs little more than a barrier; a broad, shallow, non-contained concavity supplies cells from one surface only and provides nothing to hold the graft against the flap, so it needs a slowly resorbing particulate to support the membrane, or a stiffened barrier, and it heals less predictably. Option A is the plausible alternative, because the amount of implant surface within bone genuinely predicts survival and is what most operators look at — but it describes the implant's prognosis rather than whether the graft will take. B, C and E do not describe containment. (*ITI Treatment Guide vol 7; Wang & Boyapati PASS*)

232 **E** The buccal plate of an upper incisor is thin and composed substantially of bundle bone, which depends on the periodontal ligament for its blood supply and resorbs once the tooth is removed; placing an implant into the socket does not arrest that modelling. Filling the horizontal gap with a barely resorbable

material does not prevent the resorption either — it adds volume on the buccal aspect that survives it, supporting the ridge contour and the overlying soft tissue where the plate has thinned. Option C is the seductive error, because a graft is intuitively assumed to speed healing; histologically the opposite holds, and grafted gaps contain less vital bone at reopening than ungrafted ones. A is untrue, since particulate contributes nothing to stability. (*Araújo & Lindhe; ITI Treatment Guide vol 3*)

233 **B** An apical fenestration with sound bone above it is the signature of the labial concavity that lies apical to the root eminence in the anterior maxilla. If the drill is directed along the axis of the crest rather than along the true axis of the residual bone, the tip perforates the thin labial plate at the depth of the concavity while the coronal part of the osteotomy remains fully enclosed. Reading the cross-sectional CBCT slice, not the panoramic reconstruction, is what prevents it. Option A is the distractor most candidates take, and it describes a real error with a different signature: an over-wide implant produces a crestal dehiscence, an open defect running from the bone crest downwards, not an isolated apical window with intact bone coronal to it. (*ITI Treatment Guide vol 7; Lindhe*)

234 **A** Simultaneous augmentation has two prerequisites: the implant must be placeable in the correct prosthetic position, and it must be primarily stable. Fifteen newton centimetres does not provide the immobility that both osseointegration and regeneration require, and grafting around a mobile implant risks losing both at once — the sensible course is to remove it, graft the site and return in six to nine months to place into consolidated bone. Options B and C are the same finding described two ways, and they are the archetypal indication *for* simultaneous grafting rather than against it, which is what makes them attractive. A thin residual plate is the usual state of affairs in this region, and releasing a flap to obtain closure is routine rather than a warning sign. (*ITI Treatment Guide vol 7; Wang & Boyapati PASS*)

235 **E** Three findings compound here: a deficiency in both dimensions, a span of three units, and a lip line that will display whatever soft tissue result is achieved. That combination sits in the advanced category of the SAC classification and beyond what a general dentist with a special interest should undertake; recognising it, and referring with a clear account of why, is what scores. Option A is the reasoned wrong answer and deserves credit as a thought, because horizontal augmentation with particulate under a membrane is exactly the procedure that is within competence — but it will not address 6 mm of vertical loss, and restoring the case without vertical correction produces long teeth with black triangles in a patient who shows her gingivae. C is contraindicated where height is deficient, and B cannot be done in 3 mm. (*SAC Assessment Tool; ITI Treatment Guide vol 7*)

236 **C** The ramus block is taken lateral and superior to the canal, but the canal runs closer to the buccal cortex as it passes forward through the ramus and body, and it is the depth of the inferior osteotomy cut that determines whether the nerve is reached; permanent altered sensation of the lower lip is the complication that defines this procedure's risk profile, and it is a large part of why block harvesting is referred rather than attempted in general practice. Option A is the strong alternative and it is genuinely at risk, since the lingual nerve may lie against or above the lingual crest at the third molar and can be caught during flap reflection or by a bur that perforates lingually — but the nerve the harvest itself approaches is the inferior alveolar. (*Lindhe; ITI Treatment Guide vol 7*)

237 **B** The mental foramen usually lies beneath or between the premolar apices, and the nerve frequently loops anteriorly within bone before turning back to emerge, so the trunk and its branches lie immediately deep to the periosteum in precisely the field that a releasing incision crosses. The nerve must therefore be identified before any sharp release, and the periosteum scored superficially away from it or the flap advanced by blunt dissection around it. Option A is the distractor with real substance, because the mentalis is a genuine constraint on flap design and its detachment causes chin ptosis — but that concerns the symphysis, one tooth further forward, and a muscle is a recoverable problem whereas a transected sensory nerve is not. (*Lindhe; ITI Treatment Guide vol 7*)

238 **D** The evidence supports the soft tissue and patient-centred associations rather than the skeletal one. Implants with less than about 2 mm of keratinised mucosa show more plaque, more bleeding, more brushing discomfort and more mucosal recession, and the mechanism is intuitive: a mobile, non-keratinised margin is uncomfortable to clean and is pulled away from the abutment by muscle activity. Item C is what makes option E wrong, and it is the item candidates accept because it feels like the logical endpoint of the same argument; the association between keratinised width and progressive bone loss is inconsistent across studies, and grafting has not been shown to arrest bone loss. Graft this man for comfort, cleanability and stability of the margin, and say so honestly. (*2017 World Workshop; ITI Treatment Guide vol 13*)

239 **D** Two problems must be solved: there is too little keratinised tissue, and the vestibule is too shallow to keep what there is away from muscle pull. An apically positioned split-thickness flap deepens the vestibule and the free gingival graft, carrying its own epithelium from a keratinised donor site, reliably

produces a band of keratinised, immobile tissue at the margin. Colour match is imperfect, which does not matter at a lower molar. Option A is the answer candidates reach for because connective tissue grafting is the more fashionable technique and is genuinely superior for aesthetics — but a subepithelial graft increases tissue *thickness*, and covered by mucosa it does not create a band of keratinised tissue where none exists. B and E address recession, and C is not a viable donor. (*Lindhe; ITI Treatment Guide vol 13*)

240 **A** The greater palatine vessels run forward in the groove between the alveolar process and the horizontal plate, and their distance from the gingival margin varies directly with the height of the vault — of the order of 12 mm or more in a high vault but as little as about 7 mm in a shallow one. In a flat palate the safe working zone is therefore narrow, and a harvest extended more than roughly 8 mm apical to the margin risks the artery, with brisk bleeding that is awkward to control and best managed by pressure, suturing and a stent. Option B is the attractive distractor because glandular and fatty content is a genuine determinant of graft quality — but it increases posteriorly with site, not with vault depth. (*Lindhe*)

241 **C** The overlying flap is a partial-thickness flap and survives on the vessels within it. Dissect it thinner than about a millimetre and it becomes translucent and ischaemic, sloughs in the first week, and converts a closed donor wound that heals quietly into an open, painful one that takes several weeks to epithelialise. Keeping it at 1 to 1.5 mm is the compromise between flap survival and useful graft yield. Option B is the plausible alternative and names the complication everybody fears, but the artery is avoided by controlling how far *apically* and how deep the dissection extends, not by the thickness of the superficial flap. D is not a consideration when planning a single harvest, and E is governed by the incision design. (*Lindhe*)

242 **A** Tuberosity connective tissue is dense, rich in collagen and relatively poor in fat and glandular tissue, and it is consequently more stable in volume after grafting than lateral palatal tissue — which is its principal attraction, along with a donor wound that is usually less uncomfortable. Option D is the trap and it fails on item B, which is the assumption candidates make from the bulk that a tuberosity appears to offer: the harvest is in fact limited and highly variable, depending on the size of the tuberosity and on whether the third molar is present and erupted, as it is here, so a large graft cannot be relied upon. Item C is wrong on technique, since tuberosity grafts are taken with epithelium that must be removed. (*Lindhe; ITI Treatment Guide vol 13*)

243 **E** The supracrestal soft tissue attachment around an implant requires a certain minimum vertical dimension, of the order of 3 mm, and it will establish that dimension whatever the starting tissue height. Where the mucosa is thin vertically the space is found by resorbing crest, and prospective work has repeatedly shown greater early marginal bone loss at implants placed in tissue thinner than about 2 mm, an effect that thickening the tissue at placement, or a platform-switched connection, will reduce. Option D is the attractive alternative because thin phenotype and recession are properly linked in the mind — but recession is governed principally by the buccal tissue thickness and the buccolingual position of the implant, whereas the crestal consequence is bone loss. (*Linkevicius; 2017 World Workshop*)

244 **B** The buccal volume is at its greatest immediately after the flap is raised and diminishes thereafter, so tissue added at placement is added before the modelling of the thin labial plate occurs and before the provisional begins to shape an emergence profile through it. The flap is already elevated, no additional surgical episode is imposed, and the graft matures during the integration period. Option C is the strong distractor and it is a legitimate technique — grafting at uncovering is exactly what to do when the deficiency is only recognised at that point — but by second stage some of the labial change has already happened, and you are restoring volume rather than protecting it. D and E treat an established deformity, and A grafts a site before it has been planned. (*ITI Treatment Guide vol 1; Lindhe*)

245 **D** For the first two to three days the graft has no circulation of its own: it is kept alive by plasmatic circulation, an exudate absorbed from the recipient bed that supplies enough nutrition to maintain the cells but not enough to tolerate any insult. Revascularisation by ingrowth and anastomosis follows over roughly the next week, and organic union with fibrous attachment completes the process. Everything in the technique follows from this: an immobile graft, close adaptation, firm pressure to abolish dead space, a well-vascularised bed, and a graft thin enough for diffusion to reach its full thickness. Option B is the attractive answer because inosculation is a real event, correctly named, but it belongs to the second phase, not to the first 24 hours. (*Lindhe*)

246 **D** Platelet-rich fibrin is produced by centrifuging the patient's own blood without anticoagulant, yielding a fibrin clot that concentrates platelets, leucocytes and their growth factors; it is autologous, cheap and free of the transmission and regulatory issues of exogenous materials. The evidence is best for soft tissue outcomes — palatal donor site healing and early wound closure — and is far weaker and less consistent for hard tissue gain. Item C is what makes option E wrong, and it is the claim most often made to clinicians: a fibrin matrix has no mineral scaffold and no space-maintaining capacity, so

it cannot substitute for a particulate graft in a dehiscence, and its growth factor content does not make it osteoinductive in the sense that term properly carries. (*ITI Treatment Guide vol 7*)

Answers — Block K — Restoring implants, occlusion and the laboratory (Q247–280)

247 A Every option is a true description of the closed-tray technique, which is what makes the question discriminating; only one of them generates error. In the closed tray the copings stay in the mouth when the impression is removed, so they must be unscrewed, connected to analogues and pushed back into their imprints by hand. Any rotation or incomplete reseating is transferred to the cast, and a small angular error at the platform becomes a large linear error at the occlusal end of a divergent implant. Option B is the attractive answer because elastic recovery genuinely distorts elastomeric impressions, but the deformation with a modern silicone is measured in a few micrometres and applies equally to the open tray. Divergent multiple implants call for an open-tray impression with splinted copings. (*Misch; ITI Treatment Guide vol 8*)

248 C Autopolymerising acrylic contracts as it sets, and in a bulk mass spanning an arch that contraction pulls the copings towards one another by enough to destroy the inter-implant relationship the splint was made to preserve. Sectioning the set splint releases the accumulated strain, and rejoining with a minimal volume of fresh resin means that only that tiny volume can shrink. Waiting up to 24 hours before impressing achieves something similar. Option B is the most attractive distractor because die stone expansion is a real dimensional error of the order of a tenth of a per cent, and candidates who have learned that number reach for it; it is a laboratory variable already compensated in the product and it has nothing to do with the resin. (*Misch; ITI Treatment Guide vol 8*)

249 A The jig is a physical index of the cast tried against the patient: if it seats on all five implants without rocking and without a gap when only one screw is tightened, the analogue positions are a fair reproduction of the intraoral positions, and the framework can be made with confidence. Option D is the trap and is chosen by most candidates, because verification and passivity feel like the same idea. They are not: the framework does not yet exist, it will be milled or cast with its own dimensional errors, and it must be verified separately at try-in. The vertical dimension is a wholly different record. A jig verifies the cast, not the prosthesis. (*Misch; ITI Treatment Guide vol 8*)

250 D This is the one-screw or Sheffield test, and a rocking framework with a 0.3 mm gap has failed it. Sectioning at the point of discrepancy, luting the segments together intraorally with a minimal volume of resin and returning them for rejoining is the established correction; if the discrepancy is gross, a new splinted impression or verification jig and a remake are needed. Option A is the answer that ruins frameworks and is the one taken in practice: tightening the remaining screws makes the gap disappear visually while storing that strain permanently in the screws, the abutments and the crestal bone, and the radiograph then looks reassuring. Torque cannot correct misfit, it can only conceal it. (*ITI Treatment Guide vol 8; Misch*)

251 D Intraoral scanners build an arch by stitching overlapping images, and the error of each registration accumulates along the scan path. A long edentulous span offers no stable, irregular landmarks to correct that drift, which is why full-arch implant scanning remains the least reliable application of the technology; divergence hides part of the scan body geometry in undercut, so the software has less of the cylinder from which to compute the axis. Item C is what makes E wrong, and it reverses the truth: a polished titanium surface scatters and blooms under structured light, and matt polymer scan bodies were introduced precisely to reduce that error. Splinted conventional impressions and a verification jig remain the safer route for long spans. (*ITI Treatment Guide vol 11*)

252 B In an implant impression the material has one job that outranks the others: to hold rigid metal copings in exactly the relationship they occupied in the mouth while the tray is pulled off and carried to the laboratory. A stiff material — the medium- or heavy-bodied polyether, or a rigid silicone — resists the flexure that would otherwise let a coping tip as the impression passes over adjacent teeth and undercuts, and it is the property that most distinguishes an implant impression from a crown impression. Option A is the most attractive alternative because hydrophilicity is what governs surface detail at a subgingival margin, and it is the property candidates have learned to prize; here the margin detail is carried by the machined coping, not by the material. (*Misch; ITI Treatment Guide vol 8*)

253 E A wedge-shaped gap that is open on one side and closed on the other is the classic radiographic sign of incomplete seating, usually caused by soft tissue, a fragment of provisional cement or bone trapped at the interface, or by a coping that was never fully engaged. The restoration must be removed, the interface inspected and cleaned, and the crown reseated and re-radiographed. Option D is the strong distractor, and in most clinical settings it would be the right answer, because an oblique beam readily creates an apparent gap; the stem forecloses it by stating that a paralleling technique was used with the beam perpendicular to

the implant axis, and a candidate who does not use that datum answers a different question. A cement-free screw-retained crown excludes C. (*Misch; ITI Treatment Guide vol 8*)

254 A The reviews are consistent and the pattern is worth quoting accurately: survival of the implant and of the restoration is broadly comparable, screw-retained restorations show more technical complications — screw loosening and ceramic chipping predominantly — and cement-retained restorations show more biological complications, including the ones that end in implant loss. The asymmetry matters clinically because a loose screw is retrieved and retightened at one appointment, whereas submucosal cement produces bone loss that may not be recoverable. Option E is the attractive answer because "no significant difference" is the phrase candidates associate with this comparison and because survival really is similar; it discards the complication data, which is the part that should change practice. (*Sailer et al.; Wittneben et al.; ITI Treatment Guide vol 12*)

255 D Both surviving items act on the volume of cement that can escape. Seating the crown first on an extra-oral replica expresses most of the excess before the crown ever enters the mouth, and a shallow margin means that whatever does escape is accessible to an explorer, floss and a curette. Item C is what makes E wrong, and it is the one candidates accept without examining it: a radiopaque cement changes nothing about how much is extruded, it only makes some of what remains visible, and only on the proximal surfaces that a two-dimensional radiograph can show. Detection is not prevention. Where the margin cannot be raised, the honest answer is to abandon cement retention and use a screw or an angulated screw channel. (*Linkevicius; ITI Treatment Guide vol 13*)

256 B The angulated screw channel redirects the access channel, in most systems by up to about 25 degrees, so that the screwdriver enters palatally while the screw still engages the implant along its own axis. The gain is that the two advantages usually traded against one another are kept together: the restoration remains screw-retained and therefore retrievable, and there is no cement anywhere near the mucosa. Option E is the seductive wrong answer because it appears to say the same thing, and candidates read it as a paraphrase; it is self-contradictory, since a screw-retained restoration has no cement margin to place deeply. The angled driver interface generally accepts the manufacturer's stated torque rather than a higher one, which disposes of A. (*ITI Treatment Guide vol 11*)

257 C The critical contour is the roughly one millimetre of the transmucosal surface immediately apical to the mucosal margin, and it is the part that decides where the margin sits. Making it more convex presses the margin apically; making it flatter allows the margin to creep coronally. The subcritical contour lies apical to that and governs tissue volume and papilla support, usually by being made concave. Option A is the intuitive error and the one most candidates take, reasoning that removing material must let the tissue move; it does, but coronally, which is the direction that has already caused the problem. Expect no more than about a millimetre, only in a thick phenotype, and stop if the tissue blanches for more than a few minutes. (*Su et al.; ITI Treatment Guide vol 11*)

258 E Emergence angle is measured between the transmucosal surface of the restoration and the long axis of the implant, and an angle greater than about 30 degrees has been associated with a significantly higher prevalence of peri-implantitis at bone-level implants, with the association strengthened when the profile is also convex. The mechanism is mechanical rather than mysterious: an overhanging, rapidly flaring contour creates a subgingival ledge that no brush or floss can reach, and it displaces the mucosa outwards. Option A is the best distractor because a small platform under a wide table is a genuine biomechanical mismatch, but it produces screw and bone loading problems, not a localised inflammatory lesion in a patient who cleans well. (*Katafuchi et al.; 2017 World Workshop*)

259 B This is implant-protected occlusion expressed as something you can actually test. The crown is relieved by roughly 25 to 30 micrometres relative to the natural teeth, so that at light closure the teeth hold the foil and the implant does not; under a firm clench the teeth intrude within their periodontal ligaments and the implant crown comes into contact and shares the load. Option C is the attractive answer, because permanent infraocclusion looks like the safest possible protection; it is not, since a crown that never contacts allows the opposing tooth to overerupt, loses its proximal relationships and transfers the whole load to the neighbouring teeth. Excursive contact on a posterior implant, as in E, is avoided altogether. (*Misch; ITI Treatment Guide vol 12*)

260 A The numbers make the argument. A natural tooth moves axially by something in the order of 25 to 100 micrometres under functional load, because the periodontal ligament allows it; an osseointegrated implant moves by about 3 to 5 micrometres, which is essentially the elastic deformation of bone. Contacts set equal at light closure therefore become unequal the moment the patient bites, with the implant taking a disproportionate share, and the deliberate relief simply anticipates that. Option D is a true statement and a good one — osseoperception is blunted and patients do not feel an overloaded implant as they would a

high restoration — but it explains why the clinician cannot rely on the patient's report, not why the contact is lightened. (*Misch; Lindhe*)

261 C When the implant occupies the canine position the usual rule — keep implants out of excursion — cannot be applied literally, because the tooth that would otherwise guide has been replaced. The workable compromise is to spread the guiding contact across the implant crown and its neighbours, and to make the path as flat as the overbite and the aesthetics permit, so that the horizontal component of force at the implant is minimised rather than concentrated. Option B is the distractor with real merit and deserves argument: converting to group function does unload the implant, but it loads posterior teeth laterally throughout every excursion and is rarely achievable in an intact, unworn arch without extensive adjustment. Option E abandons function altogether. (*Misch; ITI Treatment Guide vol 12*)

262 D A cantilever is a class I lever. The implant nearest the cantilever acts as the fulcrum and is driven apically into compression, while the implant furthest from it is lifted, so that its abutment screw is placed in tension — which is why the far screw is the component that loosens or fractures, and why the failure often appears at the implant that seems least involved. The magnitude scales with the ratio of the cantilever length to the inter-implant distance, so the same 8 mm cantilever is far more damaging on closely spaced implants than on widely spaced ones. Option A is the plausible error, and it is the answer a candidate gives who has thought about load sharing but not about the lever. (*Misch*)

263 A The anteroposterior spread rule is the only one of the three in common use: with a spread of 12 mm it gives a theoretical maximum around 18 mm, and most clinicians would work well inside that where the opposing arch is a natural dentition, where the patient is a bruxist, or where the implants are short or narrow. Option D is the attractive answer because implant length feels as though it ought to govern a lever, and candidates who half-remember that longer implants resist bending better select it; implant length has repeatedly failed to predict cantilever behaviour, whereas the geometry of the supporting polygon does. Occlusal table width matters to off-axis load, not to lever length, which disposes of C. (*Misch; ITI Treatment Guide vol 4*)

264 B The clinical data have been consistent and slightly counter-intuitive: studies of short implants with high crown-to-implant ratios have not shown the marginal bone loss that the biomechanical argument predicts, while screw loosening, ceramic chipping and abutment problems do rise, because the tall crown is a vertical cantilever that multiplies any off-axis force into a bending moment at the screw joint. Option A is the answer most candidates give, and it is the reasoning taught from first principles rather than from outcomes; bone appears to adapt to the load in a way that the screw joint cannot. The practical response is not to abandon the plan but to narrow the occlusal table, remove excursive contacts and warn the patient about maintenance. (*Misch; ITI Treatment Guide vol 12*)

265 E All three hold, and a candidate who stops at D has answered a slightly different question. The provisional is there to shape the tissue, not to function, so it is relieved from every contact; it is screwed rather than cemented because cement cannot be retrieved from a fresh socket wall and because the crown will need repeated removal and modification; and it is only justified at all where primary stability is sufficient to prevent micromovement during healing, conventionally an insertion torque of at least 35 Ncm or an equivalent resonance frequency value. The stem supplies 40 Ncm precisely so that the criterion can be tested against a number. Option D is chosen by candidates who read C as a surgical rather than a restorative criterion. (*ITI Treatment Guide vol 3; Chen & Buser*)

266 A The conditioned sulcus begins to collapse within seconds of the provisional being removed, so the sequence matters more than the technology: the mucosal contour is captured first, in the moments after removal, and the scan body is then seated and scanned to record the platform position and orientation, with the two data sets merged. In a conventional workflow the same problem is solved by copying the provisional's subgingival form into a customised impression coping. Option D is the attractive answer because a stock coping is what candidates associate with implant impressions; a stock coping is a circular cylinder of arbitrary diameter, and inserting one is the fastest way to obliterate the tissue architecture you spent ten weeks creating. (*ITI Treatment Guide vol 11; Su et al.*)

267 C Chipping of the veneering layer is the commonest technical complication of implant-supported fixed prostheses, and it happens because the veneering porcelain has a flexural strength around an order of magnitude below that of the framework beneath it. Removing veneering ceramic from the loaded occlusal surface — a metal occlusal table, or a monolithic ceramic — removes the site of failure altogether. Option E is the attractive error and is exactly backwards: unsupported bulk porcelain fractures more readily, not less, and uniform support by a correctly contoured framework is what protects it. A larger connector protects against catastrophic fracture through the connector rather than against surface chipping, and slow cooling rather than a lower firing temperature is what limits residual stress. (*ITI Treatment Guide vol 12; Misch*)

268 D Zirconia is far harder than titanium, and a zirconia abutment articulating directly against a titanium implant abrades the connection under cyclic load, producing wear particles, increasing microgap and micromovement, and concentrating stress at the thin zirconia collar where one-piece abutments have shown higher fracture rates. Bonding the zirconia to a titanium base restores a metal-on-metal joint of matched hardness. Option C is the most attractive distractor because shade at the margin is the reason zirconia abutments were introduced at all, and candidates reach for it automatically; the titanium base in fact makes the margin marginally greyer, which is a true statement pointing in exactly the wrong direction for the question asked. (*ITI Treatment Guide vol 11; Misch*)

269 B Only about a tenth of the torque applied to an abutment screw is converted into preload; the remaining nine tenths is spent overcoming friction under the screw head and along the threads. Because friction dominates the equation, small differences in surface condition — wear, loss of a gold or diamond-like coating, contamination, lubrication — produce large differences in the preload achieved at the same torque setting, which is the argument for a new screw at definitive fit and against improvised lubricants. Option A is attractive because screw elongation really is the mechanism that generates preload, but that elongation is elastic and is intended; the target is roughly 60 to 75 per cent of the screw's yield strength, well short of plastic deformation. (*Misch; ITI Treatment Guide vol 12*)

270 A Repeated tightening wears the thread flanks and abrades any surface coating, so the friction coefficient of a used screw differs unpredictably from new and the preload achieved at the specified torque falls; micro-deformation accumulated over eight cycles also shortens fatigue life. A new screw costs very little against a fractured fragment in the internal threads. Option D is the distractor that catches good candidates, because retorquing after about ten minutes to compensate for embedment relaxation is correct practice and should still be done — with the new screw rather than instead of it. Over-torquing, improvised lubrication and waiting for a screw to fail are all ways of managing a component that should simply have been replaced. (*Misch; ITI Treatment Guide vol 12*)

271 E Preload is stored as elastic elongation of the screw, and the manufacturer's torque value is chosen to reach roughly 60 to 75 per cent of the yield strength of that particular screw in that particular joint. Torquing half as much again takes the screw into plastic deformation: it does not recover its length, the clamping force it holds becomes unpredictable, and it is left with a shortened fatigue life under cyclic occlusal load. It should be replaced. Option A is the thoughtful wrong answer, because deeper seating of a conical abutment under load is real and is why some systems specify a second torque application; that is a designed behaviour at the specified value, not a justification for exceeding it. Implant mobility from stripped threads is not a consequence at these magnitudes. (*Misch; ITI Treatment Guide vol 12*)

272 C In an external hexagon the abutment screw is the component that resists bending, so every off-axis load is carried by the screw and loosening is common. An internal conical connection transfers bending to a long frictional interface between abutment and implant wall, so the screw is left doing what it does best — clamping — and clinical reviews consistently report markedly less screw loosening. Option D is the good wrong answer and deserves credit, because narrowing the occlusal table genuinely reduces the bending moment applied and should be done anyway; it modifies the load rather than the joint's capacity to resist it. A gold screw achieves higher preload for the same torque and is a real measure, but a lesser one. (*Misch; ITI Treatment Guide vol 12*)

273 E All three are required, and the combination is defensible item by item. Without the system, platform and connection the technician cannot select a compatible component, and a mismatched interface is a recognised cause of misfit and screw failure. Margin position is a clinical decision that cannot be delegated: the technician has no way of knowing where the mucosal margin lies or how deep you are willing to go, and an unspecified margin is how crowns arrive with a 3 mm subgingival finish line. And an item of contaminated work is a hazard to the technician, so a disinfection statement travels with it as a matter of law and of courtesy. Option D is chosen by candidates who treat cross-infection as a separate subject from prosthodontics. (*GDC Standards; Medical Devices Regulations*)

274 B An implant-borne restoration made for an individual patient is a custom-made device, and the laboratory must supply a written statement with it identifying the device, the materials used, the patient and the prescribing clinician. That statement is retained in the clinical record, and the patient is entitled to a copy on request — which is exactly the request being made here. Option D is the attractive answer because professional registration feels like the sort of thing that ought to be documented, and it is: it is a matter for the laboratory's own governance and for your due diligence in selecting it, not an item that accompanies each device. A certificate of conformity relates to CE or UKCA marked mass-produced components, not to a custom device. (*Medical Devices Regulations; GDC Standards*)

275 D Two of the three findings point one way and one points firmly the other, which is the whole point of the question. Limited interocclusal space favours a screw, because a cement-retained crown must

accommodate an abutment of sufficient height to retain it, a cement space and an adequate thickness of restorative material, and 4.5 mm does not comfortably allow all three. A margin 2.5 mm subgingival favours a screw because cement at that depth cannot reliably be removed or detected. Item C is the one that fails and makes E wrong: an unfavourable emergence is the classic indication for cement retention, or for an angulated screw channel, since a screw access opening through a buccal cusp is neither aesthetic nor mechanically sound. (*ITI Treatment Guide vol 8; Linkevicius*)

276 **A** The weakest element in this arrangement is the complete upper denture, which depends on seating forces and a peripheral seal, and any scheme that disoccludes the posterior teeth during excursion will tip it. Balanced contacts maintain simultaneous bilateral occlusion through excursions, and lingualised contact directs load palatally over the residual ridge rather than buccally. Option D is the attractive answer because mutual protection is the ideal in a natural dentition and candidates apply it reflexively; here it would unseat the denture the moment the patient moved laterally. The other point to make in planning is the risk of anterior maxillary ridge resorption when a fixed mandibular reconstruction opposes a complete denture, which is a reason for careful review. (*ITI Treatment Guide vol 4; Misch*)

277 **E** The peri-implant mucosa forms an epithelial attachment and a zone of connective tissue against the abutment surface, and each disconnection tears that seal; it re-forms, but a little further apically, and the crest follows. Trials and meta-analyses of the one-abutment-one-time approach report a small but consistent advantage of the order of a few tenths of a millimetre of crestal bone at one year. Option A is the strong distractor because it is true — a definitive contour does stabilise the margin sooner, and it is a real practical benefit — but it describes a soft-tissue convenience, whereas the question asks for the strongest argument and the biological one is stronger. Preload and component selection are not affected. (*Abrahamsson; ITI Treatment Guide vol 11*)

278 **B** A small chip with no metal exposure, no symptom and no aesthetic consequence needs the fracture made smooth and, far more importantly, needs its cause removed — here a lateral contact on a cusp of an implant-borne pontic, which is precisely the load that implant-protected occlusion sets out to avoid. Recording the change and reviewing it completes the management. Option C is the attractive error, and it is over-treatment: remaking a sound prosthesis in a stronger material without correcting the excursive interference treats the material as the problem when the occlusion is the problem, and exposes the patient to cost and risk. Option E removes the wrong contact, since the interference described is excursive rather than intercuspal. (*ITI Treatment Guide vol 12*)

279 **C** A fractured screw fragment is usually held by friction alone once the preload has gone, and gentle anticlockwise ultrasonic energy applied through a fine tip, under magnification and with good irrigation, will often walk it out without touching the threads. If that fails, a proprietary retrieval kit is the next step. The governing principle is that the internal threads are irreplaceable: damage them and the implant, though perfectly integrated, becomes unrestorable. Option A is the plausible wrong answer because slotting is exactly how a fractured post is retrieved from a root canal, and the technique transfers badly — the working space is a few millimetres wide, the bur cuts titanium threads as readily as it cuts the fragment, and there is no direct vision. (*ITI Treatment Guide vol 12; Misch*)

280 **E** The 10 micrometre figure was an engineering aspiration rather than a validated biological threshold, and it is not achieved by any routine clinical workflow; measured discrepancies in clinically successful frameworks are commonly several tens of micrometres, and no study has established a level of misfit at which bone loss or loss of osseointegration follows. What misfit does correlate with is technical failure — screw loosening, screw and framework fracture and ceramic chipping — so the discipline of minimising accumulated error remains entirely justified. Option A is the attractive answer because the number is famous and sounds authoritative; quoting it as a clinical standard leads to remaking serviceable frameworks. Verify, minimise, and judge the framework clinically. (*Misch; ITI Treatment Guide vol 8*)

Answers — Block L — Complications, peri-implant disease, maintenance and governance (Q281–300)

281 C The 2017 case definitions are built on inflammation and change, not on absolute numbers. Peri-implant health requires the absence of erythema, bleeding on probing, swelling and suppuration, and it is explicitly stated that health can exist at an implant with reduced bone support — there is no probing depth above which health cannot be diagnosed, because peri-implant probing depths depend on the depth at which the implant was placed and on the soft tissue height. Option B is the answer the numbers invite, since 5 mm and 2.5 mm of bone loss look like the peri-implantitis pattern; but the diagnosis requires bleeding or suppuration together with bone loss beyond initial remodelling, and here there is neither inflammation nor any change from the one-year baseline. Record it, and keep monitoring. (*2017 World Workshop; Berglundh et al.*)

282 D Peri-implantitis requires bone loss beyond the crestal change of initial remodelling; without it, inflammation of the peri-implant tissues is mucositis however deep the pockets or florid the bleeding, and this is why a baseline radiograph at the time of prosthesis fitting is worth taking. Her increased probing depths are explained by swelling and by loss of probe resistance in inflamed tissue. Option A is the distractor that catches candidates who have learned the definition as a list rather than a logic: suppuration appears in the case definition as "bleeding **and/or** suppuration", so its absence excludes nothing. The 6 mm figure in B belongs only to the situation where no previous records exist, C is irrelevant, and mobility signifies complete loss of integration, not early disease. (*2017 World Workshop*)

283 E In the absence of previous examination data the 2017 workshop set a deliberately conservative pair of thresholds: bleeding or suppuration on gentle probing, a probing depth of at least 6 mm, and a bone level at least 3 mm apical to the most coronal intraosseous portion of the implant. Her 7 mm pocket and 3.5 mm of bone loss satisfy both. Option D is the near miss and is chosen most often, because 2 mm is the figure candidates carry from the general statement that bone loss should not exceed about 2 mm after remodelling — a reasonable rule for a monitored implant, but not the threshold that applies when there is no baseline against which to judge remodelling, and using it would overdiagnose deeply placed implants. (*2017 World Workshop; Berglundh et al.*)

284 A The peri-implant attachment is a hemidesmosomal epithelial seal with connective tissue fibres running parallel rather than inserting, so it is genuinely more easily penetrated than a periodontal attachment — but experimental work shows it re-forms within a few days after gentle probing, and a probing force of about 0.20 to 0.25 N is both safe and necessary. Bleeding on gentle probing is the sign that identifies mucositis, which is the reversible stage. Option B is the attractive compromise, and plastic probes are widely sold for the purpose, but a metal probe does not damage a titanium surface at these forces, whereas a flexible plastic probe reads inaccurately. C inverts the sequence, since radiographic change follows clinical change, and E means finding disease late. (*2017 World Workshop; ITI Treatment Guide vol 13*)

285 E All three are defensible and belong together in the same sentence to a patient. The weighted mean patient-level figures from systematic review are of the order of 43 per cent for peri-implant mucositis and 22 per cent for peri-implantitis, which is why supportive care is presented as part of the treatment rather than an optional extra. Item C is the one candidates hesitate over, because a range that wide reads like an evasion; it is in fact the most informative of the three, since the individual studies used different case definitions, different thresholds for bone loss and different populations, and the spread is what tells you the point estimates should be quoted with caution. Option D is therefore incomplete rather than wrong. (*Derks & Tomasi; 2017 World Workshop*)

286 B A history of periodontitis, together with poor plaque control and failure to attend supportive care, is the most consistently reproduced risk factor across cohorts, and the effect is greater where the periodontal treatment has left residual pockets that bleed. The mechanism is straightforward: the same pathogens, the same host response and, often, the same behaviour. Option A is the distractor with genuine literature behind it, and it is associated with plaque accumulation, bleeding, brushing discomfort and recession — but its relationship with progressive bone loss is inconsistent, which is precisely why it ranks below periodontal history. Occlusal overload as a cause of peri-implantitis remains contested and unproven, surface effects are weak, and bisphosphonates raise a different problem altogether. (*2017 World Workshop; ITI Treatment Guide vol 13*)

287 D Radiopacity varies enormously between cements, and several widely used resin cements are barely more radiopaque than the soft tissue around them, so a clean-looking radiograph is entirely compatible with a submucosal ring of cement. This is one of the arguments for choosing a demonstrably radiopaque cement, for keeping margins at or above the mucosal level, and for screw retention where the angulation allows it. Option E is the strongest alternative and names a real limitation of two-dimensional imaging,

since buccal and palatal excess is projected over the implant body — but a radiopaque cement in that position still alters the density enough to be suspected, whereas a radiolucent one is invisible wherever it lies. At 2.5 mm submucosal, complete removal at cementation was improbable. (*Wilson; Linkevicius; ITI Treatment Guide vol 13*)

288 **B** Non-surgical therapy is the necessary first phase — it reduces inflammation, allows the patient's technique to be reassessed and prosthetic faults corrected, and identifies which sites are going to need surgery — but complete resolution of established peri-implantitis is achieved in a minority of cases, and least often at the deep, suppurating sites, because the roughened contaminated surface cannot be debrided through a pocket. Option D is the attractive choice, and item C is the reason it fails: adjunctive systemic or local antibiotics, lasers and air-polishing produce small and largely short-lived additional gains and do not substitute for surgical access. Item A is true of peri-implant mucositis, which is where the confusion arises, and not of peri-implantitis. (*2017 World Workshop; ITI Treatment Guide vol 13*)

289 **C** Mucositis is plaque-induced and is reversible, which is the whole justification for the maintenance programme, and the consensus position is that it precedes peri-implantitis in the same way that gingivitis precedes periodontitis, without every case progressing. Statement (ii) is what makes options B and E wrong, and it is the one candidates accept by analogy with experimental gingivitis: when plaque control is restored around implants the inflammatory infiltrate resolves more slowly and less completely than at teeth, with residual inflammation still demonstrable at three weeks. The clinical corollary matters — a single visit with instrumentation and instruction is not enough, and the site must be reviewed and re-treated until bleeding on probing has actually gone. (*Salvi et al.; 2017 World Workshop*)

290 **E** Defect morphology drives the surgical decision in exactly the way it does in periodontal surgery. A contained, circumferential intrabony crater with four walls holds a graft, stabilises the clot and has a good blood supply, and reconstructive therapy in this configuration produces greater radiographic defect fill and greater probing depth reduction than access surgery alone. Option B is the strong distractor, because resective surgery with implantoplasty is well supported — but for the opposite morphology, a supracrestal or horizontal defect where there is nothing to contain a graft, and in a site where the resulting recession is acceptable. Applying it to a contained crater discards a regenerable defect. C is premature at a stable implant, and D repeats therapy that has already reached its limit. (*ITI Treatment Guide vol 13; 2017 World Workshop*)

291 **A** Mobility of the implant itself, with a continuous radiolucency around the whole fixture, means complete loss of osseointegration. There is no treatment that reintegrates such an implant: it is removed, ideally with a counter-torque device rather than a trephine so that bone is preserved, the site is allowed to heal for two to three months with grafting where needed, and replacement is discussed knowing that a second failure at the same site is more likely. Option B is the answer that would usually be right and is the one to check first in practice, because a loose *restoration* from screw loosening is far commoner than a loose implant — but the stem has already distinguished the two by describing a radiolucency encircling the entire implant. (*Misch; ITI Treatment Guide vol 13*)

292 **B** The interval is the intervention. Cohort data show a clear step in the incidence of peri-implantitis once the interval between supportive visits exceeds roughly five to six months, and her fourteen months sits well outside that, with bleeding at two implants as the visible consequence. The interval is then shortened further — to three or four months — for patients with a periodontitis history, poor plaque control or smoking, none of which applies here. Option C is the attractive answer because it errs on the side of caution and sounds diligent; recall intervals are risk-based, and prescribing the shortest interval to everybody wastes the patient's money, dilutes the message and is not what the evidence supports. E abandons the principle that mucositis is detected, not reported. (*Monje et al.; EFP S3 guideline; ITI Treatment Guide vol 13*)

293 **D** Low-abrasive glycine or erythritol powders delivered by air polishing reach into threads and irregularities that no curette can follow, and they are the best-evidenced method of biofilm disruption on an exposed implant surface; curettes of titanium or a polymer such as PEEK are softer than or equal to the implant and remove deposits without altering the surface. Item C is what makes option E wrong, and it is chosen by candidates who reason that reducing the power reduces the harm: a stainless steel tip is harder than titanium, and at any power it gouges the surface, raises its roughness and leaves metal debris — producing a more plaque-retentive surface than the one you started with. Sodium bicarbonate powders are similarly abrasive at these sites. (*ITI Treatment Guide vol 13; Lindhe*)

294 **E** The determinant of resolution in mucositis is mechanical disruption of the biofilm, by the clinician and then daily by the patient; trials of adjunctive antiseptics show small, inconsistent additional benefit over good mechanical therapy alone, set against staining, calculus formation, taste disturbance and rare hypersensitivity reactions with long-term use. A short course has a place where mechanical cleaning is

temporarily impossible, such as after surgery. Option A is the answer patients like and clinicians concede, because a rinse feels like a solution to poor access — but a mouthrinse does not penetrate an established submucosal biofilm, and accepting it as a substitute leaves the actual cause untouched while both parties believe something is being done. (*ITI Treatment Guide vol 13; 2017 World Workshop*)

295 **D** Where altered sensation follows placement and the radiograph shows the implant encroaching on the canal, decompression is urgent: removing or unscrewing the implant within roughly the first 24 to 36 hours offers the best chance of recovery, and every day of continued compression makes permanent injury more likely. Candour is part of the same answer — he is told at once what has happened, what is being done and what the outlook is, and an apology is offered, which is not an admission of liability. Option C is the distractor that feels responsible and is right for a persisting injury or one you cannot manage — but referral that consumes the decompression window converts a recoverable injury into a permanent one. E documents diligently while doing nothing. (*GDC Standards; Misch*)

296 **A** Implants and their components are medical devices, and traceability is a regulatory requirement rather than a courtesy: the implant and abutment reference and lot or batch numbers, with the expiry date, are recorded in the clinical notes at the time of surgery and at fitting, usually from the peel-off labels. A recall is then a database search rather than a guess. Option D is the attractive alternative and is good practice for other reasons, but the card is held by the patient and identifies nothing at your end. C tells you what you bought and not whom you fitted it to, B narrows the field without identifying anyone, and E shows a component's outline but not its lot. (*Medical Devices Regulations; GDC Standards*)

297 **B** Every exposure must be individually justified, and a routine time-based schedule for a healthy, asymptomatic implant does not meet that test. The two films he already has do the job that baseline imaging is for: they establish the bone level at loading and the extent of initial remodelling in the first year. Thereafter the clinical parameters — bleeding on gentle probing, suppuration, increasing probing depth — change before the radiograph does, and it is those findings that justify the next film. Option A is the answer that feels conscientious and is still widely practised, which is exactly why it is the distractor to beat; annual irradiation of a healthy site cannot be justified, and D adds a far larger dose for no diagnostic gain. (*IR(ME)R; FGDP/RCR selection criteria*)

298 **A** Complaints are handled by responding promptly, listening in full, examining, and only then explaining your view and setting out the options, which may include an independent opinion, an adjustment or a remake. Nothing in that sequence can be done down a telephone or by letter, and a patient who feels heard frequently does not pursue the complaint. Option C is the sophisticated wrong answer, and it deserves comment because informing your indemnifier early is genuinely good practice and you should do it — but it is not the *first* action, and a delay in speaking to the patient while you take advice reads as defensiveness and hardens the complaint. B pre-judges, D concedes without understanding, and E outsources the conversation. (*GDC Standards*)

299 **C** Consent is a dialogue, and what evidences a dialogue is a record of what was actually said — the material risks discussed, including peri-implant disease and the maintenance it demands, the alternatives including doing nothing, and the questions this particular patient asked and how they were answered. Option D is the attractive choice and it is where most clinicians place their confidence: a signed form is evidence that a form was signed, and a leaflet is evidence that a leaflet exists, neither of which establishes that the risk was discussed with her in terms she could weigh. Since *Montgomery* the test is whether a reasonable person in her position would have attached significance to the risk, and a note written at the time is what answers it. (*Montgomery; GDC Standards*)

300 **C** Six millimetres of bone loss at two adjacent, splinted implants in the anterior maxilla, in a patient with a high lip line and almost no keratinised mucosa, is an advanced case by any classification: the decision between reconstructive surgery, resection and explantation with reconstruction turns on judgements about aesthetics and prosthetic sequencing that sit beyond a general dentist with a special interest. Option A is the reasoned wrong answer and would be right in a lower molar site, since resective surgery is well supported for defects that cannot be regenerated — but implantoplasty with an apically positioned flap here exposes threads and metal in a patient who shows his gingivae, and the aesthetic result would be indefensible. D is over-hasty, and E accepts progression without offering him a choice. (*ITI Treatment Guide vol 13; SAC Assessment Tool*)

Answers — Block M — History, risk assessment and medical considerations, extended (Q301–340)

301 E All three operate together, and each is separately established, which is what makes the combination defensible rather than a catch-all. Advanced glycation end-products cross-link collagen irreversibly, stiffening the matrix and impairing osteoblast differentiation and the remodelling of woven to lamellar bone. Neutrophil function is measurably depressed once glucose is persistently raised, which is why infection is both more likely and more destructive. And the microangiopathy that produced his retinopathy affects every capillary bed, including the periosteal and endosteal supply on which a healing osteotomy depends. Option D is the answer most candidates give, because the immunological and biochemical mechanisms are the ones taught; they omit the vascular one, even though his retinopathy is in the stem as evidence of it. (*Lindhe; NICE NG28; Misch*)

302 C The figure usually quoted as adequate control for elective implant surgery is in the region of 58 mmol/mol, and at 61 she sits above it — not dangerously, but on the wrong side of the line, with treatment newly changed and its effect not yet expressed in her HbA1c. Three months is the interval over which the new agent will show its effect and over which a repeat measurement becomes meaningful. Option D is the seductive error, and it is a threshold applied in the wrong direction: 48 mmol/mol is the value at which diabetes is diagnosed, not a target to which a diabetic patient is expected to return, and setting it would defer her treatment indefinitely. Option A invents a threshold above the accepted one. (*NICE NG28; Lindhe; ITI Treatment Guide vol 2*)

303 D Glycated haemoglobin accumulates over the lifespan of the red cell, about 120 days, so a single value is weighted towards the preceding two to three months. A test four weeks after a change of therapy is therefore dominated by the period before the change and will understate the improvement; repeating it at three months gives a usable number. Option B is the attractive distractor because candidates know the test is not a snapshot but compress the window, and a two-week figure would justify testing now — which is precisely the error the question is built around. Note also the conditions that make HbA1c misleading whatever the interval: haemoglobinopathies, recent transfusion, haemolysis and iron deficiency, none of which apply here. (*NICE NG28*)

304 B Gliclazide is a sulfonylurea: it closes the ATP-sensitive potassium channel of the beta cell and provokes insulin release whether or not the blood glucose is high, so a missed or delayed meal produces hypoglycaemia. Metformin reduces hepatic gluconeogenesis and improves peripheral sensitivity without stimulating insulin, and sitagliptin acts through incretin potentiation, which is glucose-dependent; neither causes hypoglycaemia when used alone. Option D is the commonest answer, because metformin is the drug every candidate associates with diabetes and its name arrives first; it is the wrong one to fear. The practical management is unchanged — treat early, confirm she has eaten, keep glucose to hand and recognise that the symptoms of hypoglycaemia are easily mistaken for anxiety under drapes. (*BNF; SDCEP*)

305 A The two outcomes must be separated, as they must be for every patient being consented. With control at this level, implant survival is close to that of non-diabetic patients, and it is fair to tell him so. What the cohort data show in addition is a greater susceptibility to peri-implant mucositis and peri-implantitis, mediated by the same impairment of neutrophil function and microcirculation that operates at higher glucose levels, so the recall interval is shortened and the threshold for intervening at an inflamed site is lowered. Option C is the attractive half-truth and the one that gets clinicians into trouble, because it quotes the reassuring survival figure while implying that nothing about his aftercare needs to differ. (*2017 World Workshop; Lindhe*)

306 C The accepted definition has three limbs, and all three are satisfied here: exposed bone in the maxillofacial region, or bone that can be probed through a fistula, persisting for more than eight weeks; current or previous treatment with an antiresorptive or antiangiogenic drug; and no history of radiotherapy to the jaws or obvious metastatic disease at the site. She is at ten weeks. Option D is the distractor that catches candidates who know a duration criterion exists but misremember it as three or six months, and it is worth being exact, because the label changes management: this is a referral to an oral surgery or maxillofacial unit, not a matter for further curettage in practice. (*SDCEP MRONJ; AAOMS position paper*)

307 A The indication is the single most powerful discriminator: an oral bisphosphonate taken for osteoporosis places a patient in the low-risk group, while an intravenous or subcutaneous antiresorptive given at oncology dosing raises the risk by one to two orders of magnitude. Duration of therapy, concurrent systemic corticosteroids or antiangiogenic drugs and any previous episode of osteonecrosis complete the assessment. Option D is the attractive answer and it is the one a physician may well propose: serum CTX was once promoted as a threshold test, but it varies diurnally, with feeding and with many unrelated factors, no cut-off has been shown to predict osteonecrosis, and guidance explicitly advises against using it. The T-score measures fracture risk, not jaw risk. (*SDCEP MRONJ*)

308 B Alendronate is a nitrogen-containing bisphosphonate. It binds avidly to hydroxyapatite, is released as the osteoclast acidifies the bone surface, is taken into that cell, and there blocks farnesyl pyrophosphate synthase in the mevalonate pathway. Without prenylation the small GTPases that maintain the ruffled border and the cytoskeleton cannot function, and the osteoclast loses its resorptive apparatus and undergoes apoptosis. Option D is the best distractor because it stitches together two true facts from neighbouring pharmacology: the non-nitrogenous bisphosphonates such as clodronate really do form cytotoxic ATP analogues, and sclerostin really is a drug target — for romosozumab, not for any bisphosphonate. Option A describes denosumab, the contrast the session was drawing. (*BNF; SDCEP MRONJ*)

309 D The orders of magnitude are what matter, and they are what allow an honest conversation. For patients taking an oral bisphosphonate for osteoporosis without other risk factors the reported incidence of osteonecrosis is a small fraction of one per cent, which is why guidance places them in the low-risk group and why implant treatment is not withheld from them. For patients receiving antiresorptives at oncology dosing the figure rises to around one per cent or higher, which is why elective implant placement is not offered. Option E is the evasion that feels safe and fails the consent test: the ranges are well enough established to be quoted with their caveats, and *Montgomery* requires that material risks be conveyed rather than avoided. (*SDCEP MRONJ; AAOMS position paper; Montgomery*)

310 A Implant placement with flap elevation is a higher bleeding-risk procedure, and for a direct oral anticoagulant the advice — miss the morning dose of a twice-daily drug, delay the morning dose of a once-daily one — exists to bring the procedure into the trough of the drug's activity. A patient who takes his rivaroxaban in the evening is already at trough at nine the next morning, so there is nothing to alter: treat early in the day, suture, pack and give written post-operative instructions. Option B is the attractive answer because it is the correct instruction for the morning-dosed patient and candidates apply it without reading the timing in the stem. The INR does not usefully measure rivaroxaban. (*SDCEP anticoagulants and antiplatelets*)

311 C Around four fifths of an absorbed dose of dabigatran is eliminated unchanged in the urine, far more than for apixaban or rivaroxaban, so a falling glomerular filtration rate prolongs its effect and raises its plasma concentration; the licensed dosing reflects this and the drug is avoided altogether at severe impairment. The practical consequence is a discussion with the prescriber before a higher bleeding-risk procedure, rather than an assumption that the standard advice applies. Option A is the elegant trap: dabigatran etexilate genuinely is a prodrug, converted by esterases, but that conversion happens in plasma and liver and has nothing to do with the kidney — a true fact relocated to the wrong organ. (*BNF; SDCEP anticoagulants and antiplatelets*)

312 B Warfarin's own elimination half-life is around a day and a half, and its effect depends on the clearance of clotting factors already carboxylated before the drug acted — prothrombin's half-life alone is around 60 hours. One missed dose therefore produces a trivial and unpredictable change while introducing a thromboembolic risk in a woman with a mechanical valve. Guidance is not to interrupt warfarin for dental procedures, and her INR of 3.6 is below the ceiling of 4.0, measured within 72 hours as her stability allows. Option E is the informed error: factor VII does have a half-life of a few hours, which is why the INR moves early, but prothrombin does not, and it is prothrombin that determines when haemostasis actually returns. (*SDCEP anticoagulants and antiplatelets; BNF*)

313 E Ticagrelor binds P2Y12 reversibly and non-competitively, has a half-life of about seven hours, and is given twice daily for exactly that reason; platelet function recovers over roughly three to five days as the drug is cleared, without waiting for the marrow to replace the platelet pool. Clopidogrel and prasugrel bind covalently and irreversibly, which is why their offset is measured in the seven to ten days it takes to renew the circulating platelets. None of this alters the answer to the clinical question, which is that dual antiplatelet therapy after recent stenting is not interrupted for dental surgery and bleeding is managed locally. Option A describes clopidogrel, and it is the answer given by candidates who have learned one thienopyridine story and applied it to a drug from a different class. (*BNF; SDCEP anticoagulants and antiplatelets*)

314 E All three are standard and none of them is a compromise. Aspirin acetylates cyclo-oxygenase-1 irreversibly for the life of the platelet, so stopping it a day or two beforehand achieves nothing haemostatically while removing cardiovascular protection; guidance is explicit that it is continued for dental procedures. Treating early in the day and early in the week means that any delayed bleeding declares itself while the patient can still reach you, which is advice about systems rather than about drugs and is easily forgotten. And local haemostatic measures — sutures, pressure, an absorbable dressing — are what actually control the bleeding in this group. Option D is chosen by candidates who read C as an admission that something has gone wrong; it is routine practice. (*SDCEP anticoagulants and antiplatelets*)

315 B Osteoradionecrosis is a local consequence of a local dose, and intensity-modulated planning creates steep gradients, so a site several centimetres from the target volume may have received a small fraction of the prescribed dose. The number that governs the risk at the anterior mandible is the 28 Gy that the anterior mandible actually absorbed, which is well below the region of roughly 50 to 60 Gy at which the risk becomes material. Option A is the figure everyone quotes and is the answer most candidates give, because the prescribed dose is the number that appears in the oncology letter; treating it as the site dose leads to declining treatment that could safely be provided. Obtain the dose map, and plan jointly with the oncology and restorative team. (*Misch; ITI Treatment Guide vol 2*)

316 A Permanent hyposalivation is the change that dominates the rest of her dental life: buffering, clearance and the antimicrobial constituents of saliva are all lost, and rampant cervical caries follows within months rather than years. High-concentration fluoride toothpaste, dietary advice, saliva substitutes and a short recall interval are the interventions that preserve both the natural teeth and the peri-implant tissues. Option C is the seductive answer, because hyperbaric oxygen is the intervention everyone associates with irradiated jaws; the evidence does not support it as routine prophylaxis, and it is irrelevant to a maintenance question. Option D proposes extracting teeth in an irradiated field, which is the very event that precipitates osteoradionecrosis. (*SDCEP; Misch*)

317 D Non-steroidal drugs reduce prostaglandin-mediated dilatation of the afferent arteriole and so reduce renal perfusion, and in a transplant recipient already taking a calcineurin inhibitor, itself nephrotoxic, that combination can cost graft function which cannot be recovered. Paracetamol, with codeine if needed, is the safe choice. Option A is the attractive error because immunosuppression feels as though it demands antibiotic cover; a prolonged course confers no benefit after uncomplicated placement and exposes him to Clostridioides difficile, to which he is more vulnerable than most. Note in passing the interaction that does matter if infection develops: clarithromycin and erythromycin inhibit CYP3A4 and raise tacrolimus levels sharply. Steroid supplementation is not indicated. (*BNF; SDCEP*)

318 C Implant placement is elective, and nothing is lost by waiting for the three cycles to finish; myelosuppression, mucosal fragility and the risk that an infective complication interrupts her oncological treatment are all reasons to wait rather than to be clever about timing. Option B is the distractor built for candidates who know that the neutrophil nadir falls roughly 7 to 14 days after a cycle: they have the physiology right and then operate at the worst possible moment. Option A at least gets the direction correct, which is why it is worth considering, but elective surgery still waits. Two further points belong in the assessment: ask whether she is also receiving an antiresorptive for bone protection, and provide preventive care throughout. (*SDCEP; Misch*)

319 D Prosthetic valves and previous endocarditis are both in the increased-risk group, along with acquired valvular disease with stenosis or regurgitation, hypertrophic cardiomyopathy and most structural congenital heart disease. The category that is specifically excluded is the one in the stem: isolated atrial septal defect, fully repaired ventricular septal defect and fully repaired patent ductus, together with endothelialised closure devices, because after repair there is no turbulent jet and no prosthetic surface to seed. Membership of the group does not reinstate routine antibiotic prophylaxis, which is not recommended; it means the risk is discussed, oral health is optimised before surgery, and the patient knows when to seek help. Option E is chosen by candidates who assume any congenital lesion counts. (*NICE CG64; SDCEP*)

320 B Monopolar electrosurgery is the recognised source of clinically important electromagnetic interference in dental practice, because the current travels through the patient to a remote plate and can be sensed by the device as an arrhythmia or can inhibit it. A scalpel, a laser or a bipolar unit avoids the problem entirely, and the tissue here can perfectly well be removed with a blade. Option A is the attractive over-reaction: deactivation belongs to major surgery in a monitored setting and would expose him to unprotected arrhythmia for a procedure that need not disturb the device at all. Adrenaline in dental doses is acceptable, ultrasonic scalers are not a recognised hazard, and dental magnets are too small and too distant to interfere. (*SDCEP; BNF*)

321 A A single reading taken at the moment a nervous patient sits in the chair is the least reliable measurement you will make all day, and the white-coat effect is large enough to move a patient across a decision threshold. Resting him quietly and repeating the reading costs ten minutes and frequently resolves the question. If it remains at or above 180/110 mmHg the elective procedure is deferred and his general practitioner informed, and a reading of 180/120 mmHg or more with headache, visual disturbance or chest pain requires same-day medical assessment. Option B is defensible-sounding and is the answer that scores second, but it acts on a number that has not yet been shown to be his. (*NICE NG136; SDCEP*)

322 C A pack-year is twenty cigarettes a day for a year, so the calculation is the number smoked daily divided by twenty, multiplied by the number of years: 25 divided by 20 is 1.25, and 1.25 multiplied by the 22 years from 27 to 49 gives 27.5, which rounds to the region of 30 and is the only option in that range.

Option B is the commonest error and is chosen by candidates who record the number of years and stop, ignoring that she smoked more than a pack a day; option D doubles instead of dividing. The figure matters because risk is dose-related, and because her two years of abstinence reduce but do not abolish the excess: an ex-smoker of this burden is not equivalent to a never-smoker. (*Misch; Lindhe*)

323 **E** Two things must be said and neither on its own is honest. Switching away from combustion removes tar and carbon monoxide, which is a genuine gain and worth acknowledging rather than dismissing. But nicotine itself constricts the microvasculature, impairs fibroblast and neutrophil function and reduces the perfusion on which a grafted site depends, and the published studies of peri-implant outcomes in vapers are few, small and short. The defensible position is that vaping is probably less harmful than smoking, is not risk-free, and full cessation remains the aim before augmentation. Option D is the attractive half-answer, because carbon monoxide is a real mechanism that candidates can name; it is not the only one. (*Lindhe; ITI Treatment Guide vol 7*)

324 **D** The protocol usually quoted asks for one week of abstinence before surgery, long enough for the acute effects on platelet aggregation and microvascular tone to reverse, and eight weeks afterwards, covering the period during which the initial bone response to the implant and the graft is established; smokers who complied have shown failure rates approaching those of non-smokers. Item C is what makes E wrong and it is the one candidates accept, reasoning that nicotine is the harmful agent and should therefore be avoided in every form. Nicotine replacement delivers nicotine without carbon monoxide or tar, it substantially raises the chance that she stops at all, and the balance of advice favours using it. (*Bain; ITI Treatment Guide vol 7*)

325 **B** The accepted grading has three tiers: possible bruxism rests on self-report alone, probable bruxism adds a positive clinical examination, and definite bruxism requires instrumental confirmation by polysomnography or electromyography. He has report plus signs, so he is probable, and that is enough to change the plan — a hard full-coverage splint, materials chosen to resist chipping, cantilevers avoided, occlusal tables narrowed. Option C is the attractive error, and the distinction is worth making precisely: a partner's account is still report, however convincing, because grinding noise is neither sensitive nor specific for the muscle activity that damages restorations. Sleep studies are rarely indicated for this purpose. (*Lobbezoo et al. consensus; ITI Treatment Guide vol 12*)

326 **A** A cantilever is a lever, and in a parafunctional patient it multiplies loads that are already abnormal in magnitude and duration and that occur when no protective reflex is operating; the terminal screw is placed in tension every time the cantilever is loaded, and screw loosening, screw fracture and framework failure follow. If a third unit is wanted, a third implant is the answer. The other four options are all measures that make the reconstruction more robust: a monolithic occlusal surface removes the veneering ceramic that chips, screw retention makes the inevitable maintenance possible without destroying the restoration, a narrow table reduces off-axis load, and the splint removes the prosthesis from the nocturnal load altogether. (*Misch; ITI Treatment Guide vol 12*)

327 **E** This is Lekholm and Zarb type 4 bone, where primary stability normally comes from a cortex that is barely present, and both numbers confirm it: 12 Ncm and 52 ISQ are well below the values of about 35 Ncm and 70 ISQ conventionally required before any immediate or early loading. The correct response is to leave it undisturbed and allow secondary stability to develop, commonly four to six months in the posterior maxilla, having under-prepared the osteotomy and used a tapered design to gain what stability was available. Option B is the attractive answer because a non-functional provisional feels harmless; micromovement during healing, not occlusal contact alone, is what prevents osseointegration. Option A is mechanically impossible without stripping the site. (*Misch; ITI Treatment Guide vol 2*)

328 **B** Six years of an oral bisphosphonate exceeds the five-year duration that moves a patient taking one for osteoporosis into the higher-risk group, and stopping it does not undo that: alendronate binds hydroxyapatite and remains in the skeleton for years, so the drug at her surgical site is still there fourteen months after her last tablet. Teriparatide is a recombinant fragment of parathyroid hormone and is anabolic, not antiresorptive; it is not implicated in osteonecrosis. Option D is the answer most candidates give, on the assumption that any drug prescribed for osteoporosis carries the same jaw risk, and it is the assumption this question exists to correct. Calcium and vitamin D are irrelevant. (*SDCEP MRONJ; BNF*)

329 **C** Paracetamol at normal adult doses is the analgesic of choice throughout pregnancy and is the only one of the three that can be given here without qualification. Non-steroidal drugs are avoided in the third trimester because they can cause premature constriction of the ductus arteriosus, impair fetal renal function and delay the onset of labour. Codeine close to term risks neonatal respiratory depression and withdrawal and is best avoided. Option D is the attractive answer, and it is chosen by candidates who correctly exclude the non-steroidal drug and then treat an opioid as the automatic step up the analgesic

ladder; the timing in the stem, at 34 weeks, is what makes it wrong. Definitive treatment of the source remains the priority. (*BNF; SDCEP Drug Prescribing*)

330 E Demonstrated cessation of skeletal growth is a necessary condition for placing an anterior implant, and he satisfies it — but it is not a guarantee. Continued eruption of natural teeth and slow dentoalveolar compensation go on through adult life, more markedly in patients with a long-face pattern, and discrepancies of a millimetre or more have been reported at anterior implants decades after placement. He should consent knowing that the crown may need to be replaced, or a discrepancy accepted, rather than discovering it at 35. Option A is the intuitive error and inverts the mechanism: the implant behaves as an ankylosed tooth and stays exactly where it is put, which is why everything around it appears to move. (*Misch; ITI Treatment Guide vol 1*)

331 A A cantilevered resin-bonded bridge is minimally destructive, is well tolerated over many years, maintains the space and, importantly for a high smile line, preserves the soft tissue architecture that a future implant restoration will depend on. It also allows the crown length and contour to be developed and reviewed as she grows. Option B is the reasonable alternative and the one many candidates choose, and it deserves credit as a fallback where the abutment teeth are unsuitable, but a denture is worn variably, is disliked at 14, and loads the ridge in a way that does nothing to preserve it. Option C is the answer this whole domain exists to prevent: an implant placed now will be left in infraocclusion. (*ITI Treatment Guide vol 1; Misch*)

332 D Expectations that lie outside the mouth are the classic predictor of dissatisfaction, because no dental result can meet them: the teeth are restored, the promotion does not arrive, and the failure is attributed to the treatment and to the clinician who provided it. Eliciting that belief and testing it gently — what will be different, and what will not — is the substance of the consultation, and it may be a reason to decline. Option A is the good wrong answer, and it is a useful reminder that a reference photograph is a help rather than a warning sign: it makes an expectation explicit, discussable and, if necessary, refutable. Cost and risk can both be explained; a misattributed life problem cannot. (*GDC Standards; Montgomery*)

333 B Three separate reasons converge on deferral. She is medically unstable at that body mass index, with the electrolyte and cardiac risks that accompany purging; the aetiology of her tooth surface loss is active and uncontrolled, so restorations placed now will be destroyed by the same acid; and irreversible elective treatment while an eating disorder is untreated is poor care whatever the patient asks for. Meanwhile prevention, desensitisation and a sympathetic route into medical care are exactly what you should provide. Option E is the sophisticated wrong answer, and it is worth naming: a psychiatric report is not a gatekeeping document for a scale and polish, and imposing one is a way of declining while appearing thorough. (*GDC Standards; SDCEP*)

334 C The Mental Capacity Act is explicit that a person is not to be treated as unable to make a decision merely because it is one that others would regard as unwise, and nothing in his presentation suggests a functional impairment. Option D is the important error and the one that catches thoughtful candidates, because it appears to follow logically from C: capacity confers the right to *refuse* treatment, not a right to demand it. You are not obliged to extract a restorable tooth, and doing so on request without a clinical indication is not defensible before the GDC whatever the patient signed. The correct course is to record his wishes and your advice, explore why he wants it, and be willing to decline. (*Mental Capacity Act; GDC Standards*)

335 E Under the Family Law Reform Act a young person of 16 or 17 is presumed to have the capacity to consent to treatment as an adult would, and that presumption stands unless there is evidence to rebut it. Option B is the attractive error and the one most candidates choose, because Gillick is the case they can name: Gillick competence is the route by which a child *under* 16 may consent, and applying it here demotes a young man who already has the presumption. Two separate points remain: encouraging him to involve a parent is good practice with his agreement, and implant placement will in any case be deferred until facial growth has demonstrably ceased, which at 16 it almost certainly has not. (*GDC Standards; Montgomery*)

336 D Clarithromycin is a potent inhibitor of CYP3A4, and simvastatin is metabolised by that enzyme; concurrent use raises simvastatin concentrations several-fold and has caused myopathy and rhabdomyolysis, which is why the combination is one the BNF advises avoiding. The options are to use a different antibiotic, most simply metronidazole for this presentation, or to arrange with his prescriber that the statin be withheld for the few days of the course. Option C is the well-informed distractor: clarithromycin genuinely does prolong the QT interval and that matters in other combinations, but statins do not add to it, and the candidate who chooses C has remembered a real hazard and attached it to the wrong drug. (*BNF; SDCEP Drug Prescribing*)

337 B Enough miconazole is absorbed from the oral mucosa and from swallowed gel to inhibit cytochrome P450 2C9, the enzyme that clears the more active enantiomer of warfarin, and the interaction has produced INR values above 10 and serious bleeding in reported cases; the BNF lists it among the interactions to avoid. Nystatin suspension is the straightforward alternative, alongside the measures that actually matter here — cleaning and leaving out the overdenture at night, and treating the denture surface itself. Option C is the plausible pharmacological alternative, since displacement from albumin is a real mechanism for some drug pairs, but it is not how this interaction works and it would not produce the sustained rise that is seen. (*BNF; SDCEP Drug Prescribing*)

338 D Timing and character together make the diagnosis. Urticaria and angio-oedema arising within an hour of a dose are the classic immediate, IgE-mediated presentation, and the two together in the same episode are as clear an account as a history will give; a beta-lactam should be avoided and the record should say why, in those words. Item C is what makes E wrong, and it is the item that clutters allergy records everywhere: diarrhoea from the third day of a course is a predictable pharmacological effect on the gut flora, not an allergy, and recording it as one commits the patient to second-line antibiotics for life. Distinguishing the two at the time of the history is worth several minutes of anyone's clinic. (*SDCEP Drug Prescribing; BNF*)

339 C With effective therapy, an undetectable viral load and a CD4 count in the normal range of roughly 500 to 1500 cells per microlitre, implant outcomes are comparable with those of matched patients without HIV, and there is no basis for refusing, delaying or medicalising his treatment. Standard infection control is designed to be applied to every patient regardless of known status, which disposes of D and of the thinking behind it. Option E is the distractor that looks rigorous and invents a threshold above the normal range, so that no patient could ever satisfy it. The genuine considerations are drug interactions — protease inhibitors inhibit CYP3A4, which matters for midazolam — and any oral manifestations of disease. (*SDCEP; Misch*)

340 E The three drugs attack renal perfusion from three directions at once, which is why the combination has a name in prescribing safety work. The non-steroidal drug removes the prostaglandin-mediated dilatation of the afferent arteriole, the ACE inhibitor removes angiotensin-mediated constriction of the efferent arteriole, and the diuretic reduces circulating volume; glomerular filtration falls, and in an older patient, or during any intercurrent illness with reduced fluid intake, acute kidney injury follows. Paracetamol, with codeine if needed, is the answer. Option C is the well-informed distractor and is a true statement — ACE inhibitors raise potassium and non-steroidal drugs add to that — but hyperkalaemia is a consequence of the renal impairment rather than the primary hazard. (*BNF; SDCEP Drug Prescribing*)

Answers — Block N — Examination, occlusion and space, extended (Q341–375)

341 C The measurement that governs is the one at bone level, not the one between the contact points. At least 1.5 mm must separate the implant shoulder from each adjacent root, so even a 3.3 mm implant needs 6.3 mm at the platform; the 5.0 mm available leaves about 0.85 mm on each side. Option B is the answer most candidates give, because narrowing the implant is the reflex response to a narrow site and the 8.5 mm at the contact points reads as generous — but the arithmetic defeats it, and the crestal remodelling around the implant would involve the crestal bone of both neighbours and with it the papillae. Angling the implant (D) moves the apex, not the shoulder, which is where the clearance is needed. Root uprighting converts an unfavourable site into a routine one. (*Tarnow; ITI Treatment Guide vol 1*)

342 C Tarnow's measurements between natural teeth remain the most useful predictor available and they transfer reasonably to the implant-to-tooth papilla: at 5 mm from contact point to crest the papilla is present in essentially every case, at 6 mm in a little over half, and at 7 mm in only about a quarter. At the 7 mm quoted here a black triangle is the expected outcome, and the honest response is either to lengthen the contact point on the wax-up, to accept and consent for the appearance, or to reconsider. Option B is the trap, because 6 mm and 7 mm feel interchangeable and the fall between them is steep rather than gradual. The medium phenotype and ovoid crown forms are favourable modifiers, but they do not overcome the vertical distance. (*Tarnow; ITI Treatment Guide vol 1*)

343 A Crestal remodelling around an implant extends laterally as well as vertically, by something of the order of 1.4 mm. Where two implants sit 3 mm or more apart those two zones do not overlap and the mean lateral bone loss measured between them is around 0.45 mm; where they sit 3 mm or less apart the zones coalesce and the loss roughly doubles, to about 1.0 mm, taking with it the height of bone that has to support the inter-implant papilla. Option D is the attractive alternative because the buccal plate is the tissue candidates worry about in the anterior maxilla, and it is genuinely at risk — but the specific penalty of reducing inter-implant distance is interproximal, not buccal. Primary stability and osseointegration are unaffected by the spacing. (*Tarnow; ITI Treatment Guide vol 1*)

344 **A** Only the first statement is the reason for the rule. The saucer of remodelled bone that forms around the implant neck extends roughly 1.4 mm laterally, and 1.5 mm of separation is what keeps that zone clear of the adjacent root's crestal bone — bone which, unlike the implant's, carries the attachment apparatus and supports the papilla. Option D is the commonest choice, because the 3 mm figure in statement B is familiar and is correct in a different setting: it is the minimum distance between two adjacent implants, where two remodelling zones must be separated rather than one. Statement C inverts the anatomy; the papilla is supplied by supraperiosteal vessels and by the periodontal ligament of the adjacent tooth, and an implant contributes no vascular supply of its own. (*Tarnow; Lindhe*)

345 **A** The crest is wide enough for the implant, so the coronal part is contained; the ridge narrows below it, so the defect will appear where the bone is thinnest, as a window of exposed implant surface with intact bone above and below it. That is a fenestration, and it is the signature complication of a buccal concavity in the posterior maxilla. Option B is the answer most candidates give because dehiscence and fenestration are learned as a pair and the distinction is rarely rehearsed — but a dehiscence is continuous with the crest, and here the crest has 1.2 mm on each side of the implant. Fourteen millimetres of height makes sinus perforation improbable with a 10 mm implant, and the concavity should be read as a reason to angle the implant palatally or to plan simultaneous regeneration. (*ITI Treatment Guide vol 7; Lindhe*)

346 **E** All three are legitimate, and the case is written so that each is available: crestal reduction is affordable only because 16 mm of height and a generous interocclusal dimension mean that losing 3 mm costs nothing that matters, ridge expansion suits a mandible with a cancellous core between two cortices, and staged regeneration is the slowest but most conservative route. Option D is the choice of candidates who reject regeneration as excessive for a 1.5 mm shortfall, which is a defensible clinical preference but not a reason to call it wrong. The examinable point is that the answer to a narrow crest is not automatically a narrow implant: a 3.3 mm implant in a molar-loaded premolar site trades a surgical problem for a mechanical one. (*ITI Treatment Guide vol 7; Misch*)

347 **C** Retention of a cemented crown is a function of the axial height and taper of the abutment, and about 5 mm of height is the figure usually quoted as the minimum for a molar. With 4.5 mm of total space, an abutment must also allow for the occlusal thickness of the restorative material, so the surviving axial wall is nearer 2 to 3 mm and the crown will come off, repeatedly, until it is remade. Option A is the plausible alternative, since a short crown is often assumed to concentrate load on the screw, but a low crown height actually shortens the vertical lever arm and reduces screw stress. The lesson is that short restorative space is an indication for screw retention, or for creating space before restoring. (*Misch; ITI Treatment Guide vol 12*)

348 **B** The vertical stack for a stud attachment is unforgiving: the transmucosal abutment, the metal housing, a layer of acrylic thick enough to hold it, and the denture tooth above. Something in the region of 8 to 9 mm from platform to occlusal plane is needed before the housing can be accommodated without thinning the base, and at 7 mm the acrylic over the housing is reduced to a wafer that fractures, usually in the midline. Option A is the distractor of experience, because insert wear is the commonest maintenance event with these attachments and every clinician has seen it; it is not, however, what a shortage of vertical space produces. The remedy is to measure before surgery and to consider a shorter attachment system, a different design, or reduction of the ridge. (*McGill consensus; ITI Treatment Guide vol 4*)

349 **B** Opening the vertical dimension rotates the mandible about the transverse hinge axis, so the separation achieved at any point is proportional to its distance from the condyle. The first molar sits roughly half as far forward as the incisal edge, so an increase of 4 mm at the incisors yields something in the region of 2 mm at the molars — and progressively less further back. Option C is the answer given by candidates who treat the increase as a parallel separation of the arches, and it leads directly to the clinical error of promising a posterior restorative space that never materialises. The practical corollary is that a large anterior opening is a poor way to solve a posterior space problem, and that space in the molar region is better created at the opposing tooth. (*Misch; Prosthodontics*)

350 **A** Two findings settle it. The facial height is unchanged over a decade, and the freeway space is 2 mm — at the lower end of the usual 2 to 4 mm — rather than increased. Teeth do not wear without the alveolus following them: continuous eruption carries the gingival margin and the crestal bone occlusally as the crown shortens, so the occlusal vertical dimension is preserved and the space the clinician hoped to find has already been filled by bone. Option B is the assumption almost every candidate brings to a worn dentition, and it matters because it leads to opening the bite by an arbitrary amount on the basis of tooth wear alone. Restorative space here must be made deliberately — by crown lengthening, by a Dahl approach, or orthodontically — not by presuming it has been lost. (*Prosthodontics; Misch*)

351 **D** Four millimetres is the number that decides this. Occlusal enamel is at most 2 to 2.5 mm thick over the cusps and considerably less in the fissures, so removing 4 mm within enamel is impossible and

statement A fails; enameloplasty remains the correct first-line answer for a millimetre or so of interference, which is exactly why option E attracts candidates who have learned the escalation ladder without the dimension attached to each rung. Devitalisation with crown shortening will create the space but leaves the gingival margin and bone where they are, so the emergence remains unattractive; orthodontic intrusion moves tooth, margin and alveolus together and is the more complete answer where the patient will accept it. Extraction remains a legitimate fifth option. (*Prosthodontics; Misch*)

352 **E** The mesial pocket is not simply a consequence of the tilt; it is an active periodontal lesion with a plaque-associated angular defect, and orthodontic movement through inflamed tissue accelerates attachment loss rather than resolving it. Cause-related therapy comes first, then uprighting, which will both level the defect and relocate the crestal bone that must support the papilla mesial to the LR7 — and only then is the implant placed, into a space whose dimensions are now known. Option B is the answer that scores second and is chosen by most candidates, because uprighting is correctly identified as necessary; it simply starts one step too late. Placing the implant first (A and D) fixes the distal boundary of the space before the tooth beside it has been moved. (*Lindhe; 2017 World Workshop*)

353 **A** The adjustment is a compensation for a mismatch in vertical mobility. Under functional load a natural tooth sinks within its periodontal ligament by something of the order of 25 to 100 micrometres, whereas an osseointegrated implant moves only about 3 to 5 micrometres, almost all of it bone deformation. If the implant crown were adjusted to contact equally with its neighbours at light closure, it would be the only unit still contacting once the teeth had intruded, and would take the whole load. Option B is a true statement and the reason implant patients cannot report a high crown themselves, but it explains why the clinician must measure rather than ask — not why the contact is deliberately lighter. C misstates the mechanics and D reverses the finding. (*Misch; ITI Treatment Guide vol 12*)

354 **B** The principle behind implant-protected occlusion is that lateral load should be distributed and kept as near axial as possible, not that an implant may never guide. Where the implant replaces a canine, restoring exclusive canine guidance concentrates the whole of the excursive load onto a single unit with no ligament and no proprioceptive protection; a group function in which the implant shares shallow contact with the premolars and the lateral incisor spreads it. Option D is the answer candidates carry over from the single posterior implant, where freedom from excursive contact is exactly right, and it is the strongest distractor here — but in the canine position it forces the guidance onto teeth that are poorly placed to accept it, and a canine crown entirely out of contact in excursion is difficult to make look natural. (*Misch; ITI Treatment Guide vol 12*)

355 **B** In a right lateral excursion the left side is the non-working side, and a contact there acts on a crown that is being carried away from the condyle it is loaded against — a lateral force applied at the end of a lever whose length is the crown height and whose offset is the distance from the cusp to the implant axis. The resulting bending moment opposes the preload holding the joint closed, and the screw loosens again however carefully it is torqued. Option A is the trap and it is a naming error rather than a conceptual one: candidates recognise that the contact matters but assign it to the working side, and would then adjust the wrong cusp. Occlusal correction, not repeated retightening, is the treatment. (*Misch; ITI Treatment Guide vol 12*)

356 **B** A reorganised approach is indicated when the existing intercuspal position cannot be used as the reference for the new restorations, and the commonest legitimate reason is that the vertical dimension is being deliberately changed — here to recover some of the 6 mm of freeway space and give the posterior implants the restorative height they need. Option D is the popular answer because a slide feels like a defect requiring correction; it is not. Slides of 1 to 2 mm are found in most people, and a stable intercuspal position with no symptoms is precisely the situation in which a conformative approach is safest. Statement C, if anything, argues for conforming. Whatever is chosen must be decided in advance and tested reversibly before any irreversible preparation. (*Prosthodontics*)

357 **C** An osseointegrated implant behaves as an ankylosed unit while the natural dentition continues to move throughout adult life, erupting and drifting mesially by small increments. The implant does not follow, so the contact opens; the phenomenon is common, reported in something like a third to a half of implant restorations followed for several years, and it occurs on the mesial far more often than the distal. Option A is the most attractive alternative because proximal wear is a real phenomenon between natural teeth and the timescale fits, but wear at a contact is measured in tens of micrometres over years and does not produce a gap that admits food. The management is restorative — reshaping or remaking the contact — and the patient should be warned it may recur. (*Misch; ITI Treatment Guide vol 12*)

358 **D** An angulated screw channel allows the screwdriver to enter at up to about 25 degrees to the implant axis, moving the access opening away from the functional palatal contact while keeping the restoration retrievable and free of cement. That is precisely the problem it was designed to solve. Option

B is the answer most candidates give and it is defensible in isolation, which is what makes it dangerous: cement retention in the anterior maxilla places a margin subgingivally in an aesthetic site where residual excess is both likely and difficult to detect, and it is a recognised cause of peri-implant disease. Option A moves the shoulder buccally in a thin site and invites recession, C proposes orthodontics for a problem with a restorative solution, and E leaves composite in a heavy functional contact. (*ITI Treatment Guide vol 12; 2017 World Workshop*)

359 **C** Sustained isometric contraction produces the muscular and mucosal signature — hypertrophic masseters, cheek mucosa ridged where it lies between the arches, tongue borders indented against the lingual surfaces of the teeth — without generating the eccentric excursions that abrade enamel. Matching facets are the product of grinding, and their absence alongside every other sign of heavy loading is the finding that identifies the habit as a clenching one. That distinction matters for planning: the load is axial, sustained and unguided, so occlusal design cannot dissipate it and a hard splint remains the protection. Option B is the reasoning candidates import from the worn dentition and it fails here because compensatory eruption preserves the vertical dimension while leaving the facets themselves plainly visible. (*Misch; Prosthodontics*)

360 **B** Every finding listed raises the aesthetic risk score, but only the phenotype changes what your hands do. A thin scalloped biotype dictates a more palatal shoulder position, a flapless or minimally reflected approach where possible, the addition of a subepithelial connective tissue graft, and a preference for staged rather than immediate protocols — decisions taken in the operating chair. Option A is the finding most candidates select, and correctly identifies the factor that most raises the *consequence* of an error, but a high smile line changes what you must achieve, not how you achieve it. The triangular crown form predicts where a black triangle will appear, the 7 mm gap is favourable, and unrestored neighbours are an argument for an implant rather than a bridge. (*ITI Treatment Guide vol 1; Lindhe*)

361 **C** The lip line is the filter through which every other aesthetic finding is judged. With the gingival margins and the cervical portions of the crowns covered even on full animated smile, a 2 mm asymmetry of the mucosal margin will not be seen, and a compromise that would be unacceptable in a high smile line becomes a reasonable plan honestly consented. Option B is the attractive answer because a patient's refusal genuinely constrains what you may do, but a constraint on treatment is not the same as a justification for the outcome; if the discrepancy were going to be visible, the correct response to that refusal would be to reconsider the whole plan rather than to proceed. The lip line must be assessed on animation, not at rest, and photographed. (*ITI Treatment Guide vol 1*)

362 **E** The junction between prosthesis and mucosa is the feature that betrays a fixed full-arch reconstruction, and the smile she has just demonstrated shows tissue 2 mm apical to the cervical margins of her denture teeth — exactly where that junction would fall. The choice therefore lies between an implant-supported overdenture with a flange, substantial vertical augmentation, or a soft-tissue-toned prosthesis designed and positioned so the transition is hidden, and none of those is a decision to take alone at a first consultation. Option B is the finding candidates choose, and the vertical deficiency is real and necessary to the argument — but ridge loss concealed beneath a low lip line is routinely restored with pink material and no one is any the wiser. It is the visibility, not the deficiency, that governs. (*ITI Treatment Guide vol 4; SAC Assessment Tool*)

363 **A** What the evidence supports is modest and worth stating precisely: implants with less than about 2 mm of keratinised mucosa accumulate more plaque, show more mucosal inflammation and recession, and — as here — are more often reported as uncomfortable to clean, which is itself the mechanism by which the rest follows. Option D is the choice of candidates who assume that anything associated with inflammation must eventually be associated with loss; survival rates are not demonstrably different, and to claim otherwise oversells the case for grafting. Statement C is the honest sticking point: an association with peri-implantitis is plausible and reported, but the 2017 workshop treated the evidence as inconclusive, and "established" is what makes it false. A free gingival graft is justified by comfort and control, not by survival. (*2017 World Workshop; Lindhe; ITI Treatment Guide vol 13*)

364 **B** Staging is set by severity and complexity: bone loss reaching the middle third or beyond with four or fewer teeth lost to periodontitis gives stage III, and it is tooth loss of five or more that would raise him to stage IV. Grading estimates the rate: 60 per cent bone loss divided by 45 years of age gives a ratio of about 1.3, and any value above 1.0 is grade C, with 15 cigarettes a day confirming it as a grade modifier in its own right. Option E is the near miss that most candidates take, because three lost teeth feels severe and the distinction between four and five is arbitrary until you have learned it. His normal HbA1c excludes the diabetic modifier. The grade, not the stage, is what should most concern you before placing implants. (*2017 World Workshop*)

365 D The 2017 case definition is written in terms of attachment loss, not probing depth, and offers two routes: interdental clinical attachment loss detectable at two or more non-adjacent teeth, or buccal or oral attachment loss of at least 3 mm with pocketing greater than 3 mm at two or more teeth, in either case with local causes such as a subgingival margin or a fractured root excluded. This patient satisfies the first by a wide margin, despite pockets of only 3 mm — which is exactly why the definition is written this way, since recession masks attachment loss from anyone charting depths alone. Option E fails on statement C: radiographic bone loss supports the diagnosis and is essential for staging, but the case itself is defined clinically. Option A alone is the commonest answer and is incomplete. (*2017 World Workshop; BSP*)

366 C There is no absolute probing depth that defines health around an implant, because the reading is measured from a mucosal margin whose position was determined by how deeply the fixture was placed and how thick the tissue is. A 5 mm reading around a platform buried 3 mm beneath a thick biotype is entirely compatible with health, and the findings that matter — no bleeding, no suppuration, no radiographic change from baseline — are all present. Option B is the answer that catches candidates who have memorised a number without the qualifiers: peri-implantitis requires bleeding or suppuration together with bone loss beyond initial remodelling, and increased probing depth only in comparison with a previous recording. This is why the baseline film and the baseline probing chart matter more than any single later measurement. (*2017 World Workshop; ITI Treatment Guide vol 13*)

367 E Altered sensation in the distribution of the inferior alveolar nerve is not rare in an adult population, and a good proportion of people with a minor deficit from an earlier extraction, a fracture or a block injection are unaware of it until someone tests them. Without a baseline, a deficit discovered at the one-week review cannot be attributed, and both the clinician's management and the patient's understanding are impaired. Option C is the strongest distractor and is the reason many clinicians give, but it describes a by-product rather than the purpose; a record made only to defend a claim is a poor clinical record, and the same test performed for clinical reasons defends the claim anyway. The safety margin is determined from the CBCT and the accuracy of the technique, not from sensory testing. (*GDC Standards; ITI Treatment Guide vol 11*)

368 E The neutral zone is the space in which the outward force of the tongue is balanced by the inward force of the lips and cheeks, and a prosthesis whose teeth and flanges lie outside it is displaced during function no matter how well it is retained at rest — which is exactly the pattern described, since retention is adequate when seated. A broad tongue lying over the lingual flange with the teeth set lingual to the crest is the classic arrangement for this failure. Option A is the answer suggested by experience and is the first thing most clinicians change, but the inserts were renewed a month ago and retention is not the complaint. Implants improve retention; they do not correct tooth position, and a two-implant overdenture with the teeth in the wrong place will still be unstable. (*McGill consensus; Prosthodontics*)

369 D Cheek biting is a horizontal problem, not a vertical one. In a normal buccal segment the maxillary buccal cusp overhangs the mandibular one by a millimetre or so, and that overjet holds the cheek clear of the occluding surfaces as the teeth come together. A mandibular crown built too far buccally, or an opposing cusp reduced during adjustment, abolishes the overlap and the mucosa is caught between the cusps at every closure — hence a linear ulcer at exactly the occlusal level. Option A is the answer most often given, and cusp height does contribute to how sharply the tissue is caught, but a tall cusp with adequate horizontal overlap does not bite the cheek. Correction is by reshaping the buccal contour to restore the overlap. (*Prosthodontics; Misch*)

370 D A complete upper denture opposed by natural lower anterior teeth is the setting for the changes grouped as combination syndrome: the anterior maxillary ridge is loaded by teeth that concentrate force on a small area and it resorbs, the tuberosities descend, and the denture rocks about the fulcrum this creates. Adding a posterior implant that cannot move and does not resorb makes the denture pivot more, not less, which is why the upper prosthesis must be reviewed and often remade as part of the plan. Statement C is what makes E wrong and it is the one candidates accept: maximum bite force against a complete denture is a fraction of that between natural teeth, so the implant is if anything mechanically protected. Its problem is the fulcrum it creates for the denture above. (*Prosthodontics; McGill consensus*)

371 E The soft tissue between two adjacent implants is supported only by the inter-implant crest, and its height above that crest averages about 3.4 mm — considerably less than the 5 mm or so achievable between a tooth and an implant or between an implant and a pontic, where the adjacent attachment apparatus does the supporting. Reducing the number of implants to one therefore removes the least favourable interface altogether. The implant is placed at the central, the wider and more heavily loaded unit, and the lateral is cantilevered, so the pontic papilla is supported by the crest of the UR3 on one side and by the implant on the other. Option A is the plan she has been offered and is the one candidates accept; in a 13 mm space

with a thin biotype and a high smile line it reliably produces a short, flat inter-implant papilla. (*Tarnow; ITI Treatment Guide vol 1*)

372 D A margin 4 mm subgingivally cannot be inspected, cannot be reached with an instrument and cannot reliably be cleared of cement, and the deeper the margin the greater the volume of cement left behind; residual excess is a well-documented cause of peri-implant inflammation and bone loss, and it is undetectable at the appointment where it is created. The remedy is either screw retention or an intermediate abutment that raises the margin to within about a millimetre of the mucosa. Option C is the strongest alternative and would be right in a different case — recession follows a thin buccal plate and a shoulder placed too far labially, whereas this plate is 2 mm thick and the error is one of depth, not of orofacial position. Depth also lengthens and over-contours the emergence it was meant to improve. (*ITI Treatment Guide vol 12; 2017 World Workshop*)

373 E A buccal wall under about 1 mm thick has no cancellous core and depends on a periosteal blood supply that surgery disturbs; it resorbs, and the mucosa it supported follows it apically. Mid-facial recession of a millimetre or more within the first year is the expected sequence, which is why 2 mm of buccal bone is the target in the aesthetic zone and why a thin plate is an indication for simultaneous contour augmentation rather than something to be noted and left. Option A is the answer candidates reach for because they associate any anterior implant problem with peri-implant disease, but recession from resorption of a thin wall is a modelling event, not an inflammatory one, and typically occurs with no bleeding and no suppuration. The interproximal crests, supported by the adjacent teeth, are usually preserved. (*Araújo & Lindhe; ITI Treatment Guide vol 1*)

374 A Symmetry across the midline is the single most powerful determinant of whether an anterior restoration reads as natural, and a lateral incisor 2.5 mm wider than its partner will be noticed by anyone who looks, particularly beneath a high smile line. The space must therefore be redistributed before the implant fixes it permanently — a modest orthodontic task, and the reason the case should be reassessed with the orthodontist rather than restored as presented. Option B is the pragmatic choice most candidates make, because a wider implant does fill the space and the arithmetic of clearance to the roots still works; it simply produces an asymmetrical tooth. Option C is worse: a correctly sized crown in an oversized space leaves black triangles that a thin biotype will never fill. (*ITI Treatment Guide vol 1*)

375 D Two independent arguments converge, and a candidate who offers only one has given half the answer. Shortened dental arch studies show that occlusal stability, chewing ability and patient satisfaction are maintained with premolars and first molars in occlusion, so there is no functional deficit to correct here; and with a 30 mm opening the combined height of handpiece, drill and shank simply will not fit at a second molar site, which is the practical reason such implants are attempted less often than they are planned. Statement C is what makes E wrong and it is the one candidates believe because they associate the site with difficulty: survival at second molar sites is not markedly different when placement is achievable. The honest plan is to review, not to operate. (*SAC Assessment Tool; Misch*)

Answers — Block O — Investigations, imaging, guides and the wax-up, extended (Q376–410)

376 C Justification is a decision about this patient, and it cannot be made without knowing what clinical question the scan is meant to answer — the residual bone height suspected, whether the canal can be traced on existing films, whether augmentation is contemplated. The practitioner is the person who must be satisfied of sufficient net benefit, and a bare request supplies none of the evidence needed. A telephone call to the referrer, or a short clinical assessment yourself, usually resolves it within minutes. Option B is the answer that feels responsible and is the one most candidates give: optimising an exposure that has not been justified is doing the second duty while omitting the first, and a small field of view aimed at the wrong question is still an unjustified dose. D discards a patient who probably does need the scan. (*IR(ME)R; FGDP/RCR Selection Criteria*)

377 D The test is whether the cross-sectional information will change management for that individual. An untraceable canal beneath a molar site leaves the vertical safety margin unknown, and a palpable buccal concavity in the anterior maxilla determines whether the implant can be contained, whether augmentation is needed and whether the case exceeds the operator's competence — both alter the plan before a drill is picked up. Option E fails on the third request, and it is the one candidates wave through because a CBCT before any implant has become habitual in some practices: with 12 mm of bone below the sinus, an unremarkable panoramic and a clinically adequate ridge, the scan confirms what is already known and adds dose without benefit. Habit is not justification, and the reasoning must be recorded for each patient. (*IR(ME)R; FGDP/RCR Selection Criteria; SEDENTEXCT*)

378 A Effective dose is a weighted sum across irradiated organs, so the dominant variable is not how hard the beam is driven but how much radiosensitive tissue lies inside the beam at all. Shrinking the field

of view from one that includes both jaws, the thyroid and the salivary glands to one that includes a single premolar site removes whole organs from the calculation, and the reduction is far larger than anything achievable by adjusting exposure factors. Option B is the reflex answer and is genuinely useful — reducing mA lowers dose roughly proportionally — but it is a linear saving within a fixed volume, and it is bought at the cost of noise. Reducing basis projections has the same trade-off, and increasing tube voltage lowers dose per projection while degrading contrast. (*SEDENTEXCT; IR(ME)R*)

379 **B** Each voxel collects photons, and its signal-to-noise ratio depends on how many it collects. Halving the linear dimension of a voxel reduces its volume roughly eightfold, so at unchanged exposure factors the reconstruction becomes markedly noisier; restoring the same image quality means raising the dose. The finer setting therefore buys spatial resolution the implant question does not need, at a cost the patient pays, which is why 0.2 to 0.3 mm is the usual choice for implant planning and sub-0.1 mm voxels are reserved for questions such as a fine root fracture. Option E is the strongest distractor because the two settings are indeed linked on many machines, and small fields are often the only ones offering high resolution — but that is a constraint of the equipment, not the reason for the choice. (*SEDENTEXCT; Radiology*)

380 **D** The relevant harm here is stochastic: a probability of inducing a malignancy that rises with dose and has no threshold. Two factors multiply that probability in a young patient — tissues with a higher proportion of dividing cells are more radiosensitive, and a longer remaining lifespan gives more time for a cancer with a latency of decades to declare itself. Risk coefficients for a child are conventionally taken as several times those for an adult of retirement age at the same dose. Option C is the attractive error and it matters, because it confuses the two categories of radiation effect: deterministic effects such as erythema or cataract have a threshold measured in grays and do not occur at any dental exposure, at any age. The practical consequence is that justification is stricter for the young. (*SEDENTEXCT; IR(ME)R*)

381 **C** Dividing the measured 14 mm by the magnification factor of 1.25 gives 11.2 mm, and the arithmetic must be done before the figure is used, because 14 mm and 11 mm sit on opposite sides of most planning decisions for a molar site. Option D is chosen more often than any other and is the error that puts implants into canals: the measurement is read off the film with a ruler and used as though the image were life-size. Option E multiplies rather than divides, which is the same mistake in the opposite direction. Note also that the stated factor is nominal, that it applies to vertical dimensions in the plane of the focal trough only, and that a value this close to a threshold is a reason to obtain cross-sectional imaging rather than to trust the sum. (*Radiology; FGDP/RCR Selection Criteria*)

382 **A** Panoramic imaging is a form of tomography in which the receptor and the beam move at a set speed while the object stands still; horizontal magnification therefore depends on how the object's speed relative to the receptor compares with that of the machine, and a structure lying buccal or lingual to the centre of the focal trough is stretched or compressed accordingly. Vertical magnification is comparatively stable, which is why heights are quoted with a single factor and widths are not. Option C is the strongest distractor because a calibration sphere is exactly the right answer to the vertical question and is routinely used for it; placed at a different depth from the structure being measured, it does nothing to correct horizontal distortion. Mesiodistal space should be measured on the patient or on casts. (*Radiology*)

383 **E** A ghost arises when an object lying between the source and the centre of rotation is passed through the beam twice. It is projected onto the opposite side of the film, higher than the real object because the beam is angled upwards, magnified horizontally and unsharp — every feature described here. Recognising it prevents an unnecessary referral and an unnecessary repeat, and the remedy is simply to remove metallic objects before the exposure. Option B is the alternative most candidates choose, and it is a sensible differential for a real neck opacity, but a calcified node is well defined, does not have a mirror-image partner and does not sit high on the ramus. Movement artefact affects the whole image rather than one object, and it is the operator's duty to report the finding either way. (*Radiology*)

384 **E** The inferior alveolar canal does not end at the mental foramen: a branch continues forward within the bone as the mandibular incisive canal, supplying the anterior teeth and the labial mucosa, and it is identifiable on CBCT in a large proportion of patients as a corticated canal one to two millimetres across running anteriorly at about the level described. It matters because damage during placement in the interforaminal region causes bleeding and occasionally persistent altered sensation, and because it is one of the reasons the anterior mandible is not the risk-free zone it is assumed to be. Option A is the answer most candidates give, and the anterior loop is a real and important structure — but it lies at or distal to the foramen and curves back to it, whereas this canal runs forward from it. (*Lindhe; ITI Treatment Guide vol 11*)

385 **E** All three are cross-sectional measurements and all three change the plan. Width at platform level determines diameter and whether augmentation is needed; height above the canal determines length and safety margin; and the lingual concavity determines the axis, because a ridge of adequate width at the

crest may still funnel lingually below it, and the submandibular fossa is where the drill exits into the floor of the mouth. Option D is chosen by candidates who treat canal height as a panoramic measurement already known — and it often is, approximately, but a magnified two-dimensional estimate is not what a cross-section is for, and the two should be reconciled. The examinable habit is to read every cross-section for width, height and concavity before an axis is drawn. (*ITI Treatment Guide vol 11; Lindhe*)

386 **D** A corticated border is absent beneath a substantial minority of sites, particularly where the canal runs through cancellous bone, and its absence means the canal cannot be seen — not that it is not there. The safe practice is to identify the bundle by the best available evidence on adjacent slices, measure to the most coronal point at which it can be traced, and then apply the usual safety margin from that conservative line. Option B is the tempting answer and is wrong for a reason worth knowing: cortication is a matter of tissue contrast, not of spatial resolution, so a finer voxel adds noise and dose without making an uncorticated canal appear. Option A inverts the logic of a negative finding, and E abandons planning in favour of an arbitrary rule. (*SEDENTEXCT; ITI Treatment Guide vol 11*)

387 **C** The described appearance and the drill's behaviour both indicate the softest of the bone qualities, with a thin cortex and sparse trabeculae. Primary stability in that bone comes from lateral compression of an under-prepared site — omitting the final drill, stopping short of the planned diameter, or using an implant designed for the purpose — and secondary stability takes longer to arrive, so healing is extended rather than shortened. Option A is a genuine technique for soft bone and is the strongest distractor, but a 6.5 mm ridge does not have the width to spare: a wider implant would leave less than a millimetre of bone on each side. Tapping removes the interference that generates stability, immediate loading in this bone risks micromotion beyond tolerance, and irrigation addresses heat rather than density. (*Misch; ITI Treatment Guide vol 2*)

388 **E** Beam hardening and the streaks that accompany it are generated along the lines joining dense objects, so the artefact is worst in the axial plane that contains the restorations. Tipping the occlusal plane by altering chin position moves the crowns out of the plane containing the crest of the LR5, and the site becomes readable; metal artefact reduction algorithms help further, at some cost in overall sharpness. Option A is the answer most candidates choose, because a poor image intuitively calls for more radiation, and it is precisely wrong: the streaks are a reconstruction problem arising from preferential absorption of lower-energy photons, not from insufficient signal, and raising mA gives a brighter image with the same artefact and a higher dose. Reducing voxel size and altering projections do not address the mechanism. (*Radiology; SEDENTEXCT*)

389 **A** Double contours across every interface, with no metal in the volume, are the signature of movement during acquisition; the projections no longer register with each other and the algorithm reconstructs the same border in two places. The exposure lasts many seconds in most units, which is a long time for a patient with a tremor to hold still, so the remedy is a protocol that shortens it — fewer projections or a half-rotation scan — supported by a chin rest, a head strap and a bite block. Option B is the strongest distractor because unsharpness is instinctively read as inadequate resolution: a smaller voxel usually means a longer acquisition and a noisier image, so it would make this scan worse in two ways at once. The repeat must itself be justified and the reason recorded. (*Radiology; IR(ME)R*)

390 **E** All three follow, and together they are the argument for collimating to the clinical question. A larger volume includes more radiosensitive tissue and therefore delivers a higher effective dose; every part of the acquired volume must be examined and reported, which for a large field means structures a dentist may not be competent to assess and an obligation that does not disappear because the finding was incidental; and a detector of fixed pixel count spread over a larger volume yields larger voxels, so the very detail that implant planning needs is degraded. Option D is chosen by candidates who know the dose and reporting arguments but treat resolution as an independent setting; on most machines it is coupled to the field, which is why small-volume scans are also the high-resolution ones. (*SEDENTEXCT; IR(ME)R; FGDP/RCR Selection Criteria*)

391 **C** Two millimetres is the conventional margin, and the accuracy figures quoted in the stem are the reason it survives the move from freehand to guided placement rather than being abandoned with it. Mean deviations of about 1 mm at the entry and 1.5 mm at the apex describe the centre of a distribution whose tail extends well beyond those values, and the margin must accommodate the tail, not the mean. Option B is the answer given by candidates who reason that guidance has halved the error and the margin may be halved with it — the single most dangerous inference in this field, since it consumes the whole of the reserve at the first outlier. With 13 mm available, a 10 mm implant leaves 3 mm and is entirely comfortable; the margin costs nothing here. (*ITI Treatment Guide vol 11*)

392 **A** Every other item in the list contributes a few tenths of a millimetre; compressible mucosa contributes considerably more, because the guide is designed against a surface recorded in one state and

seated against tissue that can be compressed by an unknown and uneven amount, and because a guide seated slightly deeply on one side is also rotated. That is why mucosa-supported guides are the least accurate of the three types, why fixation pins are not optional, and why the seating pressure should be reproduced and checked. Option D is the sophisticated wrong answer, and registration error is real and worth minimising — but the fiducial markers of a dual scan are rigid and radiopaque, which is precisely what makes that step comparatively reliable in an arch that offers no other landmarks. (*ITI Treatment Guide vol 11*)

393 **A** A guide that seats on bone must have bone to seat on, so the flap is necessarily wider than the surgery itself demands, with the periosteal stripping, the swelling and the crestal resorption that follow — and a shred of interposed periosteum or soft tissue will hold the guide off its seat and displace every osteotomy drilled through it. Those are the real disadvantages. Statement B is false and is what makes option D attractive: a bone-supported guide can be used in a partially dentate jaw, and indeed the remaining teeth then offer a useful check on seating. Statement C reverses the accepted ranking, in which tooth-supported guides are the most accurate, bone-supported intermediate and mucosa-supported the least. (*ITI Treatment Guide vol 11*)

394 **D** The stack of guide, sleeve, drill and handpiece is what defeats a static guide in the second molar region, and 36 mm of opening will not accommodate it. Dynamic navigation places the drill under real-time tracking with nothing interposed between the handpiece and the bone, so the only vertical requirement is the one freehand surgery already imposes — which is why posterior access is the situation in which it earns its cost. Option C is the attractive answer because tracking the patient is exactly what the reference array does, and head movement genuinely is accommodated; but that is true of the technique everywhere and is not what makes it preferable here. A CBCT and a plan are still required, and the safety margin is unchanged. (*ITI Treatment Guide vol 11*)

395 **B** Navigation works by holding a fixed relationship between the patient array and the anatomy recorded on the CBCT. If the splint moves on the teeth, that relationship is silently invalidated: the screen continues to display a plausible, confident and entirely wrong position, and nothing in the display announces the error. This is why the splint must be rigid, verified before and rechecked during the case. Option C is the answer most candidates give, and interruption of the line of sight is a genuine nuisance — but it is a self-announcing failure, since tracking simply stops and the operator waits. Head movement is precisely what the patient array exists to accommodate, and irrigation and drill speed affect the bone rather than the position. (*ITI Treatment Guide vol 11*)

396 **D** A pilot-only guide transfers the part of the plan that matters most — the entry point and the axis at the crest — and then allows the osteotomy to be completed with shorter, unsleeved drills in the space that is actually available. It is the standard solution to limited posterior access and it keeps most of the benefit of the plan. Option B is the strongest distractor because shorter drills do exist and shortening the implant does reduce the drill length required, but the handpiece head accounts for much of the 42 mm and a 12 mm site should not be restored with a short implant to solve an access problem. Option E is the dangerous one: the sleeve height is part of the calibrated offset, and grinding it changes the depth of every osteotomy. (*ITI Treatment Guide vol 11*)

397 **B** The wax-up is a prosthetic instrument: it fixes where the teeth are to be, and from that follow the cervical margin, the emergence, the occlusal scheme and the restorative space needed at each site — a dimension that can then be compared with what the mouth offers and, where it falls short, corrected before rather than after the implants are placed. Option D is the commonest answer and it confuses the two halves of the planning process: the sinus floor and the bone density are read from the CBCT, and the whole purpose of merging the wax-up with that scan, or of converting it into a radiographic guide, is that neither record answers the other's question. A wax-up made without reference to the imaging can propose the impossible; imaging read without a wax-up plans implants to bone rather than to teeth. (*Misch; ITI Treatment Guide vol 11*)

398 **E** An 18 mm vertical dimension from crest to incisal edge cannot be filled with teeth: a maxillary central incisor is about 10 to 11 mm long, so a fixed prosthesis of natural-looking teeth would leave 7 or 8 mm unaccounted for. The prosthesis must therefore carry pink material replacing the lost ridge, which is entirely compatible with a fixed reconstruction provided the junction is hidden — and his low smile line is why the stem tells you he displays no gingiva. Option D is the strongest distractor and is the right answer in a patient with the same measurement and a high smile line, where the transition would show and a flanged removable prosthesis becomes the honest choice. Option B describes the test that has already been passed. (*ITI Treatment Guide vol 4; Prosthodontics*)

399 **C** The whole purpose of the appliance is to show the prosthetic plan against the underlying bone, and a metal marker defeats it at exactly the site being examined: dense metal produces beam hardening and

streaking that spread through the crestal bone beneath the tooth being planned. Barium-loaded acrylic is radiopaque enough to be seen and segmented while remaining far less attenuating than metal, so the crest stays readable. Option A is true and is a real secondary advantage — seeing the whole tooth outline rather than a point is genuinely useful — which is why it attracts candidates who have not thought about what the metal does to the image. Statement D is simply the wrong way round, and the barium concentration should be kept to the minimum that images reliably. (*Misch; SEDENTEXCT*)

400 **B** In a fully guided protocol every vertical dimension is referenced to the upper surface of the sleeve, so a guide held off its seat by a millimetre or two — by a resin nodule, an unrelieved undercut, a proximal contact or a fragment of soft tissue — carries the whole osteotomy and the implant with it, in the coronal direction, with the axis undisturbed. That is exactly the pattern described, and it is the commonest guided error. Option C is the strong alternative and would produce an identical radiograph, which is why the stem tells you the plan was checked and found correct; an offset error is systematic and reproducible, whereas incomplete seating is a single-case accident. Omitting a drill stop would deepen, not shorten, the preparation. (*ITI Treatment Guide vol 11*)

401 **C** Surface registration works by finding the same geometry in both datasets, and the geometry it is being offered here exists accurately in only one of them: the intraoral scan records the crowns faithfully, while on the CBCT their surfaces are smeared by streak artefact and blooming. Matching an accurate surface to a distorted one produces a merge that looks convincing on screen and is wrong by an amount nobody measures, and the whole guided chain inherits it. The remedy is to register on unrestored teeth elsewhere in the arch, or to use radiopaque fiducials. Option D is the attractive answer because a badly fitting guide is the failure candidates have seen, but the intraoral scan is the reliable half of this pair; the guide will seat and will still be wrong. (*ITI Treatment Guide vol 11*)

402 **D** A CBCT is acquired with the patient in an arbitrary jaw relationship and it images bone, not function: it cannot show where the mandible closes when it is guided along its retruded arc, and it cannot be used to measure the space available between the planned platform and the opposing occlusal surface with the arches related as they will be restored. Both belong to mounted casts. Option E fails on statement C, which is the item candidates accept because the canal is undeniably three-dimensional and casts obviously cannot show it — and that is the point: the two records answer different questions, and neither substitutes for the other. Digital equivalents are acceptable, but only when the occlusal relationship has been captured and articulated rather than assumed. (*Misch; Prosthodontics*)

403 **E** All three are ordinary requirements rather than counsels of perfection. Every exposure must be justified and the justification recorded, and every radiograph must be reported on — not merely filed. The consent record must show that material risks and the reasonable alternatives, including doing nothing, were discussed, which is what *Montgomery* requires and is the specific allegation here. And the implant is a medical device: traceability of the batch or lot number is a regulatory expectation, and it is what allows a patient to be identified if a component is recalled or a connection discontinued. Option D is the answer of candidates who think of traceability as the manufacturer's business; the number belongs in the notes, along with the diameter, length and system. (*GDC Standards; IR(ME)R; Montgomery*)

404 **A** A tooth that lost its blood supply in the same injury that removed the UR2 can remain asymptomatic for years and then declare itself as a periapical radiolucency beside a new implant — where it threatens the site with infection, and where its discovery after surgery invites the reasonable but mistaken conclusion that the surgery caused it. Diagnosing and treating it beforehand removes both problems. Option E is the distractor with some merit, since an unexplained sensibility result may well prompt imaging, but the test is done for the diagnosis, not to justify a scan. The result must be recorded for each tooth with the method used, and any equivocal response repeated rather than dismissed. (*GDC Standards; ITI Treatment Guide vol 1*)

405 **D** The appearance described is that of a mucous retention pseudocyst — a common incidental finding, present in something like one in ten scans of this region, asymptomatic, non-neoplastic and of no consequence where the ostium is patent and the sinus otherwise healthy. With 9 mm of bone below the floor a standard implant can be placed without entering the sinus at all, so the lesion is not even in the surgical field. It should nonetheless be reported and recorded, and the patient told. Option A is the cautious answer and is the one most candidates give; referring every dome-shaped opacity generates anxiety, delay and cost for a finding that requires none of them. An opacity obstructing the ostium, or one with irregular margins or bone destruction, is a different matter entirely. (*SEDENTEXCT; Radiology*)

406 **B** No exposure is justified if the information can be obtained from images that already exist, and a previous scan does not become useless because it was requested for a different reason — an eight-month-old volume of a healed site is very likely to answer the question, and if the field of view or resolution turns out to be inadequate, the new request is then properly justified and documented. Obtaining previous

imaging is an explicit expectation, not a courtesy. Option A is the answer given most often, and it dresses a habit as a principle: the purpose for which an image was taken has no bearing on whether the anatomy it shows is still valid. Patient recollection (D) is not a substitute for reviewing the images. (*IR(ME)R; FGDP/RCR Selection Criteria*)

407 **A** Only the first would change the decision. Where the canal cannot be traced, the vertical margin cannot be established and a cross-section earns its dose. The other two are the reasons candidates give for scans that are not justified: 15 mm of height above a visible canal is generous for any conventional implant and confirms rather than alters the plan, and a patient's preference for a three-dimensional image is not a clinical indication — the duty to justify rests with the practitioner and cannot be delegated to the person receiving the dose. Option D is chosen by candidates who read the 15 mm as marginal; it is a comfortable measurement, and it is the figures close to a threshold, not those far from it, that generate a need for better imaging. (*FGDP/RCR Selection Criteria; SEDENTEXCT; IR(ME)R*)

408 **B** The field must be collimated to the clinical question, which here is a single anterior site, and the resolution must be adequate for the structures being assessed — a buccal plate a millimetre or so thick and the outline of the incisive canal — both of which are comfortably resolved at 0.2 mm. Option A is the answer that looks most diligent and is the commonest error at this level: an ultra-fine voxel is reserved for questions such as a hairline root fracture, and choosing it here raises noise, extends the acquisition, and either degrades the image or costs the patient dose for detail that changes nothing. The large-field options irradiate organs that have no bearing on the plan, and the exposure has already been justified, so E does not answer the question asked. (*SEDENTEXCT; FGDP/RCR Selection Criteria*)

409 **B** The mouth is the only reference that matters, because the model and the guide were generated from the same dataset and share any error in it: a guide can seat perfectly on its model and rock on the teeth it was designed for. Trying it intraorally, confirming that it is fully down through the inspection windows and that it does not move under finger pressure, is what detects the failure that produces the commonest guided error. Option D is the answer most candidates give and it is not harmful — the model check is a reasonable first look and will catch a gross manufacturing fault — but it cannot detect the error it matters most to find. Sterilisation compatibility is a genuine requirement and no safeguard against malposition. (*ITI Treatment Guide vol 11*)

410 **C** Implants are placed to restore teeth, so the prosthetic position comes first and the bone is made to fit it — the sequence that is meant by prosthetically driven planning, and the reason the wax-up is completed before the scan is read rather than after. An implant 3 mm palatal to the planned crown in a thin biotype beneath a high smile line produces a ridge-lap emergence, a plaque trap and a visible discrepancy that no ceramist can correct. Option A is the choice that reads as pragmatic and is the commonest real-world failure, and it deserves a hearing: if the augmentation needed to reach the prosthetic position is beyond you, the answer is to refer or to offer a different prosthesis — not to move the implant to the bone and restore around it. (*Misch; ITI Treatment Guide vol 1; Buser*)

Answers — Block P — Surgical anatomy, extended (Q411–440)

411 **B** The anterior loop is the length of nerve that passes forward beyond the foramen before turning back to exit, and its whole clinical importance is that it is variable — absent in many patients and several millimetres long in others — so it must be measured rather than assumed. It can only be measured on cross-sectional imaging: on a panoramic film the apparent forward extension is frequently the mandibular incisive canal, which generates false positives, and the film cannot locate anything buccolingually. Once the anterior-most point is identified, a safety margin of at least 2 mm in front of it is conventional. Option D is the attractive answer because a fixed rule of thumb anterior to the foramen is widely taught and is a reasonable default when no CBCT exists — but here the loop has been measured at 3.5 mm, and a rule of thumb should not override the anatomy you have irradiated the patient to see. (*Misch; Greenstein & Tarnow*)

412 **C** A bifid canal changes the planning arithmetic in one specific way: the structure to be avoided is whichever branch lies closest to the crest, and here that is the main canal at 13 mm, not the accessory branch 4 mm deeper. Planning to the "canal" without specifying which one is how these cases go wrong. Option A is the distractor that catches most candidates, and it is a perfectly true statement — an accessory bundle entering separately is a recognised cause of incomplete block anaesthesia — but it is an anaesthetic inconvenience, not the datum that governs implant length, and this patient has anaesthetised normally in the past. B is false, since accessory canals carry a neurovascular bundle. D reverses the hierarchy of evidence, as a panoramic film cannot confirm what a CBCT has shown. (*Misch*)

413 **A** The mentalis muscles arise from the labial surface of the symphysis close to the alveolus, and they are the only elevators of the chin pad and lower lip. A flap taken to the depth of the labial sulcus

strips them, and unless they are deliberately re-suspended at closure they heal at a lower level: the chin pad drops and flattens, the lip becomes incompetent at rest and the sulcus appears obliterated. It is a design fault, not a complication of the implants. Option B is the answer candidates give because they associate anterior mandibular surgery with the mental nerve, but the stem records normal sensation, and a neuropraxia produces altered feeling rather than altered form. C and D describe real sequelae of the incisions used, but neither flattens the chin. (*Misch*)

414 **C** The incisive canal is the direct forward continuation of the inferior alveolar canal, carrying a neurovascular bundle to the anterior teeth and the labial gingiva, and it can be traced on CBCT in the great majority of mandibles, whereas on a panoramic film it is usually invisible or mistaken for the anterior loop. Both of those facts matter, because they abolish the idea that the interforaminal region is a structure-free safe zone. Statement (iii) is what makes E and D wrong, and it is the item candidates accept without examining it: the lower lip is supplied by the mental nerve, which has already left the bone, so incisive canal injury causes bleeding into the osteotomy and, occasionally, altered sensation of the anterior teeth or a small labial gingival area — usually transient. (*Misch; Greenstein & Tarnow*)

415 **D** In a ridge with a submandibular concavity the crest and the body of the mandible are not coaxial: a drill following the apparent long axis of a lingually inclined crest runs straight into the undercut, which is why the cross-section, not the crest, must dictate the trajectory. Tilting the osteotomy buccally moves the apex away from the concavity, and because the buccal contour then limits depth, a shorter implant is the price. Option A is the distractor with genuine merit, and it is chosen by candidates who reason that what you can see you cannot damage — but a retractor protects the soft tissue and the lingual vessels behind the plate, it does not stop a drill perforating the plate itself, and elevating a lingual flap in the molar region adds lingual nerve risk. C makes perforation more likely, and E moves towards a deeper part of the fossa. (*Misch; Lindhe*)

416 **B** The mylohyoid line runs obliquely, lying high on the lingual surface in the molar region and descending towards the symphysis, so the same absolute depth below the crest lies on opposite sides of the muscle in different parts of the arch. Ten millimetres below the crest at a lower first molar is beneath the mylohyoid attachment, and blood therefore collects in the submandibular space, presenting as a swelling below the lower border. Option A is the answer most candidates give, because the airway-threatening sublingual haematoma is the story they have learned — but that is the pattern of a perforation in the canine and premolar region, where the mylohyoid line is low, and it produces elevation of the floor of the mouth and displacement of the tongue, neither of which is described here. (*Misch; Lindhe*)

417 **E** This is an expanding sublingual haematoma, and it kills by obstructing the airway rather than by blood loss: the swelling is above the mylohyoid diaphragm, the tongue is being driven up and back, and the interval between "difficulty speaking" and inability to ventilate can be very short. Everything else follows from having called for help early, while intubation is still possible. Option A is the most attractive wrong answer because bimanual pressure is a genuine and useful measure, and it belongs in the management — it is the word "observe" that makes it lethal, since local pressure will not arrest an arterial bleed in a space you cannot compress. D is described in the literature as an adjunct but does not address the airway, and B and C both accept a delay this patient does not have. (*Misch; Lindhe*)

418 **D** The two measures that work are anticipation and monitoring: a cross-section shows the depth and level of any lingual concavity and the thickness of the plate before a drill is picked up, and a finger resting on the lingual plate detects the change in feel that precedes perforation while the drill can still be redirected. Item C is what makes E wrong, and it is the item that reads like sound surgical practice: following the long axis of the crest is exactly the error that perforates lingually, because in an atrophic or lingually inclined anterior mandible the crest is angled towards the undercut and the axis of the crest is not the axis of the available bone. The safe trajectory is defined by the cross-section, which will often mean drilling more labially than the crest suggests. (*Misch; Lindhe*)

419 **A** The nasopalatine canal transmits the nasopalatine nerves and the terminal branches of the sphenopalatine vessels within loose fibrous connective tissue, and it commonly also contains fat and minor salivary tissue. An implant surface presented to that tissue has nothing to integrate with: osteogenic cells never reach it, and a fibrous interface forms along the exposed threads with progressive loss of stability. Option B is the distractor most candidates take, because nerve injury feels like the obvious hazard of drilling into a nerve canal — but nasopalatine injury produces at most a small area of transient anterior palatal numbness that patients scarcely notice, and denervation is not the mechanism of failure. C, D and E each name a real phenomenon that does not apply: the canal walls are not the problem, the bleeding is modest, and the canal does not open into the antrum. (*Misch; Lindhe*)

420 **C** Do the arithmetic before choosing the site. A 6 mm graft taken with a 2 mm coronal margin requires 8 mm of apico-coronal working height, and in this shallow vault only about 7 mm is available

before the bundle is reached; in a high vault the same measurement would be nearer 17 mm and the lateral palate would be entirely adequate. The tuberosity, which has no comparable vessel and yields dense collagenous tissue, is the site that fits. Option D is the trap and it is chosen by most candidates, because the anterior lateral palate is the donor area everyone has been taught and it is genuinely correct for grafts of ordinary height — it simply cannot deliver 6 mm here. B is worse still, since it lies directly over the bundle's course. (*Lindhe*)

421 **B** The greater palatine artery runs forward in the submucosa from its foramen, and an incision carried too far distally or too far apically will cut it. It bleeds hard, but it is a vessel lying on bone within a confined submucosal compartment, so sustained direct pressure against the palatal shelf controls the great majority of cases; five uninterrupted minutes, timed rather than guessed, is the intervention that works. Option A is the sophisticated wrong answer and is where many candidates go, because adrenaline infiltration is a reasonable second measure — but it takes several minutes to act, tissue distension makes the field worse, and reaching for a syringe before pressure wastes the window. C is the correct next step if pressure fails, D belongs to intraosseous bleeding in the lateral sinus wall, and E is not required. (*Lindhe*)

422 **E** The membrane is most tightly adherent along the crest of a septum, so elevation carried across it tents the membrane over a sharp bony edge and tears it. Where a septum divides the intended field, the standard solutions are two separate windows either side of it, or a single window modified around it, with the membrane released from each compartment before any attempt is made to lift it over the partition. Option D is the answer that reads as pragmatic and is the commonest error: a single large window gives access but delivers the elevator to the septum with the membrane still attached on both sides. A removes bone the membrane is stuck to before it has been freed. Note also that 5 mm of bone with an 8 mm septum is a fair trigger for referral rather than a first lateral window. (*ITI Treatment Guide vol 5*)

423 **A** The posterior superior alveolar artery and the infraorbital artery anastomose twice: extraosseously, in the periosteum above the window, and intraosseously, within the lateral antral wall itself. It is the intraosseous branch that a window osteotomy crosses, and it lies at a height that overlaps the upper border of a conventional window, so its position should be looked for on the CBCT before surgery. It is controlled by crushing the bone around the canal with a blunt instrument, by diathermy, or by pressure and bone wax. Option E is the good wrong answer, because the extraosseous anastomosis is real and does bleed — but that vessel lies in the soft tissue of the flap, and the stem specifies bleeding from within the bony wall as the bur passes through a canal. (*ITI Treatment Guide vol 5; Misch*)

424 **D** The nasal floor has been breached by about a millimetre, which the CBCT measurement predicted. A small perforation of the nasal mucosa is of no consequence in itself — it heals — and the correct response is to stop, shorten the plan by a drill length and place an implant that stays in bone, since it is the implant apex protruding into the nasal cavity, not the perforation, that risks contamination and mucosal irritation. Option B is the seductive one, because bicortical engagement is a genuine strategy for gaining stability in a soft maxilla and candidates recall it approvingly; here it means driving an implant through the floor into the nose. A and E abandon a site that is still usable, and C escalates a minor and self-limiting event. (*Misch*)

425 **A** Only the first is true, and it is the whole argument: the trajectory passes towards the pterygomaxillary region, where the maxillary artery and its accompanying venous plexus lie, access for haemostasis is poor, and a haemorrhage there cannot be managed in a dental chair. That, with the difficult angulation and prosthetic access, places the procedure firmly outside the scope of a general dentist with a special interest. Option D is the distractor that catches candidates who reason that a procedure they should not do must be unproven — but pterygoid implants have a substantial published literature with respectable survival, and dismissing them as unevidenced is both wrong and unhelpful when explaining the referral. C is the reverse of the truth: the tuberosity is soft, which is why the implant is angled to engage the dense pterygoid bone beyond it. (*Misch; SAC Assessment Tool*)

426 **C** Periosteum is inelastic, and it is the periosteum, not the mucosa, that tethers a full-thickness flap. Scoring it horizontally at the base, superficially and under direct vision, converts a rigid flap into an elastic one and is the single manoeuvre that delivers centimetre-scale advancement. Vertical incisions and distal extension improve access and allow the flap to be mobilised, but they do not lengthen it. Option B is the distractor that deserves argument rather than dismissal, because lingual flap release with detachment of the mylohyoid is a described technique for vertical augmentation in exactly this region — but it is an advanced manoeuvre with lingual nerve and floor-of-mouth risk, and it is not what is needed when a buccal advancement will do. E redistributes tension without abolishing it. (*ITI Treatment Guide vol 7*)

427 **A** Perfusion enters a mucoperiosteal flap through its base, so the base must be at least as wide as the free margin and the vertical incisions must therefore diverge apically; a flap that narrows towards its base is progressively starved along its most vulnerable edge, which is precisely the edge that must heal over a

graft. Item C is the one that makes D attractive, and it is subtle: raising the flap split-thickness does leave the periosteum, and its vessels, on the bone — which protects the *bone*, at the cost of a thinner and less well perfused flap. Item B inverts the rule, and it is the design a surgeon drifts into when trying to advance a flap without releasing it. Advancement should come from periosteal release, never from narrowing the base. (*Lindhe; ITI Treatment Guide vol 7*)

428 **E** In a proportion of people the lingual nerve lies at or above the level of the alveolar crest in the retromolar region, and in some it lies directly against the lingual plate, so any incision carried distally and lingually may divide it. Angling the distal extension buccally, up onto the external oblique ridge, takes the blade away from the nerve and gives better access at the same time. Option C is the distractor with a genuine following, because lingual flap retraction is taught as protective in third molar surgery — but it requires the very lingual dissection that endangers the nerve, and it is associated with more temporary lingual disturbance, not less. A and B place the incision where the nerve may be, and D sacrifices the access the case requires. (*Lindhe; Misch*)

429 **B** The description is that of a lingual bone defect containing submandibular salivary tissue — a well-corticated, asymptomatic, non-expansile radiolucency below the canal near the lower border, in the classic site. CBCT confirms it instantly, because the lesion is a concavity open to the lingual surface rather than a cavity within the bone, and once that is seen no further action is required; it does not preclude an implant placed above the canal, though its lingual concavity is worth knowing about before drilling. Option A is the cautious answer and is the one most candidates choose, but biopsy of a radiographically diagnostic developmental defect exposes the patient to surgery in the floor of the mouth for nothing. E misreads a corticated defect as a cyst, C defers a diagnosis already available, and D withdraws treatment without cause. (*Misch*)

430 **B** The conventional minimum clearance between the implant apex and the roof of the canal is 2 mm, which allows for the measurement error of the scan, the accuracy of the surgical guide or the free hand, and any overshoot of the drill beyond its marked depth. Subtracting 2 mm from 12.5 mm leaves 10.5 mm of usable bone, so the longest safe implant on the list is 10 mm. Option C is the error that produces nerve injuries: 11.5 mm leaves a single millimetre, which is less than the combined uncertainty of imaging and surgery, and candidates choose it because the arithmetic appears to fit. A is safe but discards 2.5 mm of bone in a site where length contributes to stability, and D and E would enter the canal. (*Misch; ITI Treatment Guide vol 11*)

431 **D** Where the crest is of adequate width but the labial contour falls away apically, the coronal part of the implant sits comfortably in bone and the apex emerges through the labial plate into a window with intact bone above and below it. That is a fenestration, and the distinction from a dehiscence is not semantic: a dehiscence is continuous with the crest, communicates with the sulcus and threatens the marginal tissues, whereas a fenestration is contained and is often managed simply by grafting the exposed threads at placement. Option A is the answer most candidates give because dehiscence is the term they use for any labial exposure — but the stem specifies that the concavity begins 6 mm apical to a crest of adequate width, which is exactly the anatomy that spares the crestal bone. (*Lindhe; ITI Treatment Guide vol 7*)

432 **C** A smooth dome-shaped opacity on the floor of an otherwise healthy, well-aerated sinus with a patent ostium is a mucous retention pseudocyst: a common incidental finding, present in a substantial minority of scans, with no epithelial capsule and no tendency to obstruct drainage. Provided the ostium is patent and the patient is symptom-free, it is not a contraindication to elevation, and the relevant risks are the ordinary ones of the procedure. Option A is the defensive answer and the one most candidates pick, because "cyst" sounds like something a specialist should remove — it exposes the patient to unnecessary intervention. B treats an infection that is not present, E defers without a question that the delay would answer, and D abandons the plan on a benign finding, although 5 mm of residual bone is itself a reason to consider referral. (*ITI Treatment Guide vol 5*)

433 **E** The distinction is between a random-pattern flap, which is perfused only by the small vessels entering across its base and whose safe length is therefore limited by the width of that base, and an axial-pattern flap, which contains a named artery running along its length and can be raised long and narrow because perfusion travels with the pedicle. The greater palatine artery makes the palatal flap axial, which is why pedicled palatal flaps can be rotated to cover oro-antral communications and sinus windows. Options A and B are true statements about palatal mucosa, and they are what candidates reach for because they describe the tissue's obvious qualities — but toughness and keratinisation determine how a flap handles, not how it is perfused. (*Lindhe*)

434 **D** The symphyseal donor site is defined by a set of 5 mm margins — below the root apices, so the anterior teeth are not devitalised and the incisive canal is respected; mesial to each mental foramen; and

above the inferior border, so the contour of the chin is preserved. Item C is what makes E wrong, and it is chosen by candidates who reason that a thicker block is a better block: the harvest is confined to the labial cortex, because perforating the lingual plate opens the floor of the mouth to the sublingual vessels and converts a controlled donor procedure into the haemorrhage described elsewhere in this block. The morbidity of this site — altered incisor sensation, chin paraesthesia, contour change — is itself a reason to consider the ramus first. (*ITI Treatment Guide vol 7; Misch*)

435 A Two findings in the stem point the same way: the implant is demonstrably clear of the canal, and the drill produced brisk bleeding. Together they suggest that the cortical roof of the canal was breached or the intraosseous vessels close to it were torn, and blood accumulating within a rigid bony tube compresses the bundle it contains. This is a well-recognised mechanism of neurosensory disturbance after apparently accurate placement, and it usually recovers, which is the point of recognising it rather than assuming the worst. Option B is the answer candidates jump to, and it deserves consideration — but transection would require the drill to have entered the canal, which the radiograph argues against, and it would not have produced the transient bleeding described. C, D and E do not fit the timing or the imaging. (*Misch*)

436 B Tuberosity bone is the softest in the jaws — a thin and often deficient cortex over wide-spaced trabeculae — so an osteotomy prepared to the nominal diameter engages very little mineralised tissue. Stopping short of the final drill, or omitting it altogether, leaves a smaller preparation into which the implant compresses the surrounding trabeculae and gains mechanical interlock; the same logic drives osteotome condensation. Option A is the answer that catches candidates who have learned countersinking as good practice in dense bone, where it relieves compression at the crest — here it removes the only cortical bone available and makes matters worse. D loses the additional compression a tapered body provides, E cuts a thread into bone that has none to spare, and C addresses heat, which is not the problem in soft bone. (*Misch; ITI Treatment Guide vol 3*)

437 E As the maxilla resorbs, the crest moves towards the canal, and in advanced atrophy the incisive foramen may open at or near the crest — a normal consequence of resorption rather than a pathological finding. The straightforward solution is prosthetic rather than surgical: move the implants away from the midline into the lateral incisor and canine positions, where the stem tells you there is 7 mm of width and 13 mm of height, and let the prosthesis restore the central incisors. Option D is the distractor with real substance, because enucleation and grafting of the canal is a described technique — but it is an advanced procedure with a risk of fibrous rather than bony fill and of altered palatal sensation, undertaken when there is no alternative, and here there plainly is one. (*Misch; ITI Treatment Guide vol 4*)

438 C The sublingual space is a closed compartment: the mandible laterally and anteriorly, the mylohyoid diaphragm below, and the tongue medially and above. Blood entering it cannot expand outwards, so the only direction available is upwards and backwards, and the tongue is displaced against the palate and towards the oropharynx. That is why quite small volumes cause airway obstruction and why the deterioration is rapid. Option B is the most attractive alternative, because the mucosa of the floor of the mouth is genuinely loose and does distend — but that distensibility is what makes the swelling visible, not what makes it dangerous; the danger comes from the rigid boundaries beyond the mucosa. A is a true statement about the vessels that does not explain the airway effect. (*Misch; Lindhe*)

439 E All three are measurable on the images that have already been taken, and each governs a different decision. The foramina and any loop set the mesiodistal limits of the safe zone and therefore where the implants may be placed. The lingual concavity sets the trajectory and the depth, and is the finding that prevents the floor-of-mouth haemorrhage that makes this region dangerous. The incisive canal determines whether an apparently interforaminal site is in fact free of a neurovascular structure, and it lies exactly where the anterior implants of a two-implant overdenture are placed. Option D is the answer of most candidates, who omit the incisive canal because they have been taught that the anterior mandible is a safe zone; that assumption is the one this block exists to remove. (*Misch; ITI Treatment Guide vol 4*)

440 D The buccal fat pad lies immediately deep to the periosteum in the posterior maxilla, and a periosteal releasing incision carried too deep will deliver it into the wound. It is an incidental event of no consequence provided it is recognised: the fat is gently repositioned, the flap is closed over it, and no attempt is made to excise it. Option B is the answer that a surgeon reaches for in the moment, because the herniated fat is genuinely obstructing closure and cutting it away looks like the quick solution — but excision risks haemorrhage, cheek contour deformity and loss of a tissue that is itself used as a pedicled flap for closing oro-antral defects. The lesson is that periosteal release is a superficial, controlled scoring incision, not a deep cut. (*ITI Treatment Guide vol 7; Misch*)

Answers — Block Q — Osseointegration, bone biology and healing, extended (Q441–475)

441 A Forced eruption develops bone because the force is delivered through the periodontal ligament: tension on the inserting fibres provokes osteoblastic apposition at the crest, and bundle bone together with the soft tissue attachment travels coronally with the root. A replanted tooth showing replacement resorption has no ligament left — the root is fused to bone, which is exactly what the high percussion note and the obliterated ligament space record — so force is transmitted directly to bone and the tooth does not move; sustained force simply intrudes the adjacent segment or tips the anchorage. Option C is the attractive error because candidates associate extrusion with soft tissue development and forget that the bone comes too; where a ligament exists, both move. Decoronation is the conventional alternative in a young ankylosed tooth. (*Lindhe; ITI Treatment Guide vol 1*)

442 C Bundle bone is periodontal-ligament-dependent tissue: it exists because Sharpey's fibres insert into it and it is nourished through the ligament's vascular plexus. Leaving a vital, decoronated root submerged keeps the ligament alive, so the bundle bone and with it the buccal plate and the crestal height are maintained, which is the whole rationale of root submergence and the biological basis of the socket-shield concept. Remove the root and the ligament goes with it, which is what happened at the LR4. Option D is the distractor with genuine substance, because raising a mucoperiosteal flap does add to crestal resorption and flapless extraction is worth doing — but it modifies the amount of loss rather than preventing it, and it cannot explain a ridge that has not resorbed at all. (*Araújo & Lindhe; Lindhe*)

443 B A void between implant and bone that is stable and uninfected is a healing chamber, not a defect: it fills with clot, the fibrin scaffold is retained on the roughened surface, osteogenic cells migrate onto it, and woven bone appears in the chamber within days, well before the bone in direct contact with the implant has been remodelled. This is the anatomical basis of contact osteogenesis and one reason a moderately rough implant does not need to be jammed into its site. Option D is the distractor with real mechanical logic, and it inverts the physiology: the dip is deepest where compressed and microdamaged bone must be resorbed, which is the consequence of high torque, not of 25 Ncm in a tapped site. (*Davies; Lindhe; ITI Treatment Guide vol 2*)

444 A This is a cutting cone, the advancing end of a basic multicellular unit, and it is the only route by which dense cortical bone can be remodelled: osteoclasts tunnel, capillaries follow, and osteoblasts refill the tunnel behind them. Because resorption precedes formation, a cortical graft is temporarily more porous and mechanically weaker part-way through incorporation than it was on the day of fixation, which is why fixation screws must hold and why loading is deferred. Option C is the distractor for candidates who know the vascular anatomy of cortical bone but not its turnover; a Volkmann canal is a pre-existing transverse channel, not a remodelling front, and revascularisation of a cortical block does not occur without resorption. (*Lindhe; ITI Treatment Guide vol 7*)

445 E Resorption and formation are coupled within the basic multicellular unit, and the coupling runs in one direction: as osteoclasts dissolve mineralised matrix they liberate transforming growth factor beta, insulin-like growth factors and other factors stored within it, and these recruit and activate the osteoblasts that refill the cavity. Suppress the osteoclast and fewer remodelling units are initiated, so formation falls with resorption — which is why turnover markers of both kinds decline and why the deficit produced by an antiresorptive is one of repair rather than of initial healing. Option A is the plausible error and it is a lineage mistake: osteoclasts derive from haematopoietic monocyte-macrophage precursors, osteoblasts from mesenchymal stromal cells. (*Lindhe; SDCEP MRONJ*)

446 B Denosumab is a monoclonal antibody to RANKL that is not incorporated into bone, so its action is reversible — but reversal is not benign. Stopping it is followed by a rebound overshoot of osteoclast activity, rapid loss of the density gained, and a well-described risk of multiple vertebral fractures, which is why guidance is that denosumab is not simply discontinued but followed by another antiresorptive if it is to be stopped at all. Options A, D and E are the bisphosphonate answers transplanted onto the wrong drug, and E is the most seductive because it is exactly what happens after alendronate is stopped; it is untrue of denosumab, where density is lost within months. (*SDCEP MRONJ; AAOMS position paper; BNF*)

447 C The osteocyte is the mechanosensor of bone. Loading deforms the matrix and drives interstitial fluid through the lacunocanalicular network; the shear this produces on osteocyte processes alters their signalling, notably by suppressing sclerostin — which otherwise inhibits Wnt-driven bone formation — and by modulating the RANKL and osteoprotegerin they express to control osteoclast recruitment. Unloading has the opposite effect, which is the cellular basis of disuse resorption. Option D is the distractor that catches well-read candidates, because piezoelectricity in bone is a genuine and much-quoted phenomenon; it is a physical property of collagen under strain, not the signalling pathway that recruits cells, and it has never been shown to be the operative mechanism. (*Lindhe; Frost*)

448 **B** Total stability is the sum of a falling mechanical component and a rising biological one. The bone that was compressed and surgically damaged at placement is devitalised and must be resorbed before it can be replaced, while the woven bone forming in the healing chambers has not yet mineralised enough to compensate; the resulting nadir sits at roughly two to four weeks, and it is deepest where insertion torque was highest, as here. Option A is the intuitive answer and it is wrong for a good reason: in the first week the implant still holds all of its primary mechanical stability, and the fibrin scaffold is protected by that rigidity rather than being the sole interface. (*Raghavendra et al.; ITI Treatment Guide vol 2; Misch*)

449 **A** Cone beam units do not reconstruct true attenuation values: grey values vary with the field of view, the exposure parameters, the position of the site within the volume and the amount of scattering tissue around it, so the same bone imaged twice can return different numbers and the scale is not transferable between machines. Density estimated this way is a guide, not a measurement, and the operative assessments remain the tactile feedback during drilling and the insertion torque achieved. Option C is the tempting statement because it sounds appropriately sceptical, but it is false in the opposite direction — density and insertion torque correlate reasonably well, which is precisely why the surgeon's hand is the better instrument here. (*Misch; ITI Treatment Guide vol 3*)

450 **E** All three belong to the dense-bone protocol and all three act on the same problem. Tapping cuts the thread profile in advance so that the implant is not obliged to cut its own path against an unyielding cortex, which is the single most effective way of bringing insertion torque down; countersinking relieves the crest, where the cortex cannot deform and where sustained compressive strain produces the ischaemia and microdamage that cost marginal bone; and a graded sequence of sharp drills at low speed limits both heat and the mechanical trauma of a single large-diameter cut. Candidates most often stop at D, excluding C because they file it under thermal injury rather than compression; in dense bone the two mechanisms of crestal loss run together. (*Misch; Eriksson & Albrektsson*)

451 **A** The threshold that matters is relative movement between implant and bone, in the region of 50 to 150 micrometres, above which the fibrin scaffold is disrupted and the interface repairs with collagen instead of bone. Splinting converts two independently loaded implants into one rigid unit, so a force applied over either one is shared and the tipping movement of each is greatly reduced; this is why immediate loading protocols for the edentulous mandible are built on splinted designs. Option C is the answer that sounds biologically kind and is the opposite of what is required: a resilient attachment permits movement of the prosthesis relative to the implant, which is useful for load sharing with the mucosa but does nothing to hold the fixture still. (*Szmukler-Moncler et al.; Brunski; McGill consensus*)

452 **D** Both surviving items destroy the same thing. Movement of about 200 micrometres is well above the tolerated band of roughly 50 to 150 micrometres: the fibrin scaffold and its ingrowing capillaries are torn repeatedly, and granulation tissue matures into scar. A necrotic rim of bone produced above about 47 °C must be resorbed before anything can be built, and if the volume is large the implant loses its support before secondary stability arrives. Item A is what makes E wrong, and it is the item candidates accept because they have been taught to graft every gap: marginal defects of about 2 mm or less around a stable implant in an uninfected site fill predictably with bone. (*Botticelli et al.; Szmukler-Moncler et al.; Eriksson & Albrektsson*)

453 **D** The threshold established by Eriksson and Albrektsson is 47 °C sustained for one minute, above which osteocytes die and the bone is resorbed rather than repaired, and it is a great deal lower than most clinicians assume. The relevance to a five-second exposure is that temperature at the cutting flute rises within a second or two in dense cortical bone and does not fall the moment the drill is withdrawn, so the tissue's exposure is longer than the operator's; the margin is small enough that no dry drilling is defensible. Option A is the attractive answer because 60 °C is a figure candidates half-remember from protein denaturation, and quoting it concedes his argument by setting the bar far too high. (*Eriksson & Albrektsson; Misch*)

454 **D** The Albrektsson criteria are not solely radiographic. Alongside immobility, absence of a peri-implant radiolucency, and bone loss under 1.5 mm in the first year with less than 0.2 mm annually thereafter, they require the absence of persistent pain, infection, neuropathy, paraesthesia and violation of the mandibular canal. This implant satisfies every mechanical and radiographic requirement and fails the neurological one, so it survives without being a success — a distinction that matters when quoting outcomes to patients. Option A is the commonest answer and it reveals a bank of memorised numbers unattached to the rest of the criteria; option E misapplies the numbers, since 0.9 mm is comfortably inside the first-year allowance. (*Albrektsson & Zarb; ITI Treatment Guide vol 12*)

455 **E** Of the four requirements summarised as primary closure, angiogenesis, space maintenance and stability, it is stability that fixation buys. A membrane that slides carries the graft particles and the clot with it, and the shearing that results converts an osteogenic compartment into a fibrous one — the same

mechanism as micromotion at an implant interface, on a larger scale. Option D is the distractor to argue with rather than dismiss, because tacks do contribute to holding a barrier in position, but the stem has deliberately given the space-maintaining role to the graft itself; the membrane in this case is limp and space maintenance is not what it is doing. (*Wang & Boyapati PASS; ITI Treatment Guide vol 7*)

456 **A** Regeneration is limited by the biology the recipient bed can deliver, and a dense sclerotic cortex delivers very little: it carries few vessels to the surface and no access to the marrow spaces where osteoprogenitor cells reside. Intramarrow penetration opens that supply, produces bleeding into the graft and accelerates revascularisation, which is why it is recommended where the recipient surface is corticated — the angiogenesis limb of the PASS principles. Option B is the plausible mechanical answer and it is not entirely fanciful, since perforation does key the surface a little; but interlock is not what determines whether a graft becomes bone, and a graft on an unperforated dense plate fails for want of blood supply, not for want of grip. (*Wang & Boyapati PASS; Lindhe*)

457 **C** Incorporation of cortical bone proceeds by creeping substitution, and resorption leads formation: cutting cones tunnel through the graft, so its porosity rises and its strength falls before osteoblasts refill the tunnels behind them. The graft is therefore at its most fragile part-way through incorporation, which is the reason rigid fixation must be maintained, the reason the site is protected from denture loading, and part of the reason several months are allowed before implants are placed. Option A is the intuitive answer and describes a cancellous graft rather better than a cortical one, since an open trabecular architecture is revascularised within days and gains strength progressively without the same resorptive trough. (*Lindhe; ITI Treatment Guide vol 7*)

458 **C** The incision line epithelialises within about 24 to 48 hours, which seals it against gross contamination but contributes almost nothing to strength. Strength is a matter of collagen, and during the proliferative phase fibroblasts are still depositing immature type III collagen; a week-old wound holds only a small percentage of the strength it will eventually have, which is why the sutures are still doing most of the work and why tension-free closure, not suture technique, is what saves a regeneration. Option A is the tempting answer because a raw wound looks most at risk, but the first day is protected by an intact clot and by sutures that have not yet begun to cut through. (*Wang & Boyapati PASS; Lindhe*)

459 **E** All three define the proliferative phase, which follows haemostasis and inflammation and occupies roughly the first three weeks. New capillaries sprout towards the hypoxic centre of the wound under the influence of vascular endothelial growth factor; epithelium migrates from the wound margins and re-establishes continuity; and fibroblasts, recruited during inflammation, lay down a provisional matrix rich in type III collagen. Option D is the popular answer, and candidates exclude C because they associate collagen with the remodelling phase — but remodelling is the *replacement* of that early type III collagen by orientated, cross-linked type I collagen over months, not the moment at which collagen first appears. (*Lindhe; Wang & Boyapati PASS*)

460 **A** Generalised loss of the lamina dura with a ground-glass trabecular pattern is one of the few genuinely systemic radiographic appearances in the jaws, and combined with a lytic lesion and a history of renal stones it points to hyperparathyroidism, the lesion being a possible brown tumour. That is an undiagnosed metabolic bone and calcium disorder, and the implant question is entirely secondary to it; the referral is a duty of care and the diagnosis may also explain the bone he has. Option D is the answer of a candidate who has correctly noticed abnormal bone and then filed it under a surgical classification, treating a metabolic disease as a drilling protocol. (*Lindhe; Misch*)

461 **E** Teriparatide is recombinant parathyroid hormone, and its behaviour depends entirely on how it is delivered: continuously raised parathyroid hormone, as in hyperparathyroidism, drives resorption, whereas a daily pulse stimulates osteoblasts and produces a net anabolic effect. It is therefore not an antiresorptive, it is not retained in the skeleton, and it is not the drug class associated with medication-related osteonecrosis. Option A is the answer most candidates give, because they have learned that osteoporosis drugs act on osteoclasts and apply the rule without asking which drug this is; it also describes denosumab, which is the neighbouring agent they are half-remembering. Liaison with her physician remains sensible. (*BNF; SDCEP MRONJ*)

462 **E** All three are part of the accepted explanation, and it is their combination rather than any one of them that makes the jaws unique. Alveolar bone remodels faster than most of the skeleton, so a drug that suppresses remodelling produces its largest deficit there. The bone lies beneath a mucosa a few millimetres thick that is breached by extraction, denture trauma or periodontal breakdown, so it is exposed to the mouth in a way the femur never is. And the dentition provides a permanent bacterial reservoir, which is why infection so often precedes the exposed bone. Candidates commonly stop at D, treating the bacterial challenge as a consequence rather than a contributor. (*SDCEP MRONJ; AAOMS position paper*)

463 **B** Radiotherapy produces a progressive endarteritis: vessels are lost rather than merely narrowed, and the tissue becomes hypovascular, hypocellular and hypoxic over years rather than recovering with time. Eight years therefore confers no reassurance, and elective surgery in a field that received 66 Gy remains a matter for the oncology and restorative team rather than for primary care. Item A is what makes D attractive and is the assumption most candidates carry, because the Marx protocol is widely taught; hyperbaric oxygen has not been established as effective prophylaxis before implant placement in irradiated bone and the evidence remains insufficient to rely upon. Item C reverses the natural history. (*Misch; ITI Treatment Guide vol 2*)

464 **E** Experimental work shows that some new bone can form against a previously contaminated implant surface after decontamination, but the contact achieved is incomplete and variable, and no clinical method reliably restores a surface to its original condition. What the radiograph demonstrates is bone filling a defect, which may or may not be in contact with the implant at all; the outcome to promise the patient is disease resolution and a maintainable site, not restored integration. Option A is the interpretation that both clinician and patient want to make, and it conflates two different things — mineralised tissue occupying a space, and mineralised tissue bonded to titanium. Lifelong supportive care remains the determinant of the result. (*2017 World Workshop; ITI Treatment Guide vol 13*)

465 **C** By ITI consensus terminology, loading within one week of placement is immediate, loading between one week and two months is early, and loading after two months is conventional; three weeks is therefore early loading. The objection is one of timing rather than of category: two to four weeks is precisely the period in which resorption of compressed and surgically damaged bone has reduced primary stability while new woven bone has not yet mineralised enough to replace it, so the interface is at its least able to tolerate load. Option B is chosen by candidates who know the definitions but place the dip too early, and the error is instructive, since it would license loading at exactly the wrong moment. (*ITI Treatment Guide vol 2; Raghavendra et al.*)

466 **C** Brånemark's original definition was histological — direct structural and functional connection between ordered living bone and the surface of a load-carrying implant — and it cannot be tested in a patient. The clinical definition subsequently offered by Zarb and Albrektsson is deliberately functional: a process whereby clinically asymptomatic rigid fixation is achieved and maintained in bone during functional loading. Option D is the distractor for candidates who remember the histology and forget that it requires a retrieved specimen; bone-to-implant contact can only be measured on undecalcified sections and is never complete even in ideal bone. A radiograph cannot resolve the interface, and torque and ISQ describe stability at a moment, not integration. (*Albrektsson & Zarb; Lindhe*)

467 **B** Early placement with soft tissue healing, at about four to eight weeks, buys one thing above all: an envelope of keratinised mucosa has formed over the socket, so the flap can be advanced and closed over a graft without tension, and the granulation tissue and infection have resolved. Item A is what makes D wrong, and it is the assumption candidates make because they know the buccal plate resorbs early: most of the dimensional change occurs over the first three months and continues for longer still, so at six weeks it is well under way and certainly not finished. Item C is false in the other direction — an immediate implant engaging the palatal wall and apical bone often achieves higher primary stability than one placed into a partly healed socket. (*ITI Treatment Guide vol 3; Chen & Buser*)

468 **A** Removing the mineral phase with acid leaves a collagen scaffold in which the bone morphogenetic proteins of the matrix become accessible to host cells, so demineralised allograft has osteoinductive potential — it can persuade undifferentiated mesenchymal cells to become osteoblasts — whereas the mineralised form is essentially osteoconductive. The trade is volume: demineralised material is soft, radiolucent and resorbs faster, so it is a poor choice where a dimension must be held. Option E is exactly the wrong way round and is the answer chosen by candidates who assume the more processed material must be the more durable; mineralised freeze-dried allograft is the volume-stable member of the pair. Inductive potency also varies considerably between donors and batches. (*Lindhe; ITI Treatment Guide vol 7*)

469 **D** Autogenous bone is the only graft material that is osteogenic, osteoinductive and osteoconductive at once: it transfers surviving osteoblasts and progenitor cells, and its matrix carries bone morphogenetic proteins and growth factors that recruit the host's own cells. Item B is what makes E wrong, and it is the item candidates accept without argument because autogenous bone is habitually called the gold standard: it is also the fastest of all graft materials to resorb, and where the object is to hold a contour rather than to regenerate bone it is the *worst* choice, which is why a slowly substituted xenograft is preferred for that purpose or the two are mixed. (*ITI Treatment Guide vol 7; Araújo & Lindhe*)

470 **D** The distinction is worth holding precisely. Remodelling replaces bone in place: a basic multicellular unit resorbs a packet and refills it, so the shape is preserved. Modelling changes shape, because resorption and formation act independently on different surfaces — and the loss of the buccal plate is

modelling, driven by the disappearance of the periodontal ligament that supplied its bundle bone. Option C is the distractor with a genuine idea behind it, since bone does respond to loss of load and the ridge is unloaded after extraction; but disuse cannot explain why the loss is so much greater on the buccal than the palatal aspect, and it is the bundle bone content, not the loading, that predicts the asymmetry. (*Araújo & Lindhe; Lindhe*)

471 **A** Retrospective cohorts have reported a modest excess of implant failure among users of selective serotonin reuptake inhibitors, plausibly through serotonin's role in bone metabolism, though the association is observational and confounded by depression itself, by smoking and by oral hygiene. It is worth mentioning during consent; it is not a reason to withhold treatment, and certainly not a reason to suggest stopping a psychotropic that is keeping her well. Item B is what makes D attractive, and it points the wrong way: statins have if anything been associated with better outcomes, as has antihypertensive therapy in the same literature, which is why E is wrong on two counts. (*Lindhe; BNF*)

472 **C** Experimental work following mucosal healing around transmucosal components shows that a barrier epithelium with its hemidesmosomal attachment and an underlying collagenous connective tissue cuff take on the order of six to eight weeks to become established, considerably longer than the epithelial closure of an incision. The clinical corollary is the one this stem is really about: disconnecting and reconnecting components during that period breaks the attachment and forces it to re-form more apically, at the expense of crestal bone, which is the argument for the one-abutment-one-time approach. Option A is the answer of a candidate applying soft tissue wound healing generally, and it is out by an order of magnitude. (*Berglundh & Lindhe; Abrahamsson et al.*)

473 **A** Regeneration depends on the recipient bed: a defect enclosed by several bony walls has a large area of vascularised, cell-bearing bone in contact with a small volume of graft, and it fills predictably. Building above the crest inverts that ratio — the graft sits on a single surface, is surrounded by soft tissue rather than bone, and must be held in a space that the overlying flap is constantly trying to collapse. Add the difficulty of tension-free closure over an enlarged ridge and the higher rate of exposure, and the outcome becomes far less certain. This is also the point at which the case leaves the scope of a general dentist with a special interest, and a short implant is a reasonable and evidence-supported alternative. (*ITI Treatment Guide vol 7; SAC Assessment Tool*)

474 **B** The two curves are of different natures and cross each other. Primary stability is purely mechanical and can only fall, because the bone gripping the implant at placement includes a rim that the drill devitalised and, where torque was high, a zone that was compressed beyond what it tolerates; both must be removed before they can be replaced. Secondary stability is biological, rising as woven bone forms in the healing chambers and then mineralises. Statement (iii) is what makes C and E wrong, and it is the marketing claim candidates accept: a hydrophilic surface accelerates early bone formation and makes the trough shallower and shorter, which is the basis for earlier loading, but it does not abolish the resorptive limb, which is surgical in origin. (*Raghavendra et al.; ITI Treatment Guide vol 2*)

475 **D** Insertion torque measures the friction generated as an implant is driven into an osteotomy; it is a proxy for primary mechanical stability and nothing more. Beyond the level needed to hold the implant still — commonly quoted as around 35 Ncm for immediate restoration — additional torque buys no biological advantage, and in bone that cannot deform it produces sustained compressive strain at the crest, ischaemia, microdamage and precisely the early marginal loss he is seeing. Option A is his own premise restated and is the answer most candidates take: torque reflects the mechanical grip at one moment, whereas bone-to-implant contact is earned biologically over the following weeks, and the two are not the same measurement. (*Misch; Eriksson & Albrektsson*)

Answers — Block R — Materials, implant design and surfaces, extended (Q476–515)

476 C Ti-6Al-4V contains roughly six per cent aluminium and four per cent vanadium, and the reservation about it as a bone-contacting material has always been the fate of those elements: particles and ions released by fretting at the joint, by wear during insertion or by corrosion in a crevice carry aluminium and vanadium into the tissues as well as titanium, and vanadium in particular has been questioned on cytotoxicity grounds. The alloy remains in universal use for abutment screws, where its strength is essential and the surface area is small. Option A simply reverses the facts and is chosen by candidates who assume commercially pure must mean superior; the alloy does passivate, does take a rough surface, and its modulus is close to that of the pure metal. (*Misch*)

477 B Beta-stabilised titanium alloys were developed in orthopaedics for exactly this reason and they do achieve a substantially lower elastic modulus than commercially pure titanium — very roughly half — but "lower" is not "matched". Cortical bone is an order of magnitude more compliant still, so the mismatch that concentrates functional strain where the stiff implant meets the crest is reduced rather than removed, and no clinical demonstration of less marginal bone loss follows automatically from the physics. Option A is the overstatement the claim invites and is the answer that would have you buy the product; option D is the opposite error, dismissing a real principle because it has been oversold. The oxide layer is a few nanometres thick and contributes nothing to bulk stiffness. (*Misch; ITI Treatment Guide vol 2*)

478 D Titanium is corrosion-resistant only because its oxide re-forms within milliseconds wherever oxygen is available. A narrow, fluid-filled crevice is precisely where oxygen is not available: the small volume trapped inside a connection is consumed, chloride concentrates, the pH within the crevice falls, and film that is disrupted by micromovement between the mating surfaces cannot be rebuilt — so damage accumulates internally while the well-oxygenated external surface remains passive. Option C is the distractor with real substance, since screws are commonly of a different alloy from the body, but the galvanic couple between two passivated titanium materials drives very little current; it is the crevice conditions and the fretting, not the dissimilar metals, that do the work here. (*Misch; ITI Treatment Guide vol 13*)

479 A Titanium's corrosion resistance depends entirely on its passive oxide, and fluoride at low pH is one of the few agents encountered in dentistry that attacks it: hydrofluoric acid forms in acidulated preparations and dissolves the film, roughening and discolouring exposed titanium and permitting continued corrosion. Neutral sodium fluoride does not do this, and is the appropriate choice where titanium is exposed to the preparation. Option B is the answer most candidates give, because fluoride is thought of as a benign, universally beneficial agent and the interaction is not widely taught; the practical point is that the patient still needs her fluoride, and the decision is about which preparation rather than whether to prescribe. (*Misch; ITI Treatment Guide vol 13*)

480 D A freshly prepared titanium oxide is hydroxylated and has very high surface energy; left in air it acquires a monolayer of hydrocarbon contamination from the atmosphere and loses that hydroxylated state, so the same topography that was completely wetted by blood when new becomes hydrophobic within weeks. Storing the surface under saline, or under an inert gas, is a means of preventing that ageing rather than a property of the surface itself — which is the answer to the colleague's question. Item C is what makes E wrong, and it sounds plausible because oxides do grow in air: any such growth is negligible at room temperature and would not explain a change in wettability, which is a matter of surface chemistry. (*ITI Treatment Guide vol 2; Albrektsson & Wennerberg*)

481 C Sa is the arithmetic mean deviation of the surface from its mean plane — a single height average that says nothing about the shape, spacing or sharpness of the features producing it. Two surfaces of equal Sa may present very different amounts of actual surface, and the developed interfacial area ratio, Sdr, captures exactly that: the percentage by which the true surface exceeds a flat projection of the same footprint. Since bone apposition depends on the area and geometry available to cells, Sdr often discriminates where Sa does not. Option A is the attractive near-miss, because Sa did evolve from the two-dimensional Ra; both parameters as quoted for implants are three-dimensional, and the distinction between them is what they measure, not in how many dimensions. (*Albrektsson & Wennerberg*)

482 E The argument is entirely about what happens if the crestal portion is one day exposed to the mouth, which in a treated periodontitis patient is a realistic prospect. A smooth machined surface accumulates less biofilm, is accessible to a curette and an air polisher, and can be cleaned without the surface alteration that instrumentation of a rough surface entails; the trade is a slightly lower bone-to-implant contact at the collar, which matters less than maintainability here. Option C is the true statement that argues for the opposite design and is chosen by candidates who read the question as "which surface integrates better"; roughness does raise crestal bone contact, and that is the reason the other design exists. (*ITI Treatment Guide vol 13; Albrektsson & Wennerberg*)

483 A Contact angle is a measure of wettability and therefore of surface energy. An angle near zero means a drop spreads rather than beading, so blood entering the osteotomy wets the entire topography including the depths of the pits, plasma proteins adsorb and spread, and the fibrin network formed during clot retraction stays anchored to the implant instead of pulling away from it — which is the basis for earlier migration of osteogenic cells onto the surface and for shortened healing protocols. Option B inverts the meaning of the number and is chosen by candidates who associate a "protected" surface with water repellency; a high contact angle is the hydrophobic, aged state that the packaging exists to prevent. (*ITI Treatment Guide vol 2*)

484 D Translucency in zirconia is bought by increasing the yttria content, which stabilises a greater proportion of the cubic phase. Cubic grains are optically isotropic, so light scatters less — but they do not participate in transformation toughening, and both flexural strength and fracture toughness fall accordingly. In a bruxist restoring a molar, that is the wrong trade, and the conventional grade is the safer choice. Option C is the plausible-sounding error and it points the wrong way: it is the tetragonal phase that transforms during ageing, so the more cubic, higher-yttria materials are in fact *less* prone to low-temperature degradation while being weaker — a distinction candidates frequently reverse. (*Misch*)

485 B Yttria-stabilised tetragonal zirconia is held in a metastable state at mouth temperature. The stress field at an advancing crack tip supplies the energy for those grains to convert to the monoclinic form, and because the monoclinic unit cell is about four per cent larger, the transformed grains expand against their surroundings and place the crack tip under compression, absorbing energy and arresting propagation. That is transformation toughening, and it is why zirconia's fracture toughness so exceeds that of a glass ceramic. Option D is the good wrong answer because crack deflection by a dispersed crystalline phase is a real toughening mechanism — it is the mechanism of leucite-reinforced and lithium disilicate ceramics, not of zirconia. (*Misch*)

486 C Low-temperature degradation is driven by water and heat acting on metastable tetragonal grains at the surface, and steam autoclaving is so effective at producing it that it is the standard laboratory method for accelerated ageing studies. Repeated autoclave cycles therefore spend part of the material's transformation capacity before the restoration has been loaded at all, leaving a microcracked surface layer that is rougher and weaker. Option B is the attractive alternative because annealing is a genuine heat treatment that alters ceramics, and the conclusion sounds right; but autoclave temperatures are far below anything that anneals zirconia, and the mechanism at work is hydrothermal, not thermal. (*Misch*)

487 D A flexural strength figure is a mean obtained from polished bars loaded monotonically and dry. Clinically the material is loaded millions of times in water, through a surface that has been milled, adjusted and perhaps sandblasted, and ceramics fail from the largest flaw present rather than at an average stress — which is why strength is properly described by a distribution, and why the same material fractures at widely differing loads. Item C is what makes E wrong and it is the item candidates accept because it is true of metals: ceramics have essentially no plastic range, and that absence is the whole reason surface flaws are so dangerous in them. (*Misch*)

488 E The idea that a compliant superstructure protects the interface is an old one and has never been substantiated: the force applied to a prosthesis is determined by the muscles and by the food, and the material of the framework or the occlusal surface alters the peak force transmitted to bone very little. What a low-modulus framework does reliably is flex, which threatens the passivity of the fit, the cement or screw joints and any veneering material carried on it. Option B is the seductive answer because the missing periodontal ligament is a genuine and much-discussed deficiency; but the ligament's damping is measured in tens of micrometres of physiological movement, and no prosthetic material has been shown to reproduce it. (*Misch; ITI Treatment Guide vol 12*)

489 A A shallow-angled conical connection retains its abutment largely by friction over the length of the taper, and after prolonged loading the two surfaces engage so closely that the joint behaves as though welded — the phenomenon usually described as cold welding. The screw's role is to draw the taper home rather than to hold it, so removing the screw does not release the abutment, and systems with this geometry supply an extraction instrument that pushes against the implant to break the lock. Option B is the sensible first thought and is excluded by the stem, which states that the screw came out intact; attempting to lever the abutment out before recognising the mechanism is how connections are damaged. (*Misch; ITI Treatment Guide vol 12*)

490 C A conical abutment does not reach its final position the instant it is torqued: the mating surfaces deform microscopically and the abutment sinks a small distance further into the taper over the following minutes and under subsequent loading, so its restorative platform ends up slightly more apical than when the impression was taken. Records made immediately after torquing therefore capture a position the abutment no longer occupies. Option E is the sophisticated distractor and it names a real process that is

happening at the same time — but embedment relaxation describes loss of clamping force at the asperities, which is corrected by re-torquing and does not by itself alter the vertical position enough to explain a rocking framework. (*Misch; ITI Treatment Guide vol 12*)

491 D Every implant–abutment junction harbours a small chronic inflammatory infiltrate, and crestal bone establishes itself at a distance apical to that infiltrate; the deeper the junction is placed, the further apically the bone finishes. This is the experimental basis for tissue-level designs, for placing bone-level implants no deeper than the restoration requires, and for platform switching, which moves the junction horizontally instead. Option E is the strong distractor because repeated disconnection genuinely does cost crestal bone, and it is a real and separate phenomenon — but in these studies bone loss related to the position of the junction occurred in implants that were not repeatedly disconnected at all. (*Hermann et al.; Lazzara & Porter*)

492 E All three are documented and they are consequences of one another. A screw joint is held closed by preload, and any lateral force large enough to exceed that preload on one side opens the gap there by a few micrometres and closes it again as the load passes; that cyclical opening pumps fluid and bacteria in and out of the internal cavity, which is why the space inside an implant is colonised, and the same relative movement between mating surfaces produces fretting debris. Candidates commonly stop at D, treating particle release as something that only follows instrumentation or implantoplasty; the fretting source is intrinsic to a two-piece design and is one reason conical connections, which move less, are preferred. (*ITI Treatment Guide vol 13; Misch*)

493 B The peri-implant supracrestal attachment requires a vertical dimension of roughly 3 to 4 mm, and if the soft tissue does not provide it the bone is resorbed until it does. Clinical studies of thin mucosa, conventionally defined as about 2 mm or less at placement, show marginal bone loss of this order in the first year irrespective of platform switching, whereas thickening the tissue at surgery or placing the implant deeper with a taller transmucosal component prevents it. Option E is the distractor with genuine anatomical logic, since phenotype components do correlate — but the loss described is circumferential rather than buccal, which is what identifies the attachment dimension rather than plate resorption as the cause. (*Linkevicius; Berglundh & Lindhe*)

494 E Microthreads are intended to transfer strain to the crestal bone within a physiological range and to add surface area where bone contact is most valuable, and the reported reductions in marginal bone loss are modest. The counterweight, which is the point in this patient, is what the same geometry does when threads become exposed: a finely threaded collar is markedly harder to debride than a smooth one and offers extensive shelter to biofilm, in a patient whose history already places him at raised risk of peri-implant disease. Option B overstates a real effect and is the answer the representative would prefer; crestal remodelling around a microgap is driven by the junction, not by the thread form above it. (*ITI Treatment Guide vol 13; Misch*)

495 C In a fresh extraction socket the coronal part of the implant faces a void, so all of the primary stability of a tapered design is generated in the few millimetres of bone beyond the apex and against the palatal wall. Torque rising steeply at the end of insertion is a sign that the apex is being wedged into that bone, and the hazards are perforation of the labial plate above the apex, compression necrosis in a small volume of engaged bone, and deflection of the implant out of its planned axis. The correct responses are to prepare the apical bone adequately and to accept a lower torque rather than forcing the implant. Option B is a real but secondary consideration and does not describe an insertion risk. (*ITI Treatment Guide vol 3; Misch*)

496 A Around ninety per cent of applied torque is consumed overcoming friction at the threads and under the screw head, and only the remainder becomes preload — the elastic tension in the screw that holds the joint closed. The manufacturer's torque value is calculated for clean, dry, unused mating surfaces with a defined friction coefficient, so any film of blood, saliva or debris changes the relationship between torque and preload in a direction and by an amount that cannot be predicted at the chairside. The remedy is a clean, new screw. Option D is the answer of a candidate who has learned to trust the torque wrench and has never asked what the number is for; torque is a means of generating preload, not a measure of it. (*Misch; ITI Treatment Guide vol 12*)

497 D The specified torque is derived from the screw's diameter, material, thread form and surface treatment together with the geometry of the joint it closes, and it is set so that the resulting preload uses a defined fraction of the screw's yield strength. Torquing a 20 Ncm screw to 35 Ncm may take it past that limit, so it deforms plastically instead of stretching elastically: preload is lost rather than gained, and the screw may fracture in service. Option B is the intuition behind the error and is worth stating plainly, since many clinicians believe tighter is safer; beyond yield, additional torque destroys the very tension it is meant to create. (*Misch; ITI Treatment Guide vol 12*)

498 **B** The screw is deliberately the most sacrificial part of the assembly: it is intended to loosen or fail before the abutment, the implant or the bone, so that a recoverable complication occurs instead of an unrecoverable one. A screw that fractures repeatedly under correct handling is therefore reporting a mechanical problem elsewhere, and in this patient the parafunction described is the obvious candidate, along with occlusal design. Replacing it a third time without addressing that is treating the fuse rather than the fault. Option C is the distractor with real mechanics behind it — inadequate preload does cause cyclical loading and fatigue — but the stem specifies correct torque with a calibrated driver at each delivery. (*Misch; ITI Treatment Guide vol 12*)

499 **D** Both surviving statements are true and together they give the measured answer. Galvanic corrosion requires an appreciable current between two metals in an electrolyte, and titanium's passive film is an efficient barrier that keeps that current very small; cobalt-chromium is itself a passivating alloy, and the combination is used routinely without clinical corrosion problems. What does cause metal release at these interfaces is the local environment — oxygen depletion within the joint, a falling pH under an inflammatory or biofilm-laden sulcus, and micromovement that abrades the film faster than it re-forms. Item C is what makes E wrong and is the alarming claim the concern rests on; it is not observed. (*Misch; ITI Treatment Guide vol 13*)

500 **E** Tribocorrosion describes the synergy between wear and corrosion: micromovement abrades the passive film while the surrounding medium prevents it from re-forming, so each process accelerates the other. Around an inflamed implant the medium is unfavourable — biofilm and inflammatory cells lower the pH and generate reactive oxygen species that destabilise titanium dioxide — and the result is a surface that both corrodes and sheds particles without any instrumentation having occurred. Option A is the answer most candidates reach for because two metals are undeniably present; a galvanic couple between passivated titanium components drives little current, and it would not explain damage confined to the exposed, inflamed portion of the implant. (*ITI Treatment Guide vol 13; Misch*)

501 **A** Genuine delayed hypersensitivity to titanium is described but is uncommon, and a nickel reaction does not predict it. Item B is what makes D attractive, and it is the pivot of the case: titanium salts penetrate the skin poorly, patch testing for titanium is not validated or standardised, and the commercial lymphocyte transformation assays marketed for this purpose have no established diagnostic value — so a positive result of that kind is not evidence of allergy. Item C reverses the epidemiology, since peri-implant inflammation is overwhelmingly biofilm-driven. The honest course is to explain what the test does and does not show, and to discuss a ceramic implant as a legitimate choice she may make on her own terms. (*ITI Treatment Guide vol 13; GDC Standards*)

502 **B** Milled and polished components leave the laboratory carrying machining lubricant, polishing paste, abrasive particles and handling contamination, and surface analyses of abutments delivered "ready to use" have repeatedly demonstrated them. These residues sit exactly where the connective tissue cuff must attach, and they have been shown to provoke an inflammatory response and to compromise that attachment. Sterilisation is not decontamination: an autoclave kills organisms but leaves the chemical residue in place, which is why option A is the answer that feels responsible and is insufficient. Ultrasonic cleaning in detergent followed by alcohol and steam, or an equivalent validated protocol, is what the surface requires. (*ITI Treatment Guide vol 13*)

503 **B** Beta-tricalcium phosphate is a synthetic, fully resorbable calcium phosphate that is dissolved and cell-mediated away over months, so the graft compartment converts to host bone relatively quickly — an advantage in a contained defect that will heal on its own account, and a liability where a contour must be held. Item A is what makes D tempting and it reverses the pair: it is the deproteinised bovine mineral that is substituted so slowly that particles remain identifiable years later, which is precisely why it is chosen for volume stability. Item C is untrue of every alloplast; only demineralised allograft and autogenous bone carry inductive proteins. (*ITI Treatment Guide vol 7; Araújo & Lindhe*)

504 **C** Deproteinised bovine bone mineral is the natural hydroxyapatite of bovine bone with the organic phase removed by heat and chemical treatment; it is osteoconductive, becomes intimately united with new bone, and is turned over so slowly that particles are demonstrable in biopsies many years later. Their persistence is the expected finding, not a failure, and it is the same property that makes the material the first choice where dimensional stability matters. Option A is the interpretation that a clinician unfamiliar with the material makes from the radiograph alone, and the clinical findings contradict it: fibrous encapsulation of a sinus graft does not support asymptomatic implants with stable bone levels for a decade. (*ITI Treatment Guide vol 5; ITI Treatment Guide vol 7*)

505 **A** Bioactive glasses are silicate-based synthetics that react with tissue fluid: sodium and calcium ions are exchanged, a silica gel forms, and a hydroxycarbonate apatite layer is precipitated on the particle surface, which is the layer to which bone bonds. The associated rise in local pH and osmolarity gives

the material a degree of antibacterial activity, which is one reason it has been used in defects at higher risk of contamination. Option B is the answer given by candidates who treat "bioactive" as a synonym for inductive; the material contains no protein at all, is wholly synthetic, and is therefore also free of the consent issues raised in option D. (*ITI Treatment Guide vol 7*)

506 **B** Reducing particle size increases the total surface presented to cells and to the resorptive machinery, so a fine graft is populated and turned over more quickly, and it adapts closely to an irregular defect surface. Item A is what makes D attractive and it reverses the relationship: small particles pack densely, leave smaller interparticle spaces for vascular ingrowth and offer less resistance to collapse of the overlying flap, so coarse granules are the better space maintainers in an uncontained site. Item C is false in the same direction, since fine particles are the more readily washed or pushed away during suturing. Neither grade is wrong here; the choice follows from what the defect needs. (*ITI Treatment Guide vol 7*)

507 **D** Cross-linking slows enzymatic degradation, so the barrier persists for months rather than weeks, which is a genuine advantage where regeneration is slow. The cost is biological: cross-linked membranes are less readily vascularised and integrated by the overlying flap, provoke more of a foreign-body response, and are consistently reported with higher rates of wound dehiscence and exposure — precisely the complication his tense flap already invites. Option E is the dangerous inference and it belongs to a different material: an exposed cross-linked collagen membrane still degrades and still permits bacterial ingress, whereas high-density PTFE, with a pore size below the dimension of oral bacteria, may be left exposed by design. (*ITI Treatment Guide vol 7; Wang & Boyapati PASS*)

508 **B** Aliphatic polyester membranes are broken down by hydrolysis rather than by enzymes, and the lactic and glycolic acid released lowers the local pH; where degradation is rapid or the volume of polymer is large, the result can be a sterile inflammatory reaction with swelling and occasionally exposure, weeks after apparently uneventful healing. Option A is the attractive answer because it describes a genuine degradation mechanism accurately — it is simply the mechanism of a collagen membrane, which is broken down by macrophage and neutrophil collagenase, and candidates who have learned one resorbable membrane apply its behaviour to the other. The distinction matters when a late swelling has to be interpreted. (*ITI Treatment Guide vol 7*)

509 **E** Vertical augmentation is the hardest regenerative problem in the jaws because the graft sits outside the bony envelope with the flap pressing down on it, so the barrier has to do something a collagen membrane cannot: hold a shape. Titanium reinforcement provides that rigidity, the dense PTFE resists bacterial penetration, and the whole assembly is removed at a second operation once regeneration is complete. Option A is the property patients and clinicians would prefer it to have and is the reason resorbable barriers are chosen elsewhere; it is not available in a membrane stiff enough to tent a vertical defect. This is also a procedure to refer rather than to attempt. (*ITI Treatment Guide vol 7; SAC Assessment Tool*)

510 **E** The classic design criteria are all three of these, together with clinical manageability, and holding them at once is what makes barriers difficult to design: a membrane must be tolerated and integrated by the tissues that cover it, must exclude the faster-migrating epithelial and connective tissue cells so that the slower osteogenic population can populate the defect, and must preserve a volume for that to happen in. Candidates commonly stop at D because they think of the barrier purely as a filter and assign space-making to the graft; a limp membrane collapsing onto the defect satisfies the first two requirements and regenerates nothing, which is the commonest reason a case under-performs. (*Wang & Boyapati PASS; ITI Treatment Guide vol 7*)

511 **C** Resin-modified glass ionomers contain hydrophilic monomer and take up water after setting, expanding hygroscopically over weeks to months. Inside a rigid ceramic shell that expansion has nowhere to go, so tensile hoop stresses develop in the crown and can produce delayed fracture with no obvious mechanical cause — which is why these cements are not recommended for all-ceramic restorations, however convenient their handling. Option A is the sensible first thought because incomplete seating genuinely does cause ceramic failure; but a low film thickness is what permits complete seating rather than preventing it, and an unseated crown would have been evident at delivery and on the radiograph. (*Misch*)

512 **A** A luting agent is specified to reach a film thickness no greater than about 25 micrometres precisely so that it can escape from between two closely fitting surfaces and allow the restoration to seat completely. Thickening the mix raises the film thickness and the hydraulic resistance to seating, so the crown ends high — which shows first as a premature occlusal contact and then as an open margin, an unbalanced screw joint and a deep, cement-filled space at the abutment. Option E states the belief behind the error, that cement can make up for fit; a thick cement lute is weaker, more soluble and more likely to leave a discrepancy than to fill one. (*Misch; ITI Treatment Guide vol 12*)

513 A Radiopacity is a property of composition: cements are visible on a radiograph because they contain fillers or salts of elements with a high atomic number — barium, strontium, ytterbium, zirconium in resin systems, and zinc in the traditional cements — and a resin cement formulated without them can be less radiopaque than dentine and effectively invisible against an abutment. Item B is what makes D attractive, because degree of conversion is a genuine and important property of a resin: it governs strength, solubility and biocompatibility, and has nothing to do with attenuation of x-rays. Item C is an optical property relevant to shade matching. A negative radiograph does not exclude retained cement. (*ITI Treatment Guide vol 13; Misch*)

514 D Studies of cement removal show that excess can be found and eliminated reliably only where the margin is at or close to the mucosal level, and that beyond roughly 1.5 to 2 mm subgingivally residue is left behind whatever the technique and whatever the operator's care. Three millimetres is well past that point, in a patient whose history already marks him as susceptible, and the answer is to remove the variable rather than manage it: a screw-retained crown has no cement at all. Option C is the response that feels prudent and is the commonest compromise; radiopacity helps you find residue, it does not help you remove it from a sulcus you cannot reach. (*Linkevicius; ITI Treatment Guide vol 13*)

515 D A visible gap that persists on two projections is a real discrepancy, and at 0.4 mm it is an order of magnitude beyond the tens of micrometres of an accurately made joint. The usual causes are mechanical and correctable — a tag of soft tissue, a fragment of bone or a chip of impression or temporary material trapped on the platform, or an abutment that has not rotated fully into its index — and all are found by removing the abutment and inspecting both surfaces. Option A is the instinct that turns a recoverable problem into a fracture: torquing against an obstruction cannot seat the abutment and takes the screw towards its yield point. An unseated joint leaks, loosens and loses bone. (*Misch; ITI Treatment Guide vol 12*)

Answers — Block S — Treatment planning, SAC, consent and referral, extended (Q516–560)

516 C The argument against a conventional bridge is almost always the biological cost of preparing sound teeth, and here that cost is already being paid: both potential abutments independently require full coverage, so the bridge adds a pontic to two crowns she needs anyway, at one-third of the cost and in a fraction of the time. Option A is the answer most candidates give, because "unrestored neighbours favour an implant" has been learned as a rule rather than as a reason, and the stem has removed the premise the rule depends on. It is not wrong so much as disproportionate — three restorations and a surgical episode where two restorations will do. B cannot be bonded reliably to a gold crown and a large amalgam, D is a removable solution to a fixed problem, and E leaves a functional space she has not accepted. (*Misch; GDC Standards*)

517 A The shortened dental arch is a legitimate endpoint, not a failure to treat, but it depends on the arch remaining stable, and this one is not: the unopposed upper second molars have already migrated 3 mm, exposing dentine and creating a food trap, and they will continue. That is a problem requiring management in its own right — reduction, restoration, intrusion or extraction — and it is the finding that makes "leave it alone" untenable. Option C is the distractor to beat, because it invokes the right criterion and states it falsely: he *has* four occluding premolar pairs and about twenty teeth, which is precisely the configuration the concept describes. B is true but is the physiological basis of the concept rather than an objection to it, D is duration without consequence, and E is a belief the epidemiology does not support. (*Käyser shortened dental arch; Lindhe*)

518 B An untreated apical lesion adjacent to a planned osteotomy is a reservoir of bacteria in the bone the implant must integrate with, and it is the recognised setting for retrograde peri-implantitis — an apical peri-implant radiolucency that appears within months of placement and is difficult to retrieve. The lesion is also a diagnosis in its own right: an eleven-year-old short root filling with periapical change needs orthograde retreatment, surgery or extraction, decided before anything irreversible happens next door. Option C is the sophisticated wrong answer, because a 3 mm separation sounds like a safety margin and candidates transfer the inter-implant distance rules to a situation they do not govern; distance does not sterilise the site. E defers without a plan, A places first and worries later, and D discards a tooth that has not yet been assessed. (*Misch; Lindhe*)

519 D An implant is a fixed point: it does not move, which is exactly why it makes excellent orthodontic anchorage, and exactly why it must be placed only into its final planned position. Here that position is unknown — the space is to be widened by 2.5 mm and the midline shifted by 3 mm, so the roots of the UL1 and UL3 will finish somewhere they are not now, and an implant sited today will end up mesial, angulated or too close to a root, with the interproximal crest and papilla lost. Option A is the attractive error and it inverts the biology: implants resist orthodontic force rather than succumbing to it, and that

resistance is the whole basis of using them as anchorage. B and C are stated without support from the case, and E is an inconvenience, not an objection. (*Misch; ITI Treatment Guide vol 1*)

520 D The endpoint of therapy is defined by pocket closure and the absence of inflammation, not by improvement: probing depths of 4 mm or less with no bleeding at any 4 mm site, and a full-mouth bleeding score below 10 per cent. Three sites at 5 mm that still bleed are residual active disease, and 14 per cent full-mouth bleeding is above the threshold, so both fall short and both must be addressed before an implant is placed into this mouth. Option E is the choice to beat and it fails on item C alone: a plaque score of 18 per cent is acceptable — the conventional target is under 20 to 25 per cent — and candidates reject it because they are looking for a perfect mouth rather than a defined criterion. Re-treat the three sites and re-evaluate. (*EFP S3 guideline; 2017 World Workshop*)

521 E The LR7 is hopeless — a through-and-through furcation, 70 per cent bone loss, mobility and a discharging sinus — and it is immediately distal to the site you propose to instrument, so it contaminates the field, it is destroying the bone that would form the distal wall of the LR6 osteotomy, and its removal will change the crestal contour the implant must be placed to. Removing it first and reassessing turns two entangled problems into one plannable space, and lets you decide whether the LR7 is replaced at all or the arch is shortened. Option D is the efficient-sounding answer and the one most candidates take, saving an appointment; it commits to an implant position beside a socket whose healing will alter the very anatomy that determined the position. A retains an infected tooth, B places into an unhealed socket, and C resects a tooth with no supporting bone to resect to. (*Lindhe; Misch*)

522 C The two dimensions are graded separately and they diverge here. Prosthodontically, two stud attachments retaining a mandibular overdenture against a satisfactory upper denture, with adequate restorative space, is the reference straightforward case — it is the McGill configuration. Surgically the anatomy has moved the case: 9 mm of height allows only a short implant, and a crest 3.5 mm wide will not accommodate a regular-diameter fixture without either crestal reduction, which spends the height she cannot spare, or a narrow implant, or augmentation. Option A is the answer most candidates give, applying the normative classification of "two-implant overdenture" without reading the millimetres — which is precisely the error the SAC tool exists to prevent, since general and site-specific modifiers are what convert a normative grade into a real one. (*SAC Assessment Tool; McGill consensus*)

523 E All three belong to the same category and each for a distinct reason, which is what makes the combination defensible rather than lazy. Altering the vertical dimension changes the occlusal scheme of the whole mouth and requires provisionalisation and a period of assessment before anything definitive is made. A high lip line exposes the mucosal margin and the papillae, so the aesthetic outcome depends on soft tissue that cannot be reliably rebuilt if it is lost. And an extended edentulous span in the aesthetic zone removes the natural attachment at both ends of the run, so papillae between adjacent implants are limited to a few millimetres and the tooth proportions must be created rather than copied. Option D is the popular answer, dismissing item C because "three units" sounds quantitative rather than aesthetic. (*SAC Assessment Tool; ITI Treatment Guide vol 1*)

524 A *Montgomery* requires the clinician to take reasonable care to ensure the patient is aware of material risks *and of any reasonable alternative or variant treatments* — and the duty attaches to the treatment, not to the practice. A sinus floor elevation with implant placement is a reasonable alternative for 4 mm of residual bone with an intact membrane and no septum, so she is entitled to know it exists, roughly what it involves, and that it can be obtained by referral, before she chooses a cantilever bridge. Option C is the distractor with the strongest pedigree, because it is the *Bolam* professional standard and is still how many clinicians reason; it was displaced for consent by *Montgomery*. B converts a commercial limitation into a clinical one, D would exclude alternatives that are well described but less studied, and E reverses the direction of the duty. (*Montgomery; GDC Standards*)

525 B A patient may decline information, but a waiver is not a switch that turns the duty off: the clinician should find out why — fear, trust, a wish not to be frightened — and must still convey what is needed for the decision to be his, including the nature of the surgery, the alternatives and the risks that would matter to anyone. Doing this briefly and kindly is almost always possible and takes minutes. Option A is the attractive answer because it appears to respect his autonomy and produces a tidy note, and it is the one that fails in retrospect: a patient who says "just do it" and then suffers an undisclosed complication has consented to nothing in particular. C mistakes paper for process, D punishes him for a coping style, and E discloses to someone with no standing to decide. (*Montgomery; GDC Standards*)

526 C Choosing no treatment is one of the reasonable options and it must be consented to like any other, which means the patient hears what is likely to follow: possible over-eruption of the UR6, drifting or tipping of the LR5 and LR7, the effect on the eventual difficulty and cost of replacement, and the fact that the decision can be revisited. Recording that discussion is what makes the refusal informed. Option D is

the choice that attracts the defensively minded, and item A is why it fails: a waiver of responsibility is not a recognised instrument in dentistry, it does not bind a dissatisfied patient and it cannot make inadequate information adequate. Item B invents an obligation — a review is offered, not imposed on a competent adult who has declined treatment. (*GDC Standards; Montgomery*)

527 **A** The Mental Capacity Act is explicit that a person is not to be treated as unable to make a decision merely because the decision appears unwise, and this man demonstrates the functional criteria in full — he understands the information, retains it, can articulate the trade-off he is making and communicates it. Continuing to smoke is a choice about his own risk, not evidence of a disturbance of mind. Option C is the distractor that catches candidates who have learned the opposite lesson too well: autonomy is a right to refuse, not a right to compel, so you may still decline if you judge the treatment unjustifiable — but a stable periodontium, excellent plaque control and reliable attendance make it justifiable here, with the raised risk consented and documented. D and E misstate consent, and B inverts the presumption. (*Mental Capacity Act; GDC Standards*)

528 **B** Two decisions are being conflated and they separate cleanly on the best-interests test. Removing roots that are ulcerating her lip relieves pain and prevents harm, is the least restrictive way of achieving that, and can be done under the Act after consulting carers and anyone else interested in her welfare, with the reasoning recorded. Elective implant treatment is a different proposition: it commits a woman who rejects a denture, cannot clean around abutments and has a progressive illness to lifelong maintenance she has not chosen, and it rarely survives the same test. Option A is the answer of the cautious candidate and it over-reads the law: the Court of Protection is for serious disputes and finely balanced or contested decisions, not for routine extractions to relieve pain. D abandons her; E is legally void. (*Mental Capacity Act; GDC Standards*)

529 **D** Removing four sound vital teeth to place implants in the most aesthetically demanding sites in the mouth cannot be justified: it exchanges a healthy dentition for a lifelong reconstruction with a real failure rate, an inferior soft tissue result and no way back. Consent does not make an indefensible plan defensible, and the professional obligation is to act in her interest, which here means declining and offering what could genuinely achieve what she wants — orthodontics, bonding or veneers, discussed with a mock-up. Option A is the trap and it is the modern form of the disclaimer: a meticulously documented request is still a request for harm, and the record would answer nothing at a hearing. C passes the problem sideways to someone who may say yes, B seeks a report to license the treatment rather than to help her, and E does the same thing more slowly. (*GDC Standards; SAC Assessment Tool*)

530 **E** All three belong in the document, and the third is the one that is almost always omitted. An implant restoration is a maintained prosthesis: over ten to fifteen years a substantial minority need a new screw, a recemented or remade crown, or a replaced abutment, and a patient who has been told this in advance experiences those visits as expected rather than as a betrayal. Maintenance is not an optional extra either — it is part of the treatment and the strongest determinant of the outcome, so it belongs in the price. Option D is the answer most practices actually give and it is the one that generates complaints at year six, because it quotes the cost of getting the crown and stays silent about the cost of keeping it. Give a revised estimate whenever the plan changes. (*GDC Standards; ITI Treatment Guide vol 12*)

531 **A** Do the arithmetic first: a 6 mm implant with the customary 2 mm of safety above the canal needs 8 mm, and he has 4 mm, so option D is not merely risky but impossible — it is the answer chosen by candidates who have learned that short implants solve posterior mandibular problems without checking whether this one fits. What remains is a man with occluding pairs to the second premolar, a stable unopposed-free arch, no functional complaint and no drifting after four years, which is the shortened dental arch performing exactly as described. Referral is the reflex answer and both B and C are real techniques, but nerve repositioning and vertical augmentation carry morbidity out of all proportion to a symptomless space. E destroys two sound teeth. Review the UR6 for over-eruption and leave the rest alone. (*Käyser shortened dental arch; ITI Treatment Guide vol 5*)

532 **D** A fractured full-arch framework on an unidentified system is outside the competence being examined for two separate reasons: the mechanical problem requires identifying the implant–abutment connection and remaking a framework, which is specialist restorative work, and the fracture itself is a symptom — it usually indicates a cantilever that is too long, an inadequate framework section or heavy parafunction, and remaking it without diagnosing why it broke guarantees a repeat. Option A is the strong distractor and deserves credit as an interim measure: removing a mobile cantilever does relieve the immediate problem and may well be the right thing to do today. It is not the answer to the question, which asks for the course, and left alone it becomes a permanent shortening no one has planned. B guesses at components, C leaves him without teeth, and E is not a repair. (*GDC Standards; SAC Assessment Tool*)

533 A Four millimetres of residual height with a septum dividing the sinus floor is a lateral window case, and a lateral window over a septum — where the elevating membrane must ride over a bony crest and tears at its apex — is the classic example of complexity that a general dentist with an interest recognises and refers. The other two features are real but neither is a referral threshold. Smoking is a general modifier that raises risk and demands counselling, dose reduction of surgical ambition and explicit consent; it is not a reason to send the patient elsewhere, or every smoker would be referred. A high lip line matters only where the mucosal margin is visible, and the stem states the UL6 is not. Option D is the popular answer, and it converts a risk factor into a competence limit. (*ITI Treatment Guide vol 5; SAC Assessment Tool*)

534 C With adequate bone at both sites, a low functional demand and no parafunction, one implant per missing tooth restored with individual crowns is the design that keeps the load axial, allows each unit to be cleaned and retrieved independently, and leaves the two sound neighbours untouched. Splinting is a response to a problem — poor bone quality, short implants, heavy occlusal load, cantilevers — and none of them is present, so option E buys shared load at the cost of hygiene access and retrievability. Option B is the distractor with the strongest commercial logic and the weakest mechanics: cantilevering a premolar from a single implant applies a bending moment to one screw joint every time he chews, and screw loosening and fracture are the documented consequence. A destroys two virgin teeth, and D cantilevers in the less favourable direction. (*Misch; ITI Treatment Guide vol 12*)

535 B Two constraints point the same way. A bar consumes vertical space — the bar itself, the clip housing and a thickness of acrylic over it — and typically needs something in the region of 12 to 14 mm from platform to occlusal plane; at 9 mm the denture base over a bar would be thinned to the point of midline fracture. And a bar creates a covered space beneath it that must be cleaned with floss or a small brush, which is exactly what a resting tremor prevents, whereas studs are cleaned by wiping around two separate abutments. Option D is the thoughtful wrong answer, because magnets are genuinely low-profile and forgiving of poor dexterity and insertion path; their retention is weaker and less durable, and the keepers corrode and wear, so they are a compromise reserved for patients who cannot manage anything else. (*ITI Treatment Guide vol 4; McGill consensus*)

536 D Implants are placed into a controlled oral environment, and the two items that must be settled are the two active diseases: untreated caries is a reservoir of cariogenic and periodontal organisms and a source of future emergencies mid-treatment, and untreated periodontitis identifies both an infected mouth and a susceptible host, the group in whom peri-implantitis rates are highest. Both must be treated and the result verified, not merely started. Option E fails on item C, and that is the item candidates accept because it feels morally correct: cessation must be offered, supported and recorded, and the raised risk must be consented to, but successful cessation is not a precondition of treatment, and making it one would deny implants to a large group who can be treated with modified expectations and closer maintenance. (*EFP S3 guideline; ITI Treatment Guide vol 13; GDC Standards*)

537 C The duty of candour is engaged as soon as something has gone wrong that could affect the patient, and it requires telling her promptly, apologising, explaining what it means and what can be done — before the decision about how to proceed is taken, because that decision is hers. An apology is not an admission of liability, and delaying it until the crown looks wrong converts a manageable conversation into a complaint. Option A is what many clinicians do and is the distractor to beat: a customised abutment and a graft may well be the right technical answer, and choosing it is not dishonest in itself, but choosing it silently deprives her of the choice between accepting a compromise, having the implant removed, or being referred. B compounds the concealment, D pre-empts her decision, and E puts the sequence the wrong way round. (*GDC Standards; duty of candour*)

538 E Describe what you can see and measure — implant position, margin visibility, recession, probing depths, radiographic bone levels — photograph it, and give her that record. What you cannot do is characterise another clinician's care as negligent on the basis of an outcome, because you did not see the starting anatomy, the consent discussion or the intraoperative findings, and negligence is a legal conclusion that requires all of them. Directing her to the practice complaints procedure, the Dental Complaints Service or her own advisers is the helpful and proper route. Option A is the tempting answer because the findings do look poor and she is plainly upset, and it is the one that damages both a colleague and your own credibility as a witness. C bypasses her consent to disclose, and D says in speech what A says in writing. (*GDC Standards*)

539 B Consent is a process that includes time to absorb information, to consider the alternatives, to discuss the cost with whoever else it affects and to ask the questions that only occur later; for elective surgery with a fee, a written plan and an irreversible component, a separate consenting visit is the expected standard and is the strongest evidence that the process happened. His inconvenience is real and should be met by sending written information the same day and booking the surgery promptly — not by compressing

the safeguard. Option A is the answer that reads well in a record and is the one that unravels under challenge, since a discussion and a decision taken in the same hour give no opportunity for reflection. C mistakes a signature for consent, E splits a single course of treatment artificially, and D is unnecessary. (*GDC Standards; Montgomery*)

540 A The honest figure is the one he can use: smoking approximately doubles the risk of implant failure, and it also raises the long-term risk of peri-implantitis, so his personal risk is meaningfully higher than the population figures he has read online. Item B is what makes option D the popular error, and it is attractive because dose reduction is a genuine public health good and it is what he has offered; there is no evidence that halving consumption abolishes or even mostly removes the excess risk in implant outcomes, and telling him so would be a false reassurance on which he would base his decision. Item C invents a pharmacological remedy for a vascular and cellular problem and adds resistance and adverse effects for nothing. Offer cessation support, consent him to the raised risk, and shorten his recall. (*Lindhe; ITI Treatment Guide vol 2; SDCEP Drug Prescribing*)

541 D Three millimetres is beyond what can be taken off an intact tooth: enameloplasty in this region will expose dentine over a wide area, risks the pulp and leaves a flattened, hypersensitive palatal cusp. Intrusion moves the tooth and its alveolar bone together, restores the occlusal plane, creates the restorative space and preserves an unrestored vital tooth — and, done with a segmental appliance or a temporary anchorage device, it is a defined, finite piece of treatment that must precede the implant because it determines where the crown will sit. Option A is the answer most candidates choose, and it is right for 1 to 2 mm of over-eruption, which is the situation they are remembering. B devitalises a healthy tooth to solve a space problem, C removes it altogether, and E makes an unrestorable crown to fit a space that should not exist. (*Misch; ITI Treatment Guide vol 1*)

542 C An ankylosed incisor in a growing face is a fixed point in a moving alveolus, so the infraocclusion will worsen and the local crest will be dragged with it; but the root is also the only thing maintaining the alveolar bone at that site. Decoronation removes the crown, leaves the root submerged beneath the flap to be replaced by bone as it resorbs, and preserves ridge height and width for an implant once growth is complete — while a bonded pontic keeps the appearance acceptable in the meantime. Option B is the answer that most candidates choose, because extraction feels decisive and the resin-bonded bridge is the correct interim prosthesis; it discards the very structure that is preserving the ridge, and the buccal plate collapses in the years before he is old enough for an implant. D fails because an ankylosed tooth cannot be moved, and E entrenches the discrepancy. (*ITI Treatment Guide vol 1; Misch*)

543 E The belief driving the request is a misreading of the literature: implant survival figures of around 95 per cent at ten years are quoted against the survival of teeth that were failing when they were studied, not against a symptom-free, well-root-filled, well-crowned molar with a healthy periodontium, which has an excellent prognosis and no reason to be removed. Extracting it exchanges a functioning tooth for a surgical procedure, a healing period, a lifelong maintenance commitment and a definite failure rate, and it cannot be undone. Option A is the trap for candidates who have absorbed the language of autonomy: he may refuse treatment, but he cannot require you to provide an intervention you judge to be against his interests, and consent does not convert harm into benefit. B is a disclaimer, C outsources a decision you can make, and D begins the same journey. (*GDC Standards; Pjetursson et al. systematic reviews*)

544 D Two arguments hold. A cantilever resin-bonded bridge removes almost no tooth tissue, so the classic objection to a fixed prosthesis on virgin teeth largely disappears, and it can be replaced by an implant later without having committed anything. And a centimetre of growth in a year means the face is still changing: an implant behaves as an ankylosed tooth and will be left in infraocclusion, a complication that appears years afterwards and is very difficult to correct. Option E fails on item C, and that item is the one candidates read as a limitation because they remember that narrow spaces exclude implants. Six and a half millimetres is *sufficient* — a 3.3 mm implant with 1.5 mm to each root needs a little over 6 mm — so the space is an argument for the implant's feasibility, not against it. (*Misch; ITI Treatment Guide vol 1*)

545 E A financial compromise must also be a biological one, and the test is whether each phase is a complete, stable endpoint in itself that forecloses nothing. A well-made removable partial denture restores function and appearance immediately, costs little, preserves both sound abutment teeth and leaves every implant option open for whenever she can afford it. Option B is the distractor with the most surface appeal, because it is fixed, affordable and delivered in weeks — and it spends two intact teeth permanently to do it, in a mouth where the definitive plan needs neither of them. C loads a single implant with a two-unit cantilever and is mechanically indefensible; D restores one space and leaves two; A is an affordability fiction that leaves her committed to a debt for surgery already performed. (*GDC Standards; Misch*)

546 A An implant is a lifelong commitment to maintenance and to the prompt management of things that go wrong — a loose screw, a fractured ceramic, a bleeding 6 mm pocket — and none of that can

wait ten months or be dealt with by someone who cannot obtain the components. The decision therefore turns on whether there is a competent clinician where he lives who can access the system, and if there is not, the honest conclusion may be to choose a design that is maintainable anywhere or to defer until his circumstances change. Option D is the sophisticated near-miss and is worth doing anyway: screw retention genuinely simplifies retrieval and is the sensible design response — but it answers *how* to build the restoration, not *whether* the plan is safe to embark on. (*GDC Standards; ITI Treatment Guide vol 13*)

547 **C** An impacted tooth 1.5 mm from an osteotomy is a problem in two directions. It may itself become symptomatic, develop a follicular cyst or need removal later — and removing it then would mean operating alongside an integrated implant with a real chance of destroying it. Taking it out first, allowing the socket to consolidate and returning to a healed bone bed of known quality is the sequence that keeps the options open. Option D is the efficient answer and the one candidates take, saving an appointment; the extraction of a horizontally impacted lower third molar leaves a large defect against the distal wall of the osteotomy, so primary stability is unpredictable and the graft, the implant and a fresh surgical wound all compete in the same site. A and E distort the restorative position to avoid a problem that should be removed, and B stores it up. (*Misch; ITI Treatment Guide vol 1*)

548 **D** Implants are placed to the position of the planned teeth, and the position of the planned teeth is not yet known: a 4 mm reverse overjet with a concave profile the patient has volunteered dislike of raises the question of whether the maxilla should be advanced, and orthognathic surgery moves the alveolus and everything fixed in it. Placing implants first commits the incisal position to a jaw relationship she may not keep, and no prosthetic contrivance recovers it. Option C is the attractive answer because the ridge deficiency is the visible technical problem and augmenting it feels like progress; it solves the wrong problem in the wrong order, and the graft would be placed to a target that the joint assessment may move. A entrenches the crossbite, B defers without addressing the skeletal issue, and E asks a question no one has raised. (*SAC Assessment Tool; ITI Treatment Guide vol 1*)

549 **B** Only the second statement holds, and it is the whole of the obligation: the discussion and the decision go in the record, contemporaneously, in enough detail that a reader can see what he was told and what he chose. Statement (i) is what makes option C the popular error, because it sounds like reasonable economy of effort; no treatment is one of the reasonable alternatives and *Montgomery* requires that reasonable alternatives be disclosed whether or not the patient thinks to ask, so it must be raised, with its foreseeable consequences. Statement (iii) invents a formality that does not exist — informed refusal, like informed consent, is a discussion rather than a signature, and a form adds nothing that a good contemporaneous note does not already provide. (*Montgomery; GDC Standards*)

550 **A** You cannot advise on a case you have not assessed, and nothing in the history so far tells you why the two failures occurred, whether the remaining implants are restorable, or what the bone and soft tissue will allow — and the two that remain are, on the figures given, healthy. History, examination, radiographs, mounted casts and a careful exploration of what she actually wants come before any recommendation, and they are also what will reveal whether her expectations are achievable. Option C is the reasonable-sounding alternative and deserves to be done in parallel: previous records are valuable and should be requested with her consent, but they are not a precondition of examining the patient in front of you, and waiting for them delays the assessment that will determine everything. B destroys bone and evidence, D prejudges her, and E outsources the plan. (*GDC Standards; SAC Assessment Tool*)

551 **B** Width measured at the crest is reassuring and misleading: a submandibular fossa means the ridge narrows and swings lingually below the crest, so a drill advanced along the apparent long axis will perforate the lingual plate even though 7 mm was measured at the top. That perforation is the mechanism of the floor-of-mouth haematoma, and avoiding it requires reading the cross-sectional image, angling the osteotomy more buccally or accepting a shorter implant — considerations that lift this from a routine posterior site. Option C is the answer candidates give because it is the number they were taught to check, and it is exactly the measurement the concavity renders unsafe to rely on. A is generous height, D is a genuine structure but not in the path of an LR6 osteotomy, and E is a normal functional finding. (*Lindhe; SAC Assessment Tool; ITI Treatment Guide vol 11*)

552 **C** The first two statements are the honest answer to his question, and they are the answer that protects both of you. Implants cost more at the outset, and neither option is maintenance-free: the bridge risks caries and endodontic complications at two abutments and will eventually be remade, while the implant crown accumulates screw loosening, ceramic chipping, decementation and the ongoing cost of supportive care and review radiographs. Whether either is "cheaper in the long run" depends on his mouth and his habits and cannot be promised. Statement (iii) is what makes option D and option E wrong and it is the belief patients most often arrive with — that the implant is a one-off purchase. Say clearly that it is not, and put the maintenance figures in the written estimate. (*Pjetursson et al. systematic reviews; GDC Standards*)

553 E The crowding is the decisive finding. Where a lateral incisor is absent *and* the arch is crowded, closing the space uses the deficiency to relieve the crowding, substitutes the canine into the lateral position with reshaping and bonding, and leaves her with a complete dentition and no prosthesis to maintain for the next seventy years. It is the option that avoids committing an adolescent to a lifetime of restorations before she has finished growing. Option B is the answer the family have already arrived at and is the one to argue against: even with growth demonstrated, an implant at eighteen is a prosthesis for life, with infraocclusion still reported in the third decade, and it presupposes that the space should be maintained at all. A and C are interim measures for a plan that may not be needed, and D introduces a second surgical site unnecessarily. (*Misch; ITI Treatment Guide vol 1*)

554 B The individual findings are the evidence; the prognosis is the conclusion, and it is the conclusion that changes the plan. Sixty per cent bone loss, mobility, a furcation involvement and a deep bleeding distal pocket in a mouth that is otherwise stable identify a tooth likely to be lost within the timescale of the reconstruction — so the number and position of implants must be decided now for a span that may extend to the LR7, rather than for a three-unit prosthesis that cannot be extended once it is made. Option A is the finding candidates choose because it is the most alarming single number, and it is one input to the judgement rather than the judgement itself; the same is true of C and D. Decide whether the LR7 is retained and treated, or removed and planned for, before an implant goes in. (*Lindhe; Misch*)

555 C Shared care fails at the boundaries, and the boundary that matters most is the one nobody wants: who owns a biological complication years after the surgery, who initiates treatment, and to whom the patient turns. Agreeing that in writing at the outset — with the patient told, so that he knows whom to ring — is what converts two clinicians into one pathway. Option D is the plausible alternative and is worth agreeing too, since radiographic baselines and intervals are exactly the sort of thing that falls between two practices; but a monitoring schedule without an owner simply generates images that nobody acts on. A is a technical detail resolved in the referral letter, B is a commercial matter that does not affect clinical responsibility, and E is already settled in law: whoever performs a procedure consents for it. (*GDC Standards; ITI Treatment Guide vol 13*)

556 D The Act requires that all practicable steps be taken to help a person decide before he is treated as unable to: simple language, one idea at a time, pictures and models, a second appointment, the support worker present to help him rather than to speak for him. His confusion so far is evidence about *your explanation*, not about his mind, and he has already demonstrated that he can express a preference and a reason for it. Option A is the error the Act was written to prevent, and it is the commonest one — moving from "he did not follow my explanation" to "he lacks capacity" in a single step. B is legally void without a registered authority, C quietly removes the choice he expressed, and E outsources an assessment that you are competent and obliged to attempt first. (*Mental Capacity Act; GDC Standards*)

557 E The consent argument is a distraction that the stem has set up deliberately: at seventeen she can indeed consent to her own treatment, and her parents' objection would not override her. It makes no difference, because the clinical answer is the same whoever signs. An implant behaves as an ankylosed tooth, and 1.5 cm of growth in the past year means the alveolus is still developing; placing now produces infraocclusion of the most visible tooth in her mouth, appearing years later and correctable only by removing and replacing the restoration or the implant. Option A is the answer candidates give when they spot the legal point and stop there. The right course is to say so plainly, make a provisional she is not ashamed of, and monitor growth with serial records. (*Misch; ITI Treatment Guide vol 1; GDC Standards*)

558 B The implants and the prosthesis have quite different life expectancies, and it is the prosthesis that generates the work: screws loosen and occasionally fracture, acrylic teeth wear and debond, the framework must be removed to clean beneath it and to inspect the abutments, and a full-arch reconstruction is generally expected to need substantial repair or remaking within its service life. Saying so before treatment converts a decade of appointments into an anticipated, budgeted programme. Option C is the answer most clinicians actually give, and it is the reason patients feel misled: hygiene visits are necessary but they are the smallest part of the burden, and they say nothing about retrievability or component cost. A and D deny the technical complication rate altogether, and E puts a single confident number on something that varies widely. (*ITI Treatment Guide vol 4; Pjetursson et al. systematic reviews*)

559 A A competent adult may refuse any investigation, and that refusal must be respected — but it does not oblige you to proceed without the information the procedure requires. Cross-sectional or at least current two-dimensional imaging is a prerequisite for siting an implant above the mandibular canal, and a nine-year-old panoramic film shows neither the present bone level nor the buccolingual dimension. Declining, and explaining why, is the answer that respects both her autonomy and your duty not to operate blind. Option D is the distractor with the best science and the worst ethics: the effective dose of a periapical is indeed trivial, of the order of a few microsieverts, and telling her so is a proper part of the conversation

— but taking it after she has refused all radiographs overrides a valid refusal. C substitutes surgery for imaging, and B and E find a way round the problem rather than through it. (*IR(ME)R; GDC Standards*)

560 E The unresolved question here is prosthodontic, not surgical: her lip needs flange support, which a fixed prosthesis cannot provide without a large amount of pink material in exactly that region, and until a trial set-up without a flange has shown what is achievable, no one can say how many implants are needed or where. Restorative-led assessment establishes the target and then commissions the surgery to meet it, and this arch is beyond both your surgical and your restorative experience. Option C is the tempting answer because bilateral grafting is the obvious technical requirement and referring it feels decisive; it commits to a technique before the design question is answered, which is prosthetically driven planning in reverse. B may well be the eventual answer and deserves credit, A places too few implants in the worst available bone, and D repeats what she cannot tolerate. (*ITI Treatment Guide vol 4; SAC Assessment Tool; GDC Standards*)

Answers — Block T — Prescribing, local anaesthesia, drug interactions and sedation, extended (Q561–590)

561 C Work it through in order. At 4.4 mg/kg a 58 kg woman may receive 255 mg of lidocaine. A 2 per cent solution contains 20 mg/ml, so a 2.2 ml cartridge delivers 44 mg. Dividing 255 by 44 gives 5.8, and because cartridges are used whole the answer is five, supplying 220 mg. Option E is the instructive error: nine cartridges is what 7 mg/kg would allow, and 7 mg/kg is the articaine figure, not the lidocaine one — candidates who have memorised a single number for "local anaesthetic" apply it to whichever solution is on the bracket table. Nine cartridges would deliver 396 mg to a small adult. The adrenaline is not limiting here at 27.5 micrograms per cartridge in a healthy patient, but the calculation should be done before the first injection, not after the fourth. (*BNF; SDCEP*)

562 B Felypressin is an analogue of vasopressin, and its vasoconstriction is mediated through vasopressin receptors rather than adrenoceptors — which is why it is reached for as the catecholamine-free option, and why the substitution is not the free lunch it appears. At higher doses it has a direct coronary vasoconstrictor action, so it carries its own caution in ischaemic heart disease, and it does nothing to remove the underlying problem. Option D is the distractor with real substance and should be argued: felypressin genuinely gives a poorer surgical field, which matters in flap surgery, but that is an inconvenience rather than a safety objection. The better answer for this man is lidocaine with adrenaline, aspirated carefully and dose-limited, because profound anaesthesia suppresses far more endogenous catecholamine than the cartridge contains. (*BNF; SDCEP*)

563 A A profoundly numb lip proves the block reached the main trunk, so repeating it treats a problem that does not exist. Residual sensitivity in bone after a successful block is characteristically due to accessory supply — mylohyoid nerve fibres entering through the lingual cortex are the classic culprit — or to incomplete blockade of the plexus at that site. Four per cent articaine given as a buccal infiltration diffuses through the mandibular cortical plate better than lidocaine at half the concentration, and supplemental buccal infiltration with articaine after a failed or partial block is the best-supported rescue technique. Option B is the reflex answer and the one most candidates give, adding dose and delay for no mechanistic reason. C anaesthetises buccal mucosa only, which is already numb; D repeats the same nerve by another route; E addresses the wrong side of the ridge. (*BNF; SDCEP*)

564 D A needle placed too far posteriorly passes behind the ramus and deposits solution within the parotid gland, where the facial nerve divides; the branches are blocked, giving a unilateral lower motor neurone palsy of sudden onset. The give-away is that the *lower lip is not numb* — the inferior alveolar nerve was never reached — and that the deficit is motor with no sensory or limb involvement. It is transient, resolving as the anaesthetic wears off over one to several hours. The immediate action is to explain what has happened, tape or patch the eye and provide artificial tears, because the blink reflex is lost and the cornea is unprotected, then review. Option A is the distractor that panics candidates; a stroke would not spare the forehead in this pattern, would not be instantaneous with the injection, and would have other signs. (*BNF; SDCEP*)

565 E All three operate together, which is why infected tissue is so much harder to anaesthetise than the textbook single explanation suggests. Local anaesthetics are weak bases: only the unionised free base crosses the nerve membrane, and the acid environment of inflamed tissue shifts the equilibrium towards the charged form, leaving less available to enter the axon. Inflammatory hyperaemia then clears what does get in more quickly, shortening and weakening the block. And peripheral sensitisation alters the nociceptor itself, with upregulation of tetrodotoxin-resistant sodium channels that are relatively resistant to blockade and a threshold low enough that ordinary stimuli fire them. Candidates most often reject

(iii), because pH is the mechanism they were taught and they treat it as the whole answer. The practical response is a regional block away from the inflamed field. (*BNF; SDCEP*)

566 C Two milligrams per kilogram allows 140 mg in this man, which is 28 ml of a 5 mg/ml solution — a large volume and a long way from what a dental appointment normally uses, but the reason the ceiling matters is qualitative. Bupivacaine is highly lipid-soluble and binds cardiac sodium channels avidly, dissociating from them slowly; the usual comfortable margin between the central nervous system prodrome and cardiovascular collapse is compressed, so arrhythmia and refractory cardiac arrest can occur with little warning and are notoriously difficult to resuscitate. Option B is the attractive answer because the slow onset is real and clinically obvious — up to twenty minutes for a block — but a slow onset delays surgery, it does not cause the toxicity. A describes articaine's ester side chain, D belongs to prilocaine, and E is true of every amide. (*BNF; SDCEP*)

567 A A 1:200,000 solution is 1 g in 200,000 ml, which is 5 micrograms per millilitre and therefore 11 micrograms in a 2.2 ml cartridge; three cartridges deliver 33 micrograms and 6.6 ml of solution, comfortably inside the 40 microgram ceiling. The 1:100,000 solution contains 10 micrograms per millilitre, or 22 micrograms per cartridge, so two cartridges give 44 micrograms and breach the limit — which is why B is wrong and why D, though safe, buys only 2.2 ml for a ninety-minute operation. Option C is the error that catches candidates who have grasped the principle but not finished the arithmetic: four cartridges of 1:200,000 also give 44 micrograms, exactly the same excess as two of the stronger solution. The articaine itself is nowhere near its own 7 mg/kg limit of 560 mg. (*BNF; SDCEP*)

568 B The pooled randomised evidence supports a single pre-operative dose with a small but statistically significant reduction in early implant failure and a number needed to treat of roughly twenty-five — a real effect, and one that should be quoted honestly, since twenty-four of every twenty-five patients take an antibiotic from which they derive no benefit. Option C is the answer that intuition supplies and is the one most candidates choose, because antibiotics obviously prevent infection; the outcome that reached significance in the trials was implant *failure*, while post-operative infection rates were not convincingly reduced. A overstates the effect by an order of magnitude, D describes an outcome antibiotics do not influence, and E reverses the population, since the trials were largely in straightforward placement. Prescribe once, beforehand, or not at all. (*Esposito et al., Cochrane review; SDCEP Drug Prescribing*)

569 D The two defensible measures are both given before the wound is made, which is the principle that unites them: a single high oral dose produces peak tissue concentrations at the moment the bone is instrumented and the membrane placed, and a pre-operative rinse reduces the intraoral bacterial load that the flap will be contaminated by. Option E fails on item C, and it is the item most clinicians actually prescribe, on the reasoning that a graft and a membrane are more vulnerable than a plain implant and deserve cover until the flap has sealed. There is no evidence that continuing antibiotics after surgery reduces failure, infection or membrane exposure, and the course brings gastrointestinal harm, occasional anaphylaxis and selection pressure for resistance in exchange for reassurance. (*Esposito et al., Cochrane review; SDCEP Drug Prescribing*)

570 E This history — urticaria, angioedema and wheeze within an hour, requiring emergency treatment — is a genuine immediate hypersensitivity, quite unlike the gastrointestinal upset that accounts for most "penicillin allergy" labels, and it excludes the whole penicillin group including option D, which is amoxicillin with a beta-lactamase inhibitor. A single 600 mg dose of clindamycin is the established alternative for surgical prophylaxis in penicillin allergy: it is active against the oral streptococci and anaerobes that matter and penetrates bone well. Option A is the distractor that rewards half-knowledge, and the half is correct — cross-reactivity between penicillins and later cephalosporins is genuinely low, of the order of a per cent or two — but a documented anaphylactoid reaction is precisely the situation in which that residual risk is not worth taking. Warn about *Clostridioides difficile*. (*BNF; SDCEP Drug Prescribing*)

571 A Metronidazole is reduced to its active form only in an anaerobic environment, so its spectrum is confined to obligate anaerobes and protozoa. Early spreading odontogenic and peri-implant infections are mixed, and the organisms driving cellulitis at this stage are typically viridans and anginosus-group streptococci — facultative organisms against which metronidazole has essentially no activity. Given alone it treats half the infection and leaves the spreading half untouched. In true penicillin allergy the appropriate choice is clindamycin, which covers both, alongside urgent attention to the source and, with trismus, dysphagia and fever, a low threshold for same-day referral. Option E is the distractor with a sound principle behind it — antibiotics never substitute for drainage — but this is diffuse cellulitis with no collection to drain, and the statement is offered as an absolute that is not true. (*BNF; SDCEP Drug Prescribing*)

572 C Failure to respond to an antibiotic almost always means the source has not been controlled, not that the wrong molecule was chosen, so the first action is to look again: re-examine, re-open and irrigate the drainage site, and reconsider whether the implant itself must come out. The second action is not a

prescription either — trismus at 20 mm with dysphagia and fever is a spreading infection threatening the airway, and it needs urgent referral to a maxillofacial unit the same day. Options D and E are attractive because escalating or broadening therapy feels like doing something, and it is what most clinicians reach for; changing or adding agents while the cause remains untreated delays definitive care in a patient who is deteriorating, and both proposals treat the prescription rather than the patient. (*SDCEP Drug Prescribing; BNF*)

573 **B** Lithium is a small ion handled almost entirely by the kidney, reabsorbed in the proximal tubule in competition with sodium. Non-steroidal anti-inflammatory drugs inhibit renal prostaglandin synthesis, reducing renal blood flow and increasing proximal reabsorption, so plasma lithium rises — and lithium has a narrow therapeutic index, with toxicity presenting as coarse tremor, vomiting, ataxia, confusion and, at higher levels, seizures and renal failure. Paracetamol, with codeine if more is needed, avoids the problem entirely. Option A is the most attractive alternative because protein-binding displacement is the mechanism candidates reach for whenever two drugs interact, and it cannot apply here: lithium is an ion and is not protein bound at all. C, D and E describe interactions and adverse effects that do not occur in this way. (*BNF; SDCEP Drug Prescribing*)

574 **A** Paracetamol is the analgesic that does not add to his bleeding risk, and taken regularly rather than as required it is considerably more effective than he has so far found; codeine can be added if it is still inadequate. Both of the other agents are non-steroidal anti-inflammatory drugs, and both impair platelet function and erode gastric mucosa on top of a direct factor Xa inhibitor — a combination associated with a clear increase in gastrointestinal bleeding, and one that also risks reopening a surgical site on the evening of the operation. Option D is the choice most candidates make, because ibuprofen with paracetamol is the correct default for post-surgical dental pain and has been rehearsed as a rule; the rule has an exception, and the anticoagulant is it. Aspirin at analgesic dose is the same objection magnified. (*BNF; SDCEP anticoagulants and antiplatelets*)

575 **E** Codeine is a prodrug converted to morphine by CYP2D6, and the enzyme is highly polymorphic: an ultra-rapid metaboliser produces far more morphine than expected, it passes into breast milk, and infant deaths from opioid toxicity have been reported. On that basis codeine is contraindicated in all breastfeeding women, not merely in the early weeks, which is what makes option B wrong although it captures the right anxiety. Ibuprofen is the appropriate addition here — it is poorly transferred into milk, compatible with breastfeeding, and combined with paracetamol it outperforms codeine anyway. Option D is a true statement about codeine and the one that tempts candidates who know the analgesic literature; it is an argument about efficacy, and the objection being asked for is one of safety to the infant. (*MHRA; BNF; SDCEP Drug Prescribing*)

576 **E** Implant placement with flap elevation and grafting is a higher bleeding-risk procedure, and for a patient taking a once-daily direct oral anticoagulant in the morning the guidance is to delay that day's dose rather than to omit it, taking it at least four hours after haemostasis has been secured. She is therefore uncovered for a matter of hours rather than a day, which is the whole point of the manoeuvre. Option B is the answer candidates give, and it is correct for the twice-daily drugs — apixaban and dabigatran, where the morning dose is missed and the evening dose taken as usual — so it is a rule applied to the wrong dosing schedule; missing a once-daily dose outright leaves her without anticoagulation for close to forty-eight hours. C is a frank interruption of therapy after venous thromboembolism, and A ignores the procedure category. (*SDCEP anticoagulants and antiplatelets; BNF*)

577 **C** Ticagrelor is a cyclopentyltriazolopyrimidine that binds P2Y12 reversibly and at a site distinct from the ADP binding site; it is active as given rather than a prodrug, is taken twice daily because its half-life is only about seven hours, and platelet function recovers over roughly two to three days rather than the seven to ten needed for the marrow to replace an irreversibly inhibited pool. Option A is the assumption in the stem and the one most candidates share, because clopidogrel's irreversible thienopyridine mechanism is the story they have learned and they generalise it to the whole class. None of this changes the management: he is inside the mandated dual therapy period after an acute coronary syndrome, the drugs are continued, and bleeding is controlled with local haemostatic measures. (*BNF; SDCEP anticoagulants and antiplatelets*)

578 **B** Dabigatran is a direct thrombin inhibitor excreted largely unchanged in the urine — of the order of 80 per cent — so its half-life, ordinarily twelve to seventeen hours, lengthens substantially as glomerular filtration falls, and an elderly patient with an eGFR of 34 and a six-month-old result deserves a current one before higher-risk surgery. It is the direct oral anticoagulant most dependent on renal clearance, which is why the reduced 110 mg dose has been chosen. Option C is the plausible-sounding alternative, and it is wrong in fact as well as in relevance: dabigatran is only about a third protein bound, which is precisely why it is dialysable, unlike the highly bound factor Xa inhibitors. D describes warfarin. Note that it has a specific reversal agent, idarucizumab, available in hospital. (*BNF; SDCEP anticoagulants and antiplatelets*)

579 D Nystatin is a polyene that is not absorbed from the gastrointestinal tract, acts topically on the mucosa and has no effect on warfarin metabolism, so it is the safe choice here. Every azole in the list inhibits CYP2C9, the enzyme that clears the more potent S-enantiomer of warfarin, and the resulting rise in INR can be dramatic and has caused fatal haemorrhage. Option E is the distractor that catches experienced clinicians, because a small quantity of cream on the angles of the mouth feels too trivial to matter; enough miconazole is absorbed through skin and mucosa to potentiate warfarin, and the interaction has been the subject of specific safety warnings. Treat the cause as well: denture hygiene, leaving the prosthesis out at night, and cleaning around the abutments. (*BNF; MHRA; SDCEP Drug Prescribing*)

580 E All three are real and they compound one another, which is what makes a long-term steroid more significant here than the bisphosphonate she is also taking. Glucocorticoids inhibit fibroblast proliferation and collagen synthesis and blunt the inflammatory phase of healing, so wounds close more slowly and dehisce more readily. They are also one of the named concomitant factors that moves a patient taking an oral bisphosphonate for a non-malignant indication out of the low-risk MRONJ group — the point candidates most often miss, because they assess the two drugs separately instead of together. And they suppress neutrophil function and cell-mediated immunity, raising infection risk. Option D is the popular answer, dropping item C on the reasoning that immunosuppression at 15 mg is modest; it is enough to matter in a surgical wound. (*SDCEP MRONJ; BNF*)

581 A Metoclopramide is a dopamine D2 antagonist, and blockade of nigrostriatal D2 receptors produces acute dystonia: oculogyric crisis, torticollis, trismus and tongue protrusion, coming on within minutes to hours of a single dose. Young women are the group most at risk, which is why the stem gives her age and sex, and it is the reason the drug's licence has been restricted to short-term use. She is fully conscious throughout, which distinguishes it at the bedside from a seizure. Treatment is to stop the drug and give procyclidine 5 mg intramuscularly or intravenously, with a prompt response. Option E is the distractor with local plausibility, since trismus after a block is common and she has had injections — but a medial pterygoid haematoma does not deviate the eyes or stiffen the neck. (*BNF*)

582 C Glucagon 1 mg intramuscularly does not supply glucose; it mobilises it, by stimulating hepatic glycogenolysis and gluconeogenesis. That mechanism requires a liver with glycogen in it, and this man has three reasons to have little — prolonged fasting, chronic alcohol excess, which also directly inhibits gluconeogenesis, and hepatic impairment — so the response may be absent or transient. Give it, because it is what you have without intravenous access, but call an ambulance at the same time rather than waiting to see whether it works, and give oral carbohydrate as soon as he is safely able to swallow. Option B is the attractive answer because glucagon is genuinely slower than intravenous glucose, taking around ten minutes; slow is not the same as ineffective, and speed is not what defeats it here. (*Resuscitation Council UK; BNF*)

583 B A convulsion continuing beyond five minutes is treated, and buccal midazolam 10 mg is the preparation held in the dental emergency drugs box for exactly this; it is placed into the buccal sulcus between the gum and the cheek and absorbed across the mucosa, with a second dose considered after ten minutes if the seizure persists and help has not arrived. Option D is the choice most candidates make, and item C is why it fails: hypoglycaemia is a recognised cause of seizures and giving glucose feels like covering the possibility, but nothing whatever should be placed in the mouth of an actively convulsing patient, and glucose gel is for the drowsy, cooperative hypoglycaemic. Item A belongs to anaphylaxis; adrenaline has no place in a seizure and would be actively harmful. (*Resuscitation Council UK; BNF*)

584 D Sildenafil inhibits phosphodiesterase type 5, preventing the breakdown of cyclic GMP; nitrates act by generating nitric oxide, which raises cyclic GMP. Given together the vasodilatation is potentiated to the point of profound, refractory hypotension, and the combination is contraindicated within twenty-four hours of sildenafil, or longer for tadalafil. So the spray is withheld, the emergency services are called, he is kept sitting up and reassured, and aspirin 300 mg is chewed unless it is contraindicated. Option A is the reflex and the reason the interaction is asked about at all: giving a nitrate for chest pain is so automatic that the history volunteered in the stem is easily skated over. B withholds the one drug that improves survival in an infarct, C is outdated for a patient who is not hypoxic, and E is for anaphylaxis. (*BNF; Resuscitation Council UK*)

585 D Standard conscious sedation techniques are narrowly defined: nitrous oxide with oxygen by inhalation, a single drug given intravenously — in practice titrated midazolam — and oral or transmucosal midazolam. Anything else is an advanced technique requiring additional training, a different environment and, usually, an anaesthetist. Adding an opioid to a benzodiazepine is a combination technique and is advanced, both by definition and because the respiratory depression of the two is synergistic. Intravenous sedation for a child under twelve is likewise advanced; inhalation sedation is the standard route at that age. Option E is the choice made by candidates who assume that intravenous sedation is

itself the boundary, when a single-drug titration in a fit adult is exactly what the standard technique describes. (*IACSD standards*)

586 A Oximetry measures oxygenation, not ventilation, and the two separate as soon as supplemental oxygen is given: a patient breathing enriched gas maintains a saturation of 99 per cent for a long time while carbon dioxide accumulates, so the number that is meant to warn you is the last thing to change. The observations that matter are the ones the stem describes — respiratory rate, depth and pattern, chest movement and the level of response — and a falling rate with shallow breathing in an over-sedated patient calls for stimulation, an airway manoeuvre and a pause in titration, not reassurance from the monitor. Option E is the near-miss, and it is half true: there is a genuine delay of tens of seconds between alveolar change and probe reading, particularly at the finger, but that is a lag, whereas oxygen removes the signal altogether. (*IACSD standards*)

587 C Long-term benzodiazepine use produces receptor tolerance, which is why she needed 12 mg, but the same adaptation means that competitive antagonism at the GABA-A receptor can strip the drug's effect from a physically dependent patient abruptly, precipitating a withdrawal state in which seizures are the feared event. Flumazenil must therefore be reserved for genuine over-sedation with respiratory compromise, not used as a routine reversal or to speed her out of the chair, which is why option A is the answer to reject. D is dangerous to believe: tolerance to the hypnotic effect does not confer proportionate tolerance to respiratory depression, and having given 12 mg you should watch her more closely, not less. B ignores the accumulated dose, and E abandons a technique that has worked. (*IACSD standards; BNF*)

588 B Consent is the discussion and the decision, not the signature: it was validly obtained three weeks ago by a patient with capacity who had time to reflect, and what is missing is documentation. That can be written up now and dated, recording clearly when the discussion took place and what was covered, with the missing entry explained rather than backdated. Option A is the answer that instinctively appeals because it produces the document everyone wants, and it is the one that would be worthless: a signature obtained after a sedative agent has been given carries no weight, and it would actively undermine the valid consent that already exists. C uses an antagonist with real hazards to solve a paperwork problem, D wastes a properly consented appointment, and E has no legal basis for an adult. (*IACSD standards; GDC Standards*)

589 D Nitrous oxide is around thirty times more soluble in blood than nitrogen, so at the end of the procedure it leaves the blood and pours into the alveoli far faster than nitrogen can be taken up in exchange. The alveolar gas is diluted, the partial pressure of oxygen falls, and a transient hypoxia can follow in the first few minutes — the reason a period of 100 per cent oxygen is given through the same hood before it comes off. Option B is a real and important consideration and is the one candidates choose, because occupational exposure is drummed into every sedation course; but staff protection is achieved by active scavenging and ventilation throughout, and it is not what the terminal oxygen is for. A misunderstands an irreversible reaction, C describes a different phenomenon, and E is not desirable. (*IACSD standards; BNF*)

590 E Tongue involvement is the key: the lingual nerve is nowhere near an LR5 osteotomy, so this is a block-related injury rather than an implant one, and the radiographic clearance of 3 mm supports that. Nerve injuries are managed by objective documentation from the outset — map the area, record two-point discrimination and the quality of the sensation, photograph the map — because that baseline is what allows recovery or deterioration to be judged, and because a burning dysaesthesia carries a worse prognosis than numbness alone. Review at short intervals and refer if there is no meaningful improvement over the following weeks. Option A is the answer given from kindness and is the one that causes harm: recovery is not certain, six months is far too long to wait before referral, and reassurance without a record leaves nothing to compare against. (*BNF; SDCEP*)

Answers — Block U — Surgical placement, extended (Q591–650)

591 B The 1.5 mm clearance required between an implant surface and an adjacent root is a bone-level dimension, and it exists because crestal remodelling extends laterally about 1.3 to 1.4 mm from the implant surface; encroach on it and the interproximal crest of the neighbouring tooth is lost along with the papilla it supports. A 3.3 mm platform therefore needs 3.3 plus 1.5 plus 1.5, that is 6.3 mm, and this site offers 5.5 mm — 1.1 mm each side. Option A is the measurement most candidates take, and it is the one that misleads: 7.0 mm at the contact points looks ample and says nothing about converging roots, which is why the assessment must be made on a radiograph at crestal level. Orthodontic root divergence, not a narrower implant, is the answer here. (*Tarnow; ITI Treatment Guide vol 10*)

592 C The mucosa re-forms a supracrestal attachment of characteristic dimension around the transmucosal component whatever depth the platform is placed at, and where the platform is sunk beyond about 3 mm the bone recedes to make room for it — deep placement buys a deep sulcus and pays for it in marginal bone. Option E is the thoughtful wrong answer and deserves argument: sinking the implant 2 mm does lengthen the clinical crown and therefore the vertical lever, and crown height is a genuine mechanical risk factor, but 2 mm is a modest change and screw fracture is not the expected outcome at 35 Ncm in intact bone. A describes the opposite error, shallow placement. B is a real nuisance at depth but is a procedural inconvenience rather than a consequence to the tissues. (*Buser et al.; ITI Treatment Guide vol 10*)

593 D A buccal wall of 0.4 mm is essentially bundle-free cortical plate with no cancellous reserve and a periosteal supply that surgery has interrupted; it resorbs, and in a thin phenotype with a high smile line the mucosa follows it, exposing the metal shoulder. Nothing the laboratory does repairs that, and the remedy is removal of the implant and reconstruction — which is why orofacial position is the least forgiving of the three planes. Option A is the distractor that catches candidates because a facial screw channel looks like the disaster in the photograph, and it is a serious error; but it is soluble, by cementation, by an angled screw-channel abutment, or by a custom abutment. B, C and E are all compensations the restoration can make, badly but visibly. (*Buser et al.; ITI Treatment Guide vol 10*)

594 E The three planes each have a zone within which the implant may be placed without prejudicing the restoration, and a zone beyond which the error cannot be retrieved. Apicocoronally the shoulder sits about 3 mm below the planned mucosal margin, roughly 1 mm apical to the cemento-enamel junctions of the neighbours; deeper than that is a danger zone of progressive bone loss and an uninspectable sulcus, shallower is an abrupt emergence. Orofacially anything buccal to the emergence line risks mid-facial recession, which is why the shoulder is set about 1 mm palatal to it. Mesiodistally the 1.0 to 1.5 mm nearest each root belongs to that tooth's crestal bone. Candidates most often reject (i), having learned "3 mm" as a target without realising it is also a limit. (*Buser et al.; ITI Treatment Guide vol 1*)

595 A Crestal remodelling around each implant extends laterally as well as vertically, so when two platforms sit closer than about 3 mm the two resorption zones overlap and the interimplant bone is lost rather than merely reduced; the papilla it would have supported collapses, and the interimplant papilla is in any case limited to a mean of roughly 3.4 mm against about 4.5 mm beside a tooth. In a high smile line that black triangle is displayed. Option C is the attractive answer because it is entirely true and is exactly how the case is made to look good — an ovate pontic developed against a supported ridge gives the best papillae available in this situation — but it describes the benefit obtained, not the reason for abandoning the two-implant plan. E is true and trivial. (*Tarnow; ITI Treatment Guide vol 10*)

596 B The abutment–implant interface is never perfectly sealed: it harbours bacteria, and a persistent inflammatory cell infiltrate sits around it. Where the abutment matches the platform, that infiltrate lies immediately against the crestal bone and the bone withdraws from it, classically to the first thread. Moving the junction horizontally inward by a few tenths of a millimetre relocates the infiltrate over the platform surface rather than over bone, and the crest is spared. Option C is the strong distractor because joint micromovement genuinely contributes to crestal loss and is a real mechanism — but a narrower abutment does not raise preload, and no clinically achievable preload abolishes micromovement. A confuses a small change in diameter with a change in lever arm; D and E describe consequences that do not follow from the mismatch. (*Lindhe; ITI Treatment Guide vol 12*)

597 C Where the site is wide enough to place without augmentation, the papillae need not be entered at all: a crestal incision confined between the line angles of the neighbours gives sufficient access to a healed ridge, leaves the interdental tissue and its blood supply undisturbed, and avoids elevating periosteum from the thin buccal plates of the adjacent roots. Option A is the design most candidates draw, because sulcular access with reliefs is the standard flap of oral surgery and it does expose the field beautifully; in a thin phenotype it also strips the periosteum over two roots that have no bone to spare and reliably costs a fraction of a millimetre of marginal recession. B removes the reliefs but keeps the sulcular incisions,

D places a relief where a scar and recession will show, and E is an endodontic and regenerative design borrowed to no purpose. (*Lindhe; ITI Treatment Guide vol 7*)

598 D A mucoperiosteal flap is an axial-pattern flap perfused from its base by supraperiosteal vessels running apico-coronally, so the base must be the widest part and any relief must diverge apically; and the incision line must not sit over the barrier, since the suture line is the weakest part of the closure and a membrane has no blood supply of its own to resist exposure. Item C is what makes option E wrong, and it is the item candidates accept because it is what gains the best view: a relieving incision taken down the mid-buccal of a tooth crosses the thinnest bone over the root prominence, heals with a visible scar and predictably produces marginal recession. Reliefs belong at line angles, over interproximal bone, away from the graft. (*Lindhe; ITI Treatment Guide vol 7*)

599 A The greater palatine artery leaves its foramen medial to the third molar and runs anteriorly in the submucosal layer of the palatal vault, deep to the donor tissue and roughly parallel to the gingival margin. Its distance from that margin depends on the height of the vault: in a high vault it is comfortably remote, in a shallow vault it may lie only a few millimetres apical to the safe harvest zone, so a flat palate demands a shorter apical extension and a thinner graft. Option B is the plausible alternative because the nasopalatine bundle is genuinely at risk in palatal surgery, but it is an anterior midline structure and irrelevant to a premolar donor site. C and D are real vessels in the wrong region, and E is not a haemorrhagic hazard here. (*Lindhe; ITI Treatment Guide vol 7*)

600 E Every technical refinement in the other four options is a way of cutting more carefully in a place you cannot see; the measure that actually protects the nerve is to see it. Reflecting the flap until the foramen and the emerging bundle are visible, and noting that the loop runs 4 mm further mesially than the foramen itself, converts a blind manoeuvre into a directed one, and the periosteal release is then made where the bundle demonstrably is not. Option A is the distractor with the strongest technical pedigree, and it is genuinely the correct blade orientation for a periosteal release — but a shallow blade held over an unidentified bundle at 9 mm from the crest still divides it. B abandons the access the case requires, and C and D reduce rather than remove the hazard. (*Lindhe; ITI Treatment Guide vol 7*)

601 C A buccal plate of 0.6 mm has almost no cancellous component and depends on its periosteal supply; strip that periosteum and the plate undergoes surface resorption, and in a thin phenotype the mucosal margin follows it down. Leaving the periosteum in place over the UR2 — a split-thickness dissection within the mucosa — preserves that supply while still allowing the flap to be reflected where bone must be exposed. Option D is the answer most candidates give, and its reasoning is respectable, since a distal relief keeps the incision away from the mid-buccal prominence and off the display zone; but it is still a full-thickness reflection over the very root the question is about. A states a general rule correctly and applies it where the exception exists, and E has no useful access. (*Lindhe; ITI Treatment Guide vol 7*)

602 A Only the first is a test of passivity. The flap must reach *beyond* where it will be sutured, and reach there under no traction at all, because a wound closed against elastic recoil is a wound whose margins are held apart by the tissue and together by the suture; capillary perfusion pressure at the margin is low, and sustained tension makes it ischaemic. The conventional figure is 3 to 4 mm of passive overlap. Item B is what makes option D attractive, and it sounds like a reasonable test until you notice the word traction: if force is needed to reach the line, the flap is short. Item C is the same error in a more dangerous form — blanching is not a small imperfection to be tolerated, it is the visible sign of the ischaemia that produces dehiscence and membrane exposure. (*Wang & Boyapati PASS; ITI Treatment Guide vol 7*)

603 B Wound strength develops behind epithelial closure, and for the first two weeks the incision line is the part of the flap most likely to open; site it over a non-resorbable membrane and any opening exposes a device that has no vascular supply, cannot be defended by the host, and becomes colonised — which is the commonest way a regenerative case is lost. Displacing the incision palatally, or off the graft altogether, keeps the two apart. Option A is the attractive distractor because it is a true and important statement about suture materials, and it is the reason a monofilament is chosen over a graft; but it is an argument about what to close with, not about where to place the line, and it is answered by choosing polytetrafluoroethylene rather than by moving the incision. (*Lindhe; ITI Treatment Guide vol 7*)

604 D A tissue punch does not displace mucosa, it removes a disc of it, and where only 2 mm of keratinised tissue remains, a 4 mm punch takes essentially all of it and leaves the transmucosal component surrounded by mobile alveolar mucosa. The correct manoeuvre in this situation is a small crestal incision with the keratinised tissue split and repositioned, which costs almost nothing in morbidity and preserves what cannot be replaced without a graft. Option A is the objection candidates reach for, and it is the standard criticism of flapless surgery — but the stem has answered it: 9 mm of width with no concavity on cross-section is precisely the anatomy in which not seeing the plate is acceptable. B is a genuine limitation but is manageable with a guide, and E is untrue. (*Misch; ITI Treatment Guide vol 11*)

605 **A** This is the characteristic error of flapless surgery and it is entirely predictable: with the bone hidden, the only visible landmark is the mucosal surface, and every depth judged from it is displaced apically by the thickness of the mucosa. Four millimetres of mucosa and a platform 3 mm too deep is that arithmetic. The remedies are to sound the bone under anaesthesia and record the thickness before drilling, or to use a template whose depth stops are referenced to the teeth rather than to the tissue. Option B is the plausible mechanical alternative and it does happen in type 4 bone, but it produces an osteotomy deeper than planned by a fraction of a millimetre, not by the exact thickness of the overlying mucosa. (*Misch; ITI Treatment Guide vol 11*)

606 **E** All three, and the third is the one that turns a fashion into a discipline. Flapless placement is placement without information, so the information must be obtained beforehand: a cross-section, not a panoramic film, showing width, height and the presence or absence of a submerged concavity; enough keratinised tissue that removing a disc of it still leaves a cuff; and a site good enough that you will not find yourself needing to graft a defect you can neither see nor reach. Option D is chosen by candidates who read item C as a counsel of perfection rather than a condition, and it is the most attractive answer because augmentation feels like something one decides at the time. In flapless surgery that decision has already been made, by default, before the first drill. (*Misch; ITI Treatment Guide vol 11*)

607 **B** Two clocks run independently and both are being tested. Placement is timed from the extraction: same-day placement is type 1. Loading is timed from placement, and a prosthesis connected within one week is immediate; whether it constitutes loading or merely restoration turns on one thing only, occlusal contact in intercuspation. Light contact is contact. Option A is the error most candidates make, and the reasoning behind it is worth naming: they take "light contact, no excursive interference" to be the definition of an immediate restoration, when in fact that phrase describes a carefully managed immediate *loading*. Excursive relief and a provisional material reduce the load; they do not change the category. E compounds the error by making provisionality the criterion, which it never is. (*Gallucci et al.; ITI Treatment Guide vol 2*)

608 **C** Type 3 is early placement with partial bone healing, conventionally twelve to sixteen weeks from extraction, and the radiographic description in the stem — partial fill, socket outline still visible — is exactly what defines it; type 4 requires a healed ridge, which usually means six months or more. Loading at fourteen weeks after placement is well beyond the two-month boundary and is therefore conventional. Option D is the answer most candidates give, and the mistake is instructive: fourteen weeks *feels* like a healed site, and the word "healed" is being supplied by the interval rather than by the radiograph. The classification is defined by the state of the tissues, not by the diary, which is why the stem describes what the film shows. (*Hämmerle, Chen & Wilson consensus; Gallucci et al.*)

609 **A** Type 2 placement is defined as early placement with soft tissue healing, and the soft tissue is the whole point: four to eight weeks converts a socket orifice covered by nothing into a site covered by matured mucosa, which can be raised as a flap, advanced without tension over a contour graft, and closed — the manoeuvre that immediate placement into an open socket cannot offer. Option C is the plausible alternative and it is chosen by candidates who confuse type 2 with type 3: appreciable bone fill has not occurred at six weeks, and primary stability at that stage still comes from the palatal and apical bone, exactly as it would on the day of extraction. B is false, since resorption of the bundle bone is still in progress at six weeks and continues for months. (*Chen & Buser; ITI Treatment Guide vol 3*)

610 **D** The two real gains are a healed mucosa that can be advanced over a contour graft, and a site in which an infected socket has been debrided and allowed to settle before a titanium surface is introduced. Item C is what makes option E wrong, and it is the item candidates accept most readily because it is what the delay appears to be for: the buccal wall of an anterior socket is largely bundle bone, a periodontal-ligament-dependent tissue that begins to resorb the moment the tooth is removed and does most of its resorbing in the first weeks. Waiting six weeks does not preserve it — it means the loss has occurred before you operate, where you can see it and graft it, rather than afterwards. (*Araújo & Lindhe; Chen & Buser; ITI Treatment Guide vol 3*)

611 **C** Every published protocol for immediate loading requires a site that needs no simultaneous augmentation, and the reason is not the implant but the graft: a particulate graft under a membrane depends on mechanical quiet for revascularisation, and loading transmits micromovement to the crestal region where the defect lies. The numbers in the stem are all favourable — 38 Ncm clears the customary 35 Ncm threshold and ISQ 73 clears 70 — which is precisely the trap, because a candidate who has memorised those two figures will find both satisfied and answer yes. Option D is the strong distractor since splinting is a genuine criterion in multi-unit work, but single unsplinted immediate loading is well documented where stability is high. Fit a healing abutment and load conventionally. (*Gallucci et al.; ITI Treatment Guide vol 2*)

612 E Immediate loading of a cross-arch splinted fixed prosthesis in the edentulous mandible has one of the better evidence bases in implant dentistry, and the honest answer is not that it is unsound but that it is difficult: it requires assessment of a periodontally destroyed arch, alveolectomy to a predictable prosthetic platform, angulation of the distal implants, immediate conversion of a prosthesis and management of the complications that follow, and it sits beyond what a general dentist with a special interest should undertake. Option B is the answer that reveals the misconception this question exists to correct — cross-arch splinting distributes load across four fixtures and in that sense is *more* forgiving than a single immediately loaded implant, so the risk profiles are not interchangeable. A overstates, C invents a figure and D is false. (*Gallucci et al.; SAC Assessment Tool; ITI Treatment Guide vol 4*)

613 B The loading vocabulary is defined by intervals measured from placement, and they are narrower than most clinicians assume: immediate loading means within one week, early loading from one week to two months, conventional loading beyond two months. Ten days is therefore early loading, and the fact that it feels immediate is the point of the question. Option A is the commonest error and it arises from treating "immediate" as a description of intent rather than a defined window; a candidate who answers A will mis-describe every protocol falling between the seventh day and the eighth week. D and E confuse the loading axis with the restoration axis, which is settled by occlusal contact alone — this crown is in full contact, so it is loading whatever else is true of it. (*Gallucci et al.; ITI Treatment Guide vol 2*)

614 E All three are conditions rather than preferences, and a case qualifies only when every one is met. Adequate primary stability is what holds interfacial movement below the threshold — figures around 100 micrometres are quoted — beyond which the healing interface differentiates into fibrous tissue rather than bone. Uncontrolled bruxism supplies exactly the wrong loading history, unguided and at night, to an interface with no bone-to-implant contact yet. And a graft under a membrane demands the mechanical quiet that loading removes. Option D is the answer of candidates who treat augmentation as a separate topic from loading; it is the criterion most often forgotten, and forgetting it is how a well-selected immediate case is lost through a dehiscence rather than through the implant itself. (*Gallucci et al.; ITI Treatment Guide vol 2*)

615 D The two indices measure different quantities. Insertion torque is the rotational friction generated as the implant threads cut and compress the osteotomy wall; pre-cutting those threads with a tap, as one properly does in type 1 bone, removes that interference and the torque reading collapses. Resonance frequency analysis measures how stiffly the implant is held by the surrounding structure, and a thick cortex enclosing dense trabeculae is very stiff whether tapped or not. Low torque with high ISQ is therefore a coherent finding here, and the implant is stable. Option B is the attractive answer because countersinking does remove crestal bone and candidates associate that with lost stability; but it removes a collar at the platform, not the engagement along the body. C inverts the known directional difference, buccolingual readings being the lower. (*Misch; Osstell*)

616 C An ISQ below about 60 is conventionally the low band, 60 to 70 the medium band and above 70 high, and 30 Ncm sits below the 35 Ncm customarily required before any accelerated protocol is contemplated. Both indices point the same way, and the correct response to low primary stability is not to intervene but to wait, allowing woven bone to form in the healing chambers so that secondary stability replaces a mechanical stability that was never adequate. Option B is the answer that attracts the thoughtful candidate, because a repeat reading before loading sounds like disciplined use of the technology — but early loading at six weeks is a plan committed to now, and the stability dip lies in between. E discards a site that will very probably integrate perfectly well. (*Osstell; ITI Treatment Guide vol 2*)

617 A Resonance frequency analysis measures the stiffness of the implant within its supporting bone, and stiffness is governed by how much of the implant stands unsupported above the first point of bony contact — the same physics as a cantilever. Lose 2.5 mm of crest and the effective lever lengthens, the resonant frequency falls and the ISQ falls with it. This is why a declining ISQ in a comfortable implant should prompt a radiograph rather than reassurance, and why the technique's value lies in serial rather than single readings. Option D is the distractor with clinical appeal, and inflammation is indeed the underlying disease in many such cases, but the stem specifies healthy mucosa without bleeding, and soft tissue inflammation does not itself alter measured stiffness. (*Osstell; 2017 World Workshop*)

618 E All three, and together they are the reason a torque figure must never be read as a measurement of bone. Undersizing leaves more bone for the implant to compress and cut, so friction rises; a tapered body with deep, aggressive threads generates a rising torque curve through the last millimetres of seating by design; and an unrelieved cortical collar produces a sharp terminal spike as the platform is forced through it. The identical reading of 45 Ncm can be obtained in type 2 bone with a parallel implant and a full drill sequence, or in type 4 bone with all three devices applied. Option D is chosen by candidates who

accept the first two as technique but read the countersink as affecting only the crestal contour; it affects the number the motor prints. (*Misch; ITI Treatment Guide vol 2*)

619 B The weakest element in the chain at this moment is not the bone but the driving interface: the internal hexagon, octagon or conical seat of the implant and the post engaging it are dimensioned for seating torques of the order of 35 to 50 Ncm, and torque beyond that rounds the connection, shears the post, or leaves a deformed seat that no abutment will subsequently fit accurately. The correct manoeuvre is to reverse the implant out, relieve the crest with a countersink and reseat. Option D is the attractive alternative because candidates associate excessive torque with damage to bone — but bone threads strip in soft type 4 bone at low torque, in an implant that spins; dense cortical bone at 80 Ncm does not strip, it resists, which is why the reading is 80. (*Misch; ITI Treatment Guide vol 2*)

620 C Heat generated at a cutting interface is work divided by time, and rotational speed is the dominant term: for a given feed rate, doubling the revolutions roughly doubles the sliding distance of each cutting edge against bone and the frictional energy delivered to it. Bone held at 47 degrees Celsius for one minute is irreversibly injured, which is why osteotomy drilling is conventionally done in the high hundreds to around 1500 revolutions per minute rather than the thousands. Option D is not wrong so much as secondary — debris clearance genuinely worsens at speed, and packed flutes convert a cutting drill into a rubbing one — but it describes a contributory mechanism, whereas the speed itself is the primary fault. E is true and is an appeal to authority rather than a reason. (*Eriksson & Albrektsson; Misch*)

621 A Only the first works, and it works for a reason worth stating: in a deep narrow osteotomy the coolant cannot reach the apical cutting surface, which is exactly where friction is greatest and where the drill is enclosed by bone on every side. Withdrawing the drill every few millimetres allows saline to flood the depth of the preparation and flushes the chips that would otherwise pack the flutes. Item B is what makes option D attractive, and it is what most operators actually do — but coolant delivered to the shank cools the part of the drill that is not cutting, which is why internally irrigated drills exist. Item C is the error of the previous generation of teaching: a faster pass raises, not lowers, the heat delivered, and the 47 degrees Celsius threshold is reached sooner. (*Eriksson & Albrektsson; Misch*)

622 D Bone taken beyond about 47 degrees Celsius for a minute dies, and a continuous deep pass through cortical bone with interrupted irrigation is the classic way to produce it. The dead cylinder immediately around the osteotomy cannot revascularise; it demarcates, and at three to six weeks the radiograph shows precisely what is described — a lucent separation line with the sclerotic sequestrum inside it, around an implant that is still mechanically held and therefore not yet mobile. Pain at this stage in an otherwise quiet site is the clinical clue. Option E is the strong distractor because compression necrosis is also a thermal-sounding, torque-related crestal disaster, and candidates conflate the two — but it produces saucerisation confined to the crest, not a halo along the whole length of the implant. (*Eriksson & Albrektsson; Misch*)

623 B Undersizing does one thing: it leaves more bone for the implant to compress and cut, so the frictional resistance to rotation rises and the implant is held tightly at time zero. That is genuinely valuable, and it is the correct manoeuvre in this bone. What it does not do is change the material it is working in — a thin cortex over sparse trabeculae remains a compliant structure, which is why the resonance frequency reading in such a site is often disappointing beside a torque of 45 Ncm. Option D is the seductive answer and is the reason this question exists: torque above 35 Ncm is a necessary condition for immediate loading, not a sufficient one, and the posterior maxilla in type 4 bone is where that distinction is most often ignored. (*Misch; ITI Treatment Guide vol 2*)

624 C This is type 1 bone, and its problems are the mirror image of the maxilla's: stability is never in doubt, but the bone is unyielding, poorly vascularised and slow to remodel, so the risks are thermal injury during preparation and crestal compression necrosis at seating. Relieving the cortical collar with a countersink and pre-cutting the threads with a tap allow the implant to seat at a physiological torque instead of crushing the crest. Option A is the answer candidates give when they have learned one protocol and apply it everywhere: undersizing is the correct response to soft bone and precisely the wrong one here, where it would drive insertion torque towards the levels that produce marginal bone loss. D compounds the same error mechanically, and B increases the heat. (*Misch; ITI Treatment Guide vol 2*)

625 A The sleeve is a closed metal tube surrounding the drill along the very stretch where coolant would otherwise enter the osteotomy, so external irrigation is deflected and the cutting tip runs comparatively dry — the reason guided protocols specify intermittent withdrawal of the drill, generous flow directed at the sleeve entry, and internally irrigated drills where the system offers them. Item B is what makes option D attractive, and it is a real phenomenon that patients occasionally notice as warmth; but heat conducted into mucosa is heat leaving the bone, not heat delivered to it, and it is not the mechanism of osseous injury. Item C is simply untrue: guided drills are longer, which is a matter of mouth opening, but the recommended speeds are unchanged. (*ITI Treatment Guide vol 11; Eriksson & Albrektsson*)

626 B The drill has passed through the sinus floor a millimetre earlier than the measurement predicted, and the membrane above it is of unknown integrity. A shorter implant seated entirely within the residual bone leaves the sinus undisturbed, and 6 mm implants in the posterior maxilla have accumulated a respectable body of evidence; the site still offers 8 mm of native bone, which is ample for one. Option D is the answer that appeals because the situation superficially resembles the start of a transcrestal lift, and candidates reason that having entered the sinus one may as well graft it. Grafting through a floor that has been perforated by a rotating drill, with no way of confirming the membrane, risks displacing particulate material into the antrum. C and E abandon a site that is straightforwardly treatable. (*ITI Treatment Guide vol 5; Misch*)

627 E An implant that spins has no primary stability, and nothing that happens subsequently creates it: the interface will move, and movement beyond about 100 micrometres produces fibrous repair rather than bone. The usual rescue is a wider implant, which the stem removes — a 6.5 mm ridge would not safely accept one in any case — so the honest course is to abandon, fill the over-prepared osteotomy with graft and return to a healed site. Option A is the attractive alternative and it is the one that does real harm: 10 mm of bone above the canal, less the 2 mm safety margin, leaves 8 mm, and an 8.5 mm implant driven a further 1.5 mm apically would sit within a millimetre of the nerve. C leaves an open oversized osteotomy for a week. (*Misch; ITI Treatment Guide vol 11*)

628 C A fenestration is a window in the plate with intact bone above and below it, and an implant that is stable, correctly positioned and crestally enclosed does not need to be removed for one — the exposed surface is covered with particulate graft and a barrier, the flap closed passively, and the defect regenerates. That is a procedure comfortably within general practice competence. Option D is the distractor with some intellectual support, since a fenestration is not in communication with the oral cavity and the periosteum will lay soft tissue over the threads; but soft tissue coverage of a roughened surface is not bone, and it leaves an area of the implant permanently unsupported and vulnerable if marginal bone is later lost. A discards a well-placed implant, and B creates a supracrestal platform. (*ITI Treatment Guide vol 7; Lindhe*)

629 E The fragment is visible, the flap is already raised, the field is dry and the patient is anaesthetised: these are the best conditions that will ever exist for retrieval, and a retained instrument fragment is a foreign body that will complicate every subsequent radiograph, invite medico-legal criticism and may prevent proper seating of the implant. A trephine or a slightly wider drill around the fragment usually releases it, and the site can then be grafted or re-prepared. Option A is the seductive answer because drills are made of stainless steel or titanium alloy and inert retained metal is tolerated elsewhere in surgery; but inertness is an argument for not panicking, not an argument for leaving a removable object where it should not be. C converts a chairside problem into a hospital referral. (*Misch; GDC Standards*)

630 D Angled abutments are supplied to about 30 degrees and angled screw-channel systems correct roughly 25 degrees, so a 40 degree error is outside the range of both; the crown could only be made by cantilevering the occlusal table well off the implant axis, which loads the components eccentrically and defeats the purpose of a molar implant. The honest options are removal and replanning, or a frank discussion about a restoration that will be mechanically compromised. Option A is the answer most candidates give, because angled screw channels have made a great many awkward implants restorable and it is easy to assume they solve everything; the numbers matter, and this is the question that tests whether the candidate knows them. B leaves the same axis with cement added. (*ITI Treatment Guide vol 12; Misch*)

631 A Only the gap dimension is a determinant, together with the aesthetic context — a horizontal defect of 2 mm or more at the shoulder is grafted, and in a thin phenotype with a high smile line a smaller one is grafted too, because the object is contour rather than osseointegration. Items B and C are the distractors, and they are attractive because both are genuinely important numbers that a candidate has just been asked about elsewhere: insertion torque governs whether the implant may be provisionalised, and implant length contributes to primary stability, but neither says anything about whether the space between the implant and the inner surface of the buccal plate should be filled. Option D is chosen by candidates who cannot bring themselves to leave torque out of an implant answer. (*ITI Treatment Guide vol 10; Araújo & Lindhe*)

632 B The colleague's premise is correct and his conclusion is wrong. Small horizontal defects do fill with bone, which is why the survival of such implants is unaffected; but the question in the anterior maxilla is never survival, it is the profile of the ridge in a high smile line. A 0.7 mm buccal plate is almost entirely bundle bone and it will resorb whatever fills the gap behind it, so a slowly resorbing xenograft is placed to occupy the volume and resist the collapse. Option A is the attractive answer because it appears to contradict the colleague most directly, and candidates prefer to win the argument on his own ground; it is simply false, and adopting it teaches the wrong biology. (*Araújo & Lindhe; ITI Treatment Guide vol 10*)

633 **E** A healing abutment must be taller than the tissue it passes through, or the mucosal margins simply close over its occlusal surface within a few days and the whole purpose of the one-stage approach — a transmucosal component from the outset — is lost, along with the shape it was there to develop. The patient then needs a second surgical exposure that the protocol was designed to avoid. With 4 mm of mucosa, a 5 mm abutment is required, and the tray should be checked before the flap is closed. Option C is the sophisticated distractor and rewards a candidate thinking about where the seal forms, but a buried abutment does not relocate the attachment to the platform; it simply becomes inaccessible. (*ITI Treatment Guide vol 2; Lindhe*)

634 **C** Coronal advancement moves everything coronally, including the mucogingival junction; the keratinised band that was 3 mm wide is dragged over the crest and what remains buccally is mobile alveolar mucosa in an already shallow vestibule. That tissue cannot be regained without a graft, and the implant that follows will be surrounded by it. A punch graft closes the orifice without moving anything. Option A is the answer many candidates give, and tension is certainly the commonest cause of a failed closure — but it is a technical failure that a proper periosteal release avoids, whereas the mucogingival displacement is intrinsic to the manoeuvre and happens even when it is done perfectly. B is not a feature of a maxillary premolar site. (*ITI Treatment Guide vol 7; Lindhe*)

635 **D** A transmucosal healing abutment beneath a mucosa-borne saddle is loaded at every meal, and the load falls exactly where the wound is; submerging the implant removes the component from the equation, and the denture is then relieved to an unmistakable gap over the site and lined with a soft material that is reviewed as swelling subsides. Item C is what makes option E wrong, and it is what is most often done in practice: rebasing in hard acrylic reproduces intimate contact with the healing tissue, which is precisely what must be avoided, and the exotherm and unreacted monomer of a chairside self-cure material over a fresh wound make it worse. A bonded pontic, where the dentition allows one, is better than any denture. (*Misch; ITI Treatment Guide vol 7*)

636 **A** Two decisions are being made. The needle for oral mucosa is reverse-cutting, its apex on the outer convexity, so that the tissue between the track and the wound margin is left uncut and thin mobile mucosa does not tear through under tension; a taper point will not penetrate keratinised tissue cleanly. The suture over a graft must be monofilament, because a braid wicks saliva and bacteria along its interstices directly onto a device with no blood supply, and it must hold its strength for ten to fourteen days. Option C is the answer most candidates give and it gets the needle right, which is what makes it attractive; polyglactin 910 is a braided multifilament whose coating does not abolish capillarity, and 3/0 is unnecessarily coarse in thin tissue. (*Lindhe; ITI Treatment Guide vol 7*)

637 **C** The properties that make polytetrafluoroethylene the material of choice over a graft — smooth, inert, non-capillary, comfortable against the mucosa — are the same properties that make its knots insecure. A knot holds by friction between the loops, and a surface with a very low coefficient of friction slips; the remedy is simply to place more throws than one would with a braided suture, commonly four or five, and to seat each square. Option D is the attractive alternative because memory is a genuine handling nuisance in monofilaments, most obviously in nylon, and candidates who have wrestled with a springy suture recognise it; memory makes a strand awkward to tie but it does not undo a properly thrown knot. B and E describe true properties with no bearing on knot security. (*Lindhe; ITI Treatment Guide vol 7*)

638 **B** A sling suture takes its anchorage from the adjacent teeth rather than from the opposing flap, so the buccal flap can be suspended at whatever coronal level the graft requires while the papillae are neither pierced nor squeezed between two tissue edges — precisely what is wanted in a thin phenotype where every millimetre of interdental tissue is aesthetically load-bearing. Option A is the reflex closure and it is the one that costs papillae: a suture through thin interdental tissue, tied to a palatal flap that pulls in the opposite direction, produces exactly the localised ischaemia and post-operative shrinkage the design was meant to avoid. D carries tension well but sets the flap flat rather than coronally, and C and E have no vertical control. (*Lindhe; ITI Treatment Guide vol 7*)

639 **A** Comparative work on straightforward implant placement has not demonstrated a difference in survival between full sterile technique and a clean technique using sterile instruments and standard cross-infection control — which is unsurprising, since the operative field is an oral cavity that cannot be sterilised and the principal inoculum is the patient's own flora. What has shown a small but real reduction in early failure is a single preoperative dose of amoxicillin. None of this is licence to be careless: sterile instruments, a sterile irrigant, sterile gloves and an uncontaminated implant surface remain mandatory. Option B is the answer that feels safest and most professional, which is exactly why it is the distractor; it asserts an effect size that the comparative literature does not support. (*Esposito et al., Cochrane review; SDCEP*)

640 **E** Dental unit waterlines are narrow-bore tubes with laminar flow and long stagnation periods, and they support a biofilm containing environmental organisms including *Legionella* and *Pseudomonas* species.

That is tolerable when the water is rinsing an enamel surface and unacceptable when it is being pumped into a fresh bony wound that is about to receive a permanent implant. A sterile irrigant delivered through a single-use line is one of the few genuinely non-negotiable elements of implant asepsis. Option C is the plausible-sounding alternative and it borrows a real idea — chlorhexidine and hypochlorite are cytotoxic to osteoblasts in culture — but treated waterlines do not deliver disinfectant at the point of use, and the hazard is biological rather than chemical. (*HTM 01-05; SDCEP*)

641 **B** Only the single preoperative dose has randomised evidence behind it: systematic review supports 2 g of amoxicillin shortly before placement, producing a small but statistically significant reduction in early failures with a large number needed to treat. Item A is what makes option D attractive, and a preoperative chlorhexidine rinse is a reasonable, harmless habit that demonstrably lowers the salivary bacterial count — but a reduction in bacterial load is not the same as a demonstrated reduction in implant failure, and the trial evidence for that endpoint is not there. Item C is the practice most widely followed and the one with the clearest verdict against it: continuing antibiotics after surgery adds no benefit while adding resistance, gastrointestinal harm and occasional anaphylaxis. (*Esposito et al., Cochrane review; SDCEP Drug Prescribing*)

642 **A** Bleeding of this character at this depth in this position means the osteotomy has entered the incisive canal and lacerated the nasopalatine vessels. The bleeding itself is easily controlled with pressure and a haemostatic dressing, and it is not an emergency; the important consequence is positional, because an implant placed into the fibrous and neurovascular contents of the canal will not integrate. The osteotomy must therefore be re-established buccal to it, which in this instance means correcting an axis that had drifted 2 mm palatally. Option B is the intuitive wrong answer — the implant does compress the bleeding point — and it trades a minute's haemostasis for a fixture in soft tissue. C leaves non-resorbable wax at the interface. (*ITI Treatment Guide vol 10; Misch*)

643 **C** The vessel is the intraosseous anastomosis between the posterior superior alveolar and infraorbital arteries, which runs in the lateral antral wall commonly around 15 to 20 mm above the crest and is visible on cross-sectional imaging in most patients — which is why it should have been identified in planning. Bleeding from it is alarming rather than dangerous, and it is controlled locally: direct pressure, crushing or burnishing the bony canal closed around the vessel, bone wax at the bony margin, electrocautery or a haemostatic dressing. Option D is the answer the competence rule tempts candidates into, and it is worth naming: this is a haemorrhage that stops, and transferring a patient with an open antrostomy is an over-reaction. The referral question in this case arose before the operation, not during it. (*ITI Treatment Guide vol 5*)

644 **D** Pressure applied for long enough is the first and most often neglected measure, and a resorbable haemostatic dressing placed into the osteotomy will control intraosseous bleeding from a nutrient canal within a few minutes and is then removed or left, since it resorbs. Item C is what makes option E wrong, and it is the one candidates accept because bone wax is a genuine haemostatic agent with a long surgical history: it works by mechanical occlusion, it does not resorb, it impairs osteogenesis and increases infection rates, and packing it into a site that is about to receive an implant places an inert barrier between the implant and the bone it must integrate with. It has a limited place at a bony margin, not in an osteotomy. (*Misch; Lindhe*)

645 **D** A root against the plate with an apical perforation is the sagittal position least suited to immediate placement: there is no palatal corridor to steer into, the apical bone that would supply primary stability is breached, and the plate that remains is 0.6 mm of bundle bone in a patient displaying it. Waiting four to eight weeks gives a matured soft tissue envelope that can be advanced over a contour graft, with the buccal resorption already declared. Option C is the defensible alternative and deserves credit — it is safe and many clinicians would choose it — but it adds a stage, and the soft-tissue advantage of early placement is obtained without waiting six months. A is the plan the anatomy specifically excludes, and E escalates a case that is manageable with simultaneous contour augmentation. (*Chen & Buser; ITI Treatment Guide vol 10*)

646 **E** All three, and the case fails if any one is missing. Torque of the order of 35 Ncm, achieved in palatal and apical bone, is what keeps interfacial movement below the threshold at which fibrous repair replaces bone. An intact plate that the implant does not touch is what allows the buccal contour to be maintained by a graft rather than lost by pressure resorption. And the provisional must be out of all contact, which is what makes it an immediate restoration rather than immediate loading — the manoeuvre exists to support the mucosa and the papillae during healing, not to chew with. Option D is chosen by candidates who treat occlusal relief as an aftercare instruction rather than a condition of the technique; it is the item most often assumed and least often checked in excursion. (*ITI Treatment Guide vol 10; Gallucci et al.*)

647 **D** Two processes are running and the slower governs. The implant would be ready at around three months in the maxilla, but the regenerated compartment beneath a non-resorbable membrane matures far more slowly, and re-entry is conventionally at about six months — the interval at which the membrane is removed and the augmented contour assessed before it is exposed to the oral environment. Option C is the answer most candidates give, and it is right for the wrong site: it applies the healing time for an ungrafted maxillary implant to a case whose limiting factor is the graft, not the implant. Uneventful healing and 40 Ncm are reassuring but they describe the implant's situation, not the graft's. (*ITI Treatment Guide vol 7*)

648 **E** The deficiency here is thickness and contour, not width of keratinised tissue, and the roll flap solves it with the tissue that is already present: a palatal flap raised at second stage, de-epithelialised and rolled buccally beneath the flap, adds connective tissue volume exactly where the concavity is, keeps its own blood supply as a pedicle, and requires no second surgical site. Option A is the attractive alternative because a free graft would certainly add bulk and candidates associate soft tissue augmentation with palatal harvesting; but a free epithelialised graft brings its own keratinised surface, heals as a visible patch of mismatched tissue in the aesthetic zone, and adds donor morbidity that the pedicled technique avoids. B and D address width, which is already adequate at 5 mm. (*Lindhe; ITI Treatment Guide vol 7*)

649 **B** Crepitus is the diagnosis. Compressed air driven into an open bony wound tracks along loose connective tissue and fascial planes, producing surgical emphysema; the swelling is dramatic and the crackling under the fingers is pathognomonic. It resolves over days, but the air carries oral flora into the deep tissues, so antibiotics are given and the patient reviewed for spread towards the neck, dysphagia or airway symptoms, which warrant admission. The lesson is prevention: never direct an air syringe or an air-driven handpiece into an implant osteotomy. Option A is the distractor that matters clinically, because rapid facial swelling does suggest angio-oedema or anaphylaxis — but there is no urticaria, no wheeze, no hypotension, and above all no crepitus in an allergic reaction. (*Misch; SDCEP*)

650 **C** The object must be localised, and only imaging of the chest and abdomen can do it. An aspirated foreign body may be entirely asymptomatic once the initial cough has settled, with a clear chest and normal saturations, because it has lodged distally in a bronchus — usually the right main bronchus — and it will then cause obstruction, infection or abscess if it is left; retrieval is bronchoscopic. A swallowed component almost always passes uneventfully, but that reassurance may only be given once it has been shown to be in the gastrointestinal tract. Option A is the answer that is right in the great majority of cases and wrong in the one that matters, which is exactly why it is dangerous; the coughing episode is the reason it cannot be assumed. (*Misch; SDCEP*)

Answers — Block V — Minor hard and soft tissue augmentation, extended (Q651–700)

651 **A** The material is bovine cancellous bone from which every organic component has been removed by heat and chemical treatment, and that processing enlarges the apatite crystallites while reducing the specific surface area. The result is a mineral of very low solubility whose cell-mediated turnover is close to negligible, so particles remain identifiable years later, osseointegrated and acting as a permanent filler — which is precisely why the material is chosen where contour must survive. Option B is the attractive error, because the phrase "natural bone mineral" implies a retained biological matrix and some products do incorporate collagen; deproteinisation by definition removes the organic phase, so it is mineral chemistry rather than any protein that resists degradation. C is untrue, since osteoclasts do attach but resorb very slowly, D inverts the solubility of apatite, and a collagen membrane has gone within months. (*ITI Treatment Guide vol 7; Lindhe*)

652 **E** A substantial residual particle fraction at six months, osseointegrated, quiescent and provoking no foreign-body response, is the histological signature of a low-substitution xenograft: the particles behave as an inert scaffold that new bone grows onto and around rather than replaces. Option B is the strong alternative, because beta-tricalcium phosphate also leaves visible particles at six months — but its resorption is a cellular and dissolution process characteristically accompanied by multinucleate cells at the particle surface, which is exactly what the specimen lacks, and most of it has gone by this stage. Autogenous chips are remodelled and become indistinguishable from host bone; calcium sulphate disappears within weeks; and demineralised allograft is soft and resorbs early. The clinical corollary is that persistence buys contour at the cost of a lower vital bone fraction. (*Lindhe; ITI Treatment Guide vol 7*)

653 **A** Allograft is processed human donor tissue, and both the law on human tissue and the duty to obtain valid consent require that the patient is told what is being implanted, because some will decline on religious, cultural or personal grounds whatever the processing. Item C is false and is the supplier's own error: donor consent, licensed tissue banking and traceability from donor to recipient are the defining regulatory features of allograft, not optional extras. Item B is what makes options D and E wrong, and it is the claim candidates most readily accept: only autogenous bone is osteogenic, mineralised allograft

is essentially osteoconductive, and such osteoinductive potential as exists belongs to the demineralised preparation and varies considerably between donors and batches. (*Human tissue legislation; GDC Standards; Lindhe*)

654 D The two autogenous forms behave differently and the difference is one of density. Cancellous bone has a vast surface area, marrow content and abundant cells, so it revascularises quickly and is remodelled early — excellent where rapid biological turnover is wanted, poor where a contour must be held. Dense cortical shavings revascularise slowly through creeping substitution and persist correspondingly longer, which is what a labial contour needs. Option A is the inversion that catches candidates, because rapid revascularisation intuitively reads as the better property and is genuinely an advantage of cancellous bone in a contained defect. B is also true of cancellous rather than cortical bone. C is wrong for both, since each is osteogenic, osteoinductive and osteoconductive, and E is untrue: the external oblique ridge is a second surgical site. (*Lindhe; ITI Treatment Guide vol 7*)

655 E All three hold, and together they are the whole argument for extending the unloaded healing period at an augmented site. Regenerated bone at six to eight months is still maturing: it carries more woven and less organised lamellar bone than the native ridge beside it, and remodelling into load-bearing lamellar bone continues for many months afterwards. Experimental and histological work is consistent that bone-to-implant contact in grafted bone is lower than in native bone, which is why insertion torque frequently disappoints in these sites. And the xenograft particles that gave the contour are, by design, barely substituted, so they occupy volume that is not vital bone. Candidates most often reject statement (iii), assuming that a graft which has "worked" must have been replaced. (*ITI Treatment Guide vol 7; Lindhe*)

656 E On contact with tissue fluid a bioactive glass exchanges ions, forms a silica-rich gel at its surface and then precipitates a hydroxycarbonate apatite layer through which it bonds chemically to bone. That surface reactivity is what distinguishes it; both materials are alloplasts and both are osteoconductive only. Option B is the conflation that catches most candidates: surface bioactivity is repeatedly marketed in language that sounds osteoinductive, but osteoinduction means recruiting host mesenchymal cells and directing them to become osteoblasts, and no synthetic ceramic does that. A misstates the behaviour of beta-tricalcium phosphate, which dissolves and is cell-resorbed over months rather than persisting; C is false of any particulate; and D describes coralline hydroxyapatite, a different material altogether. (*Lindhe; ITI Treatment Guide vol 7*)

657 B The protein does exactly what is claimed of it: it recruits host mesenchymal cells and drives their differentiation into osteoblasts, so it induces bone without supplying either cells or a mineral scaffold. The problem is the dose needed and the response to it — marked, dose-related oedema and seroma formation, which in the posterior mandible and floor of mouth is a swelling that needs a team able to manage a compromised airway. Option D is the attractive distractor because it sounds like sensible graft biology and mirrors the composite-graft argument; it is precisely wrong, since recruiting host cells is what osteoinduction means. A reverses the material's defining property, C is untrue, and E confuses a recombinant protein with a tissue-derived product. Vertical augmentation of this magnitude is in any case a referral. (*ITI Treatment Guide vol 7*)

658 C This is the A of PASS made operative. A dense buccal cortex is a poor source of the two things a graft needs — capillaries and cells — and everything that forms bone in the graft space must come from the host bed. Perforating the cortex opens the marrow compartment beneath it, so vessels and osteoprogenitor cells reach the graft from the depth of the ridge rather than only from its periosteal surface, and revascularisation is the rate-limiting step in incorporation. Option B is the sophisticated wrong answer and is genuinely part of the rationale often quoted for decortication; the evidence that any clinically meaningful quantity of growth factor is liberated is weak, and it is vascular and cellular access that determines the result. A is a trivial gain, D is not why it is done, and E is irrelevant six months later. (*Wang & Boyapati PASS; ITI Treatment Guide vol 7*)

659 D Space maintenance in a non-contained defect is achieved either by making the barrier stiff enough to hold its own shape, or by filling the space with something that supports it from beneath; a titanium-reinforced membrane fixed at its margins does the first and a low-substitution particulate does the second, and in practice the two are usually combined. Item C is what makes option E wrong, and it is the item candidates accept because tension sounds like the opposite of collapse: a native collagen membrane has essentially no stiffness when wet, and pulling it tight over an unsupported defect draws it down onto the cortex, obliterating the very space regeneration requires. The volume finally regenerated is the volume maintained, and nothing about the graft material alters that. (*Wang & Boyapati PASS; ITI Treatment Guide vol 7*)

660 D The finding to reason from is not the reduced volume but the nature of what has formed. Soft tissue that curettes off the cortex is fibrous connective tissue, and fibrous tissue in place of bone is the

signature of micromovement: mesenchymal differentiation within a healing wound is strain-dependent, so a graft that is repeatedly loaded through an unrelieved denture and is held by nothing at its margins produces fibroblasts rather than osteoblasts. Option C is the strong alternative, and denture pressure certainly does reduce the space available — but a graft that loses space still regenerates bone, only less of it, whereas an unstable graft regenerates the wrong tissue altogether. A is excluded by the stem, B would be compromised by decortication failure rather than by these findings, and asepsis is not one of the four principles. (*Wang & Boyapati PASS; Lindhe*)

661 **A** The open-barrier technique abandons the first letter of PASS on purpose, and it can do so only because the material makes it safe: dense polytetrafluoroethylene has a pore size well below one micrometre, far smaller than a bacterium, so the barrier itself performs the function that primary closure would otherwise perform — keeping the graft space separate from the oral environment. The membrane is lifted off at three to four weeks without a flap, because tissue does not grow into it. Option B is the plausible alternative and contains a true statement, since intact socket walls do maintain space; but space maintenance is not the principle being waived, and identifying which one is waived is the whole question. Option E is the trap for the antibiotic-minded: a prescription substitutes for nothing here. (*Wang & Boyapati PASS; ITI Treatment Guide vol 7*)

662 **C** Cross-linking retards enzymatic degradation, so the barrier is still present at four to six months rather than a few weeks, which is what a large staged augmentation needs. The price is biological: the cross-linked material integrates less readily with the overlying tissues, is less well tolerated when it becomes exposed, and exposures are more likely to become chronic, to suppurate and to require removal, whereas an exposed native collagen membrane will often granulate over and heal. Option A is the assumption most candidates make, because cross-linked membranes feel stiffer in the hand; they are stiffer, but nowhere near stiff enough to tent a non-contained defect, and space must still come from a particulate or from titanium reinforcement. B is untrue of both, D reverses the effect, and E belongs to dense polytetrafluoroethylene. (*ITI Treatment Guide vol 7; Lindhe*)

663 **B** The expanded form of polytetrafluoroethylene has an open microstructure with pores large enough for bacteria to enter and colonise the full thickness of the material. Once it is exposed it cannot be cleaned and it cannot be salvaged: the colonised membrane becomes a reservoir that infects the graft beneath it, so it is removed promptly and whatever bone has formed in the first four weeks is kept. Option A is the answer that would be right for a native collagen membrane, where exposure without infection is managed expectantly and the site usually epithelialises over, and candidates who have learned that rule apply it to the wrong material. C will not hold on ischaemic margins at four weeks, D treats the operator's anxiety, and E grafts onto a contaminated barrier. (*ITI Treatment Guide vol 7*)

664 **D** Tacks are driven into the buccal cortex over the premolar region, which is exactly where the mental nerve emerges and where an anterior loop may run forward beneath the periosteum before turning back, so the foramen must be identified and the tacks placed clear of it. Roots are the other real hazard: the buccal plate is thin, an adjacent root may lie closer to it than the flap suggests, and a tack driven at the wrong angulation enters the periodontal ligament or the root itself. Item C is what makes option E wrong, and it is chosen by candidates who recall that the lingual plate is the dangerous surface in the mandible: it is, but during osteotomy preparation, not during buccal fixation, since a tack of a few millimetres seated in the buccal cortex never approaches it. (*ITI Treatment Guide vol 7; Lindhe*)

665 **B** A barrier works by sealing the regenerating compartment off from the fibrous tissue and epithelium of the flap, and the seal is made at the periphery, where the membrane lies against intact bone. Trim it to the defect outline and the margin of the graft sits at the edge of the barrier, so connective tissue creeps underneath and colonises the periphery — producing exactly the pattern described, bone centrally with soft tissue at the rim. Overlapping sound bone by two to three millimetres, and fixing the membrane there, prevents it. Option C is the plausible alternative, and premature resorption is a genuine failure mode of native collagen; but it would produce a uniform reduction in regenerated volume rather than a peripheral band with a sound centre. (*ITI Treatment Guide vol 7*)

666 **C** Do the arithmetic first. Five millimetres above the canal, less the 2 mm safety margin conventionally kept above it, leaves 3 mm of usable height, and the ridge is 3.5 mm wide — a deficiency in both dimensions with no implant of any length that fits. Option B is the reasoned alternative and deserves credit, because short implants are the best-evidenced way of avoiding vertical augmentation and would be the answer with 8 mm above the canal; here the number does not reach. A is the procedure with the highest complication rate in regenerative surgery, and mesh exposure in an augmented mandible is not a generalist's problem to manage. D corrects one dimension and leaves the other, and E is a legitimate non-implant alternative that should be discussed alongside the referral rather than instead of it. (*ITI Treatment Guide vol 7; SAC Assessment Tool*)

667 D Containment is the variable that governs regeneration. A fenestration is a window in the cortical plate surrounded on all sides by bone, so the graft is held where it is placed, the clot is stabilised, and osteoprogenitor cells and vessels enter from every margin; it is the most predictable defect in guided bone regeneration. A dehiscence communicates with the crest, is open coronally, and depends on the barrier and the flap to contain what would otherwise displace, which is why the coronal portion regenerates least reliably. Option E is the attractive alternative because its conclusion is right and its reasoning sounds plausible — the apical position does reduce the consequence of a dehiscence of the flap — but distance from the mouth is not what determines the result; the number of bony walls is. (*ITI Treatment Guide vol 7; Wang & Boyapati PASS*)

668 B The comparison has been made directly in randomised trials and in systematic review, and it is unusually clean: implants of about 6 mm in the residual posterior ridge achieve survival comparable with longer implants placed in vertically augmented bone, while the augmented arm accumulates the complications — dehiscence, membrane or mesh exposure, infection, graft loss, donor-site morbidity, longer treatment and greater cost. Option C is the honest-sounding trade-off that most candidates choose, because it feels as though something must be sacrificed for the shorter implant; the evidence does not show the survival penalty it assumes. A overstates the predictability of vertical gain, which is the least reliable of all augmentation procedures, D is untrue, and E invents a restriction that does not exist. (*ITI Treatment Guide vol 7; Misch*)

669 A A buccal plate of 0.6 mm is essentially all bundle bone: it is the tissue into which the periodontal ligament fibres insert and it is supplied through that ligament, so when the tooth is removed the plate loses its blood supply and resorbs. Where the wall is thin the whole thickness is bundle bone and the crest collapses vertically as well as horizontally, whereas a plate over about 1 mm retains an outer layer of lamellar bone with a periosteal supply and loses width while keeping much of its height. That threshold is why this socket is a candidate for ridge preservation or early placement rather than unassisted healing. Option D is the near miss, because the coronal wall does resorb most; the loss is not confined there. (*Araújo & Lindhe; Chen & Buser*)

670 A Grafting a socket limits physiological resorption but cannot abolish it, because the loss of bundle bone follows removal of the periodontal ligament whatever is placed inside; the consistent finding across trials and systematic reviews is that roughly half of the horizontal change is prevented, leaving a residual reduction of the order of one and a half to two millimetres. Items B and C are the overclaims, and B is the one patients hear from marketing and candidates repeat: preservation is a preservation technique, not a regenerative one, and promising an unchanged ridge sets up a conversation you cannot honour when the site needs contour grafting at placement. Option E compounds both errors. Say what it does — it reduces the loss and often avoids further augmentation later. (*Avila-Ortiz; Lindhe; ITI Treatment Guide vol 10*)

671 B This is the socket with an intact soft tissue envelope over a missing labial plate — the configuration in which the mucosa flatters the site and conceals the defect beneath it. The soft tissue tells you nothing about the bone, and the probe does. In a thin phenotype with a high smile line the safe route is to reconstruct the wall first and place later, so that the implant goes into a rebuilt envelope rather than being used to hold up a graft. Option A is the seductive answer and is genuinely practised by experienced operators in this configuration, which is what makes it dangerous here: immediate placement with simultaneous grafting into a two-wall-deficient socket in a thin, high-smiling patient has a real incidence of recession that cannot be undone. C misreads the soft tissue, D ignores the probe, and E defers without a plan. (*Elian et al.; ITI Treatment Guide vol 3*)

672 B Early placement at four to eight weeks is chosen for two reasons, both of them soft-tissue and biological rather than skeletal: the socket has epithelialised and gained a thickness of keratinised tissue that makes flap advancement and primary closure far easier than in a fresh socket, and any apical or periodontal infection that led to the extraction has resolved, so the implant is not placed into a contaminated field. Statement (iii) is what makes options C, D and E wrong, and it is the belief that makes clinicians complacent about the labial wall: modelling of the buccal plate is well under way but nowhere near finished at eight weeks, most of the dimensional change occurring over the first three months and continuing thereafter, which is why simultaneous contour grafting is still usually required. (*Chen & Buser; ITI Treatment Guide vol 3*)

673 C Simultaneous augmentation has two prerequisites and this site meets neither: the implant cannot be placed in the prosthetically correct position with primary stability, and the defect that would result is a large, non-contained dehiscence. When both fail, the honest course is to abandon placement, graft the ridge as a staged procedure and return to consolidated bone — a decision that is far easier to take before the implant is in than afterwards. Option A is the commonest real-world error and the most attractive answer here, because 35 Ncm feels like success: it buys stability at the cost of a palatally placed, palatally

angled implant whose restoration will be a ridge-lap with a buccal void. B grafts around an implant too mobile to integrate, D does not address the dehiscence, and E restores a malposition. (*ITI Treatment Guide vol 7; Wang & Boyapati PASS*)

674 D The two conditions that govern simultaneous augmentation are both about the implant rather than the graft: it must be placeable where the restoration requires it, and it must be immobile, because osseointegration and regeneration have the same requirement for stability and grafting around a mobile implant risks losing both at once. Item C is what makes option E wrong, and it is chosen by candidates who reason that needing a release signals a defect too large to graft simultaneously: a periosteal releasing incision is a routine part of obtaining tension-free closure over any augmented ridge, it is used in the great majority of these cases, and treating it as a contraindication would rule out procedures that are entirely predictable. What matters is that closure is passive, not that it was easy. (*ITI Treatment Guide vol 7*)

675 E The healing period is set by how long the regenerate takes to become drillable, load-bearing bone, and that is governed principally by the graft. Autogenous bone is revascularised and remodelled comparatively quickly and can be re-entered at around four months, whereas a graft dominated by a low-substitution xenograft matures far more slowly and is conventionally left for six to nine months, longer for larger volumes. Option B is the plausible alternative, because a cross-linked membrane really does persist longer and candidates reason that the site cannot be opened while a barrier remains; barrier duration is chosen to match the graft, not the other way round, and a resorbing membrane does not delay re-entry. Fixation method, flap release and jaw of treatment do not change the biology of maturation in this way. (*ITI Treatment Guide vol 7*)

676 C Soft, fibrous crestal tissue is not immature bone that will improve once an implant is in it; it is the part of the regenerate that failed, and placing an implant through it puts the coronal threads — the ones that determine crestal bone level and the seal — into tissue that will never support them. The disciplined course is to remove it, re-graft the crest and come back, accepting a delay of a few months against a compromised implant for a decade. Option A is the attractive answer because under-preparation is the correct response to genuinely soft *bone* and every operator has used it; it raises insertion torque by compression, which fibrous tissue cannot provide. B buries the platform to chase stability, D hopes healing will remedy a defect that is already mature, and E gives up too soon. (*ITI Treatment Guide vol 7*)

677 A Ridge expansion depends on the buccal plate hinging outwards while remaining attached to its periosteum, and that is mechanically possible only where a layer of cancellous bone separates the two cortices and can be split and compressed. Where the cortices are in direct contact — the knife-edge ridge, and the dense mandibular body — the plate fractures completely instead of bending, loses its blood supply and is lost. Option C is the attractive distractor because it sounds like sound anatomy and the periosteal supply is genuinely the point; the mobilised segment does not retain a meaningful endosteal supply once it is out-fractured, and survival depends entirely on the periosteum, which is why a partial-thickness or minimal flap is used. B is untrue, D overstates maxillary healing, and E is precisely the technique to avoid. (*ITI Treatment Guide vol 7; Lindhe*)

678 B The technique needs something to split. Two cortices in contact give the osteotome nothing to advance into, so instead of a controlled greenstick outfracture the plate shears off completely, becomes a free fragment dependent on the periosteum alone, and is commonly lost — taking with it the width the procedure was meant to create. Option A is the number candidates reach for, and 3 mm is genuinely marginal; but width alone is not the determinant, and a 3 mm ridge with a cancellous interface can sometimes be expanded whereas a 5 mm knife-edge with fused cortices cannot. C is a real contributory factor, since the mandibular cortex is denser and less forgiving, and it is a reason to consider a two-stage corticotomy; it is not the primary reason. D is favourable, and E is a soft tissue matter. (*ITI Treatment Guide vol 7*)

679 D A segment that has kept its periosteal attachment is still vascularised and can survive if it is made immobile, so it is repositioned, secured with a screw or with tacks through the overlying membrane, the gap between the plates is filled with particulate, and the site is closed and left to consolidate. Placing implants at the same visit is what must be given up. Option B is the manoeuvre that tempts every operator at that moment, because the implant looks like the obvious means of fixation; in reality the implant would be engaging a mobile fragment on one side and would itself have no primary stability, so both the segment and the implant are lost. A discards viable bone, C abandons the buccal contour, and E leaves a mobile fragment over an ungrafted defect. (*ITI Treatment Guide vol 7*)

680 C Work the width out. A 4.1 mm implant in 4.5 mm of ridge leaves about 0.2 mm on each side, which is no bone at all; a 3.3 mm implant leaves roughly 0.6 mm, below the 1 mm minimum usually quoted and far below the 2 mm of buccal bone wanted in an aesthetic site with a thin phenotype and a high lip line, where any further modelling shows as a grey shadow or recession. Staging the augmentation

and placing into a rebuilt ridge is the answer that respects the aesthetic risk. Option A is the attractive shortcut, because narrow-diameter implants are perfectly appropriate for an upper first premolar and the site "looks placeable"; the diameter is not the problem, the envelope is. B is arithmetically impossible, D lacks the cancellous interface, and E is a legitimate alternative to offer rather than the plan. (*ITI Treatment Guide vol 7; ITI Treatment Guide vol 1*)

681 **A** Pulsatile bleeding at the apical limit of a palatal harvest means the greater palatine artery or a substantial branch has been cut, and the sequence that controls it is mechanical: sustained, uninterrupted pressure long enough for a clot to form, then a suture placed apical to the bleeding point to compress the vessel against the bone of the palate, followed by a haemostatic dressing and a stent. Option E is the answer that sounds definitive and is the one candidates choose when frightened: blind exploration in a bleeding palatal field enlarges the wound, is technically difficult under local anaesthesia and is unnecessary, because the vessel lies against bone and can be compressed. Cautery in a pool of blood is ineffective and damages the bed, C treats a venous ooze, and D buys minutes before rebound bleeding at home. (*Lindhe*)

682 **B** The greater palatine nerve accompanies the artery forward from its foramen in the groove between the alveolar process and the horizontal plate, supplying the palatal mucosa from the canine region backwards — precisely the distribution described — and it or its branches are within reach of a harvest that extends too far apically. The reassuring part is the prognosis: these are usually traction or partial injuries, and sensation returns over weeks to a few months, though the patient should be told, the finding recorded, and the area mapped and reviewed. Option A is the anatomical error that catches candidates, since the nasopalatine nerve supplies only the mucosa behind the incisors as far back as the canine, which is the territory that is *not* numb here. C, D and E do not fit either the field or the time course. (*Lindhe*)

683 **E** Two different contractions act on a free gingival graft and candidates routinely merge them. Primary contraction happens the instant the graft is freed and is caused by recoil of the elastic fibres within it, so it is greater in thicker grafts that carry more lamina propria. Secondary contraction happens over the following weeks as the graft is revascularised and the underlying bed remodels, is driven by the recipient site rather than the graft, and is greater in *thin* grafts — which is why a graft intended to widen a keratinised band is taken at around 1.5 mm rather than as thin as it will cut. Option A is the exact inversion and is the most attractive distractor, because it names a real phenomenon correctly and applies it to the wrong thickness. (*Lindhe*)

684 **D** The composition of a palatal graft depends on the depth from which it is taken. The superficial lamina propria immediately beneath the epithelium is dense, highly collagenous and contains little fat or glandular tissue, so it maintains volume well; the deeper submucosa that a trap-door dissection inevitably includes is progressively more fatty and glandular and shrinks more. Harvesting a full-thickness strip and de-epithelialising it extra-orally therefore yields the most stable tissue, at the price of an open donor wound. Option A is the strong alternative and the more familiar technique, with a closed, less painful donor site, which is exactly why it is chosen; its yield is less predictable in composition. B is not autogenous, C would leave a visible patch, and E offers useful but limited volume. (*Lindhe; ITI Treatment Guide vol 13*)

685 **A** The honest claim is a trade of morbidity against result. A collagen matrix is a resorbable scaffold that the patient's own fibroblasts and vessels colonise, so there is no palatal wound, no risk to the greater palatine vessels, less post-operative pain and a shorter operation — a genuine advantage in an anxious patient, in one taking anticoagulants, or where several sites are being treated. Item B is what makes options D and E wrong and is the claim most often implied to clinicians: volume gain with a matrix is consistently somewhat less, and less predictable, than with autogenous connective tissue, which remains the reference standard. Item C misunderstands the technique, since a matrix used to widen keratinised tissue is still placed beneath an apically positioned flap; it replaces the graft, not the flap. (*ITI Treatment Guide vol 13; Lindhe*)

686 **A** The evidence for widening keratinised mucosa is consistent for the soft tissue and patient-centred outcomes and inconsistent for the skeletal one, and the consent conversation must reflect that difference. Implants with less than about 2 mm of keratinised mucosa show more plaque, more bleeding, more brushing discomfort and more recession, and a graft that produces an immobile, keratinised band predictably improves all four — a good enough reason to operate. Option B is the attractive overstatement and the answer she would prefer to hear: the association between keratinised width and progressive bone loss is not consistent across studies, and grafting has not been shown to arrest bone loss. C promises what no procedure delivers, D describes a connective tissue graft, and E is the opposite of the truth. (*2017 World Workshop; ITI Treatment Guide vol 13*)

687 **D** Root coverage procedures work at teeth because the connective tissue attachment, the periodontal ligament vasculature and the interdental bone provide a receptive bed; around an implant the supracrestal fibres run parallel to the surface rather than inserting into it, there is no ligament and no periodontal

vascular plexus, and the underlying cause here — a buccally positioned implant — has not gone away. Partial improvement in thickness and some coronal gain are realistic; complete, stable coverage is not, and promising it is how a manageable complaint becomes an unmanageable one. Option B is the attractive answer because intact papillae genuinely are the best predictor at teeth, and candidates transfer the rule; the analogy fails at the attachment. C, A and E all overstate what is achievable. (*ITI Treatment Guide vol 13; Lindhe*)

688 **D** A stent shields the raw donor bed from the tongue, from food and from the negative pressure of swallowing, supports any haemostatic dressing and reduces both pain and secondary bleeding, and it is the single most useful thing to provide. Harvest technique matters as much: a single-incision approach leaves a closed wound that heals by primary intention with markedly less discomfort than an open wound left to epithelialise, whereas a trap-door or a full-thickness de-epithelialised strip leaves a larger raw area. Item C is what makes option E wrong, and it is the reflex that many clinicians share: donor-site pain is a wound-healing phenomenon, not an infective one, palatal wounds rarely become infected, and an antibiotic course adds resistance, gastrointestinal upset and occasional anaphylaxis for no analgesic benefit. (*Lindhe; SDCEP Drug Prescribing*)

689 **A** The roll flap solves exactly this problem: the thick palatal tissue immediately adjacent to the implant is de-epithelialised, raised as a pedicle that keeps its own blood supply, and rolled buccally beneath the flap at uncovering, adding a millimetre or two of contour with no second wound and no extra healing time. It is limited to small deficiencies and depends on the palatal tissue being thick, both of which the stem supplies. Option E is the answer candidates reach for because tuberosity tissue is dense and volume-stable and is genuinely the best free graft available — but it is still a separate surgical site, which the patient has declined, and its yield is limited and unpredictable. B would leave a pale patch, C adds a material where the patient's own tissue is at hand, and D redistributes without adding volume. (*ITI Treatment Guide vol 13; Lindhe*)

690 **B** Advancing a flap does not create tissue; it moves it. Six millimetres of coronal advancement drags the mucogingival junction and the keratinised band with it, so the crest of the healed ridge is covered by mobile alveolar mucosa and the vestibule is shallower than it was — which is why second-stage surgery at such a site is planned as an apically positioned flap, often with a free gingival graft, rather than a simple punch. Item C is the mechanically intuitive item that makes options D and E wrong: the flap does lie on the graft, but a passively closed flap exerts no meaningful compression, and it is tension, not contact, that causes trouble. Item A is untrue — the periosteal release divides an inelastic layer, and the flap's supply is within the mucosa, which is why it survives. (*ITI Treatment Guide vol 7; Lindhe*)

691 **C** The measurement is the single most useful predictor available and the figures are worth carrying: where the contact point lies 5 mm or less above the bone crest the papilla is present essentially every time; at 6 mm it is present in a little over half of sites; at 7 mm in only about a quarter. At 7 mm, therefore, the honest answer is that an open embrasure is the likely outcome, and the conversation must happen before treatment rather than after the crown is fitted. Option B is the near miss and is chosen most often, because 56 per cent at 6 mm is the figure candidates half-remember and they apply it one millimetre out. D reverses cause and effect, and E discards the best predictor there is. (*Tarnow; ITI Treatment Guide vol 1*)

692 **D** Both surviving statements come from the same body of work and both should temper enthusiasm for adjacent anterior implants. Where the platforms are separated by less than about 3 mm of bone the lateral remodelling around each implant overlaps and crestal bone loss between them is roughly doubled — of the order of a millimetre rather than half a millimetre — and the interimplant bone crest is what supports the papilla. The mean soft tissue height that can be maintained between two implants is around 3.4 mm, appreciably less than between a tooth and an implant, which is why two adjacent implants in the aesthetic zone is an advanced case. Item C is what makes option E wrong: splinting simplifies the prosthesis and may help load distribution, but it adds no tissue height. (*Tarnow; ITI Treatment Guide vol 1*)

693 **C** Surgical reconstruction of an interdental papilla is among the least predictable procedures in the mouth: the recipient bed is tiny, poorly vascularised and mobile, the tissue has nothing to be supported by once the bone crest is 7 mm from the contact, and reported gains are small and frequently lost. The reliable solution is prosthetic — lengthen the contact area apically so that the distance from contact to crest falls towards 5 mm, and the space closes. Option A is the answer that addresses the deficiency directly and is therefore the most attractive, and it is exactly the wrong instinct here. D would work biologically but means moving a sound tooth for an aesthetic problem soluble in the laboratory. E overstates spontaneous fill, which is real adjacent to single implants but modest and slow. (*Tarnow; ITI Treatment Guide vol 1*)

694 **B** A transcrestal elevation is a blind procedure in which the membrane is lifted by an osteotome pushing a plug of bone ahead of it, and it works when the floor is flat and the membrane rises evenly. Where the floor slopes, the instrument tends to deflect down the incline, the membrane is stretched acutely

over the sloping wall on one side while it is barely raised on the other, and it tears at the point of maximum stretch; a narrow angle between the medial and lateral walls concentrates that stretch further. Option C is the item candidates pick because thickening sounds pathological: a membrane of about 1.5 mm is at the upper end of normal and, in any case, it is the thin membrane, under about a millimetre, that perforates most readily. A residual height of 6.5 mm is favourable. (*ITI Treatment Guide vol 5*)

695 **C** Through a crestal osteotomy a few millimetres across you cannot see the tear, cannot judge its size and cannot confirm that anything placed over it has sealed it — so any graft introduced now may be delivered straight into the antrum, where particles obstruct drainage and cause sinusitis or, occasionally, a fungal ball. Stopping, closing and allowing the membrane to heal costs the patient two months and nothing else. Option D deserves serious credit and is what an experienced sinus surgeon would do, since direct vision allows the perforation to be repaired and the case completed; that is precisely why it is the distractor to beat, because a lateral window is not a procedure to attempt for the first time as a rescue. A is the seductive middle course and is unverifiable, B abandons the plan, and E grafts blind through a hole. (*ITI Treatment Guide vol 5; SAC Assessment Tool*)

696 **C** Sinus floor elevation depends on an antrum that drains freely through its ostium, because the graft raises the floor and reduces the volume available for mucociliary clearance. Gross mucosal thickening with an obstructed ostium, even in a patient who feels well, is a reason to have the sinus assessed and, if necessary, treated before any elevation — and to accept that the answer may be that this sinus should not be grafted at all. Option A is a true statement recruited to the wrong argument, and it is the most attractive distractor because thick membranes really are more robust; resistance to tearing says nothing about drainage. B and E proceed while naming the risk they are creating, and D changes a variable that does not address the ostium. (*ITI Treatment Guide vol 5*)

697 **A** Anything that raises pressure within the antrum acts directly on the elevated membrane and the graft beneath it: forceful nose blowing and a closed-mouth sneeze can both displace the membrane, disperse the particulate and produce surgical emphysema, and both are entirely avoidable if the patient is warned. The instruction costs nothing and is the one that maps onto the specific mechanical risk of this operation. Option E is the strongest alternative and is sensible general advice after any oral surgery, which is what makes it attractive; head elevation reduces oedema but does nothing about intra-antral pressure. B is useful adjunctive hygiene, C is not required for that duration, and D reflects a misunderstanding, since an unloaded submerged implant is not being chewed on in any case. (*ITI Treatment Guide vol 5*)

698 **A** Material origin is information that a reasonable person in the patient's position would attach significance to — on religious, cultural, dietary or simply personal grounds — and *Montgomery* makes disclosure of such material considerations a duty that does not wait for the patient to ask. She is told what the materials are, where they come from, that processing removes the organic phase and leaves a transmission risk that is theoretical and vanishingly small, and what the alternatives are. Option C is the answer many clinicians give themselves and is the most attractive, because the factual claim within it is correct; it converts a decision that belongs to the patient into a technical judgement made by the clinician. B describes a record-keeping duty, D reverses the burden, and E overstates the evidence. (*Montgomery; GDC Standards*)

699 **B** This is the combination the decision trees explicitly divert to a lateral window: with about 5 mm or less of residual height, or a floor that is not horizontal, blind osteotome elevation loses both the primary stability that a simultaneous implant needs and any control over how the membrane rises, and the lateral window that follows is not a generalist procedure. Option A is the answer candidates most often mark as beyond scope, because 5 mm sounds like a large defect; a dehiscence at a stable implant in the correct position is a contained, simultaneous augmentation and is the archetype of what is *within* competence. C, D and E are likewise routine for an experienced practitioner. Referral is a judgement about the specific anatomy, not about the name of the operation. (*ITI Treatment Guide vol 5; SAC Assessment Tool*)

700 **A** Grafting is a means to an end, and the end is an implant in the position the restoration requires. Here 8 mm above the canal, less the 2 mm safety margin, leaves 6 mm of usable height in a 7 mm wide ridge — a well-evidenced indication for a short implant, so augmentation would add a second surgical site, months of treatment, cost and a real complication rate in exchange for nothing the patient can be shown to gain. Option B is the belief being tested and the one many clinicians still hold: beyond a certain length, implant survival does not track length, and short implants match augmented sites. C makes the patient's willingness the justification for an unnecessary operation, D applies a tooth-derived ratio that has not been shown to predict implant failure, and E overstates. (*ITI Treatment Guide vol 7; Misch; GDC Standards*)

Answers — Block W — Restoring implants, occlusion and the laboratory, extended (Q701–760)

701 A Accuracy has two components and the question separates them. Trueness is how close the mean of the measurements lies to the true value; precision is how close repeated measurements lie to one another. Scanner A is consistently 61 micrometres out but repeats itself within 12, so it is precise and untrue; scanner B averages 24 micrometres but any single scan may be 88 micrometres from its neighbour, so it is true and imprecise. Option D is the answer most candidates give, because a smaller mean deviation reads as better performance — but you take one scan on one patient, and with scanner B that single scan may be the outlier. For a full-arch framework, reproducibility is what you are buying; a systematic offset that is the same every time is at least a known quantity. (*ISO 5725; ITI Treatment Guide vol 11*)

702 B The software does not see the implant; it fits a stored library geometry to whatever part of the scan body's surface has been captured, and from that fit it computes the platform position and the long axis. When only 3 mm of a cylinder is available, a surface deviation of a few tens of micrometres at each end swings the calculated axis by several degrees, and that angular error is then projected up the height of the crown to displace the access channel by millimetres. Option A is the strong distractor, and mucosa lying against the scan body genuinely does obscure geometry — but it obscures it by shortening the usable length, which is the mechanism in B rather than a separate one. The remedy is a taller scan body, or gentle retraction to expose more of it. (*ITI Treatment Guide vol 11*)

703 D Splinting exists to stop copings moving independently of one another, so its benefit rises with the number of implants, the length of the span and the degree of divergence — all three of which describe the mandibular case and none of which describes the single premolar. Item C is what makes E wrong, and it is accepted uncritically by candidates who have learned splinting as a virtue rather than as a solution to a defined problem: with one implant there is no inter-implant relationship to preserve, the adjacent teeth index the impression, and a resin splint adds a polymerising mass, an extra clinical stage and a further source of distortion for no gain. Splint where relationships must be held; do not splint reflexively. (*Misch; ITI Treatment Guide vol 8*)

704 C An open-tray coping in a molar site projects well above the occlusal plane, and the tray must be brought over it, seated, and then lifted off vertically with the coping still inside. The clearance required is the coping height plus the tray thickness plus the unscrewing of a long guide screw, which in a posterior site comfortably exceeds 26 mm. Option A is the attractive answer and it names a real difficulty — access for a screwdriver at the back of the mouth is awkward — but the screw can be shortened or a short coping used, whereas the vertical envelope cannot be created. A closed-tray transfer in a rigid custom tray is the pragmatic answer for a single implant, and the accuracy penalty of repositioning one coping is small. (*Misch; ITI Treatment Guide vol 8*)

705 D Photogrammetry computes the three-dimensional relationship of coded targets from overlapping images, and because it does not stitch a continuous surface it escapes the cumulative registration error that limits intraoral scanning across an edentulous arch. What it returns, however, is a set of coordinates and nothing else: no mucosal contour, no opposing arch, no vertical dimension. A separate intraoral scan or conventional record is therefore always needed, and the two datasets merged. Option C is the seductive wrong answer because span length is exactly what degrades an intraoral scan, and candidates transfer the rule; the whole point of the method is that inter-marker distance does not degrade it. Splinting is unnecessary, and divergence is handled by the targets' own orientation. (*ITI Treatment Guide vol 11*)

706 B Geometric distortion from an oblique beam produces a *wedge*, open on one side and closed on the other; a gap of even width across the whole interface means the coping is genuinely held off the platform, almost always by interposed mucosa, a fragment of provisional cement or bone. Unscrew it, inspect and clean the platform under magnification, reseat and confirm radiographically. Option C is the error that does real harm and is the one taken under time pressure: tightening drives the coping onto whatever is trapped, may deform the coping or the platform, and produces a cast that reproduces the error faithfully, so that every subsequent stage is built on a false position. A misfit detected before the impression costs five minutes; detected at framework try-in it costs a remake. (*Misch; ITI Treatment Guide vol 8*)

707 C The comparative literature is consistent and it splits by span. For single crowns and short-span prostheses on two or three implants, scanning matches conventional impressions and is quicker and more comfortable, which is why it has displaced them for most restorative work. Across a full arch, particularly an edentulous one, deviations reported for scanning exceed those for a rigid splinted open-tray impression, because registration error accumulates along a path with few stable landmarks. Option E is the attractive answer, and it has a respectable argument behind it — no threshold of misfit has been shown to cause biological failure — but misfit does correlate with technical failure, and telling a colleague that accuracy does not matter is not the lesson. Verify long-span work whichever route is taken. (*ITI Treatment Guide vol 11; Misch*)

708 E All three contribute and the case contains all three, which is why it failed. Mobile mucosa is not a landmark: it deforms between passes, so the software has nothing rigid on which to correct drift across the edentulous span. Fluid on the scan body alters the recorded surface, and since the library file is matched to that surface, the deviation is transferred straight into the calculated implant position. And every interruption requires the scanner to re-register against previously captured data, which adds a fresh registration error at each restart. Candidates most often exclude item C, treating a pause as neutral; it is not, and the discipline of a single uninterrupted pass in a dry, retracted field is what separates a usable full-arch scan from an unusable one. (*ITI Treatment Guide vol 11*)

709 D Angled multi-unit abutments exist precisely to convert divergent implant axes into a set of parallel prosthetic platforms with a single path of insertion and screw access relocated to the occlusal or cingulum surface. Without them, a one-piece framework on implants diverging by 30 degrees simply cannot be seated. Option A is the distractor with real merit and it is true — recording at abutment level means the abutments stay undisturbed, the mucosal seal is not repeatedly torn, and the one-abutment-one-time benefit of a few tenths of a millimetre of crestal bone is preserved. It is a genuine advantage, but it is available in any abutment-level case; the divergence in this stem is what makes the answer D. Prosthetic screws are torqued lower, not higher, which disposes of E. (*ITI Treatment Guide vol 4; Misch*)

710 A Only the first item helps. A base screwed to the implants cannot rock, cannot sink into resilient mucosa and cannot flex under closing pressure, so the relationship recorded is the one that will exist when the prosthesis is fitted. Item B is the one candidates accept most readily, since softened wax obviously permits full closure — but wax deforms under load and then rebounds elastically, and the record it yields is displaced by an amount nobody can quantify; a rigid base with a low-distortion registration paste is what is wanted. Item C imports an error rather than reducing one: a worn denture of unknown age is not evidence of the correct vertical dimension, which must be determined afresh from rest position, speech and appearance. (*Misch; ITI Treatment Guide vol 4*)

711 B The failure mode in cross-arch scanning is accumulated registration error across a span offering no rigid landmarks. Connecting the scan bodies with a rigid resin bar converts five isolated cylinders into one continuous, irregular, fixed object; the software then has a large stable geometry to register against along the whole arch, and reported deviations fall substantially. Option A is the sensible-sounding answer and a continuous sweep is indeed better than an interrupted one, so it is worth doing — but the stem has already excluded the obvious technique faults, and scan strategy alone does not solve the absence of landmarks. Larger scan bodies help marginally; merging separate scans adds registrations rather than removing them. If the bar is used, the conventional splinted impression is the natural next step. (*ITI Treatment Guide vol 11*)

712 A Only the one-screw, or Sheffield, test interrogates fit, because it removes the ability of multiple screws to disguise a discrepancy by pulling the framework down onto the abutments. Item B is the trap and it is the reason misfitting frameworks are delivered every week: a framework that does not fit still reaches 15 or 20 Ncm, because torque measures the resistance of the screw joint and not the accuracy of the casting, and reaching the set value is therefore evidence of nothing. Item C fails for a related reason — misfit is characteristically silent, and by the time a patient reports discomfort the strain has already been locked into the screws, the abutments and the crestal bone. Radiographs supplement the one-screw test but do not replace it. (*ITI Treatment Guide vol 8; Misch*)

713 C The clinical work on this question compared patients with and without a history of periodontitis in whom cement remnants were subsequently found, and the difference was stark: peri-implant disease developed in almost all of the periodontitis group and in a minority of the periodontally healthy. Cement is the foreign body; the host response and the pathogenic flora are what determine whether that foreign body is tolerated or destroys the site. A periodontitis history is therefore an indication for screw retention or for a supramucosal margin, not merely a reason for more care. Option B is a true statement about radiographic detection and is the one most candidates pick, but detection is not causation, and it does not explain why two patients with the same cement had different outcomes. (*Linkevicius; ITI Treatment Guide vol 13*)

714 D The redirection available is finite — commonly to around 25 degrees, occasionally a little more, and always system-specific — and it is achieved by a screw whose head accepts a driver entering off-axis, so the correct screw and its matching instrument are not optional extras but part of the design. Item C is what makes E wrong, and it is chosen by candidates who reason that an engineered angled interface must be a stronger one: the angled driver engages a smaller effective contact area and transmits torque less efficiently, so manufacturers specify the same value as the straight channel or occasionally a lower one, never a higher. Exceeding it risks rounding the drive interface, which converts a retrievable restoration into an unretrievable one. (*ITI Treatment Guide vol 11*)

715 **A** A definitive crown held by a soluble cement does not usually fall off; it washes out at one part of the margin while the rest still holds, so the crown feels firm, the patient reports nothing, and a plaque-filled space communicates with the peri-implant sulcus for months. The clinical consequence is mucositis and bone loss at a site that appears intact on examination, and it is the insidiousness that makes it the strongest objection. Option D is the answer many candidates choose and the risk is real and occasionally serious, but complete decementation announces itself and is dealt with at one appointment. If retrievability is the objective, it is obtained by a screw or an angulated channel, not by deliberately weakening a lute. (*ITI Treatment Guide vol 13; Misch*)

716 **B** The number in the stem is 3.5 mm, and it decides the question. Cement removal is unreliable beyond about a millimetre below the mucosal margin and effectively impossible at three and a half; a crown bonded to its titanium base on the bench, cleaned and polished extra-orally and then delivered as a single screw-retained unit, brings no cement near the tissues at all. Option C is a true and important statement and is the reason titanium bases are preferred over one-piece zirconia abutments generally, which is why candidates reach for it — but it is an advantage over a *zirconia* connection rather than over cementation, and the stem's deep margin is what is being tested. E is self-contradictory, since the restoration is screw-retained. (*ITI Treatment Guide vol 11; Linkevicius*)

717 **B** One item survives, and it is the one that fits the case: an access channel through a functional cusp tip weakens the restoration, is difficult to seal and looks poor, and cementing on an abutment moves the problem away. Item A is the half-truth that makes D wrong: a cement lute of a few tens of micrometres takes up a *minor* discrepancy, but nothing about cementing corrects a framework that is genuinely misfitting, and repeating the claim excuses casting error. Item C reverses the position altogether — retrievability is the defining advantage of a screw, and a cemented crown usually has to be cut off. Candidates who choose E have collected every favourable statement they have heard about cement without testing any of them. (*ITI Treatment Guide vol 8; Wittneben et al.*)

718 **C** The access channel communicates with the internal cavity of the implant, which is already a reservoir with a microgap at the abutment interface; a moist, absorbent, protein-retaining material sitting in it is a culture medium, and the malodour on retrieving such restorations is familiar. Polytetrafluoroethylene tape is inert, hydrophobic, does not support growth, fills the drive slot completely and lifts out in one piece. Option D is the most attractive alternative because protecting the drive mechanism really is one of the two jobs the filling material does, and cotton does it imperfectly — a pellet packs poorly into a hexagonal or star-shaped socket, and composite finds its way in. But an obstructed driver is an inconvenience at retrieval, whereas a contaminated internal cavity is a biological problem. (*ITI Treatment Guide vol 12; Misch*)

719 **A** A conical, or Morse-type, connection seats the abutment against several millimetres of machined implant wall under preload, and the resulting frictional interface leaves a microgap orders of magnitude smaller than a butt joint; leakage studies and clinical inflammation data both favour it. Option B is the answer candidates give most often because platform switching is the phrase they associate with crestal bone preservation, and it is a genuine and useful design feature — but it works by moving the inflammatory infiltrate horizontally away from the crest, not by sealing the joint, and the microgap remains. Raising the torque beyond specification takes the screw into plastic deformation, sealing rings degrade, and an external hexagon is the design with the largest leakage. (*Misch; ITI Treatment Guide vol 12*)

720 **A** For a papilla between an implant and a natural tooth the supporting structure is the tooth's own attachment, and the governing measurement is the distance from the contact point to the interproximal bone on that tooth — a papilla fills reliably at about 5 mm and becomes progressively less predictable beyond it, so 7 mm predicts exactly the deficit she has. Item B is what makes D wrong and it is the one candidates accept without argument: the bone level on the implant side governs papillae *between two implants*, where both supports are prosthetic, but next to a tooth the implant contributes little to the height. Item C describes a different tissue dimension altogether. The remedy here is prosthetic — lengthening the contact — not surgical. (*Tarnow; ITI Treatment Guide vol 1*)

721 **B** In the cohort that established this figure the increased prevalence of peri-implantitis at emergence angles above about 30 degrees, particularly where the profile was also convex, was found at bone-level implants; tissue-level implants, whose polished collar carries the transition through the mucosa and places the restorative margin at or near the mucosal level, did not show it. The mechanism explains the difference: at a bone-level implant the flare happens submucosally, creating a subgingival ledge no brush can reach. Option E is the thoughtful distractor, since convexity does modify the association and the two variables travel together — but it was reported as an additional factor, not as a replacement for the angle. Healthy tissues over four years are not an indication to remake anything. (*Katafuchi et al.; 2017 World Workshop*)

722 **C** The vertical distance between the platform and the mucosal margin is the running room in which the restoration must widen from the platform diameter to the crown diameter. Here the restoration must

gain about 2.75 mm on each side within 1 mm of height, which forces an emergence angle far beyond anything cleansable and produces exactly the overhanging submucosal ledge associated with peri-implantitis. Option A is the good wrong answer and is a genuine consequence of shallow placement — retention for a cemented crown really is compromised — but it is solved by choosing screw retention, whereas no retention mode rescues a contour that cannot be cleaned. The lesson is surgical: platform depth and diameter are restorative decisions taken at placement. (*Su et al.; ITI Treatment Guide vol 11*)

723 **E** All three follow, and stating them together is what the consent conversation requires. Cement removal is unreliable beyond about a millimetre, so at 4 mm the case is effectively a screw-retained one. Peri-implant probing depth is a function of where the platform sits, not of disease, so this implant will read 5 or 6 mm in health — which is precisely why the 2017 definitions rest on inflammation and change rather than on absolute depth, and why a baseline chart matters here more than usual. And the abutment or crown must traverse that full height, which is where the emergence profile is either developed carefully or bungled. Candidates most often omit item B, having learned probing depth as a disease measure rather than an anatomical one. (*2017 World Workshop; ITI Treatment Guide vol 11*)

724 **D** Tissue conditioning is finished when the tissue has stopped moving. Two signs report that: a mucosal margin whose position and zenith are the same at consecutive appointments several weeks apart, and blanching on reseating that resolves within a few minutes rather than persisting, which indicates that the contour is supporting the tissue rather than compressing it. Option A is the answer most candidates give because a time interval is easy to apply, and three months is a reasonable expectation — but time is a proxy for stability and not a substitute, and a profile still migrating at fourteen weeks will migrate after the definitive crown is cemented too. Papilla fill and patient satisfaction are outcomes worth having but neither tells you the tissue has stabilised. (*Su et al.; ITI Treatment Guide vol 1*)

725 **E** The whole technique depends on the coping occupying exactly the orientation the provisional occupied, because the contour being copied is not rotationally symmetrical — a labial concavity and a palatal convexity are only in the right places if the coping is indexed as the provisional was. A coping seated a quarter-turn out reproduces the profile faithfully and puts it on the wrong surfaces, which is precisely the pattern described. Both the analogue in the putty and the coping must engage the same anti-rotational feature, and the index should be marked. Option B is the reasonable alternative, and putty does distort a little on withdrawal — but distortion produces a generally under-contoured coping rather than a profile that is correct in one direction and wrong in the other. (*ITI Treatment Guide vol 11; Misch*)

726 **D** The critical contour — roughly the first millimetre of restorative surface apical to the mucosal margin — is a controlled way of displacing the free margin, and in a thick phenotype about a millimetre can be gained. A thin, scalloped phenotype has neither the volume nor the vascularity to tolerate that pressure, and sustained compression produces ischaemia and recession rather than repositioning. The greyness in the stem is the *result*: as the mucosa thins and recedes, the metal beneath becomes visible. Option A is therefore the attractive answer, and it is a correct description of what is being seen, but it names the appearance rather than the cause, and a candidate who chooses it will treat the discoloration and repeat the error. Reduce the contour and consider grafting. (*Su et al.; ITI Treatment Guide vol 1*)

727 **D** A convex intaglio, resting passively against the mucosa without pressure, allows a superfloss or interdental brush to be passed straight through beneath the prosthesis; screw retention allows you to remove the whole thing at a maintenance visit and clean what the patient cannot reach, which in a full-arch case is a substantial proportion of the surface. Item C is what makes E wrong, and it is chosen by candidates who associate a ridge lap with aesthetics and with keeping food out: a concave surface saddling the ridge traps biofilm against the mucosa, cannot be reached by any aid, and is one of the commonest design faults in hybrid prostheses. Design for the brush first, and let aesthetics accommodate it. (*ITI Treatment Guide vol 4; ITI Treatment Guide vol 13*)

728 **B** The experimental work comparing abutment materials found that a mucosal attachment comparable to that against titanium formed against zirconia and against densely sintered alumina, but that gold alloy and dental porcelain did not support it: the barrier epithelium and connective tissue relocated apically and marginal bone followed. It is one of the few materials findings in implant dentistry that translates into a simple clinical rule — the transmucosal surface should be titanium or zirconia. Option D is the attractive answer because surface topography genuinely matters at the micrometre scale and candidates have learned that roughness governs plaque retention; it does, but the studies here compared machined surfaces of different composition and still found a difference, so composition is not reducible to roughness. (*Abrahamsson; Lindhe*)

729 **C** Articulating paper 40 micrometres thick will mark any surface it is dragged across, and the size and darkness of a mark reflect contact area, surface texture and the amount of ink transferred, not load; a broad mark on a flat implant crown is entirely compatible with a contact lighter than its neighbours.

Shimstock at 12 micrometres is the discriminating test, and the finding described — held at the teeth, free at the implant in light closure — is precisely the relationship implant-protected occlusion sets out to create. Option A is the answer that leads to serial adjustment of a correct occlusion until the crown is out of contact altogether and the opposing tooth begins to overerupt. Confirm by asking the patient to clench firmly, when the crown should then hold the foil. (*Misch; ITI Treatment Guide vol 12*)

730 **B** A 6 mm overbite with a 1 mm overjet describes a very steep guidance path, and the steeper the path the larger the horizontal component of every excursive force. That horizontal force acts at the end of an 11 mm vertical lever, so the bending moment at the screw joint is large, and the consequences appear as screw loosening, ceramic chipping and, over time, screw fracture. Option C is the strong distractor because deep bites genuinely do restrict palatal restorative space and it is a real planning problem, but a shortage of space is solved at the laboratory bench by material selection, whereas a steep guidance angle is a load applied at every excursion for the life of the restoration. Flatten the path as far as aesthetics allow and share guidance with the adjacent teeth. (*Misch; ITI Treatment Guide vol 12*)

731 **A** The contact described is a non-working, or balancing, interference: the mandible moves to the left, and the LR6 on the opposite side is caught during the excursion. Two things make it dangerous. The force is almost entirely horizontal, and horizontal force at the end of a crown-height lever is what loosens screws and fractures ceramic. And the implant has no periodontal mechanoreceptors, so the reflex that would normally interrupt a damaging contact on a natural tooth does not fire — her report that it "catches" is unusually helpful. Option C is the answer of a candidate who has not worked out which side is working; a non-working interference is the least axial contact in the whole occlusion. Eliminate it and re-check with shimstock. (*Misch; ITI Treatment Guide vol 12*)

732 **B** The 25 to 30 micrometre relief exists for one reason: to compensate for the fact that an implant crown sits among natural teeth that intrude under load while it does not, so that contacts set equal at light closure become unequal in function. Where the entire arch is implant-borne there are no neighbouring teeth to protect and nothing to equalise, and relieving every unit simply means the prosthesis has no occlusion until the patient clenches. The correct prescription is even, simultaneous contacts throughout, with shallow anterior guidance and no posterior excursive contacts. Option C is the answer that most nearly gets there and is chosen by careful candidates, but it inverts the rule: the relief is needed when the *opposing* considerations involve natural teeth adjacent to a single implant, not when the opposing arch is prosthetic. (*Misch; ITI Treatment Guide vol 12*)

733 **C** The evidence separates the mechanical from the biological. Bruxists show more screw loosening, more ceramic chipping and more component fracture, consistently enough that it belongs in the consent discussion and in the design, and a hard full-coverage splint worn at night is the standard means of removing the restorations from the parafunctional load — it must be planned and made, not offered after the first fracture. Statement (ii) is what makes B and E wrong, and it is the one candidates assume: reviews have not established a clear reduction in implant survival or osseointegration in bruxists, and the failures are of hardware rather than of the bone–implant interface. That distinction shapes the plan: strengthen the prosthesis and protect it, rather than declining treatment. (*Misch; ITI Treatment Guide vol 12*)

734 **E** The shock-absorber argument has been tested repeatedly and has not survived. The compliance of a few millimetres of acrylic is negligible beside the elastic behaviour of bone and the duration of a masticatory contact, and neither laboratory nor clinical work shows a meaningful reduction in the force or the strain reaching the implant when the occlusal material is changed. What does reduce load is occlusal design — narrow tables, contacts centred over the axis, no excursive contacts — and, in a parafunctional patient, a splint. Option D is the distractor with real substance and is a genuine practical advantage of acrylic, along with ease of repair, which is why acrylic is still widely and reasonably used; but it is a maintenance argument, not the biomechanical one being advanced. (*Misch; ITI Treatment Guide vol 12*)

735 **D** An osseointegrated implant behaves like an ankylosed tooth: it does not migrate, whereas the natural dentition continues to erupt and the alveolus continues to develop, slowly and for far longer than most clinicians expect — into the third decade in many patients and beyond it in some men. The implant crown is therefore left progressively in infraocclusion, with the soft tissue margin appearing to creep coronally because the neighbouring tissues have moved. It is the reason growth must be shown to have ceased before placing an implant in a young patient, and the reason this outcome should have been discussed at 20. Option A is the answer candidates give when they look for a process acting on the crown rather than on everything around it. (*Misch; ITI Treatment Guide vol 1*)

736 **B** Loss of the proximal contact next to an implant restoration is common — reported at around a third of restorations in longer cohorts — and it is consistently more frequent on the mesial side, which is why the LR5 contact in this case is the expected one to fail. Statement (iii) is what makes C and E wrong, and it inverts the mechanism that candidates half-remember: the implant is the fixed point. The

natural dentition continues to drift mesially and to erupt, moving away from an ankylosed restoration that cannot follow. The practical consequences are that the problem tends to recur after correction, that it must be looked for at every review rather than waited for, and that patients should be told to expect it. (*ITI Treatment Guide vol 12; Misch*)

737 **C** A bending moment is the product of the force and the perpendicular distance from its line of action to the point about which it acts, so with the force held constant the moment is directly proportional to the offset. Reducing the offset from 3 mm to 1 mm leaves one third of the original moment, a fall of two thirds. That arithmetic is the whole justification for narrowing occlusal tables and centring contacts over the implant axis, and it explains why the gain is so much greater than clinicians expect from what looks like a minor adjustment. Option E is chosen by candidates who reason that reducing the width of a table does not change how hard the patient bites, which is true and irrelevant; it is the lever arm, not the force, that has been altered. (*Misch*)

738 **A** Discrimination of a high contact depends chiefly on the mechanoreceptors of the periodontal ligament, which detect displacement in the order of tens of micrometres. An implant has none, and the residual sensation — osseoperception, mediated by receptors in bone, periosteum, muscle and joint — is markedly less acute, so the interocclusal thickness a patient can detect at an implant is several times that at a natural tooth. She will not feel a contact you can measure with foil. Option D is the interesting wrong answer and contains a real phenomenon, since the opposing tooth does intrude under load, but that intrusion is what *creates* the overload of the implant rather than masking it, and it does not explain her lack of awareness. Adjust on the foil, not on the report. (*Misch; Lindhe*)

739 **B** With an anteroposterior spread of 8 mm, the commonly used guide of not more than one and a half times the spread gives a ceiling of about 12 mm, and 15 mm exceeds it against a natural opposing dentition — a demanding combination. Ending at the first molar region, or accepting a premolar occlusion, is a legitimate and well-tolerated outcome that should be discussed before surgery rather than after. Option C is the trap and it is chosen by candidates who have learned that more implants are better: an additional *anterior* implant lies within the existing polygon and adds nothing to the spread, which is increased only by placing or tilting implants further posteriorly. Narrowing the table reduces off-axis load but not the lever length. (*Misch; ITI Treatment Guide vol 4*)

740 **A** A pontic between two supports is loaded in bending across a beam and transmits compressive force to both implants; a cantilever is a class I lever, which drives the near implant into compression and lifts the far one, placing its abutment screw in tension. Tension is what opens screw joints, and the far screw is the one that loosens and fractures, often at the implant that looks least involved. Given free choice of sites the cantilever has no advantage and should not be created. Option D is the sensible-sounding alternative and shortening a cantilever is always right when you are stuck with one, but the question offers a design in which no cantilever need exist at all. Rigid splinting distributes the moment; it does not abolish it. (*Misch*)

741 **C** The tall crown is a vertical lever, and a lever only produces a moment when something pushes sideways on it. The horizontal force in function comes overwhelmingly from excursive contacts, so removing them removes the input the 13 mm arm multiplies; intercuspal contacts centred over the axis generate very little. Option B is the good wrong answer and should also be done — narrowing the table shortens the horizontal offset of the axial contacts and, as the arithmetic of moments shows, the gain is proportionally large — but the offsets involved are two or three millimetres against a lever of thirteen. Splinting to a natural tooth introduces a different problem altogether, and the crown material and screw torque affect fracture and preload rather than the applied moment. (*Misch; ITI Treatment Guide vol 12*)

742 **D** Comparisons of splinted and unsplinted adjacent implant crowns have not shown a consistent difference in marginal bone level or in survival, so the routine claim in the stem is not supported. What splinting does is share load and resist bending, which is why it is chosen selectively — short or narrow implants, an unfavourable crown-to-implant ratio, parafunction, poor bone quality, or a cantilever — and it is bought at the price of a more difficult interproximal embrasure and a restoration that must be sectioned to retrieve one unit. In this case, two 8.5 mm implants in good bone in a non-bruxist, separate crowns are reasonable. Option C is the overstatement that catches candidates who have learned cleansability as an absolute; splinted units can be cleaned with the right embrasure design. (*ITI Treatment Guide vol 12; Misch*)

743 **E** The tripod or staggered-placement argument is intuitively appealing and has been examined in strain-gauge and finite element work without producing the benefit it promises; the reductions found are small, inconsistent and swamped by the effects of cantilever length, crown height and occlusal contact position. Meanwhile the cost is real and immediate: an implant displaced buccally in the arch forces the crown to emerge from an eccentric position, distorts the contour, creates a plaque-retentive profile on one side and a ridge-lap on the other, and may thin the buccal plate. Option C is the thoughtful distractor,

since splinting genuinely does distribute load, and candidates reason that the two ideas must combine; splinting works whether or not the implants are offset. Place implants where the teeth should be. (*Misch; ITI Treatment Guide vol 12*)

744 B Zirconia has a low thermal conductivity, so during cooling the surface of the veneering porcelain loses heat while the bulk adjacent to the framework remains hot; the layers then contract at different times and residual tensile stress is locked into the veneer, where it later releases as cohesive chipping under load. Slow, controlled cooling after the final firing is the specific remedy, and it is why manufacturers publish cooling protocols for zirconia that differ from those for metal-ceramic. Option D is the answer candidates give because expansion mismatch is the classic explanation for porcelain failure on metal, and it is a genuine mechanism — but it is designed out by matching the veneer to the framework system, whereas cooling rate is a variable in the technician's hands on every case. (*ITI Treatment Guide vol 12*)

745 C Translucency in zirconia is bought by increasing the yttria content, which stabilises more of the material in the optically isotropic cubic phase. Cubic grains do not undergo the tetragonal-to-monoclinic transformation that gives 3 mol% zirconia its transformation toughening, so flexural strength and fracture toughness fall substantially as translucency rises. In a bruxist's second molar, where nobody will see the crown, the trade is plainly the wrong way round. Option E is the sophisticated distractor and is chosen by candidates who have read about ageing in zirconia; it is in fact reversed, since low-temperature degradation is a phenomenon of the tetragonal phase and the more cubic materials are less susceptible. Aesthetics do not arise here, and strength does. (*ITI Treatment Guide vol 12*)

746 A The bond between zirconia and a titanium base is an adhesive joint, and its strength scales with the area over which the two are in contact and with the resistance form that height provides against the tipping moment applied by every off-axis contact. A base of 2.5 mm gingival height offers little of either, and taller bases, or a base with an anti-rotational and retentive geometry, are the recognised answer where a debond has already occurred. Options B and C are the attractive distractors precisely because they are correct procedure — alumina abrasion of the titanium and an MDP-containing resin are exactly what should have been done — and a candidate who is looking for the technician's error will pick one of them rather than the dimension in the stem. (*ITI Treatment Guide vol 11*)

747 D The three findings together exclude almost everything else. A tight screw that reaches full torque excludes loosening and, since a fractured screw cannot be torqued at all, excludes fracture; a closed interface on a properly angled periapical excludes an unseated abutment; and an unchanged bone level with a mobile crown excludes loss of integration, which in any case produces mobility of the whole assembly with a continuous radiolucency. What remains is the adhesive joint between the zirconia and the titanium base, which is the only interface in the assembly that a radiograph cannot show. Option A is the answer most candidates give because a mobile screw-retained crown means a screw problem in the majority of cases, but the stem has already tested and excluded it. (*ITI Treatment Guide vol 11; Misch*)

748 E All three are recognised, and together they account for most of the maintenance burden of a metal-acrylic reconstruction. Denture teeth fracture, particularly on cantilevers and in patients with parafunction; acrylic occlusal surfaces wear measurably over years, altering the occlusal scheme and the vertical dimension; and the acrylic-to-metal junction is a mechanical bond that can fail, allowing a segment to lift. None of these is implant failure, and that distinction is the substance of the conversation: the implants may survive faultlessly while the prosthesis needs periodic repair, and a patient consented only on implant survival figures will feel misled. Candidates most often omit item C, having thought about the teeth and the wear but not about the interface holding them. (*ITI Treatment Guide vol 4; Pjetursson et al.*)

749 B Immediate full-arch provisionals fracture at the point of least cross-section, characteristically just distal to the terminal cylinder where the cantilever begins, and the failure is one of flexural strength across a section of acrylic. Incorporating a rigid metal bar joining the cylinders converts the provisional from a flexing acrylic beam into a reinforced one and is the established answer where the arch is long, the cantilevers are significant or the patient is a bruxist. Option C is the strong distractor and milled PMMA is genuinely denser, better polymerised and stronger than a chairside mix, so it should be used where available — but it improves the material rather than the structure, and the structural problem is a beam under bending. Shortening the cantilevers is also worth doing. (*ITI Treatment Guide vol 4; Misch*)

750 C Torque elongates the screw elastically, and that elongation clamps the abutment against the implant with a force — the preload — which is what holds the joint closed. Any external load with a tensile component at the interface, produced by an off-axis or eccentric contact, works against that clamping force; while it stays below the residual preload the joint remains closed and the screw sees no cyclic loading, and once it exceeds it the joint opens and closes with every cycle, which is how screws loosen and eventually fatigue. Option E is the good wrong answer and describes the right variable in the wrong degree: settling

removes some preload, typically a few per cent, which is why re-torquing is worthwhile, but it does not remove all of it. (*Misch; ITI Treatment Guide vol 12*)

751 D The prosthetic screw is of smaller diameter and lower load-bearing capacity than the abutment screw, and the hierarchy is deliberate: if something in the assembly is going to give under an unforeseen load, the designers want it to be the small screw at the top, which is retrieved and replaced in ten minutes, rather than the abutment screw seated within the implant, whose fracture may leave a fragment in the internal threads and the implant unrestorable. Torque values follow the size and the intended role of each component. Option C is the attractive answer because it contains a real idea — the multi-unit abutment does carry load away from the abutment screw — but that is a reason the abutment screw survives, not a reason its specified torque is higher. (*Misch; ITI Treatment Guide vol 4*)

752 D A beam wrench works because a known force applied at a known distance deflects a calibrated beam, and the scale is read against that deflection. A finger resting anywhere along the beam changes the effective lever arm and shares the applied load, so the beam deflects less than the true torque would produce and the operator keeps turning: the delivered torque is higher than the reading, sometimes considerably. The wrench must be held only at the designated grip. Option A is the answer many candidates choose, and applying load smoothly rather than jerkily is indeed correct practice, since a sudden application can overshoot on a spring-style wrench; but the stem describes it being done correctly. Sterilisation cycles justify periodic recalibration, which the stem has covered. (*Misch; manufacturer's instructions*)

753 E The elongation produced by torquing a screw is a fixed quantity for a given torque and joint. In a passive joint all of it is available as clamping force between two surfaces already in contact. Where the framework must first be drawn 60 micrometres down onto the abutment, part of that elongation is consumed in deforming the framework and pulling it into place, and only the remainder is left as preload — so the joint that most needed clamping force has the least. That is the mechanical reason misfit produces screw loosening and screw fracture, and why closing a gap by tightening is a concealment rather than a correction. Option C reaches the right answer by the wrong route: the surfaces do end up in contact, which is why the radiograph looks reassuring. (*Misch; ITI Treatment Guide vol 8*)

754 E All three are present here and all three are established. A screw that loses preload is subjected to cyclic tensile loading at every masticatory stroke instead of resting under static clamp, which is the classic route to fatigue failure; retightening without finding the cause simply resets the clock. A cantilever converts occlusal force into a bending moment that places the screw in tension. And a screw that has been torqued repeatedly has worn thread flanks, an abraded surface coating and accumulated microdeformation, so it achieves an unpredictable preload and has a shortened fatigue life. Candidates most often omit item C, since a screw that looks undamaged appears serviceable. A new screw at each definitive fit costs almost nothing. (*Misch; ITI Treatment Guide vol 12*)

755 C Ultrasonic counter-rotation succeeds in many cases and is rightly the first attempt, but when it fails the next step is the purpose-made kit: a guide that centres the instrument in the implant, and a reverse-cutting screw or tap that engages the fragment and backs it out along the thread. The governing principle throughout is that the internal threads are irreplaceable — damage them and a perfectly integrated implant becomes unrestorable. Option A is the plausible wrong answer because slotting is precisely how a fractured post is retrieved from a root canal; transferred here it means running a bur in a channel two millimetres wide, without direct vision, where titanium threads cut as easily as the fragment. Explantation, in D and E, abandons a sound implant far too early. (*ITI Treatment Guide vol 12; Misch*)

756 B A restoration made for one named patient to a clinician's prescription is a custom-made device, and the manufacturer must supply a written statement identifying the device and the materials used, naming the prescribing clinician, carrying data by which the patient can be identified, and declaring conformity with the relevant requirements. That statement is retained with the clinical record and the patient is entitled to a copy, which is what she is asking for. Item A is what makes D wrong and it is the commonest misconception in this area: custom-made devices are specifically exempt from CE or UKCA marking, which applies to mass-produced devices such as the implant and the stock components. The technician's registration is a matter for the laboratory's governance. (*Medical Devices Regulations; GDC Standards*)

757 B You are responsible for what you place in a patient, for telling her what it is, and for being able to identify it years later if a component fails or a field safety notice is issued. A substitution made without your knowledge defeats all three: the record is wrong, the consent was given on a different basis, and traceability is broken. The remedy is a prescription that specifies components explicitly and a laboratory that confirms what it used. Option C is the strongest alternative and names a real hazard, since interface tolerances and torque specifications are validated for the original component and a third-party copy may achieve a different preload — but it is one consequence of the substitution, whereas not knowing is the problem that generates all of them. (*GDC Standards; Medical Devices Regulations*)

758 **A** Restorative space governs this decision before anything else. A stud attachment needs of the order of 8 to 9 mm from the platform to the occlusal plane to accommodate the abutment, the matrix housing, the acrylic around it and the denture tooth; a bar needs appreciably more, commonly 12 mm or beyond once the bar, the clip and the acrylic thickness above it are added, and forcing one into 9 mm produces a base that fractures. Two separate studs also suit her dexterity, since each abutment is wiped clean individually. Option C is the answer candidates give when they equate a bar with better retention; retention is not the limiting factor here, and a bar in this space is a base fracture waiting to happen. E is not a defensible design on two implants. (*ITI Treatment Guide vol 4; McGill consensus*)

759 **E** All three are ordinary events rather than failures, and saying so before treatment is what prevents the patient later believing something has gone wrong. The nylon or plastic insert is a deliberate sacrificial component that wears with insertion and removal and is exchanged chairside, often within the first year or two. The prosthesis remains implant-retained and tissue-supported, so the posterior ridges continue to resorb beneath it and relining is periodically required. And the base is at its thinnest and most highly stressed directly over the housings, where fracture is a recognised event that may prompt a metal-reinforced remake. Candidates most often omit item B, forgetting that only the anterior segment is supported by the implants. (*ITI Treatment Guide vol 4; McGill consensus*)

760 **C** Stud attachments accommodate only a limited deviation between the implant axis and the path of insertion — of the order of 20 degrees each in most systems — and beyond that the matrix engages eccentrically, wears rapidly on one side and loses retention within months. A bar solves the problem by taking the divergence into the casting or milling and presenting the denture with one common path, which is its classic indication and the reason 14 mm of restorative space is worth having here. Option D is the answer most candidates give, and a bar can be made more retentive — but retention is adjustable in both systems and is not what has failed. Maintenance is not reduced by a bar; a bar simply changes the kind of maintenance required. (*ITI Treatment Guide vol 4*)

Answers — Block X — Complications, peri-implant disease, maintenance and governance, extended (Q761–800)

761 **C** Peri-implant health is defined by the absence of clinical signs of inflammation — erythema, swelling, bleeding on gentle probing and suppuration — and it is expressly stated that health may exist at an implant with reduced bone support. There is no probing depth above which health cannot be diagnosed, because peri-implant probing depth reflects the depth at which the implant was placed and the thickness of the mucosa, which is exactly why this stem supplies both. Option A is the answer most candidates choose, since an increasing probing depth is a genuine warning and is part of the peri-implantitis definition; but it must be accompanied by inflammation, and an increase alone, in the absence of bleeding, does not exclude health. Plaque scores and keratinised mucosa are risk considerations, not diagnostic criteria. (*2017 World Workshop; Berglundh et al.*)

762 **D** Bleeding and suppuration are required for the diagnosis, but they are equally present in mucositis, so neither can be the discriminating finding; what converts mucositis into peri-implantitis is radiographic bone loss beyond the crestal change of initial remodelling, and 2.5 mm measured against a film taken at fitting comfortably exceeds it. This is the argument for taking that baseline film at every fit appointment: without it the diagnosis rests on the far cruder thresholds used when no previous data exist. Option A is the distractor that catches candidates who have learned the case definitions as a list of numbers: the 6 mm figure belongs only to the no-baseline situation, and probing depth around an implant is largely a function of placement depth. (*2017 World Workshop; Berglundh et al.*)

763 **B** Initial remodelling — the crestal change that follows placement, abutment connection and the first months of loading — is essentially complete within the first year, which is why a radiograph taken at or around one year is an accepted surrogate baseline and is in some respects more useful than the fit-time film, since it already excludes physiological remodelling from the comparison. Three millimetres of further loss against that reference, with bleeding and suppuration, is peri-implantitis. Option A is the attractive answer because the phrase "no baseline" triggers the conservative thresholds; but a baseline of a sort does exist here, and applying the no-data rule discards better information. The surgical note and a pre-operative scan describe bone before the implant, not bone level around it. (*2017 World Workshop; Berglundh et al.*)

764 **D** The case definition requires "bleeding **and/or** suppuration on gentle probing", so statement (i) is wrong as written: a site that suppurates but does not bleed satisfies the inflammatory criterion, and suppuration alone is if anything the more sinister of the two signs. Increasing probing depth against previous records is explicitly part of the definition where baseline data exist, so (iii) stands. Candidates who select E or C have learned the definition as a list of features to be collected rather than as a logical structure, and

the error matters clinically: insisting on bleeding before acting will delay treatment at exactly the sites that need it most. Suppuration should be sought deliberately by running a finger along the mucosa towards the margin. (*2017 World Workshop; Berglundh et al.*)

765 A The retrospective cohorts that have plotted bone loss over time around affected implants describe a pattern quite unlike periodontitis: an early onset, then an accelerating, often exponential course, so that a lesion producing 1 mm of loss in its early years may produce several millimetres in a comparable later period. That is the argument that defeats the annual review — the interval that seems adequate at the start of the process is far too long once it is under way, and the disease is detected clinically, by bleeding and increasing probing depth, before it is detected radiographically. Option E is the evasion that sounds properly cautious; the cohorts are imperfect but consistent enough to change practice, and citing uncertainty to justify longer intervals inverts the precautionary argument. (*Fransson; Derks et al.; 2017 World Workshop*)

766 C Before the 2017 workshop there was no agreed case definition, and studies set their own thresholds — 1 mm of bone loss in one, 3 mm in another, some requiring bleeding and some not, some counting any bone loss at all. Prevalence is exquisitely sensitive to where that line is drawn, and the enormous range reported is largely an artefact of definition rather than a description of biological variation. It is also why the range is worth quoting to a patient alongside the weighted means of roughly 43 per cent for mucositis and 22 per cent for peri-implantitis: the point estimates are useful, and the spread tells her how much confidence to place in them. Option D is a genuine and frequently overlooked source of variation, and it is the best of the alternatives, but it accounts for a fraction of a range this wide. (*Derks & Tomasi; 2017 World Workshop*)

767 D Where serial radiographs have allowed the onset of bone loss to be dated, it clusters early — commonly within the first two to three years of function, well before the point at which most maintenance programmes intensify. Combined with the accelerating course the disease then follows, that finding rewrites the surveillance argument: the years immediately after loading are when the disease starts and when it is cheapest to intercept, not a settled period to be got through before vigilance begins. Option A is the intuitive answer and is how most clinicians think about implant complications, by analogy with the wear and fatigue of the prosthesis, which genuinely do accumulate with service. Biological failure and technical failure have different time courses. (*Derks et al.; 2017 World Workshop*)

768 A Only the periodontal history qualifies. A history of periodontitis, together with poor plaque control and failure to attend supportive care, is the group of factors reproduced consistently across cohorts, and the association strengthens where treatment has left residual bleeding pockets. Item B is what makes D wrong, and it is the one candidates accept because it appears in every list: absence of keratinised mucosa is associated with plaque accumulation, brushing discomfort, bleeding and recession, but its relationship with progressive bone loss is inconsistent, and the consensus regards the evidence as insufficient. Item C is contested and has never been shown to cause peri-implantitis; overload damages screws, ceramic and components. Her worn dentition is a mechanical problem, not a biological one. (*2017 World Workshop; ITI Treatment Guide vol 13*)

769 B The consensus separated factors with strong and consistent evidence — a history of periodontitis, poor plaque control, and lack of regular supportive care — from those where the data conflict, and placed both smoking and diabetes in the second group. That is a narrower claim than most clinicians make, and it needs stating precisely: it does not mean smoking is harmless, since its effect on early failure, on wound healing and on augmented sites is not in doubt, only that its specific association with peri-implantitis is inconsistent across studies. Option C is the answer most candidates give, because smoking is the risk factor they have rehearsed in every other context. Counsel both patients energetically, but describe the evidence honestly. (*2017 World Workshop; ITI Treatment Guide vol 13*)

770 E In the endoscopic series most often cited, excess cement was found at around four fifths of implants showing clinical or radiographic signs of peri-implant disease, and in the majority of those the signs resolved after the cement was removed. Two conclusions follow, and both matter: cement is a very common cause rather than a fashionable one, and it is a *reversible* cause if found early, which is why the search for it comes before any decision about surgery. Option C is the assumption that keeps cement undiagnosed, and it is wrong in two ways — many cements are barely radiopaque, and a two-dimensional image cannot show buccal or palatal excess superimposed on the implant. A clean radiograph excludes nothing. (*Wilson; Linkevicius; ITI Treatment Guide vol 13*)

771 D Everything in this history points at cement — a submucosal margin at a depth from which complete removal is improbable, localised suppuration, good plaque control and a contour that does not explain the lesion — and a radiograph that shows nothing does not exclude it, because many cements are insufficiently radiopaque and excess on the buccal or palatal aspect is projected over the implant. Cause-related therapy

here means taking the restoration off and looking. Option A is the answer that feels methodical and is what most clinicians do first; debriding around a retained foreign body treats the consequence and leaves the cause in place, and a month later the site will be no better and a month will have been lost. A scan adds dose and metal artefact without answering the question. (*Wilson; ITI Treatment Guide vol 13*)

772 **B** Only the prosthetic correction reliably changes the outcome, and it is the step most often omitted because it costs a remake. A contour that the patient cannot clean guarantees recolonisation whatever is done to the biofilm today, and no amount of instrumentation or chemotherapy compensates for it. Items A and C are what make D and E wrong, and they are chosen because both are widely used: adjunctive systemic and local antimicrobials produce small, short-lived improvements in bleeding and probing depth over debridement alone in established peri-implantitis, do not achieve resolution of deep lesions, and carry the usual costs of antibiotic use. Correct the prosthesis, debride, teach, then reassess and decide about surgical access. (*2017 World Workshop; ITI Treatment Guide vol 13*)

773 **C** Series following surgically treated peri-implantitis for five years describe a successful outcome — absence of bleeding and suppuration, no further bone loss, probing depths controlled — in something close to half of implants, with recurrence and further treatment common in the rest and a minority lost altogether. That is a worthwhile intervention and an honest one to describe, but it is not a cure, and the patient's expectations must be set accordingly, together with the recession and exposed threads that surgery frequently produces. Option A is what patients assume and what clinicians imply by silence; consenting a man on those terms and reoperating at year three is precisely the sequence that generates complaints. The surgical outcome depends heavily on what happens at maintenance afterwards. (*Carcuac & Berglundh; ITI Treatment Guide vol 13*)

774 **A** Implantoplasty converts a threaded, microrough surface that no brush or instrument can follow into a smooth, polished cylinder that the patient and the hygienist can actually clean; it does not regenerate bone and it does not reattach anything. The price is inherent in the technique: the implant body is left exposed, so recession and visible metal follow, which is why the aesthetic assessment in the stem is decisive. Option D is the distractor with real substance and deserves credit — removing material does reduce fracture resistance, and the concern is genuine in narrow implants — but it is a caution to weigh rather than the definition of the procedure, and the stem's implant is 4.8 mm. Grafting a dehiscence with no walls is unpredictable. (*ITI Treatment Guide vol 13; 2017 World Workshop*)

775 **D** Defect morphology decides the surgical technique in the same way it does in periodontal surgery. A buccal dehiscence has no containing walls, so there is nothing to hold a graft, stabilise a clot or protect a membrane, and reconstructive attempts in this configuration are unpredictable; the exposed threads are instead smoothed and polished and the flap positioned apically, leaving a cleansable surface in a site where recession is acceptable. Option A is the strong distractor because grafting feels like the more ambitious and more curative option, and it would be correct for a contained circumferential crater — the exact opposite morphology. Explantation is premature at a stable implant with intact bone on three aspects, and access surgery alone leaves the threads uncleanable. (*ITI Treatment Guide vol 13; 2017 World Workshop*)

776 **B** Reconstructive therapy in a contained defect produces measurable radiographic fill and greater probing depth reduction than access surgery alone, and it is the right choice for this morphology. What it does not reliably produce is a new bone-to-implant contact on a surface that has been contaminated by biofilm: histological work in animals shows re-osseointegration is possible but partial and unpredictable, and no clinical decontamination protocol has been shown to achieve it consistently. The honest account is therefore of a defect filled and an implant made maintainable, not of an implant restored to its original state. Option A is what patients hear when the word "regeneration" is used, and it is the reason the term should be handled carefully in the consent discussion. (*2017 World Workshop; ITI Treatment Guide vol 13*)

777 **E** All three are necessary and the operation is the part most likely to be delivered in isolation. Decontamination is what the surgery is for, since the graft is placed against a surface carrying biofilm. A prosthesis that cannot be cleaned will recontaminate whatever has been achieved, and modifying or remaking it is not an optional refinement but a condition of success. And recurrence after surgical treatment is common and is predicted above all by non-attendance and poor plaque control, so a short-interval programme is part of the treatment rather than a separate offer. Candidates who choose D have treated maintenance as aftercare; this patient's four-year absence is a large part of why she is having surgery at all. (*ITI Treatment Guide vol 13; Monje et al.*)

778 **C** The test for keeping an implant is whether any available treatment can leave it cleansable, restorable and aesthetically acceptable. Bone loss, suppuration and a deficient soft tissue phenotype are all potentially modifiable by treatment; a malposition is not. An implant 3 mm labial to where the root should be will emerge through the labial surface of the restoration whatever is done to the defect, cannot be given a cleansable contour, and will not support a margin at the level of its neighbours in a patient who shows

them. Option A is the answer candidates give because the number is the largest in the stem, but extensive bone loss around a well-positioned implant may still be treatable. Removal and reconstruction here is a complex anterior case and the referral discussion should say so. (*ITI Treatment Guide vol 13; SAC Assessment Tool*)

779 A A counter-torque device works by applying reverse rotation until the shear strength of the remaining bone-to-implant interface is exceeded, and the implant then unscrews along its own thread path, taking almost no bone with it. Where much of the surface has already been lost to disease the required torque is modest and the technique is elegant; where a long apical portion remains densely integrated, particularly in the mandible, the torque needed may exceed what the connection can transmit, and the implant or the instrument fractures instead. Option C is the good wrong answer, since the connection does set a ceiling on transmissible torque and a worn internal hexagon may strip — but that is a constraint on the instrument, whereas the resistance to be overcome is the variable being tested. Have a trephine available. (*ITI Treatment Guide vol 13*)

780 E All three, and the visit is worth little if any is omitted. Probing with gentle force and a full record of bleeding is the only way mucositis — the reversible stage — is detected, and a chart that can be compared with the last one is what identifies change. The prosthesis is examined for screw loosening, for open contacts, for chipping and above all for whether cleaning aids can still reach the sites, since the design deteriorates as tissues recede and contacts open. And reinforcement is the intervention with the largest effect on recurrence, which is why it is repeated at every visit rather than delivered once. A radiograph is taken when these findings indicate it, not by the calendar. (*ITI Treatment Guide vol 13; EFP S3 guideline*)

781 B Low-abrasive powders of glycine or erythritol, delivered subgingivally through a fine nozzle, disrupt biofilm within threads and surface irregularities that no curette can follow, and they do so without measurably roughening titanium — which is the whole point, since any increase in surface roughness produces a more plaque-retentive site than the one you started with. Option C is the answer that appeals to anyone who has struggled to remove a deposit, and alumina certainly does remove it; it also cuts titanium, and it is a laboratory abrasive rather than a clinical one. Sodium bicarbonate is likewise too abrasive for these surfaces despite its familiarity in supragingival use. Titanium or polymer curettes and ultrasonic tips with appropriate sleeves complete the armamentarium. (*ITI Treatment Guide vol 13; Lindhe*)

782 D The therapeutic endpoint in mucositis is the absence of bleeding on probing, not an improvement in it, because residual bleeding marks residual inflammation at exactly the sites that go on to develop peri-implantitis. Resolution around implants is slower and less complete than resolution of gingivitis at teeth, so a single episode of instrumentation and instruction frequently does not achieve it, and the correct response is to repeat both and review again. Option A is the comfortable answer and it half-quotes a true finding — inflammation does settle more slowly around implants — turning an observation about time course into permission to stop. Chlorhexidine adds little to genuinely effective mechanical disruption; surgery has no place in mucositis; a radiograph is not indicated by these findings. (*Salvi et al.; ITI Treatment Guide vol 13*)

783 C The two radiographs exclude the mechanism candidates reach for first: with 4 mm of bone between the apex and the canal roof, neither the drill nor the implant has entered it, and options A and B are foreclosed by data in the stem. What remains, in a ninety-minute procedure with a flap raised over the foramen and a retractor held against it, is a traction or compression injury of the mental nerve — a neurapraxia, in which recovery is usual over weeks to a few months. Option D is the strongest alternative and is a real cause of altered sensation, but it typically follows a block with immediate electric pain on injection. She should be seen, examined, mapped and reviewed, and told today what has happened. (*Misch; ITI Treatment Guide vol 11*)

784 A The professional duty of candour requires that a patient is told promptly when something goes wrong with their care, offered an apology, given an explanation of what happened and what will be done, and that the conversation is recorded — and it applies whether or not anyone was at fault. Option E is the belief that causes most of the harm in these situations and it is legally wrong: an apology is an expression of regret, not an admission of liability, and legislation makes that explicit. Option C is the sophisticated error, because informing your indemnifier promptly is entirely correct and you should do it; what you must not do is postpone the conversation with the patient while you take advice, which reads as evasion and hardens a recoverable situation into a complaint. (*GDC Standards; Compensation Act 2006*)

785 D Serial mapping is what demonstrates whether an injury is recovering or static, and the description of unpleasant sensation on touch indicates a dysaesthetic element that is unlikely to settle without help. An injury unchanged at four weeks, and particularly one with neuropathic features, should be referred without waiting further, because the interventions that exist are more effective early and recovery after several months is unusual. Item C is what makes E wrong and it is chosen by candidates who feel

that more imaging is more diligence: the anatomical relationship has already been established twice, a repeat examination cannot change the management, and under IR(ME)R an exposure that cannot alter the outcome cannot be justified. (*IR(ME)R; Misch*)

786 **B** A foreign body in the maxillary sinus obstructs mucociliary clearance and is a recognised cause of chronic sinusitis and of fungal colonisation, so it is retrieved rather than observed, and retrieval is usually by a lateral antrostomy or endoscopically — a procedure beyond a general dentist with a special interest, undertaken promptly and with antibiotic cover while it is arranged. Option C is the reflex of the operator standing at the chair and it is the manoeuvre that turns a retrievable problem into a difficult one: suction through a 4 mm crestal window has a low chance of engaging the implant, may push it further, and risks tearing the membrane more widely. Tell the patient what has happened and why she is being referred. (*ITI Treatment Guide vol 5*)

787 **E** Purulent unilateral discharge with facial pain and fever at six days is acute bacterial sinusitis complicating the graft, and it is treated actively: an antibiotic with appropriate cover, a decongestant to restore ostial drainage, analgesia, and specialist assessment because a graft that becomes infected may need removal and because the sinus itself needs evaluation. Option C is the strong distractor, and removal of the graft is sometimes exactly what happens — but that is a decision taken by the team assessing him, after imaging and a trial of medical management, not an immediate response at the chair, and a stable implant may well be retained. Reassurance and a decongestant alone under-treat an established infection in a closed cavity. (*ITI Treatment Guide vol 5*)

788 **B** The first two are the figures worth carrying into a consent discussion: abutment or screw loosening heads the list for single crowns, at something of the order of one in ten by five years, and chipping of the veneering layer is the commonest technical event for fixed prostheses more generally. Statement (iii) is what makes C, D and E wrong, and it inflates a rare event by an order of magnitude: fracture of an abutment or of a screw is reported at well under one per cent over five years, which is why it appears in the literature as a notable complication rather than a routine one. Getting the magnitudes right matters, because a patient warned that a screw may loosen is prepared, and one told a component will probably break is not. (*Jung et al.; Pjetursson et al.; ITI Treatment Guide vol 12*)

789 **A** The mandibular two-implant overdenture has excellent implant survival and transforms function, and it is also the implant prosthesis with the highest ongoing prosthetic maintenance: the majority of patients need attention to the attachments within the first five years, most commonly exchange of a worn retentive insert, along with relining as the posterior ridges continue to resorb and occasional repair of the base. None of this is failure, and the McGill position that this is the appropriate standard of care stands. Option D is the answer that seems intuitive from mechanical wear generally and is the one that produces a disappointed patient at eighteen months, when the retention has gone and she believes something is wrong with her implants. (*McGill consensus; ITI Treatment Guide vol 4*)

790 **D** A screw that fractures twice is reporting a mechanical problem that has not been addressed, and the stem names two candidates: a 6 mm distal cantilever converting occlusal force into a bending moment, and an excursive contact supplying horizontal load. Restoring again without shortening the cantilever, eliminating the excursion and reviewing the crown height and occlusal table simply schedules a third fracture, with the added risk that the next fragment will not come out and a sound implant will be lost. Option A is the answer he is asking for and is the one that gets given under time pressure; it is a repair rather than a treatment. Cement retention would make the next fracture harder to manage, and explantation discards an integrated implant for a prosthetic fault. (*Misch; ITI Treatment Guide vol 12*)

791 **E** All three, and the third is the one most often left undone. Traceability exists precisely so that a notice of this kind converts into a database search rather than a guess, and the search is only useful if the patients it identifies are actually contacted, told plainly why, and seen. Recording the notice, the search, the patients identified and what was done for each is what allows the practice to demonstrate that it responded, and it is the documentary counterpart of the regulatory duty; without it the work has been done and cannot be shown. Candidates who choose D have thought clinically and not organisationally, which is the commonest gap in governance questions. Any adverse incident found during those reviews is separately reportable. (*Medical Devices Regulations; GDC Standards*)

792 **B** An implant is a medical device, and the failure of a device in a way that has caused or could cause harm is an adverse incident reportable to the regulator, in the United Kingdom through the MHRA's reporting system, as well as to the manufacturer, who is required to investigate. The fractured components are retained rather than discarded, because they are the evidence, and the patient is told what has happened and what is being done. Option E is a good practice that should also happen and is the answer many candidates give, since a practice audit feels like the natural home for a failure; but an internal record

informs only your own practice, whereas reporting is what allows a pattern across many practices to be detected. (*Medical Devices Regulations; MHRA*)

793 C Beam hardening and streak artefact from a titanium implant degrade the image in precisely the few millimetres you want to see, so a cone beam examination frequently cannot resolve the buccal plate adjacent to the implant and may report a defect that is not there or miss one that is. Since the defect morphology that determines the surgical technique is in any case read directly at flap elevation, the additional dose usually buys nothing that changes management, and under IR(ME)R that is the test. Option B is what candidates assume from every other application of the technology, and it is true of bone away from metal; next to an implant the artefact is the dominant limitation. A periapical with a paralleling technique remains first line. (*IR(ME)R; FGDP/RCR selection criteria*)

794 A The film taken at prosthesis fitting does a job no other film does: it records the marginal bone level at the moment loading begins, against which every future radiograph is compared, and without it the diagnosis of peri-implantitis later rests on the crude thresholds used when no baseline exists. It must be taken with a paralleling technique, with the beam perpendicular to the implant axis, so that the threads are resolved and the comparison is valid, and it should be filed as the baseline rather than as one film among many. Option C is the answer that sounds like good radiation practice and it is why baseline films are so often missing; justification here rests on the future diagnostic value, which is real and specific. (*IR(ME)R; 2017 World Workshop; FGDP*)

795 D The record must let another clinician reconstruct both the diagnosis and the reasoning. The full chart with bleeding and suppuration is the clinical evidence, and the explicit comparison against the baseline is what converts inflammation into a diagnosis of peri-implantitis rather than mucositis; both must appear, together with the risk factors identified, the prosthetic assessment, the options discussed with their adverse effects, and the plan agreed. Item C is what makes E wrong, and it is attractive because insertion torque is diligently recorded at surgery and feels like part of the implant's history: it is a surgical figure with no diagnostic value years later, and in a transferred patient it will not exist at all — which cannot be allowed to prevent a diagnosis. (*GDC Standards; 2017 World Workshop*)

796 B A clinical record is a contemporaneous account and its evidential value depends on not being rewritten afterwards. Where a patient disputes an entry, the entry stands and her account is added as a dated, signed addendum, so that the record shows both what was written at the time and what she says about it. Option C is the sophisticated error and it is chosen by candidates who know that inaccurate data should be corrected: amending the original obscures what was actually recorded, and the right of rectification is satisfied by the addendum where the underlying fact is disputed rather than demonstrably wrong. Deleting the entry, with or without a hidden copy, would be worse than either and would be indefensible if the record were later examined. (*UK GDPR; GDC Standards*)

797 E Every practice must have a written complaints procedure that patients can see, with defined timescales for acknowledgement and response, and the correct handling is to use it: acknowledge quickly, offer to see her and listen, investigate, then explain your findings and set out the options, which may include an independent opinion, an adjustment or a remake. Option A is the sophisticated wrong answer, and taking indemnity advice is entirely right — you have already done it — but waiting for approval before acknowledging a complaint introduces exactly the delay that converts a disappointed patient into a determined one. C concedes before understanding, D is indefensible, and B invents a regulatory timescale that does not exist in that form. (*GDC Standards*)

798 C The professional duty of candour rests on every registered dental professional individually, and it is not gated by a harm threshold: if something goes wrong with a patient's care, they are told promptly, given an apology, given an explanation of what has happened and what will be done, and the conversation is recorded. Option E is the belief in the stem and it confuses two things that coexist — a *statutory* duty does sit on registered providers and is triggered by notifiable safety incidents at defined levels of harm, but it sits alongside the professional duty rather than replacing it. B is wrong in law, since an apology is not an admission of liability, and D confuses recording the event with telling the patient, which is the part that matters to her. (*GDC Standards; Duty of candour guidance*)

799 A Significant event analysis is a structured, non-punitive team review of a single event: what happened, why it happened, what went well, and what will change — here, most obviously, a written protocol for retraction and appointment length near the mental foramen, and a plan for periodic breaks in long procedures. Item B is what makes D wrong, and it is the instinct that destroys the process: framing the review as a judgement on an individual guarantees that the next event is not reported, which is precisely how systems fail to learn. Item C misunderstands the regulator's role, since the Council is not a repository for adverse events; referral concerns fitness to practise, which is not in question in a recognised complication managed openly. (*Clinical governance; GDC Standards*)

800 **D** Continuity of care for an implant patient depends on information a new clinician cannot generate for herself: the system, platform and connection, the abutment and screw references with their lot numbers and specified torque values, the retention mode and how to retrieve the prosthesis, the occlusal scheme, the maintenance interval, the history of mucositis, and — most importantly — the baseline and serial radiographs against which future bone levels will be judged. Option A is the answer that captures the single most valuable item and is therefore the most attractive, but a radiograph without the component data leaves the next clinician unable to order a screw, and a docket is not a clinical summary. She is also entitled to all of this under a subject access request. (*GDC Standards; ITI Treatment Guide vol 13*)

Answer key grid

Mark a block quickly against this grid, then read the explanations above for every question in it.

#	Ans	#	Ans	#	Ans	#	Ans	#	Ans	#	Ans	#	Ans	#	Ans
1	A	101	B	201	C	301	E	401	C	501	A	601	C	701	A
2	B	102	C	202	D	302	C	402	D	502	B	602	A	702	B
3	A	103	E	203	A	303	D	403	E	503	B	603	B	703	D
4	D	104	B	204	B	304	B	404	A	504	C	604	D	704	C
5	B	105	A	205	E	305	A	405	D	505	A	605	A	705	D
6	A	106	E	206	C	306	C	406	B	506	B	606	E	706	B
7	A	107	A	207	D	307	A	407	A	507	D	607	B	707	C
8	A	108	B	208	A	308	B	408	B	508	B	608	C	708	E
9	D	109	C	209	B	309	D	409	B	509	E	609	A	709	D
10	C	110	D	210	C	310	A	410	C	510	E	610	D	710	A
11	C	111	E	211	D	311	C	411	B	511	C	611	C	711	B
12	D	112	C	212	A	312	B	412	C	512	A	612	E	712	A
13	E	113	A	213	A	313	E	413	A	513	A	613	B	713	C
14	C	114	D	214	B	314	E	414	C	514	D	614	E	714	D
15	B	115	B	215	C	315	B	415	D	515	D	615	D	715	A
16	D	116	E	216	E	316	A	416	B	516	C	616	C	716	B
17	C	117	A	217	E	317	D	417	E	517	A	617	A	717	B
18	B	118	C	218	D	318	C	418	D	518	B	618	E	718	C
19	A	119	D	219	C	319	D	419	A	519	D	619	B	719	A
20	C	120	B	220	E	320	B	420	C	520	D	620	C	720	A
21	A	121	A	221	A	321	A	421	B	521	E	621	A	721	B
22	B	122	B	222	B	322	C	422	E	522	C	622	D	722	C
23	D	123	E	223	D	323	E	423	A	523	E	623	B	723	E
24	E	124	A	224	A	324	D	424	D	524	A	624	C	724	D
25	C	125	C	225	B	325	B	425	A	525	B	625	A	725	E
26	E	126	D	226	C	326	A	426	C	526	C	626	B	726	D
27	B	127	B	227	B	327	E	427	A	527	A	627	E	727	D
28	E	128	E	228	A	328	B	428	E	528	B	628	C	728	B
29	B	129	A	229	E	329	C	429	B	529	D	629	E	729	C
30	D	130	E	230	C	330	E	430	B	530	E	630	D	730	B
31	A	131	C	231	D	331	A	431	D	531	A	631	A	731	A
32	C	132	D	232	E	332	D	432	C	532	D	632	B	732	B
33	E	133	D	233	B	333	B	433	E	533	A	633	E	733	C
34	A	134	D	234	A	334	C	434	D	534	C	634	C	734	E
35	E	135	A	235	E	335	E	435	A	535	B	635	D	735	D
36	B	136	B	236	C	336	D	436	B	536	D	636	A	736	B
37	C	137	C	237	B	337	B	437	E	537	C	637	C	737	C
38	D	138	B	238	D	338	D	438	C	538	E	638	B	738	A
39	A	139	E	239	D	339	C	439	E	539	B	639	A	739	B
40	E	140	C	240	A	340	E	440	D	540	A	640	E	740	A
41	D	141	D	241	C	341	C	441	A	541	D	641	B	741	C
42	C	142	A	242	A	342	C	442	C	542	C	642	A	742	D
43	B	143	E	243	E	343	A	443	B	543	E	643	C	743	E
44	A	144	B	244	B	344	A	444	A	544	D	644	D	744	B
45	B	145	D	245	D	345	A	445	E	545	E	645	D	745	C
46	D	146	A	246	D	346	E	446	B	546	A	646	E	746	A
47	E	147	C	247	A	347	C	447	C	547	C	647	D	747	D
48	C	148	E	248	C	348	B	448	B	548	D	648	E	748	E

#	Ans	#	Ans	#	Ans	#	Ans	#	Ans	#	Ans	#	Ans	#	Ans
49	B	149	B	249	A	349	B	449	A	549	B	649	B	749	B
50	C	150	D	250	D	350	A	450	E	550	A	650	C	750	C
51	D	151	A	251	D	351	D	451	A	551	B	651	A	751	D
52	A	152	C	252	B	352	E	452	D	552	C	652	E	752	D
53	E	153	E	253	E	353	A	453	D	553	E	653	A	753	E
54	B	154	B	254	A	354	B	454	D	554	B	654	D	754	E
55	D	155	C	255	D	355	B	455	E	555	C	655	E	755	C
56	D	156	A	256	B	356	B	456	A	556	D	656	E	756	B
57	B	157	B	257	C	357	C	457	C	557	E	657	B	757	B
58	C	158	A	258	E	358	D	458	C	558	B	658	C	758	A
59	A	159	A	259	B	359	C	459	E	559	A	659	D	759	E
60	E	160	C	260	A	360	B	460	A	560	E	660	D	760	C
61	B	161	A	261	C	361	C	461	E	561	C	661	A	761	C
62	E	162	D	262	D	362	E	462	E	562	B	662	C	762	D
63	A	163	A	263	A	363	A	463	B	563	A	663	B	763	B
64	C	164	B	264	B	364	B	464	E	564	D	664	D	764	D
65	A	165	C	265	E	365	D	465	C	565	E	665	B	765	A
66	E	166	E	266	A	366	C	466	C	566	C	666	C	766	C
67	B	167	B	267	C	367	E	467	B	567	A	667	D	767	D
68	C	168	C	268	D	368	E	468	A	568	B	668	B	768	A
69	D	169	D	269	B	369	D	469	D	569	D	669	A	769	B
70	A	170	D	270	A	370	D	470	D	570	E	670	A	770	E
71	D	171	B	271	E	371	E	471	A	571	A	671	B	771	D
72	C	172	E	272	C	372	D	472	C	572	C	672	B	772	B
73	B	173	C	273	E	373	E	473	A	573	B	673	C	773	C
74	E	174	A	274	B	374	A	474	B	574	A	674	D	774	A
75	E	175	B	275	D	375	D	475	D	575	E	675	E	775	D
76	A	176	E	276	A	376	C	476	C	576	E	676	C	776	B
77	C	177	D	277	E	377	D	477	B	577	C	677	A	777	E
78	B	178	C	278	B	378	A	478	D	578	B	678	B	778	C
79	A	179	E	279	C	379	B	479	A	579	D	679	D	779	A
80	B	180	A	280	E	380	D	480	D	580	E	680	C	780	E
81	D	181	C	281	C	381	C	481	C	581	A	681	A	781	B
82	A	182	A	282	D	382	A	482	E	582	C	682	B	782	D
83	E	183	B	283	E	383	E	483	A	583	B	683	E	783	C
84	A	184	D	284	A	384	E	484	D	584	D	684	D	784	A
85	C	185	E	285	E	385	E	485	B	585	D	685	A	785	D
86	D	186	A	286	B	386	D	486	C	586	A	686	A	786	B
87	A	187	C	287	D	387	C	487	D	587	C	687	D	787	E
88	B	188	D	288	B	388	E	488	E	588	B	688	D	788	B
89	C	189	B	289	C	389	A	489	A	589	D	689	A	789	A
90	D	190	E	290	E	390	E	490	C	590	E	690	B	790	D
91	E	191	C	291	A	391	C	491	D	591	B	691	C	791	E
92	A	192	D	292	B	392	A	492	E	592	C	692	D	792	B
93	C	193	E	293	D	393	A	493	B	593	D	693	C	793	C
94	D	194	A	294	E	394	D	494	E	594	E	694	B	794	A
95	B	195	B	295	D	395	B	495	C	595	A	695	C	795	D
96	A	196	C	296	A	396	D	496	A	596	B	696	C	796	B
97	C	197	D	297	B	397	B	497	D	597	C	697	A	797	E
98	E	198	A	298	A	398	E	498	B	598	D	698	A	798	C
99	B	199	B	299	C	399	C	499	D	599	A	699	B	799	A
100	D	200	A	300	C	400	B	500	E	600	E	700	A	800	D

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